Farmer's Handbook on Basic Agriculture









A holistic perspective of scientific agriculture

A joint initiative to impart farmers with technical knowledge on basic agriculture.

german cooperation
DEUTSCHE ZUSAMMENARBEIT
giz
DESAI FRUITS & VEGETABLES
MANAGE मेनेज
develoPPP.de

Disclaimer:

The opinions expressed provided in this publication are those of the authors and do not necessarily reflect those of GIZ . The designations employed and the presentation of material in this publication do not imply the expression of any opinion whatsoever on the part of GIZ concerning the legal status of any country, territory, city or area, or concerning the delimitation of its frontiers or boundaries.

Farmer's Handbook on Basic Agriculture

Prepared & compiled by

Dr. P. Chandra Shekara
National Institute of Agricultural Extension Management (MANAGE)
Ministry of Agriculture, GoI
Hyderabad, Andhra Pradesh
India

Dr. N. Balasubramani
National Institute of Agricultural Extension Management (MANAGE)
Ministry of Agriculture, GoI
Hyderabad, Andhra Pradesh
India

Dr. Rajeev Sharma

Dr. Chitra Shukla
Desai Fruits & Vegetables Pvt. Ltd.
Navsari, Gujarat
India

Dr. Ajit Kumar
Desai Fruits & Vegetables Pvt. Ltd.
Navsari, Gujarat
India

Bakul C. Chaudhary
Desai Fruits & Vegetables Pvt. Ltd.
Navsari, Gujarat
India

Mr. Max Baumann
Planning Officer
Section "Agricultural Production & Resource Use"
Division 45 - Rural Development and Agriculture
Deutsche Gesellschaft für Internationale Zusammenarbeit (GIZ) GmbH
Germany
Max.Baumann@giz.de

Financed by

Desai Fruits & Vegetables Pvt. Ltd.
Navsari, Gujarat, India

German Federal Ministry for Economic Cooperation and Development (BMZ)

Published by

Desai Fruits & Vegetables Pvt. Ltd.
Navsari, Gujarat
India

Second Edition: August 2016

The Authors acknowledge the contribution of following experts/professionals in developing the Handbook.

Mr. Max Baumann, Planning Officer, GIZ, Germany

Mr. Fredrick Oberthur, Planning Officer, GIZ, Germany

Mr. Ajit Kumar Desai, Chairman, DFV, Navsari, Gujarat

Dr. Sashidhar, Professor, University of Agriculture and Horticultural Sciences, Shimoga, Karnataka

Dr. Syed Ahmed Hussain, Professor, ANGRAU, Hyderabad, Telangana

Mr. V.Gunasekaran,Agricultural Officer, Pesticide Testing Laboratory, Dharmapuri, Tamilnadu

Dr. M.V.Shantharam, Former Dean, ANGRAU, Hyderabad, Telangana

Dr. K. Kareemulla, Principal Scientist, NAARM, Hyderabad, Telangana

Dr. K.V. Jayaraghavendra Rao, Principal Scientist, NAARM, Hyderabad, Telangana

Mr. Thomas A Vivian, Assistant Professor, College of Agriculture, Dhule, Maharastra

Prof. T. M. Bahale, Professor of Agronomy, College of Agriculture, Dhule, Maharastra

Dr. R. K Rahane, Professor of Agricultural Economics, College of Agriculture, Dhule, Maharastra

Dr. G. D. Patil, Professor of Soil Science & Agricultural Chemistry, College of Agriculture, D hule, Maharastra

Dr. D. N. Padule, Professor of Plant Pathology, College of Agriculture, Dhule, Maharastra

Dr. A. R. Pathak, Vice Chancellor, Navsari Agricultural University, Navsari, Gujarat

Dr. J. B. Patel, Professor, Anand Agricultural University, Anand, Gujarat

Dr. Bhaskar Gaikwad, Programme, Coordinator, KVK, Babhaleshwar, Maharashtra

Mr. Mahendra Dhaibar, CEO, Sustainable Agricultural Development Foundation, Pune, Maharashtra

Dr. R. M. Pankhaniya, Associate Professor, Department of Agronomy, NM College of Agriculture, Navsari Agricultural University, Navsari, Gujarat

Dr A. M. Bafna, Principal & Dean, Aspee Agri-Business Management Institute, Navsari Agricultural University, Navsari, Gujarat

Dr G. G. Radadia, Professor & Head, Department of Entomology, NM College of Agriculture, Navsari Agricultural University, Navsari, Gujarat

Dr B. P. Mehta, Professor, Department of Plant Pathology, Aspee College of Horticulture and Forestry, Navsari Agricultural University, Navsari, Gujarat

Dr. L. J. Desai, Associate Professor, Department of Agronomy, NM College of Agriculture

Navsari Agricultural University, Navsari, Gujarat

Dr. N. S. Manohar, Associate Professor, Department of Agricultural Economics, Aspee College of Horticulture and Forestry, Navsari Agricultural University, Navsari, Gujarat

Dr O P Sharma, Associate Professor & Head, Department of Extension Education, College of Veterinary Science & AH, Navsari Agricultural University, Navsari, Gujarat

Dr. Bruno Schuler, Advisor and Planning Officer, Sustainable Agriculture-Rural Development, GIZ, Germany

Mr. Rajiv Ahuja, Technical Expert, Natural Resource Management, GIZ, India

Mr. Akhil Dev, Junior Technical Expert, Natural Resource Management, GIZ, India

Acknowledgement

Higher demand for agricultural raw material is now anticipated and agriculture is not any more about producing farm products and selling them exclusively at the local market. Instead farmers today have a world market to serve. But the new chances bring new challenges. Farmers and agricultural enterprises, willing to be part of the new expanding world market, not only have to take into consideration customers' preferences whom they want to serve, but also adhere to international trade regulations set by WTO and comply with high production and quality standards required by the importing countries.

Agriculture contributes around 17% to GDP and continues to be among the most important and successful sectors in India. Around 58% of the Indian population depend on agriculture for their livelihood. Apart from delivering the local industries with top quality raw materials for processing, agriculture provides almost 10% of total export earnings. However, to support the impressive Indian economic growth in the coming years, agriculture will have to contribute more towards value addition, productivity enhancement, high quality products and trained manpower to successfully tackle these challenges.

The states of Gujarat and Maharashtra have competitive advantages for the production of several commodities. However, productivity and competitiveness remains low. Rising quality requirements of export and domestic markets require an up-scaling of the production which is only feasible with educated farmers and skilled workers.

Desai Fruits and Vegetables (DFV)in cooperation with the Deutsche Gesellschaft für Internationale Zusammenarbeit(GIZ) GmbH on behalf of the German Federal Ministry for Economic Cooperation and Development (BMZ) takes up the existing education gap by implementing a development partnership called "Partnership Farming India".

The goal of Partnership Farming India (PFI) is to enable farmers to be self-sufficient decision-makers, "agripreneurs", which allows for a more flexible production system and highlights farming as profession by choice and not by inheritance.

Furthermore, PFI strengthens farmers' and farm workers' management skills to adopt modern agricultural practices and concepts and enhance the international competitiveness of smallholders' agricultural produce by giving farmers and workers in Gujarat and Maharashtra access to practical agricultural education. Therefore, DFV and GIZ in close cooperation with the National Institute of Agricultural Extension Management (MANAGE, an organization of Ministry of Agriculture, Government of India) developed the training material on basic agricultural knowledge and skills.

The states of Gujarat and Maharashtra will serve as an example on how to establish long term successful and trustful business relationships by combining small scale production in the field with large scale processing and marketing. I am confident that this effort will serve the Indian agriculture as a replicable model make lasting contributions towards sustainable agriculture and prosperous farmers.

I would like to express my sincere gratitude to the people and institutions namely MANAGE, DFV and GIZ,which supported this project and enabled making information available. This is a useful source of information for farmers, trainers, and other interested persons to improve not only the agriculture but also the livelihood of the farming community.

Mrs. Sabine Preuss
GIZ-India Programme Director
Natural Resource Management

Preface

Agriculture is an important sector of Indian Economy as more than half of its population relies on Agriculture as principle source of income. Research and Extension systems play major role in generation and dissemination of Agricultural technologies aiming at enhancing the income of farmers. The extension system adopts series of extension methods such as Training, demonstration, exposure visit to transfer the technologies from lab to land. Majority of these extension efforts mainly focus on location and crop specific technologies, and mostly on solution to problem basis. However, there is a need for equipping the farmers with Basic knowledge of Agriculture in order to create a better knowledge platform at farmer level for taking appropriate farm management decisions and to absorb modern technologies.

In view of this, Desai Fruits and Vegetables Pvt. Ltd. (DFV), India, in cooperation with the Deutsche Gesellschaft für Internationale Zusammenarbeit (GIZ) GmbH on behalf of the German Federal Ministry for Economic Cooperation and Development (BMZ) in close cooperation with National Institute of Agricultural Extension Management (MANAGE- An Organization of Ministry of Agriculture, Government of India) brought out Farmer's Handbook on Basic Agriculture to impart technical knowledge on Basic Agriculture to farmers to provide holistic perspective of scientific Agriculture.

This handbook is a product of series of brainstorming workshops and consultative meetings with various stakeholders such as Researchers, Academicians, Extension Functionaries, Agripreneurs, Master Trainers and Farmers. Based on the identified needs, the topics were prioritized and contents were developed with the help of experts. The farmer-friendly content has been written in simple language, using more pictures with practical examples for the benefit of farmers.

The book contains six chapters, each focusing on a particular topic. The first chapter, "General conditions for cultivation of crops", talks about the basic needs of farmers and farming sector, by providing basic knowledge on Good Agricultural Practices (GAP), enhancing the awareness of farmers on critical factors in selection of crops and cropping patterns, judicious use of natural resources such as soil and water, and emphasizing the importance of mechanization in the field of agriculture.

The second chapter "Soil and Plant Nutrition" is aimed at increasing the awareness and understanding of farmers about soil, it's structure, physical, chemical, biological properties, soil fertility and managing the soil fertility in an economically and environmentally sustainable manner. It also focuses on the need for soil testing, plant nutrition requirement, organic & inorganic fertilizers, and Integrated Nutrient Management (INM) for efficient, economic and sustainable production of crops.

The third chapter of the book is about Pest Management, and focuses on enhancing the awareness of and understanding among farmers about the crop pests, diseases and weed management through Integrated Pest Management. It also aims at sensitizing farmers on safe handling of chemicals and plant protection equipments as also elaborated further in the fifth chapter on "Occupational health and safety of farmers". It creates awareness about causes, preventive measures of health hazards, risks & fatalities in agriculture, and use of first aid in emergencies. It further includes safety tips and care to reduce the risk of injuries and fatalities while handling machineries and pesticides by farmers.

Time and resources management is an integral part of each and every activity, be it service sector, business or day-to-day activities of life. Farming sector too has not remained untouched by it. Therefore, the fourth chapter of the book is devoted to "Farm Management". It is to educate and equip the farmers to make proper plans, take appropriate decisions and also to take advantage of the improved technologies to increase production, assure food security for the family and market opportunities to increase income considering available resources, anticipated risks, including market fluctuations.

"Farmer's access to services", the last chapter of the book, aims at enhancing awareness among farmers about sources of extension, information and services, public and private extension services, agricultural credit, insurance and legal aspects through Information & Communication Technologies. The content is useful not only for farmers but also for other stakeholders involved in farm advisory services such as Agri input dealers, Agripreneurs, Kisan Call Centers and extension functionaries working at grass roots level.

We trust that this Handbook will benefit maximum number of farmers to make farming economically and environmentally more sustainable.

B. Srinivas, IAS
Director General
MANAGE

Contents

1.	General Conditions for Cultivation of Crops	1-32
2.	Soil and Plant Nutrition	33-70
3.	Plant Protection	71-96
4.	Categories of Pesticides and Precautions	97-98
5.	Farm Management	99-120
6.	Occupational Health and Safety	121-130
7.	Farmer's Access to Service	131-136

1. General Conditions for Cultivation of Crops

1.1. Objectives of the session

- To enhance awareness of farmers on critical factors in selection of crops and cropping patterns.
- To create an understanding on judicious use of natural resources such as soil and water.
- To provide basic knowledge on seed and cropping systems.
- To emphasize the importance of mechanization.
- To sensitize the farmers on Good Agricultural Practices (GAP).

1.2. What do we know at the end of the session

- Critical factors in selection of crops and cropping patterns
- Judicious use of natural resources such as soil and water
- Basic knowledge on seed
- Cropping systems
- Mechanization
- Good Agricultural Practices (GAP)

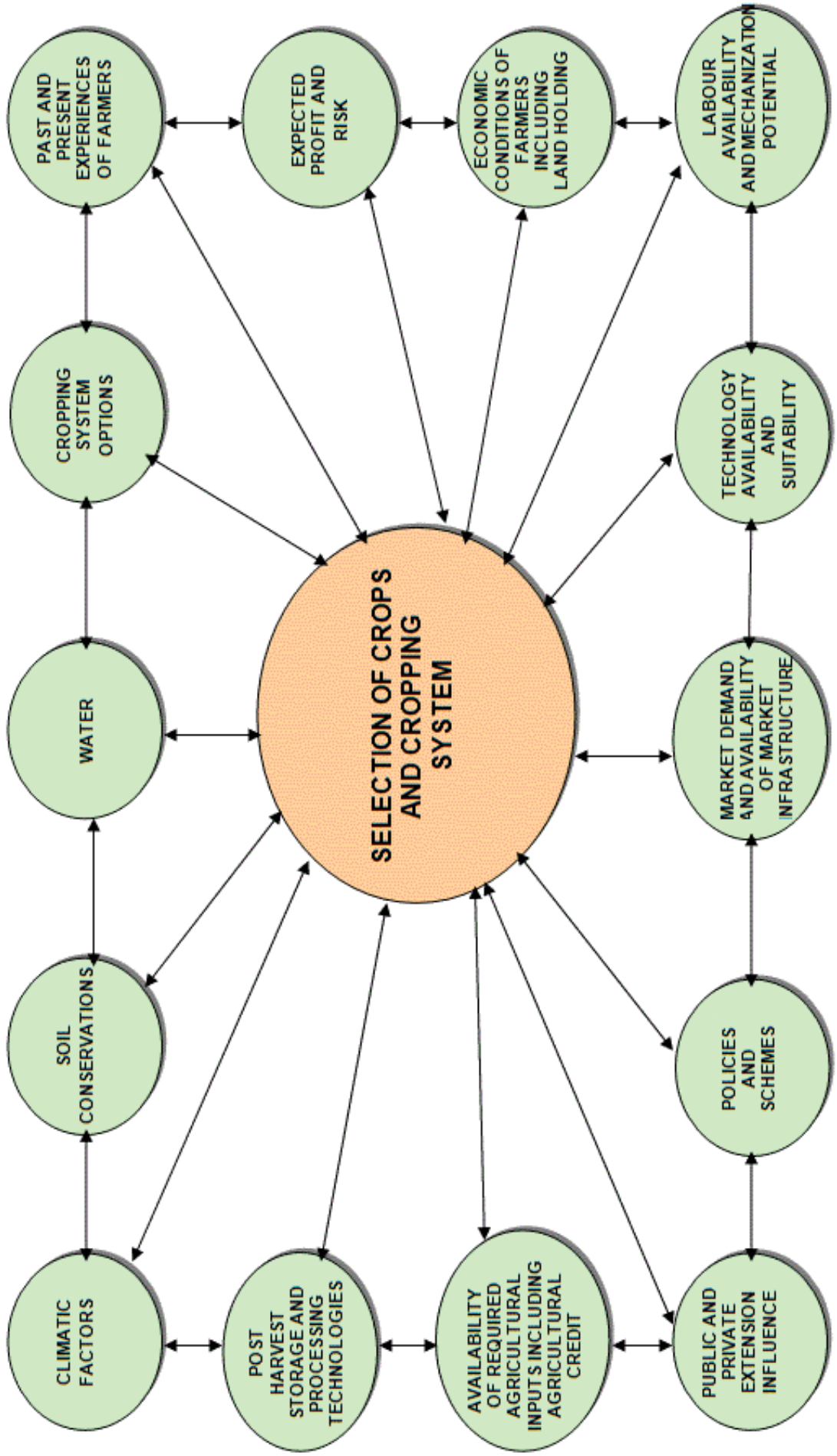
Agricultural Universities, Research Institutes, Krishi Vigyan Kendras have been generating ample technologies to improve the productivity and profitability of the farmers. How many of these technologies are reaching the farmers? Baseline Situation Assessment conducted by Partnership farming in India, in Gujarat and Maharashtra, clearly indicated that farmers with access to technical knowledge on agriculture realized better income compared to others. Fifty one percent of sample farmers who were part of partnership farming India had knowledge of soil testing compared to only 28% of control group. Mulching and intercropping as a practice were not widely adopted by control group of farmers. They were less aware about other organic fertilizers. Fertigation as a method of application of fertilizers was not widely practiced by control group. Farmers who accessed information from agricultural universities and magazines were less in number in the control group. The treatment farmers had an average yield of 35.65 tons and control farmers had yield of 22.36 tons of banana per acre. The average net income of the treatment farmers was Rs 93,822 and for the control farmers Rs.81,659. More than 85% of the farmers wanted basic education on agriculture and crop production and ready to pay for undergoing such basic education and training. There was a clear interest in the farmers to improve their skill and knowledge and they were ready to pay for the service.

The above study clearly indicates that the knowledge gap is prevailing among farmers and those who have access to knowledge harvested better profits.

Increase in productivity and profitability can be achieved through:

- Blending practical knowledge with scientific technologies
- Efficient use of natural resources
- Adopting time specific management practices
- Giving priority for quality driven production
- Adopting suitable farming systems
- Adoption of location specific technology
- Market demand driven production
- Adopting low cost and no cost technologies

Factors influencing decisions on the selection of crops and cropping system



1.3. Factors influencing decisions on the selection of crops and cropping system

Climatic factors

Is the crop/cropping system suitable for local weather parameters such as temperature, rainfall, sun shine hours, relative humidity, wind velocity, wind direction, seasons and agro-ecological situations?

Soil conditions

Is the crop/cropping system suitable for local soil type, pH and soil fertility?

Water

- Do you have adequate water source like a tanks, wells, dams, etc.?
- Do you receive adequate rainfall?
- Is the distribution of rainfall suitable to grow identified crops?
- Is the water quality suitable?
- Is electricity available for lifting the water?
- Do you have pump sets, micro irrigation systems?

Cropping system options

- Do you have the opportunity to go for intercropping, mixed cropping, multi-storeyed cropping, relay cropping, crop rotation, etc.?
- Do you have the knowledge on cropping systems management?

Past and present experiences of farmers

- What were your previous experiences with regard to the crop/cropping systems that you are planning to choose?
- What is the opinion of your friends, relatives and neighbours on proposed crop/cropping systems?

Expected profit and risk

- How much profit are you expecting from the proposed crop/cropping system?
- Whether this profit is better than the existing crop/cropping system?
- What are the risks you are anticipating in the proposed crop/cropping system?
- Do you have the solution? Can you manage the risks?
- Is it worth to take the risks for anticipated profits?

Economic conditions of farmers including land holding

- Are the proposed crop/cropping systems suitable for your size of land holding?
- Are your financial resources adequate to manage the proposed crop/cropping system?
- If not, can you mobilize financial resources through alternative routes?

Labour availability and mechanization potential

- Can you manage the proposed crop/cropping system through your family labour?
- If not, do you have adequate labours to manage the same?
- Is family/hired labour equipped to handle the proposed crop/cropping system?
- Are there any mechanization options to substitute the labour?
- Is machinery available? Affordable? Cost effective?
- Is family/hired labour equipped to handle the machinery?

Technology availability and suitability

- Is the proposed crop/cropping system suitable?
- Do you have technologies for the proposed crop/cropping system?
- Do you have extension access to get the technologies?
- Are technologies economically feasible and technically viable?
- Are technologies complex or user-friendly?

Market demand and availability of market infrastructure

- Are the crops proposed in market demand?
- Do you have market infrastructure to sell your produce?
- Do you have organized marketing system to reduce the intermediaries?
- Do you have answers for questions such as

where to sell? When to sell? Whom to sell to? What form to sell in? What price to sell for?
- Do you get real time market information and market intelligence on proposed crops?

Policies and schemes

- Do Government policies favour your crops?
- Is there any existing scheme which incentivises your crop?
- Are you eligible to avail those benefits?

Public and private extension influence

Do you have access to Agricultural Technology Management Agency (ATMA)/ Departmental extension functionaries to get advisory?

- Do you know Kissan Call Center?
- Do you have access to KVKs, Agricultural Universities and ICAR organizations?
- Do you subscribe agricultural magazines?
- Do you read agricultural articles in newspapers?
- Do you get any support from input dealers, Agribusiness Companies, NGOs, Agriclinics and Agribusiness Centers?

Availability of required agricultural inputs including agricultural credit

- Do you get adequate agricultural inputs such as seeds, fertilizers, pesticides, and implements in time?
- Do you have access to institutional credit?

Post harvest storage and processing technologies

- Do you have your own storage facility?
- If not, do you have access to such facility?
- Do you have access to primary processing facility?
- Do you know technologies for value addition of your crop?
- Do you have market linkage for value added products?
- Are you aware about required quality standards of value added products of proposed crops?

Farmers need to answer all the above questions while making decisions for choosing a crop/ cropping pattern. During this decision making process, farmer cross check the suitability of proposed crop/cropping systems with his existing resources and other conditions. Thereby, they justify choosing or rejecting a crop/cropping systems. This process enables the farmers to undertake a SWOT analysis internally which in turn guides them to take an appropriate decision.

1.4. Climatic factors

Climate and agriculture

- Monsoon is a key source of water in agriculture
- Most of our rivers are seasonal fed by the monsoon; even irrigated agriculture depends on monsoon.
- Cropping pattern has evolved over years based on climate.
- Market forces influence cropping patterns in recent times.

Climatic factors and crops

- Rainfall drives water availability and determines sowing time (rainfed crops).
- Temperature drives crop growth, duration and influences milk production in animals.
- Temperature and relative humidity influence pest and diseases incidence on crops, livestock and poultry.
- Wet and dry spells cause significant impact on standing crops, physiology, loss of economic products (e.g. fruit drop).
- Extreme events (e.g. high rainfall, floods, heat / cold wave, cyclone, hail, frost) cause enormous losses of standing crops, livestock and fisheries.

Climate and seasons

- Rainy (June-September) season also known as Kharif, supports most of the rainfed crops (coarse cereals, pulses, oilseeds, etc.).
- Post-rainy (October-February) season also known as Rabi, supports the irrigated or stored moisture grown crops (wheat, mustard, chickpea, etc.).
- Summer season (March-May) supports short duration pulses and vegetables.
- Rabi production is more assured, has a higher yield and reduces pest and disease related problems.
- Over time, with irrigation development, the contribution of Kharif is declining and Rabi is increasing.

Climate, cropping pattern and agricultural production issues

- Cropping patterns based on climate and land capability are sustainable but market forces and farmers' aspirations are forcing unsustainable systems.
- Farmers must innovate in producing more even from less endowed areas by adopting suitable technologies to cope with changing climate.
- Climate change will likely to cause further problems in our crop production and is likely to become the most important environmental issue in the 21st century.

Important agricultural related factors responsible for climate change

- Deforestation and forest degradation
- Burning of fuel and farm waste
- Water logged condition
- Excessive use of external input
- Large-scale conversion of land for non-agricultural purpose

Impact of climate change in India

- **Rainfall:** No long-term trend noted. However, regional variations seen, increased summer rainfall and less number of rainy days.
- **Temperature**: About 0.6 °C rise in surface temperature during 100 years. Projected to increase 3.5 to 5 °C by 2100.
- **Carbon dioxide:** Increasing at the rate of 1.9 ppm per year and expected to reach 550 ppm by 2050 and 700 ppm by 2100.
- **Extreme events:** Increased frequency of heat wave, cold wave, droughts and floods observed during last decade.
- **Rising sea level:** Rise of 2.5 mm/year since 1950.
- **Glaciers:** Rapid melting of the glaciers in the Himalayas.
- **Rainfall distribution:** Shift in peak rainfall distribution also noticed in some parts of country.

Expected impact of climate change on agriculture

- Due to increase in temperature, crop may require more water.
- Yield may be reduced in cereal crops especially in Rabi; i.e. wheat.



Impact of Drought



Impact of Flood



Heat Wave on Maize



Cold wave damage to chana harvest

Change in pest and disease scenario due to climate change

- **Due to increase in rainfall:** Pests like bollworm, red hairy caterpillar and leaf spot diseases may increase.Due to increase in temperature: Sucking pests such as mites and leaf miner may increase.
- **Due to variation in rainfall and temperature:** Pest and diseases of crops to be altered because of more enhanced pathogen and vector development, rapid pathogen transmission and increased host susceptibility. Sometimes a minor pest may become a major pest.
- Agricultural biodiversity is also threatened by decreased rainfall and increased temperature, sea level rise and increased frequency and severity of drought, cyclone and flood. Quality of farm products such as fruits, vegetables, tea, coffee, aromatic and medicinal plants may be affected.

Water

- Demand for irrigation to increase with increased temperature and higher amount of evapo-transpiration. This may result in lowering of groundwater table at some places.
- The melting of glaciers in the Himalayas will increase water availability in the Ganga, Brahmaputra and their tributaries in the short-run but in the long-run the availability of water will decrease considerably.
- A significant increase in runoff is projected in the rainy season, however, may not be very beneficial unless storage infrastructure could be vastly expanded. This extra water in the rainy season, on the other hand, may lead to increase in frequency and duration of floods.
- The water balance in different parts of India will be disturbed and the quality of ground water along costal track will be more affected due to intrusion of sea water.

Soil

- Organic matter content, which is already quite low in Indian soil, would become even lower. Quality of soil organic matter may be affected.
- Reduction in rate of decomposition and nutrient supply.
- Increase in soil temperature may reduce Nitrogen availability due to volatilization and denitrification.
- Change in rainfall volume and frequency as well as wind may alter the severity, frequency and extent of soil erosion.
- Rise in sea level may lead to salt water entry in the coastal lands turning them less suitable for conventional agriculture.

Livestock

- Affect feed production and nutrition of livestock. Increased temperature would reduce digestibility. Increased water scarcity would also decrease the food and fodder production.
- Major impacts on vector-borne diseases through expansion of vector populations during rainy years, leading to large outbreaks of diseases.
- Increase water, shelter, and energy requirement of livestock for meeting projected milk demands.
- Climate change is likely to aggravate heat stress in dairy animals, adversely affecting their reproductive performance.

Fishery

- Increased sea and river water temperature is likely to affect fish breeding, migration and harvest.
- Impacts of increased temperature and tropical cyclonic activity would affect capture, production and marketing costs of the marine fish.

Coping options for farmers

Access to information

- Progressive Farmers
- ATMA extension functionaries – Block Technology Manager, SMS, farmer friend, Farm School
- Trained input dealers
- Agri Clinics and Agribusiness Centers
- KVK
- Agricultural Research Stations
- Agricultural Universities
- ICAR Organisations
- Kissan Call Centers (Toll free no.1551 or 1800 – 180 – 1551)
- Concerned NGOs
- Agribusiness Companies
- Radio, TV, Agricultural Magzines, Community Radio, Newspapers, Agricultural Websites etc.

Coping options for farmers



Enlarging the Food Basket

- Diversifying the livelihood sources.
- Changing cropping patterns.
- Increased traditional coping strategies.
- Change to a mixed cropping pattern.

E.g: Crop Mixture-Nutri Millets, Pulses and Oilseed

Integrated Farming System

- Increased share of non-agricultural activities

  E.g: **Type of Integrated Farming Systems**

  Agriculture +vegetable cultivation

  Agriculture + animal husbandry





Neem, Mulberry & Cowpea

- Planting more drought tolerant crops and increased agro-forestry practices.
- Agro-forestry systems to provide more stable incomes during years of extreme weather events.

Mango, Pumpkin, maize mixed cropping





Mixed farming/Multi level farming

Coping options for farmers continued....



Lucerne & Sunhemp for green manuring & fodder

Farm Pond





Conservation Furrow

- Improved on-farm soil & water conservation.
- Adopting scientific water management, nutrient management and cultural practices.

Vegetative Barriers





Percolation Tanks

Coping options for farmers continued....



Contour trenching for runoff collection

Conventional Raised Bed Planting

- 20-25% Saving in irrigation water





Shelterbelts

- Shelterbelts reduce wind velocity.
- Moderate temperature.
- Reduce evaporative loss and conserve soil moisture.

Straw Thatching

- Protecting young seedlings against cold by covering with straw thatching.





Frost Protection

1.5. Soil and Water Conservation

Soil and water are our precious heritage. Hence, it is obligatory on our part to protect and hand over these resources to further generations. It is estimated that about 50% of the cultivated area in India suffers from severe soil erosion and requires remedial measures.

- Water resources are essential for increasing and stabilizing crop production.
- Wind erosion has been responsible for destroying the valuable top soil.

1.5.1. Degradation of soil and water takes place with water and wind erosion

- The main cause of water erosion is unmanaged runoff.
- Runoff is the portion of the rainfall or irrigation water applied which leaves a field either as surface or as subsurface flow.

Several factors are responsible for runoff

- **Climatic factors:** Precipitation characteristics - duration, intensity, distribution, direction, temperature, humidity, wind velocity.
- **Watershed characteristics:** Geological shape of the catchments, size and shape of the catchments, topography, drainage pattern.
- **Barren land without vegetation**
- **Soil types:**
  - **Sandy soil:** Average rain – no problem of erosion. High intensity – More serious of less binding material i.e. fine soil particle.
  - **Clay soil:** Ordinary rain – more runoff in moderate and steep slopes but high water holding capacity.
  - **Silt loam, loamy and fine sandy loam:** More desirable soils from the point of view of minimizing soil erosion.

How vegetation reduces runoff

- Interception of rainfall
- Root structure
- Biological influences
- Transpiration effects
- Intercept, absorb the impact of raindrop
- Hindrance to runoff water slows down the rate at which travels down the slope
- Knitting and binding effect aggregates the soil into granules
- Die and decay increase pore space and water holding capacity
- One cubic meter of soil has several kilometres of root fibre
- More vegetative cover, most active soil fauna, channels of earth worm, beetles and other life
- Vegetation increases the storage capacity of the soil for rainfall by the transpiration of large quantities of moistures from the soil

Soil erosion

Soil erosion is the detachment and transportation of soil material from one place to another through the action of wind, water in motion or by the hitting action of the rain drops.

- When the vegetation is removed and land is put under cultivation the natural equilibrium between soil building and soil removal is disturbed.
- The removal of surface soil takes place at a much faster rate than it can be built up by the soil forming process.

Erosion by water: Known as water erosion, is the removal of soil from the lands surface by water in motion.

Sheet erosion: The removal of a thin relatively uniform layer of soil particles by the action of rainfall and runoff.

- Extremely harmful
- Usually so slow that the farmer is not conscious of its existence
- Common on lands having a gentle uniform slope
- Results in the uniform removal of the cream of the top soil with every heavy rain
- Shallow top soil overlies a tight sub soil are most susceptible to sheet erosion
- Movement of soil by rain drop splash is the primary cause of sheet erosion
- Sheet erosion has damaged millions of hectares of slopping land throughout the India

Rill erosion is the removal of soil by running water with the formation of shallow channels that can be smoothed out completely by normal cultivation.

- There is no sharp lines of demarcation where sheet erosion and rill erosion begins but rill erosion is more readily apparent than sheet erosion.



Sheet and Rill Erosion



Gully Erosion



Landslide



Shelterbelts for Moderating microclimate

- Rills develop when there is a concentration of runoff water which, if neglected, grow into large gullies.
- More serious in soils having a loose shallow top soil.
- Transition stage between sheet erosion and gullying.

Gully erosion: Removal of soil by running water with the formation of channels that cannot be smoothed out completely by cultivation.

- Advance stage of rill erosion.
- Any concentration of surface runoff is a potential source of gullying.
- Cattle paths, cart tracks, dead furrows, tillage furrows or other small depression down a slope favour concentration of flow.
- Unattended rills deepen and widen every year and begin to attain the form of gullies.
- Unattended gullies may result over a few years for an entire landscape to be filled with a network of gullies.
- More spectacular than other type of erosions.

Stream channel erosion: Erosion caused by stream flow.

- Closely resembles rill erosion.
- Intensive channel erosion areas are on the outside of lands where flow shear stresses are high.

Mass movement: Enmass movement of soil.

- Landslides, land slips, soil and mudflows are various forms of mass movement.

Wind erosion: Movement of soil particles is caused by wind force exerted against or parallel to surface of the ground.

1.5.2. Conservation

Conservation is the utilization without wastage of resources is required to ensure a high level of production.

Important soil conservation measures are

- Conservation Tillage
- Minimum tillage
- Zero tillage
- Stubble mulching
- Trash farming

Conservation farming

- Farming across the slope
- Strip cropping
- Rotations
- Mixed cropping and intercropping
- Surface mulching
- Timely farm operations
- Improved water user efficiency
- Land levelling
- Providing safe drainage
- Intermittent terraces
- Growing vegetation on the bunds

Vegetation and vegetative management

- Strip cropping
- Stubble mulching
- Mulching

Wind erosion management

- Protect the soil surface with a cover of vegetation or vegetative residues.
- Produce or bring to the surface soil aggregates or clods which are large enough to resist the wind force.
- Roughen the land surface to reduce wind velocity and trap drifting soil.
- Establish barriers or trap strips at intervals to reduce wind velocity and soil drifting.

Best practices to control soil blowing

- Deep ploughing
- Summer ploughing
- Surface roughness
- Conserving moisture
- Wind breaks and shelterbelts
- Mechanical or vegetative barriers

For instance: Shelterbelts for moderating microclimate

- Shelterbelts reduce wind velocity
- Moderate temperature
- Reduce evaporative loss and conserve soil moisture

Water erosion can be managed by

- In situ water harvesting
- Summer ploughing

Overland flow management

- Contour bund
- Graded bund
- Broad based bund
- Bench terrace
- Water harvesting and recycling

Zero tillage

- Several practices are in use such as zero tillage, minimum tillage and direct seeding.
- Planting crops in previously untilled soil by opening a narrow slot, trench or band only of sufficient width and depth to obtain seed coverage. No other soil tillage is done.

Advantages of zero tillage farming

- Erosion control: Retained stubble and crop residue reduces soil erosion and enhances soil fertility
- Moisture conservation: Stubble traps water, reduce runoff water, better infiltration leading to improved soil moisture condition
- Higher nitrogen availability
- Seedling protection: Stubbles protects young seedling from wind and heat
- Crop yields will be on par with traditional tillage system. However good yield can be harvested during dry years
- Reduce labour and save time
- Savings on equipment cost
- Savings on oil/fuel cost

Mulching: Benefits of crop residue mulching are

- Increased availability of water and organic matter
- Less erosion
- Environment protection

Additional benefit to farmers

- Less drought susceptibility
- Improved soil quality and fertilizer efficiency
- Minimises long term dependency on external inputs

1.6. Irrigation

An adequate water supply is important for plant growth. When rainfall is not sufficient, the plants must receive additional water from irrigation.

Points consider for irrigation decisions

- Land suitability for irrigation like slope
- Effective rainfall: Part of the total rain is useful for crop production
- When to irrigate: Decide based on soil, crop and climatic condition
- How much to irrigate: Decide based on crop water requirement
- How to irrigate: Select appropriate method for irrigation
- Quality of irrigation water

1.6.1. Various methods can be used to supply irrigation water to the plants

- Surface irrigation:
  - Basin irrigation
  - Furrow irrigation
- Sprinkler irrigation
- Drip irrigation

Surface Irrigation

Surface irrigation is the application of water by gravity flow to the surface of the field.

- Either the entire field is flooded (Basin Irrigation) or the water is fed into small channels (furrows) or strips of land (borders).

Basin Irrigation

- Basins are flat areas of land, surrounded by low bunds.
- The bunds prevent the water from flowing to the adjacent fields.
- Basin irrigation is commonly used for rice grown on flat lands or in terraces on hillsides. Paddy grows best when its roots are submerged in water. Hence, basin irrigation is the best method to use for this kind of crop.
- Trees can also be grown in basins, where one tree is usually located in the middle of a small basin.
- In general, the basin method is suitable for crops that are no affected by standing in water for longer periods.
- Basin irrigation is suitable for many field crops.
- Crops suitable for basin irrigation include pastures, citrus, banana and crops that are broadcasted such as cereals and to some extent row crops such as tobacco.
- Basin irrigation is generally not suited to crops, which cannot stand in wet or waterlogged conditions for periods longer than 24 hours; eg: potatoes, beet root and carrots
- The flatter the land surface, the easier it is to construct basins.
- It is also possible to construct basins on sloping land, even when the slope is quite steep. Level basins, called terraces, can be constructed like the steps of a staircase.
- Soils suitable for basin irrigation depend on the crop grown.

Basin should be small if the:

- Slope of the land is steep
- Soil is sandy
- Stream size to the basin is small
- Required depth of the irrigation application is small
- Field preparation is done by hand or animal power

Basin can be large if the:

- Slope of the land is gentle or flat
- Soil is clay
- Stream size to the basin is large
- Required depth of the irrigation application is large
- Field preparation is mechanized
- The land slope, the soil type, the available stream size, the required depth of the irrigation application and farming practices mainly determine the shape and size of basins

- If the land slope is steep, the basin should be narrow; otherwise too much earth movement will be needed to obtain level basins.
- Three other factors, which may affect basin width, are depth of fertile soil, method of basin construction, agricultural practices.
- **There are two methods to supply irrigation water to basins:** (i) The direct method: Irrigation water is led directly from the field channel into the basin through siphons,

or bund breaks. (ii) The cascade method: irrigation water is supplied to the highest terrace, and then allowed to flow to a lower terrace and so on.

Maintenance of basins

- Bunds are susceptible to erosion. This may be caused by, for example, rainfall, flood or the passing of people when used as foot-paths.
- Rats may dig holes in the sides of the bunds.
- Therefore, it is important to check the bunds regularly, notice defects and repair them instantly, before greater damage is done.

Advantages of basin irrigation

- Conservation of rainfall and reduction in soil erosion.
- High water application and distribution efficiencies.
- Useful in leaching of salts.
- Suitable to all close growing crops, row crops and orchards.



Basin Irrigation

Furrow irrigation

- Furrows are small channels, which carry water down the land slope between the crop rows.
- Water infiltrates into the soil as it moves along the slope.
- The crop is usually grown on the ridges between the furrows.
- This method is suitable for all row crops and for crops that cannot stand in water for long periods. Crops such as maize, sunflower, sugarcane, and soybean can be irrigated by furrow irrigation.
- Crops that would be damaged by inundation, such as tomatoes, vegetables, potatoes, beans; fruit trees like citrus and grape as well as broadcasted crops like wheat.
- Irrigation water flows from the field channel into the furrows by opening up the bank of the channel or by means of siphons or spiles.
- Furrows must be on consonance with the slope, soil type, stream size, irrigation depth, cultivation practice and field length.
- Uniform flat or gentle slopes are preferred for furrow irrigation.
- On undulating land, furrows should follow the land contours.

Advantages of furrow irrigation

- Suitable for row crops and vegetables.
- Suitable for soils in which the infiltration rates vary between 0.5 and 2.5 cm/hr.
- Ideal for slopes varying from 0.2 to 0.5 per cent and a stream size of 1-2 liters/sec.
- In areas requiring surface drainage or prone to temporary water logging, furrows are very effective.
- In areas where water for irrigation purposes. is scarce, the practice of alternate or skip furrow irrigation can save considerable quantity of water without significantly affecting yields.



Furrow Irrigation

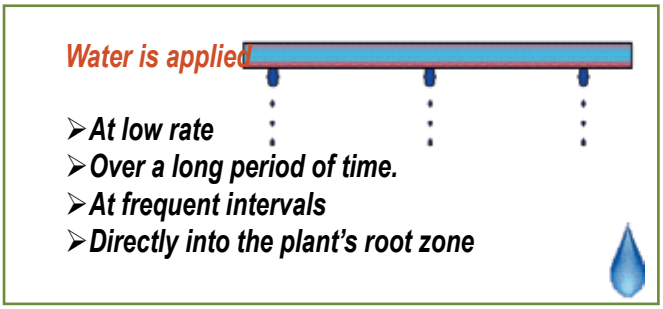
Sprinkler irrigation

Water is pumped through a pipe system and then sprayed onto the crops through sprinkler heads.

Advantages

- Water conservation
- Soil conservation
- Efficient use of water
- Saving of labour
- Early seed germination
- Fertigations
- Soil amendments
- Frost protection

- Cooling of crops
- Higher pro+ductivity of crops



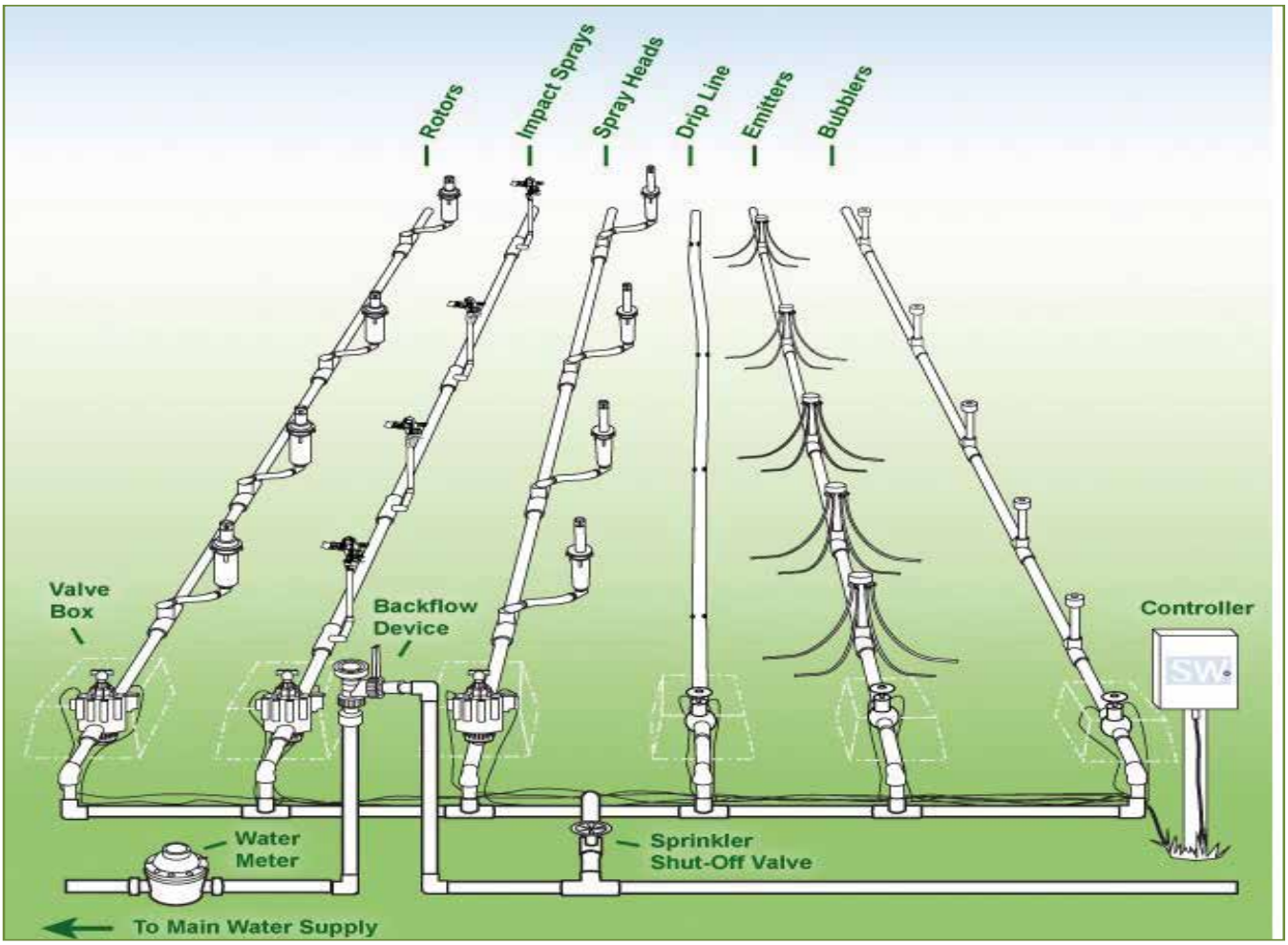
Drip Irrigation

Use of Sprinklers for different crops

Crop Type	Crop Example
Cereals	Maize, Sorghum, Wheat, Jowar
Flowers	Carnation, Jasmine, Marigold
Oilseeds	Groundnut, Mustard, Sunflower
Vegetables	Onion, Potato, Radish, Carrot
Fodders	Asparagus, Pastures
Pulses	Gram, Pigeon pea, Beans
Plantation	Coffee, Rubber, Tamarind
Fibre	Cotton, Sesame
Spices	Cardamom

Response of different crops to sprinkler irrigation

Crop	Water saving (%)	Yield increase (%)
Bajra	56	19
Barley	56	16
Bhendi	28	23
Cabbage	40	3
Cauliflower	35	12
Chillies	33	24
Cotton	36	50
Cowpea	19	3
Fenugreek	29	25
Garlic	28	6
Gram	69	57
Groundnut	20	40
Jowar	55	34
Lucerne	16	27
Maize	41	36
Onion	33	23
Potato	46	4
Sunflower	33	20
Wheat	35	24



Lay out of Sprinkler Irrigation System

Drip irrigation

Water is conveyed under pressure through a pipe system to the fields, from where it is discharged slowly or at a pre designed rate. The latter can be matched to the soil infiltration capacity through emitters or drippers that are located close to the root zone of the plants.

A typical drip irrigation system consists of the following components:

- Pump unit
- Control unit
- Filtering unit
- Mainline and sub mainlines
- Laterals
- Emitters



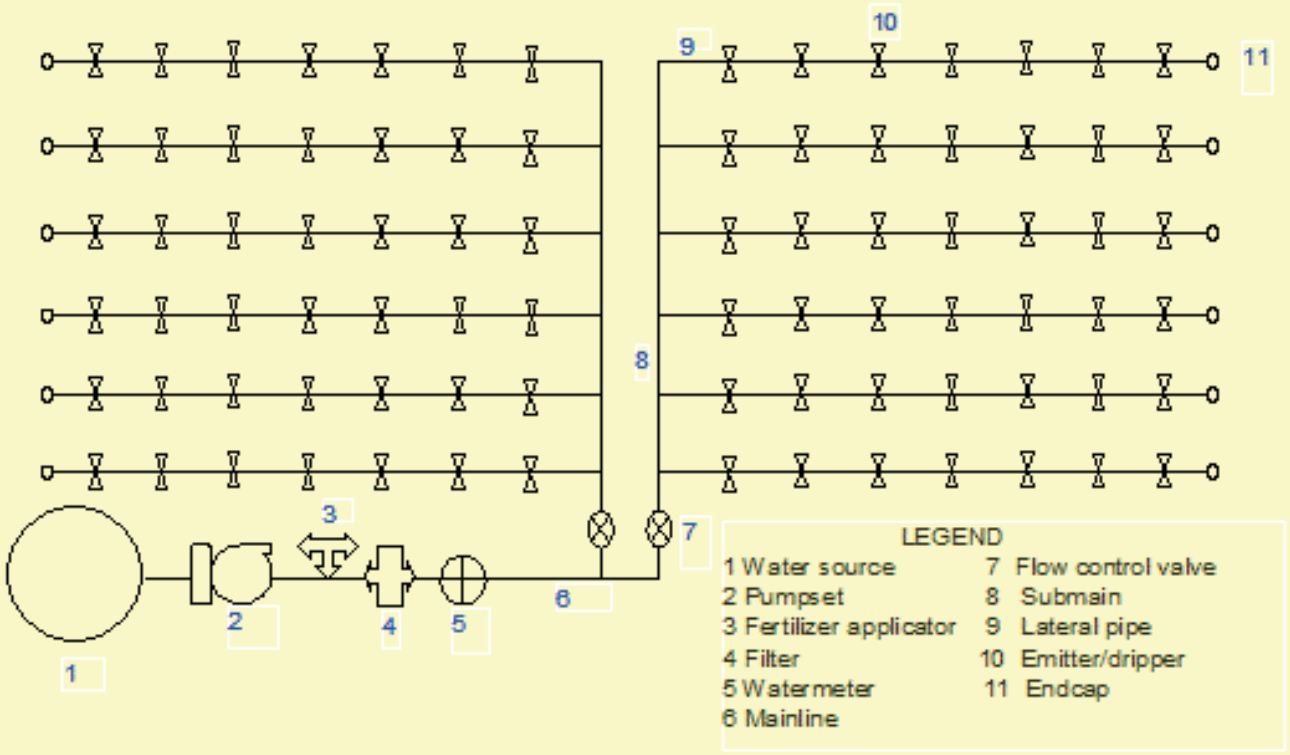
Head Control Unit

Critical stages for irrigation in different crops

Name of the Crop	Critical Stages
Cereals	
Rice/Paddy	Tillering, Panicle Initiation, Heading and Flowering
Wheat	Crown Root Initiation, Tillering to Booting
Sorghum	Booting, Blooming and Milky Dough Stage
Maize	Silking and Tasseling to Dough Stage
Pearl millet	Heading and Flowering
Finger millet	Primordial Initiation and Flowering
Pulses	
Chickpea	Late Vegetative Phage
Black gram	Flowering and Pod Setting
Green gram	Flowering and Pod Setting
Beans	Flowering and Pod Setting
Peas	Flowering and Early Pod Formation
Alfalfa	After Cutting and Flowering

Name of the Crop	Critical Stages
Oil Seeds	
Ground nut	Flowering, Peg Formation and Pod Development
Sesame	Blooming to Maturity
Sunflower	Pre-flowering to Post-flowering
Soybean	Blooming and Seed Formation
Vegetables	
Onion	Bulb Formation and Pre-maturity
Tomato	Flowering and Fruit Setting
Chilies	Flowering and Fruit Setting
Cabbage	Head Formation
Potato	Tuber Initiation to Maturity
Carrot	Root Enlargement
Others	
Cotton	Flowering and Boll Formation
Citrus	Flowering, Fruit Setting and Fruit Enlargement
Mango	Pre-flowering and Fruit Setting

Layout of micro irrigation system



Benefits of drip irrigation over surface irrigation

Crop	Yield increase (%)	Water saving (%)
Mango	80.0	34.8
Banana	52.0	45.0
Grapevine	23.0	48.0

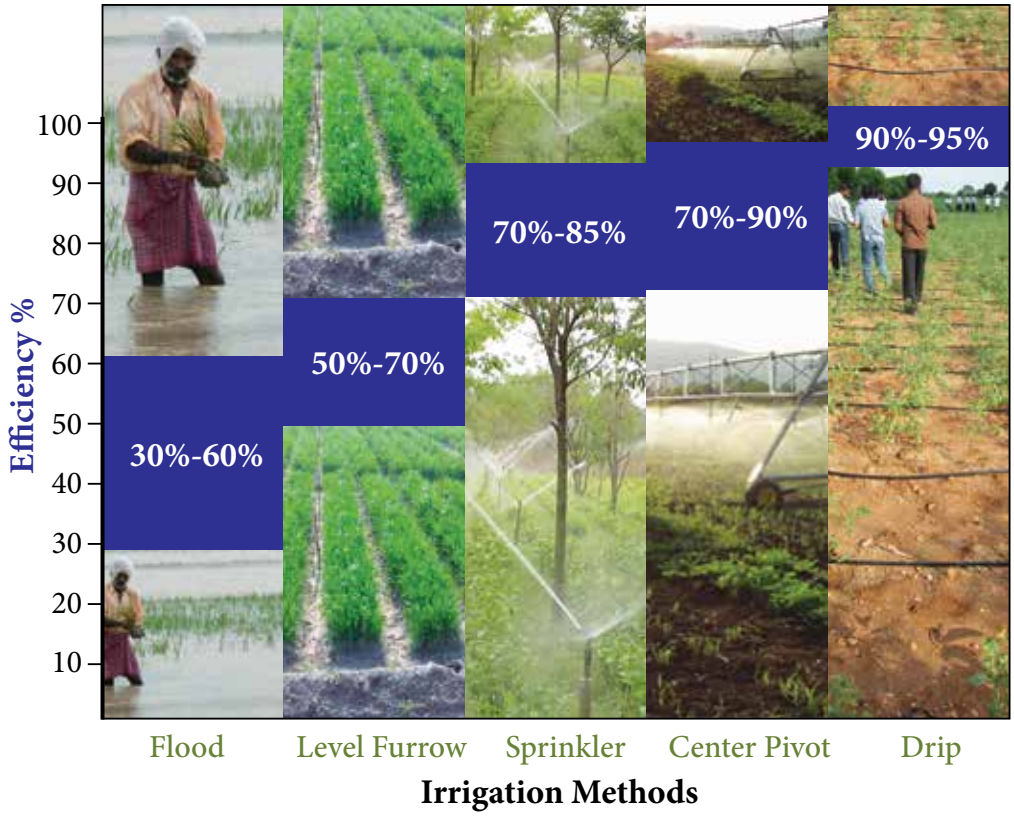
Crop	Yield increase (%)	Water saving (%)
Pomegranate	98.0	45.0
Tomato	50.0	39.0
Watermelon	88.0	36.0

Benefits of drip irrigation over surface irrigation continued....

Crop	Yield increase (%)	Water saving (%)
Lady's finger	16.0	40.0
Brinjal	14.0	53.0
Chillies	44.0	62.0
Papaya	75.0	68.0

Crop	Yield increase (%)	Water saving (%)
Sugarcane	133.3	49.3
Cotton	88.0	46.6
Onion	53.8	46.1
Potato	79.5	54.1

On-farm irrigation efficiency of different irrigation methods



1.6.2. Centrally sponsored micro irrigation scheme

It is clear from the above diagram that drip irrigation is the most efficient irrigation in terms of water use efficiency compared to all other methods. Flood irrigation method is found to be the most uneconomical irrigation method in terms of water use efficiency when compared to all other methods.

In order to popularize micro irrigation, the Govt. of India is implementing the Micro Irrigation Scheme through which interested farmers be supported. The farmers can approach nearest extension functionary. The details are as follows:

Name of Scheme — Micro Irrigation

Type — Centrally Sponsored Scheme (CSS)

Year of Commencement — 2005-06

Objectives — To increase the area under efficient methods of irrigation viz. drip and sprinkler irrigation as these methods have been recognized as the only alternative for efficient use of surface as well as ground water resources.

Salient Features

- Out of the total cost of the Micro Irrigation (MI) System, 40% will be borne by the Central Government, 10% by the State Government and the remaining 50% will be borne by the beneficiary either through his/her own resources or soft loan from financial institutions.
- Assistance to farmers will be for covering a maximum area of 5 hectare per beneficiary family.
- Assistance for drip and sprinkler demonstration will be 75% of the cost for a maximum area of 0.5 ha per beneficiary, which will be met entirely by the Central Government.
- The Panchayati Raj Institutions (PRIs) will be involved in selecting the beneficiaries.
- All categories of farmers are covered under the Scheme. However, it needs to be ensured that at least 25% of the beneficiaries are small and marginal farmers.
- The scheme includes both drip and sprinkler irrigation. However, sprinkler irrigation will be applicable only for those crops where drip irrigation is uneconomical.
- There will be a strong HRD input for the farmers, field functionaries and other stakeholders at different levels.
- Moreover, there will be publicity campaigns, seminars/workshops at extensive locations to develop skills and improve awareness among farmers about importance of water conservation and management.
- The Precision Farming Development Centres (PFDCs) will provide research and technical support for implementing the scheme.
- Supply of good quality system both for drip and sprinkler irrigation having BIS marking, proper after sales services to the satisfaction of the farmer is paramount.

Subsidy Pattern: Assistance is provided @ 50% (40% by the Government of India and 10% by the State Government) for drip/sprinkler Irrigation System. Assistance to the extent of 75% of the cost of demonstration is provided up to a limit of 0.5 ha.

Structure of Scheme

- At the National level, National Committee on Plasticulture Application in Horticulture (NCPAH) will be responsible for coordinating the Scheme, while the Executive Committee of NCPAH will approve the Action Plan. At the State level the State Micro Irrigation Committee will coordinate the programme, while at the District level the District Micro Irrigation Committee will oversee the programme.
- The Scheme will be implemented by an Implementing Agency (IA), appointed by the State Government, which will be the District Rural Development Agencies (DRDAs) or any identified Agency, to whom funds will be released to directly on the basis of approved district plans for each year.
- The IA shall prepare Annual Action Plan for the District which will be forwarded by the DMIC and SMIC for approval by the Executive Committee (EC) of NCPAH.

Funding Pattern

80:20 by the Centre and States

Eligibility As indicated in column 5 above.

Area of Operation The focus will be on horticultural crops being covered under the National Horticulture Mission in 24 States/UTs. A cluster approach will be adopted. The focus has also been extended to non horticultural crops.

Procedure to Apply Project proposals are submitted through the State Government for release of assistance.

1.6.3. Drainage

Drainage it is a removal of water from the field as a moisture control mechanism.

- Drainage and irrigation are important aspects to be understood by the farmers
- Drainage provides desirable environment in the crop root zone
- Necessity of drainage is felt when there is excess water in root zone
- Source of excess water are
- Uncontrolled irrigation
- Seepage loss from an unlined channel
- Ground water moving from a shallow aquifers
- Non maintenance of natural drainage system

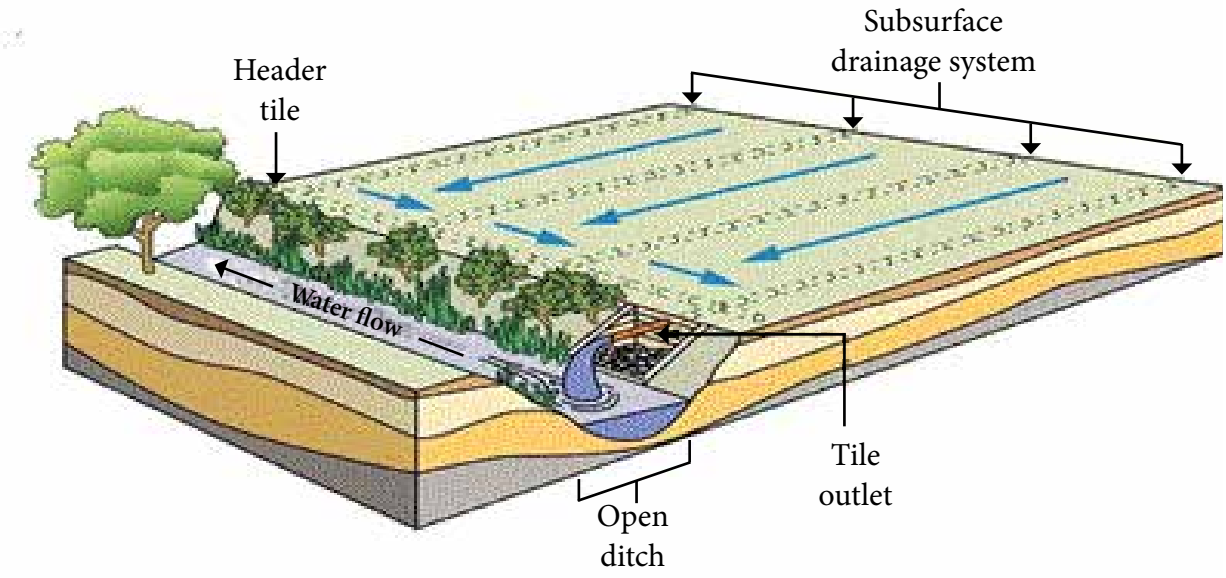
Generally two types of drainage systems are adopted based on techno-economic feasibility:

Surface drainage: Can be achieved by following any one of the below method based on the need and intensity of the problem.

- Land forming
- Land smoothening
- Land grading or levelling
- Bedding system
- Open ditches

Sub surface drainage: Can be achieved by following any one of the below method based on the need and intensity of problem.

- Horizontal sub surface drains
- Vertical drainage
- Other methods like
  - Mole drainage
  - Seepage intercepting farm pond
- Bio drains



1.7. Seed

A 'seed' (in some plants, referred to as a 'kernel') is a small embryonic 'plant' enclosed in a covering called the seed coat, usually with some 'stored food'. Seeds fundamentally are a means of reproduction and most seeds are the product of 'sexual reproduction', which remixes genetic material and 'phenotype variability' that 'natural selection' acts upon.

The seed is the basic input in agriculture upon which other inputs are applied. A good vigorous seed utilizes all the resources and realizes a reasonable output to the grower. It is wealth to the farmer since yesterday's harvest is tomorrow's hope. Good seed in good soil realizes a good yield. Moreover, it is the link between two generations.

Functions of seeds

- Nourishment of the embryo
- Dispersal to a new location
- Dormancy during unfavourable conditions

Characteristics of good seed

- Genetically pure
    - Breeder /Nucleus - 100%
    - Foundation seed - 99.5%
    - Certified seed - 99.0%
- Required level of physical purity for certification
    - All crops - 98%
    - Carrot - 95%
- High pure seed percentage
    - Bhendi - 99.0 %
    - Sesame, soybean & jute - 97.0 %
    - Ground nut - 96.0 %
- Free from other crop seeds
- Free from designated diseases like loose smut in wheat
- Free from objectionable weed seed like wild paddy in paddy
- Have good shape, size, colour, etc. according to specifications of variety
- Have high physical soundness and weight
- Posses high physiological vigour and stamina
- Posses high longevity and shelf life
- Have optimum moisture content for storage
    - Long term storage: 8% and below
    - Short term storage: 10-13%
- Have high market value

Seed types and characteristics

Seed Type	Characteristics	Genetic Purity	Tag Colour
Nucleus Seed	Produced by the breeder and it is genetically pure seed	100%	-
Breeder Seed	Produced by the breeder from nucleus seed	100%	Yellow
Foundation Seed	Produced by the breeder seed under the supervision of the concerned seed certification agency	99.5%	White
Certified Seed	Certified seed is the progeny of foundation seed and its production is supervised and approved by certification agency. The seed of this class is normally produced by the State and National Seeds Corporation and Private Seed Companies on the farms of progressive growers. This is the commercial seed which is available to the farmers.	99.0%	Azar Blue

Seed treatment

Seed treatment is usages of specific products and specific techniques to improve the growth environment for the seed, seedlings and young plants. It ranges from a basic dressing to coating and pelleting.

Seed dressing: This is the most common method of seed treatment. The seed is dressed with either a dry formulation or wet treated with a slurry or liquid formulation. Dressings can be applied at both, the farm and industries. Low cost earthen pots can be used for mixing pesticides with seed or seed can be spread on a polythene sheet. The required quantity of chemical can be sprinkled on the seed lot and mixed mechanically by the farmers.

Seed coating: A special binder is used with a formulation to enhance adherence to the seed.

Seed pelleting: The most sophisticated Seed Treatment Technology changes the physical shape of a seed to enhance pelletability and handling. Pelleting requires specialized application machinery and techniques and is the most expensive application.

The farmer must take care of the following while buying the seeds

- When purchasing the seed farmer should obtain a bill/cash memo wherein the lot number and seed tag number is mentioned.
- After purchasing the seed, empty bag/packet (pouches) and receipt should be kept safely.
- Out of purchased seed, 100 seeds are taken from each purchased variety to test them for germination before sowing in the field. Knowing the germination percentage, the farmer can decide the seed rate when sowing in the field.



Seed Dressing



Pelleted Onion Seed

Recommendation of seed treatment for different crops contiued...

Name of Crop	Pest / Disease	Seed Treatment	Remarks
Sugarcane	Root rot, wilt	Trichoderma spp. 4-6 gm/kg seed	For seed dressing metal seed dresser/earthern pots or polythene bags are used.
Rice	Root rot disease other insects /pests Bacterial sheath blight	Trichoderma 5-10 gm/kg seed (before transplanting) Pseudomonas flourescens 0.5% W.P. 10 gm/kg.	For seed dressing metal seed dresser/earthern pots or polythene bags are used.

Recommendation of seed treatment for different crops contiued...

Name of Crop	Pest / Disease	Seed Treatment	Remarks
Chillies	Anthracnose spp. Damping off Soil borne infection of fungal disease Jassid, aphid, thrips	Seed treatment with Trichoderma viride 4g/kg Trichoderma viride @ 2 gm/kg. seed and Pseudomonas flourescens@10gm/kg Captan 75 WS @ 1.5 to 2.5 gm a.i./litre for soil drenching. Imidacloprid 70 WS @ 10-15 gm a.i./kg seed (To be used in proper doses under guidance of an agriculture expert)	For seed dressing metal seed dresser/earthern pots or polythene bags are used.
Pigeon pea	Wilt, Blight and Root rot	Trichoderma spp. @ 4 gm/kg. Seed	For seed dressing metal seed dresser/earthern pots or polythene bags are used.
Pea	Root rot White rot	Seed treatment with 1. Bacillus subtilis 2. Pseudomonas fluorescens Soil application @ 2.5 – 5 kg in 100kg FYM	For seed dressing metal seed dresser/earthern pots or polythene bags are used.
Bhendi	Root knot nematode	Paecilomyces lilacinus and Pseudomonas fluorescens @ 10 gm/kg as seed dresser.	For seed dressing metal seed dresser/earthern pots or polythene bags are used.
Tomato	Soil borne infection of fungal disease Early blight Damping off Wilt	T. viride @ 2 gm/100gm seed. Pseudomonas fluorescens and V. clamydosporium @ 10gm/kg as seed dresser.	For seed dressing metal seed dresser/earthern pots or polythene bags are used.
Sunflower	Seed rot Jassids, Whitefly	Trichoderma viride @ 6 gm/kg seed. Imidacloprid 48FS @ 5-9 gm a.i. per kg. Seed (To be used in proper doses under guidance of an agriculture expert) Imidacloprid 70WS @ 7 gm a.i. per kg. Seed (To be used in proper doses under guidance of an agriculture expert)	For seed dressing metal seed dresser/earthern pots or polythene bags are used.

1.8. Cropping systems

Farmers resort to cultivation of a number of crops and rotate particular crop combinations. More than 250 cropping systems are being followed in India, of which 30 cropping systems are more prevalent. Some of the important cropping systems are:

1. Sequential cropping system:

Growing crops in sequence within a crop year, one crop being sown after the harvest of the other. For example, rice followed by pigeonpea, pigeonpea followed by wheat.

2. Intercropping System:

Growing more than one crop in the same area in rows of definite proportion and pattern.



Cereals + Legumes

The following intercropping practices were found to be remunerative in India's groundnut growing states.

State	Crop combination
Maharashtra	Groundnut + Red gram (6:1/4:1)
	Groundnut + Soybean (6:2)
	Groundnut + Sunflower (6:2/3:1)
Gujarat	Groundnut + Castor (9:2/3:1)
	Groundnut + Sunflower (3:1/2:1)
	Groundnut + Red gram (4:1)

Alley cropping

Is an agroforestry practice in which perennial, preferably leguminous, trees or shrubs are grown simultaneously with an arable crop. The trees, managed as hedgerows, are grown in wide rows and the crop is planted in the interspace or 'alley' between the tree rows.

During the cropping phase, the trees are pruned. Prunings are used as green manure or mulch on the crop to improve the organic matter status of the soil and to provide nutrients, particularly nitrogen, to the crop.



Alley Cropping and Silvipasture

a. Season based cropping system

i. Kharif rice based cropping system
ii. Kharif maize based cropping system
iii. Kharif sorghum based cropping system
iv. Kharif millet based cropping system
v. Kharif groundnut based cropping system
vi. Winter wheat and chickpea based crop ping system
vii. Rabi sorghum based cropping system

b. Mixed cropping

In order to minimise the risk and uncertainty of mono cropping and to have sustainable yield and income, farmers are advised to go for mixed cropping.



Mixed Cropping

Integrated farming System (IFS)

To feed ever-increasing population of the country, extensive cropping system give ways to intensive cropping which are exploiting natural resources. Therefore in future more thrust will be on efficient natural resource management and sustainable production system. This encompasses an animal component, an perennial and annual crop component, aqua culture, agro based production and processing units. Integrated farming system typically involves:

- Many enterprises including animal component
- Planning is based on resource available
- It is purely location specific/farmer/holding specific activity plan
- Very high resource use efficiency
- Sustainable farming



IFS - Duck & Fish rearing

Objectives of IFS

- To compliment and maximize use of by products
- To provide useful employment to all the family members
- Maximizing land use
- Value addition
- Self sustainability
- Less dependence on external resources

Crop production in IFS

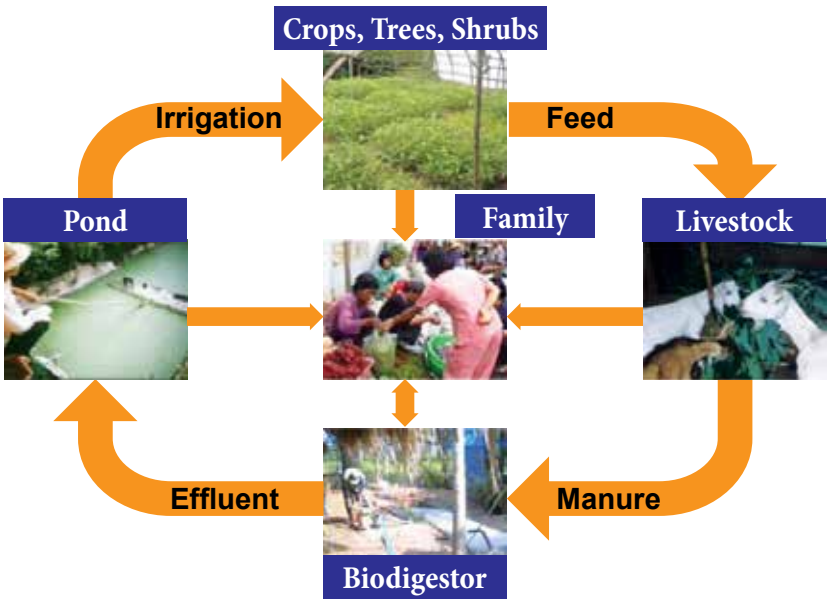
- Food crop should find a place
- Family food requirement should be planned
- Fodder production to meet the demand of animal component
- Specific enterprise based crops; e.g. mulberry/ sunflower linked to honey
- Infrastructure based cropping
- Sufficient employment to family members

Animal component in IFS

- One or more animal components or combination of animal component may be planned
- Complimentary enterprises should be identified
- Composting should be the interface between animal and crop enterprises
- Market should be considered before hand
- Need based demand driven enterprises should be prioritized

Allocation of resource in IFS

- List the resources available and required
- Prioritize the resources based on scarcity
- Resource demand will be prioritized based on economic impact and sustainability
- Scares resource on the farm should be allocated for the most important activity
- Recycling of resources should be planned
- Resource based contingent plan should be prepared in advance. This will serve as a security and sustainable alternative in case of crisis



1.9. Mechanization

Modernization of agriculture requires appropriate machinery for ensuring timely field operations, effective application of agricultural inputs and reducing drudgery in agriculture.

Advantages of mechanization

- Increase cropping intensity
- Ensure large area coverage and timeliness
- Increasing farm labour productivity
- Increases crop productivity and profitability

First step in mechanization

- Get good hands on training
- Read manufacturer information
- Give attention to maintenance
- Understand do's and don'ts with respect to equipments and machinery used
- Take utmost care in following safety tips given in the manufacture information booklet.

Selection of farm machinery

- Select based on holding size
- Economic feasibility
- Availability of skilled labour to operate
- Workout the feasibility of hiring v/s owning
- Decide between universal equipment v/s crop specific equipment when multiple crops are grown
- If the initial investment is huge, think of community ownership/custom hire centres, etc.

Benefits of Agricultural Mechanization

Benefits	Value, %
Saving in seed	15-20
Saving in fertilizer	15-20
Saving in time	20-30
Reduction in labours	20-30
Increase in cropping intensity	5-20
Higher productivity	10-15
Substantial reduction in drudgery of farm workers especially that of women	

Farm Mechanisation Potential

Land Preparation



Wooden Plunk



Bullock Drawn Country Plough



Laser Guided Land Leveller



Field Operation of Tractor Drawn Disc Plough

Seeding and Planting Machinery



CRIDA 2 Row Planter



Seed Treating Drum



Field of Operation of Yanji Transplanter for SRI



Tractor Drawn CRIDA 9 Row Planter

Inter-Cultivation Equipments



Grubber Weeder
Cost savings of up to 60% are possible at the early stages of crop growth.



Cono Weeder
Weeding under wetland paddy cultivation



Tractor – Operated Cotton Weeder



Wheel Hoe
Reduces the cost of weeding up to 50%



B.D. 3 Tyne Cultivator

Plant Protection Equipments



Knapsack Power Sprayer



Tree Sprayer



Blower Sprayer



Power Tiller Mounted Sprayer

Harvesting Equipments



Coconut Tree Climber

- Used for picking of coconuts
- Average time taken for climbing up and down is about 6.30 min for a 13 m tree and time for fixing and removing the device on the tree is 4 minutes.



Groundnut Digger



Banana Clump Remover



Austoft Chopper Harvester



Cotton Stalk Puller

Threshing Equipments



Groundnut Pod Stripper

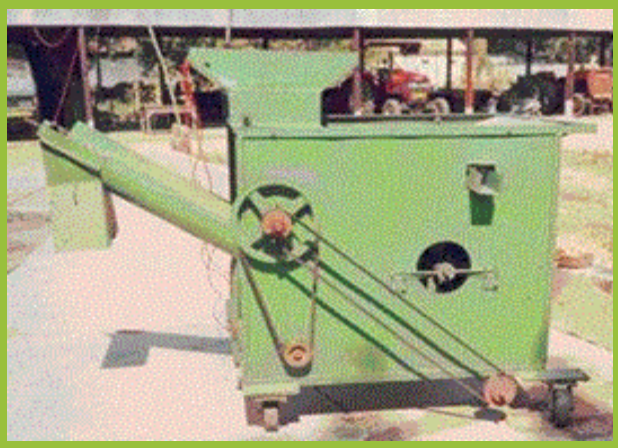


Castor Sheller

Winnowing and Clearing Equipments



Winnowing Fan



Seed Cleaner

1.10. Good Agricultural Practices (GAP)

Good Agricultural Practices (GAP) are *"practices that address environmental, economic and social sustainability for on-farm processes, and which result in safe and quality food and non-food agricultural products".*

What are GAP codes, standards and regulations?

Good Agricultural Practices (GAP) codes, standards and regulations are guidelines which have been developed in recent years by the food industry, producers' organizations, governments and NGOs aiming to codify agricultural practices at farm level for a range of commodities.

Why do GAP codes, standards and regulations exist?

These GAP codes, programmes or standards exist because of:

- Growing concerns about food quality and safety worldwide.
- Fulfilment of trade and government regulatory requirements.
- Specific requirements especially for niche markets.

Objectives

- Ensuring safety and quality of produce in the food chain.
- Capturing new market advantages by modifying supply chain governance.
- Improving natural resources used, workers' health and working conditions to creating new market opportunities for farmers and exporters in developing countries.

The benefits of GAP codes

- Standards and regulations are numerous, including food quality and safety improvement.
- Facilitation of market access.
- Reduction in non-compliance risks regarding permitted pesticides, Maximum Residue Limits (MRLs) and other contamination hazards.

GAP related to crop protection

- Use resistant cultivars and varieties.
- Crop sequences, associations and cultural practices.
- Biological prevention of pests and diseases.
- Maintain regular and quantitative assessment of the balance status between pests and diseases and beneficial organisms of all crops.
- Adopt organic control practices where and when applicable.
- Apply pest and disease forecasting techniques where available.
- Determine interventions following consideration of all possible methods and their short and long-term effects on farm productivity and environmental implications. This will allow the minimizing of agrochemicals, in particular, to promote Integrated Pest Management (IPM).
- Store and use agrochemicals according to legal requirements of registration for individual crops, rates, timings, and pre-harvest intervals
- Ensure that agrochemicals are only applied by specially trained and knowledgeable persons.
- Ensure that equipment used for the handling and application of agrochemicals complies with established safety and maintenance standards.
- Maintain accurate records of agrochemical use.
- Identify the GAP in each protection method.

Crop rotation systems

- Sequence crops by selecting pest host relation.
- Selected crop for rotation in order to break the life cycle of pest (Jowar should be rotated with pulses to combat striga weed).
- The selected crop for rotation should not be the food of previous crop pest.
- To select appropriate crops for rotation:
  - Analyze the pest habitat
  - Follow forecasts
  - Monitor pest and natural enemies

Privilege resistant species

- Cultivate plant varieties which are less prone to pest attack.
- The resistant varieties reduce production cost.
- Pest resistant transgenic crops developed for specific pest can be used. This is new avenue for reducing pesticide load.

Seeding techniques

- Depth of placement
- Method of placement
- Time of placement
- Seed treatments
- Managing the above based on pest nature will give good results

Promote useful animals

- Keep good predator population.
- Promote growth of beneficial insects.
- Create an environment congenial for predators; e.g. keeping bird perch in the field.
- Identify the useful animals and study their habitat for providing the required environment.

Observe and control populations

- Follow forecast-short term and long term.
- Study habitat of pest and congenial weather.
- Accordingly take necessary precautions to manage pest.

Give priority to mechanical and biological measures (instead of chemical)

- Get the full knowledge about botanical pesticides.
- Get the knowledge on available parasites and predator/friendly insects and pests.
- Accordingly develop action plan for mechanical and biological measures.

- **Use non cash inputs:** Saves money.
- **Use information on plant protection:** Analyze spatial and temporal distribution and trend analysis.

Monitoring of performance through taking notes each year/season.

- Keep the pest management record along with season, weather and other agriculture activity.
- Document the pest load and control achieved
- Use this experience for future planning.

Precision farming: Use precision farming modules and apply Information Technology (IT) to economize and for effective monitoring.

Good Agriculture Practices help the farmers to make use of the opportunities available in International Markets for selling their products and realising better farm profits.

1.10. Lessons Learnt

1. Critical factors to be considered while deciding the crops and cropping pattern are climatic factors, soil conservation, water, cropping system options, past and present experiences of farmers, expected profit and risk, economic conditions of farmers including land holding, labour availability, mechanization potential technology availability and suitability, demand and availability of market policies and schemes, public and private extension influence, availability of required agricultural inputs including agricultural credit and post harvest storage and processing technologies.
2. Soil, water and wind erosion may be managed through various recommended practices.
3. Method of irrigation has to be decided considering the quantity of water available and crop to be grown.
4. Recommended certified seeds may be used.
5. Mechanisation enhances quality of agricultural operations and minimises the cost and dependence on labour.
6. Good Agricultural Practices (GAP) may be considered essentials to enhance the price and market competitiveness of the produce.

2. Soil and Plant Nutrition

2.1. Objectives of the session

- To increase the awareness and understanding about the soil, its structure, physical, chemical and biological properties and soil fertility.
- To strengthen the farmer's knowledge to manage the soil fertility in an economically and environmentally sustainable manner.

2.2. What we know at the end of the session

- Soil composition
- Physical, chemical and biological characteristics of soil
- Soil testing
- Plant nutrition requirement
- Organic and inorganic fertilizers
- Integrated Nutrient Management (INM) for efficient, economic and sustainable production

Know your Soil

2.3. What is Soil?

Soil is a thin layer of earth's crust, which serves as natural medium for the growth of plants. Rocks are the important sources for the parent materials over which soils are developed.

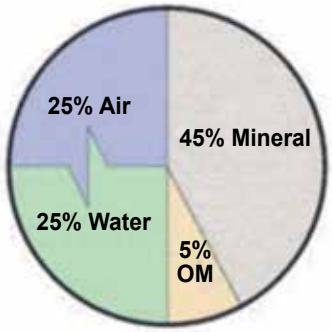
Soil Constituents



Rocks, the source of parent matererial

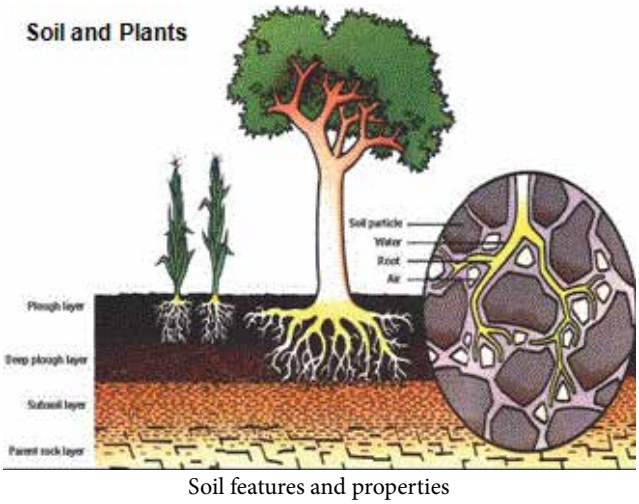
Soil is a dynamic medium made up of minerals, organic matter, water, air and living creatures including bacteria and earthworms.

It was formed and is forever changing due to 5 major physical factors: parent material, time, climate, organisms present and topography. The way in which we manage soil is another major factor influencing the character of the soil.



Soil Constituents

Soil features, properties and their importance



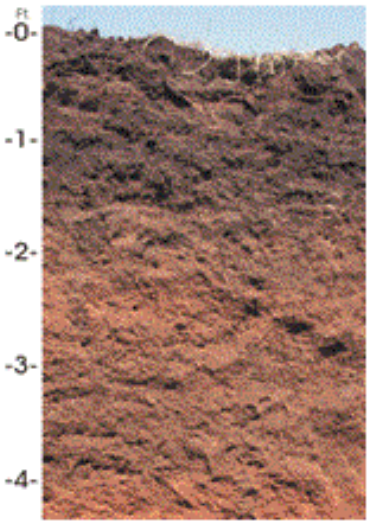
Soil features and properties

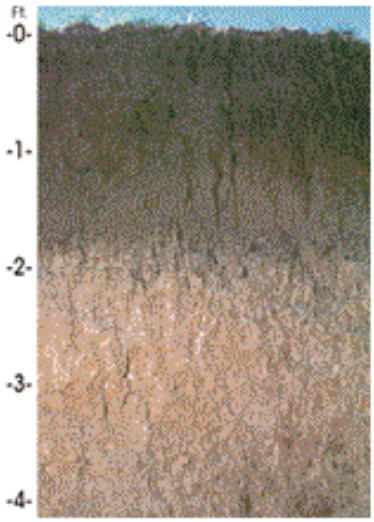
Soil colour

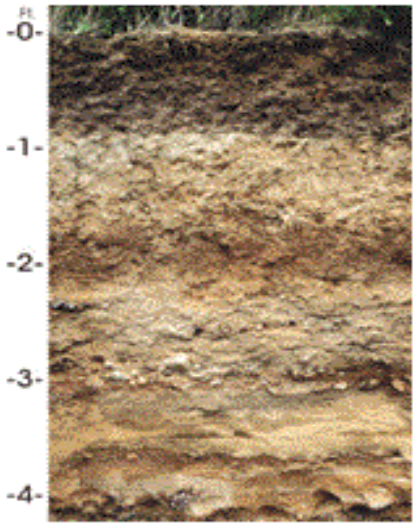
- Dark colour indicates usually medium to high fertility due to high amount of organic matter. These soils have usually high amount of nutrients, good water holding capacity and structure and are well aerated.
- Light colour indicates medium to low fertility. These soils may have leaching issue (water makes organic matter and other nutrients move downward faster).



Soil

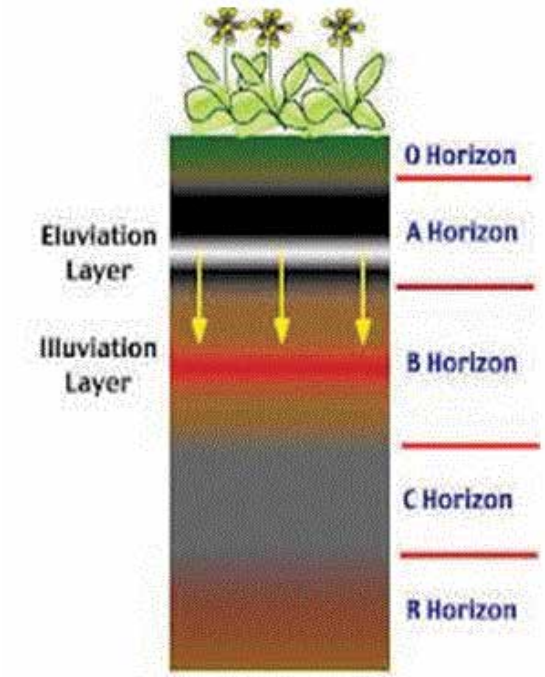






Soil depth

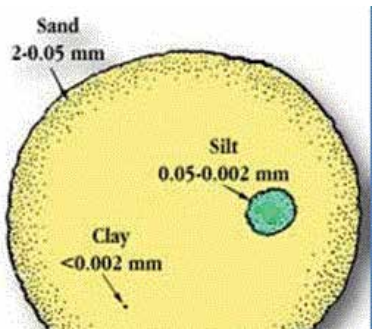
- The depth of soil to which the roots of a plant can readily penetrate to in order to reach water and nutrients.
- Minimum of 3-5 feet is desirable, deeper soils are better because they can hold more nutrients and water.



Soil Depth

Soil texture

- Texture refers to relative proportion of mineral particles (sand, silt, and clay) in soil. Many properties of soils; e.g. drainage, water holding capacity, aeration and the nutrient availability; depend largely on soil texture.
- **Sandy:** Low fertility and water holding capacity but good aeration.
- **Loamy:** Medium fertility and good aeration.
- **Clayey:** High fertility and poor aeration, hard to plough.

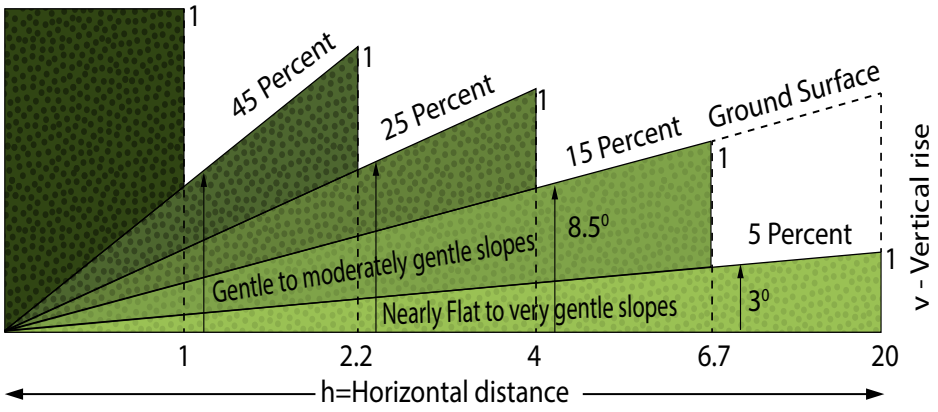


Soil Constituents

- Farmers may refer to their soil as heavy or light, illustrating the ease of working. The heavy soils are usually hard to plough and require much more effort than light soils. Organic matter may be added to improve the soil texture.

Land slope

- Soil gradient is the angle of inclination of the soil surface from the soil.
- It is expressed in percentage, which is the number of feet raise or fall in 100 feet from the horizontal distance.
- Mild gradient up to 1% is desirable.
- Higher gradients are not desirable as it leads to soil and water erosion.
- Perfect levelling is required only for paddy crop.

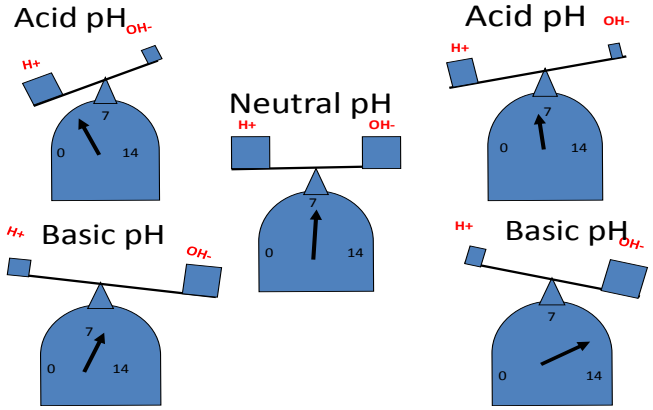


Land Slope

Soil pH

- Soil pH is of utmost importance in plant growth as it influences nutrient availability, toxicities and the activity of soil organisms.

PH Range	Soil Reaction Rating
<4.6	Extremely acid
4.6-5.5	Strongly acid
5.6-6.5	Moderately acid
6.6-6.9	Slightly acid
7.0	Neutral
7.1-8.5	Moderately alkaline
>8.5	Strongly alkaline



Tips for soil pH management

- Acid soils are to be corrected by using lime, quantity of lime application is as per soil test report.
- Alkali soils are to be corrected by gypsum/sulphur, quantity of application is as per soil test report.
- Saline – alkali soils should be treated with gypsum and improved drainage.

Soil organic matter

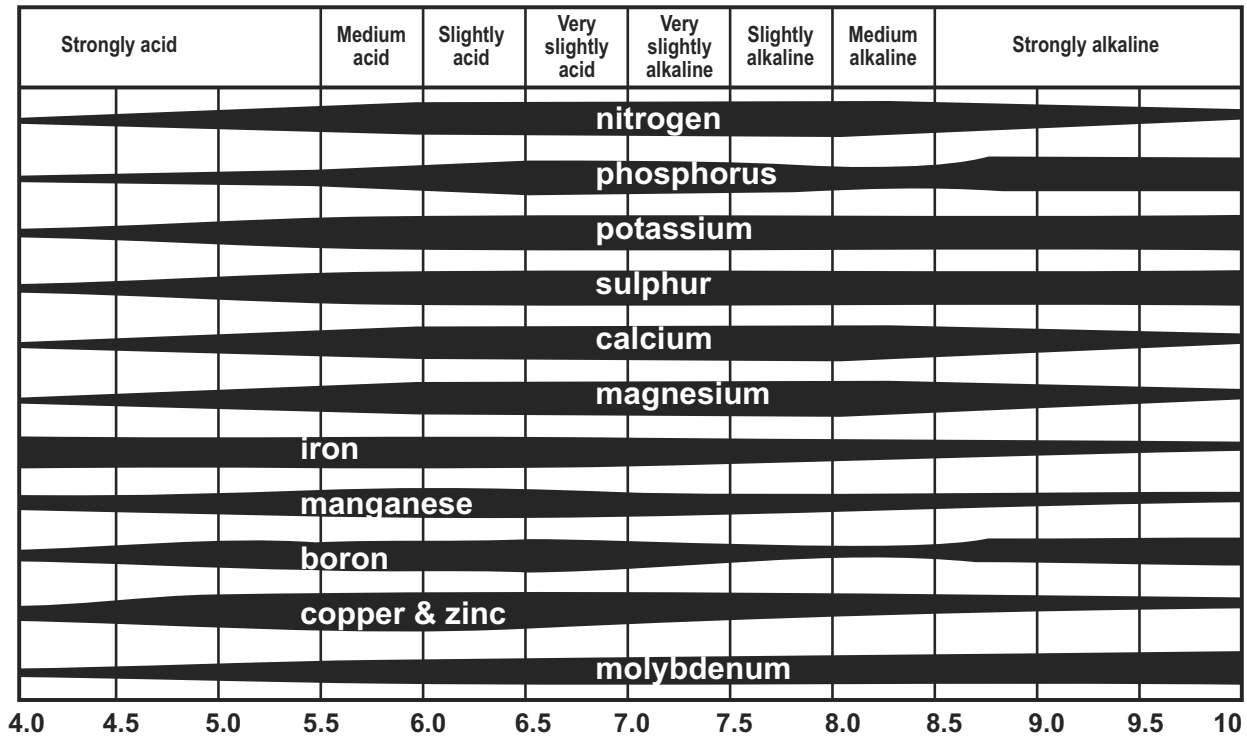
- Soil organic matter is the mix of plant and animal matter in different stages of decay.
- Soil organic matter plays a key role in biological, physical, and chemical function in soil.

Soil organic matter helps by:

- Providing nutrients for soil organisms
- Acting as major reservoir of plant nutrients
- Making nutrient exchange between soil and root of the plants easier
- Improving soil structure
- Influencing soil temperature
- Reducing the risk of soil erosion
- Increasing water holding capacity

Soil organic matter can be improved through:

- Recycling the crop residue back to field without wasting and burning
- Applying compost



Effect of Soil pH on Nutrient Availability

- Applying various organic manures
- Mulching organic wastes
- Using green manures and cover crops
- Suitable crop rotation
- Reducing soil tillage
- Avoiding soil erosion

Carbon - Nitrogen Ratio (C:N Ratio)

There are chemical elements in the organic matter, which are extremely important, especially in their relation or proportion to each other. They are Carbon and Nitrogen. The relationship is called Carbon - Nitrogen Ratio (C:N Ratio). For example, composed manure has 20:1 and sawdust has 400:1 of carbon and nitrogen. Generally speaking, the legumes are highest in nitrogen and have low C:N Ratio, which is highly desirable. Farmers can use blood meal, bone meal, poultry manure, cottonseed meal and soybean meal and other nitrogen rich material as organic matter, which enhance the decomposition.

Electrical Conductivity (EC): EC is normally considered to be a measurement of the dissolved salts in a solution.

General interpretation of EC values

Through application of zypsum, the saline/sodic soils can be amended. The quantity of zypsum to be applied is decided by EC value. Farmers having problem of saline/sodic soils can go for soil testing and approach extension officials for further guidance.

Soil	EC(mS/cm)	Crop reaction
Salt free	0 - 2	Salinity effect negligible, except for more sensitive crops
Slightly saline	4 - 8	Yield of many crops restricted
Moderately saline	8 - 15	Only tolerant crops yield satisfactorily
Highly saline	> 15	Only very tolerant crops yield satisfactorily

Soil fertility

- Soil fertility is generally defined as "ability of soil to supply plant nutrients". Soil structure, soil texture, temperature, water, light and air also play an important role in maintaining soil fertility.
- Plant nutrients which are often scarce in soil are nitrogen, potassium and phosphorus since plants use large amounts for their growth and survival.
- Important nutrients, their function and deficiency symptoms are described below.

2.4. Deficiency Symptoms of Nutrients in Plants



Nitrogen (N) – deficiency symptoms

Nitrogen (N) – deficiency symptoms

1. Stunted growth.
2. Appearance of light green to pale-yellow colour on the older leaves, starting from the tips. This is followed by death and/or dropping of the older leaves depending upon the degree of deficiency.
3. In acute deficiency, flowering is greatly reduced.
4. Lower protein content.



Phosphorous (P) – deficiency symptoms

Phosphorous (P) – deficiency symptoms

1. Overall stunted appearance, the mature leaves have characteristic dark to blue-green colouration, restricted root development.
2. In acute deficiency, occasional purpling of leaves and stems; spindly growth.
3. Delayed maturity and poor seed and fruit development.



Potassium (K) – deficiency symptoms

Potassium (K) – deficiency symptoms

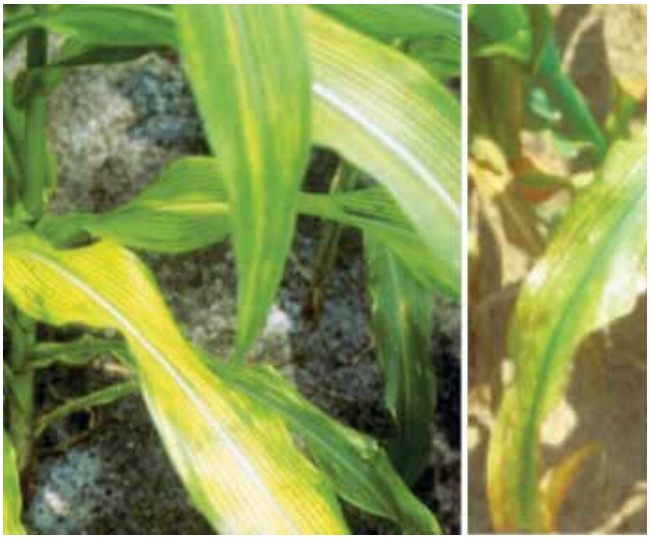
1. Chlorosis along the leaf margins followed by scorching and browning of tips of older leaves. These symptoms then gradually progress inwards.
2. Slow and stunted growth of plants.
3. Stalks weaken and plant lodge easily.
4. Shrivelled seeds of fruits.



Calcium (Ca) – deficiency symptoms

Calcium (Ca) – deficiency symptoms

1. Calcium deficiencies are not seen in the field because secondary effects associated with high acidity limit growth.
2. The young leaves of new plants are affected first. These are often distorted, small and abnormally dark green.
3. Leaves may be cup-shaped and crinkled and the terminal buds deteriorate with some breakdown of petioles.
4. Root growth is markedly impaired; rooting of roots occurs.
5. Dessication of growing points (terminal buds) of plants under severe deficiency.
6. Buds and blossoms shed prematurely.
7. Stem structure weakened.



Magnesium (Mg) – deficiency symptoms

Magnesium (Mg) – deficiency symptoms

1. Interveinal chlorosis, mainly of older leaves, producing a streaked or patchy effect; with acute deficiency, the affected tissue may dry up and die.
2. Leaves usually small, brittle in final stages and curve upwards at margin.
3. In some vegetable plants, chlorotic spot between veins, with tints of orange, red and purple.
4. Twigs weak and prone to fungus attack, usually premature, leaf drop.



Sulphur (S) deficiency symptoms

Sulphur (S) deficiency symptoms

1. Younger leaves turn uniformly yellowish green or chlorotic.
2. Root growth is restricted, flower production often indeterminate.
3. Stems are stiff, woody and small in diameter.



Zinc (Zn) deficiency symptoms

Zinc (Zn) deficiency symptoms

1. Deficiency symptoms mostly appear on the 2nd or 3rd fully mature leaves from the top of plants.
2. In maize, from light yellow striping to a broad band of white or yellow tissue with reddish purple veins between the midrib and edges of the leaf, occurring mainly in the lower half of the leaf.
3. In wheat, a longitudinal band of white or yellow leaf tissue, followed by interveinal chlorotic mottling and white to brown necrotic lesions in the middle of the leaf blade; eventual collapse of the affected leaves near the middle.
4. In rice, after 15-20 days of transplanting, small scattered light yellow spots appear on the older leaves which later enlarge, coalesce and turn deep brown, the entire leaf becomes rust-brown in colour and dries out within a month.
5. In citrus, irregular interveinal chlorosis; terminal leaves become small and barrowed (little-leaf); fruit-bud formation is severely reduced, twigs die back plants.



Copper (Cu) - deficiency symptoms

Copper (Cu) - deficiency symptoms

1. In cereals, yellowing and curling of the leaf blades, restricted ear production and poor grain set, indeterminate tillering.
2. In citrus, die back of new growth; exanthema pockets of gum develop between the bark and the weed; the fruit shows brown spots.



Iron (Fe) - deficiency symptoms

Iron (Fe) - deficiency symptoms

1. Typical interveinal chlorosls; youngest leaves first affected, points and margins of leaves keep their green colour longest.
2. In severe case, the entire leaf, veins and interveinal area turn yellow and may eventually become bleached.



Boron (B) - deficiency symptoms

Boron (B) - deficiency symptoms

1. Death of growing plants (shoot tips).
2. The leaves have a thick texture, sometimes curling and becoming brittle.
3. Flowers do not form and root growth is stunted.
4. "Brown heart" in root crops characterized by dark spots on the thickest part of the root or splitting at centre.



Manganese (Mn) - deficiency symptoms

Manganese (Mn) - deficiency symptoms

1. Chlorosis between the veins of young leaves, characterized by the appearance of chlorotic and necrotic spots in the interveinal areas.
2. Greyish areas appear near the base of the younger leaves and become yellowish to yellow orange.
3. Symptoms of deficiency popularly known in sugarcane as "streak" disease.



Molybdenum (Mo) - deficiency symptoms

Molybdenum (Mo) - deficiency symptoms

1. Chlorotic interveinal mottling of the lower leaves, followed by marginal necrosis and in folding of the leaves.
2. In cauliflower, the leaf tissues wither leaving only the midrib and a few small pieces of leaf blade ("whiptail").
3. Molybdenum deficiency is markedly evident in leguminous plants.

Some common deficiency symptoms are:





Chlorosis - It is the loss of chlorophyll leading to yellowing in leaves. It is caused by the deficiency of elements like K, Mg, N, S, Fe, Mn, Zn and Mo.



Necrosis are death of tissues, particularly leaf tissue is caused by deficiency of K, Ca, Mg

Inhibition of cell division is caused due to lack or deficiency of N, K, S and Mo.



Premature fall of leaves and buds

- deficiency of K and P.

	**Stunted/Retarded plant growth** caused by the deficiency of N, P, K, Zn, Ca.
	**Delay in flowering** due to deficiency of N, S and Mo.

In case the farmers observe the above symptoms, farmers are advised to consult the local extension worker for remedies.

2.5. Different fertilizers and their nutrient content

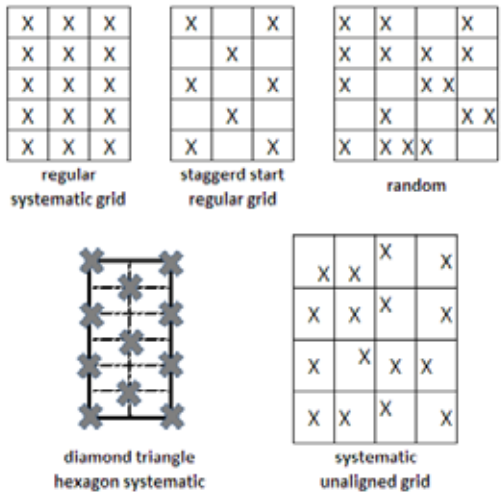
Important chemical fertilizers are the source of major nutrients. Different fertilizers and their nutrient content are illustrated in the table below:

Fertilizer	Nutrient content (%)		
	N	P*	K
Single nutrient fertilizers			
Ammonium sulphate	20	0	0
Urea	46	0	0
Calcium ammonium nitrate[1]	28	0	0
Single super phosphate	0	7	0
Triple Super Phosphate	0	20	0
Potassium sulphate	0	0	40
Muriate of Potash[2]	0	0	48
Double fertilizers			
Ammonium Phosphate	11	23	0
Diammonium Phosphate[3]	18	20	0
Complete Fertilizers			
Sampurna[4]	19	19	19
Vijaya Complex[5]	17	17	17
IFFCO Grade I[6]	10	26	26

A. Soil Analysis: Key to a Successful Nutrient Management Plan

Higher crop yields and quality of the crops depend largely on the efficient supply of nutrients. Soil provides not only the medium but also functions as the source of these nutrients for the plants. Soil resources get depleted with every harvest and need to be replenished for every crop. However, one must know which nutrients are to what extent depleted and what addition of fertilizers should be planned accordingly. Soil analysis, in this regard, helps in determining the level of nutrients and in deciding the required amount of fertilizer application.

Accuracy of soil analysis is directly related to the quality of the soil sample taken. Application of appropriate fertilizers with the proper nutrient mix will help not only to increase the productivity and farm income but also provide a more realistic chance to obtain the desired yield. There are various methods for taking a soil sample from a field. The right method of sampling may be decided by consulting local extension officer or your DESAI-trainer.



Soil Sampling Methods

[1]***Calcium ammonium nitrate***- *To be used with caution.*
[2]***Muriate of Potash***- *To be used with caution and protective equipment (Respirator)*
[3]***Diammonium Phosphate***- *To be used with caution.*

[4]***Sampurna- Compilation Source***- *Manufacturer (Zuari Agro Chemicals).*
[5]***Vijaya Complex***- *Indigenous Complex Fertilizer.*
[6] ***IFFCO Grade I***- *Compilation Source- Manufacturer*
(To be used with caution.)

icrisat.org

Soil sampling: an illustration



Selecting sampling spot

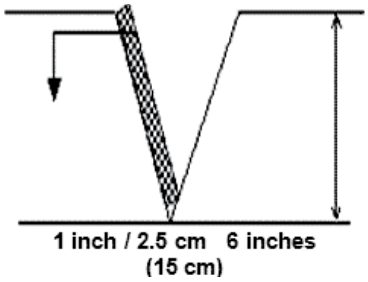


Remove the surface litter at the sampling spot



Make a 'V' shaped cut to a depth of 15 cm in the sampling spot





Collect soils - V shaped cut:
Remove thick slices of soil from top to bottom of exposed face of the 'V' shaped cut and place in a clean container



Mix the samples thoroughly



Quartering is done by dividing the thoroughly mixed sample into four equal parts



Two opposite quarters are discarded and the remaining is mixed



Collect the sample in a clean cloth or polythene bag



Process of collection



Label with required information:

- Name of the farmer
- Location of the farm
- Survey number
- Previous crop grown, present crop
- Crop to be grown in the next season
- Date of collection
- Name of the sampler, etc



Places for soil testing:

- Krishi Vigyan Kendra (KVK)
- State/district agricultural labs
- Agriculture University, Research Stations
- ATMA Officials/Agri-preneurs

B. Plant Analysis

Plant Analysis is the second tool, after soil testing that is critical to improve crop nutrition and yield. Only plant analysis can identify the actual nutrient status of a plant or crop. While soil testing identifies the nutrients offered to the crop or plants, Plant analysis identifies how well the plants utilizes the soil nutrients and applied nutrients. Plant analysis allows the plant to tell us what nutrients it needs.

It is critical that the correct plant part and stage of growth be sampled. The normal nutrient concentration differs between the various plant parts. Also, the normal nutrient concentration of each plant part changes as the plant matures. It is important to keep these factors in mind to assess the nutrients status of plants. The plant parts required to be collected for different crops are as follows.

Crop	Growth stage	Plant part to be sampled	Quantity
Cotton	(a) Seedling, 6" to 12" tall	Entire above ground portion of plant.	15 –20 plants
	(b) Prior to or at first bloom	Youngest fully mature leaves from the main stem of plant. Discard the petioles.	15 – 20 leaves
	(c) When first squares appear		
Sugar Cane	2 Months to mature	Second fully mature leaf without sheath.	15 to 25 leaves
Sunflower	(a) Seedling stage	Entire above ground portion of plant.	15 to 20 plants
	(b) Vegetative to full bloom	Youngest fully mature leaf. No petiole.	15 to 20 leaves

Nutrient requirement and productivity

Keeping soil fertility and soil nutrients at optimum level helps increasing the productivity of the soil. Although the ability of the soil to supply the plants with the required nutrients depends also on the soil condition like i.e. soil texture, soil structure and soil organic matter. Therefore, soil fertility management does not only include nutrient management but also soil condition management.

Apart from other soil management practices, soil fertility focuses on:

- Maintain a balance between nutrient uptake and nutrient application.
- Adequate fertility for the plants at the specific growth stages.
- Soil fertility and organic matter maintenance
- Minimizing the nutrient loss by avoiding excess application.

C. Irrigation Water Analysis

Irrigation water, irrespective of its source, always contains some soluble salts. The suitability of waters for a specific purpose depends upon the types and amounts of dissolved salts. Some of the dissolved salts or other constituents may be useful for crops but the quality or suitability of water for irrigation purposes is assessed in terms of the presence of undesirable constituents. Some of the dissolved ions such as NO3 are useful for crops.

The most important characteristic that determine the quality of irrigation water are:

- pH
- Total concentration of soluble salts are judged through Electrical Conductivity (EC).
- Relative proportion of sodium to other cations such as Ca and Mg are referred to as Sodium Adsorption Ratio (SAR).
- Concentration of boron or other elements that may be toxic to plants.
- Concentration of carbonates and bi-carbonate as related to the concentration of calcium plus magnesium are referred to as Residual Sodium Carbonate (RSC).
- Content of anions such as chloride, sulphate and nitrate.

Methods of water sample collection

As water quality and suitability plays an important role in deciding production and productivity of crops, farmers are advised consult extension worker to test the water.

The soil nutrient demand usually is based on:

- Soil nutrient level
- Crop variety and yielding ability
- Soil moisture
- Targeted yield

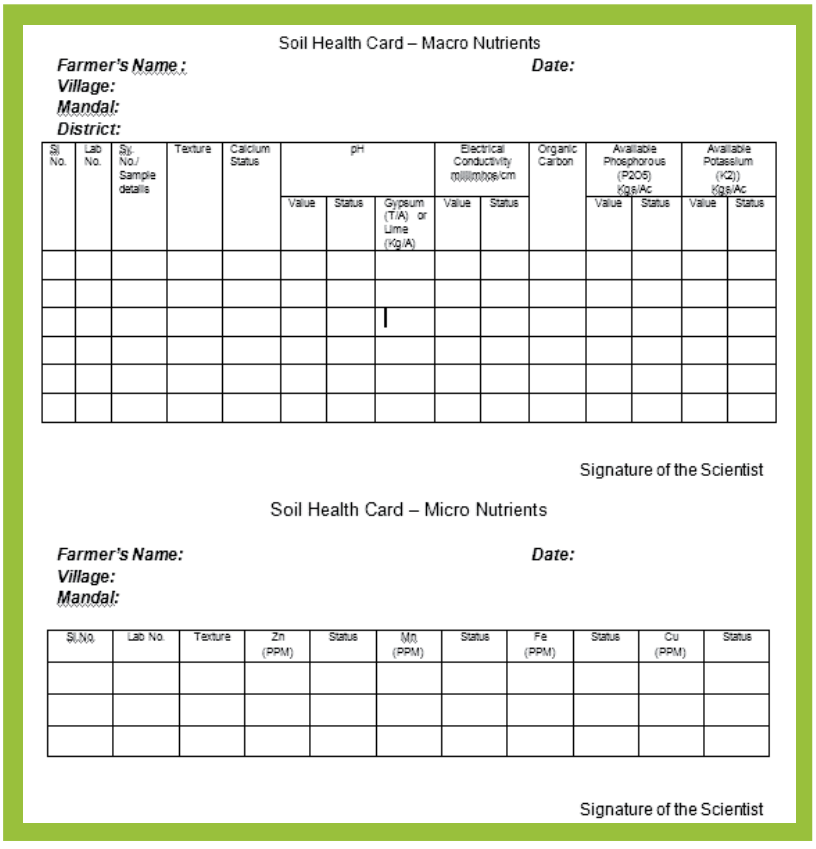
Soil analysis direct the farmer on quantity and quality of fertilisers to be used. Farmers can get the crop specific nutrient (fertilizer) recommendations after consulting the local extension service or their DESAI-trainer and discussing with them the soil analysis report.

C. Recommended Fertilizer Dose for Important Crops

General considerations before fertilizer application:

- Nutrients and not fertilizers should be bought. (Think nutrients, not fertilizers).
- Each nutrient applied as fertilizer should give a desired production response.
- The cost of fertilization must be calculated on the basis of applied plant nutrients per unit area of land.
- Calculate one nutrient at a time, considering available sources, prices and feasibility of using.
- A sample Soil Health Card, issued to the farmers based on the soil test results, should be used by the farmer to calculate the quantities of fertilizers required:

	N (kgs/ ha)	P2 O5 (kgs/ha	K2O (kgs/ha	Remarks
Banana	110	35	330	Apply 50% extra fertilizers at 2nd , 4th, 6th & 8th months after planting for tissue culture banana
Cotton	120	60	60	(TCHB – 213)
Citrus (sweet orange)	0.6 kgs	0.2 kgs	0.3 kgs	From 6th year onwards
Mango	1.0 kg	1.0 kg	1.5 kg	Kg of NPK/tree for 6th year onwards
Sugar cane	275	-	112.5	
Sun flower	60	90	60	Irrigated Hybrid
	40	50	40	Rainfed/Varieties



Soil Health Card – Macro Nutrients

Farmer's Name : *Date:*
Village:
Mandal:
District:

Sl. No.	Lab No.	Sl. No./ Sample details	Texture	Calcium Status	pH			Electrical Conductivity (millimhos/cm		Organic Carbon	Available Phosphorous (P2O5) Kgs/Ac		Available Potassium (K2)) Kgs/Ac	
					Value	Status	Gypsum (T/A) or Lime (Kg/A)	Value	Status		Value	Status	Value	Status

Signature of the Scientist

Soil Health Card – Micro Nutrients

Farmer's Name: *Date:*
Village:
Mandal:

Sl.No.	Lab No.	Texture	Zn (PPM)	Status	Mn (PPM)	Status	Fe (PPM)	Status	Cu (PPM)	Status

Signature of the Scientist

If the samples are all above the critical level, there is no deficiency of any element.

Nutrient	SL, SCL	CL
Zn	0.65	0.70
Cu	0.20	0.30
Fe	4.00	6.00
Mn	2.00	3.00

Fertilizers: Chemical or Organic

Chemical fertilizers

Advantages:

- Nutrients are immediately available for plant uptake.
- Price is lower as compared to organic fertilizer.
- Small quantities are required because they are nutrient rich.

Disadvantages:

- Over application usually results in economic and environmental losses.
- Over supply makes plant tissues soft and vulnerable to diseases and pathogens.
- Increased rate of soil organic matter decomposition resulting in soil degradation.
- Many nutrients applied are easily lost through different chemical reactions.

Organic fertilizers

Advantages:

- Balanced nutrient supply.
- Enhance the soil biological activity.
- Help in improving soil structure.
- Increase the organic matter content.
- Slow release of nutrients makes soil on the long run fertile.
- Help in combating plant diseases.

Disadvantages

- Low nutrient content.
- Only effective in the long run.
- It may not supply all the nutrients required for plant growth.
- High cost.
- Bulkiness.

Efficient fertilizer use

Good knowledge and management practices can improve the fertilizer use efficiency.

- Select the crops and varieties that suit the locality and have best fertilizer response.
- Select right kind of fertilizer according to crop and soil.
- The fertilization should be planned for the cropping pattern and not for single crops.
- Fertilization application rate should be decided only after discussing your soil analysis report with your local extension officer or Desai-Trainer.
- Balanced fertilization should be practiced.
- Crops should only be sown at the locally recommended periods.
- Maintain optimum plant population and proper plant spacing.
- Effective control of pests and diseases will help in maximizing the fertilizer efficiency.
- To maximize the yield increase through fertilizer, all other growth critical factors must also be optimum e.g. crop must be irrigated at critical growth stages.

Fertilizer application methods

Broadcasting: fertilizer is distributed manually over the cropped field.

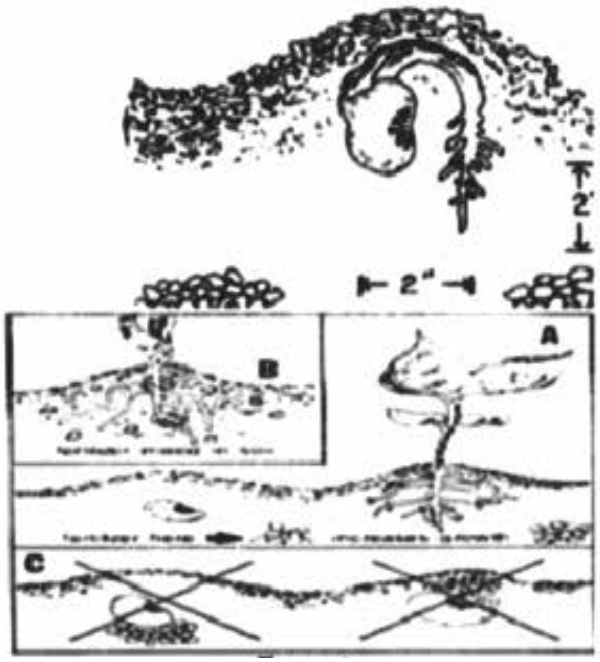
- The most common fertilizer application method.
- Highly inefficient method
- High economic and nutrient losses



Broadcasting

Placement: application in band or packets near the plants.

- Two sub-types:
  - i. Band application
  - ii. Spot Application
- The fertilizer use efficiency is high.
- Labour intensive.
- Efficient method but with high labour input.



Placement

Ring application: Spread the fertilizer around the tree at a distance of about one meter.



Ring Application

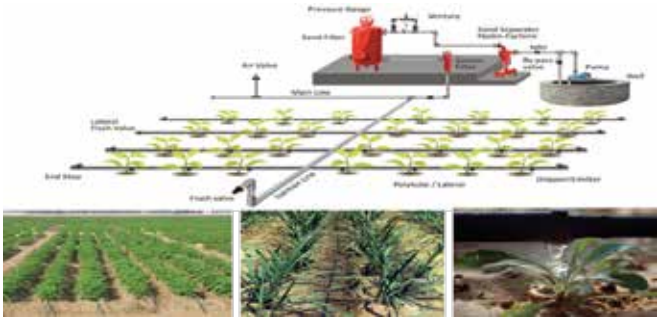
Foliar application: liquid fertilizers are sprayed on the crops.



Foliar Application

- Highly efficient
- Special equipment required
- High cost
- Only selected fertilizers can be applied

Fertigation along with irrigation



Fertigation

Method of application and nutrients

The method of application should be chosen according to the nutrient, crop, soil and cultivation method.

- **Nitrogen** application should be applied in splits and slow release mode to minimize loss.
- Frequent application in small quantity through foliar application is most efficient and results in quick recovery of crops.
- Use slow release nitrogen for plantation crops and long duration crops.
- Under puddle condition, use coated urea-neem oil. Coal tar sulphur coating will make urea to release nitrogen slow to match the uptake pattern.
- Urea can be cured with soil (1 part urea with 5-10 parts soil) to reduce the losses.
- Phosphate should be placed 4 to 6 cm below and 4 to 6 cm away from the seeds to ensure maximum availability.
- Phosphatic fertilizers give better response when placed in bands near the plant rows.
- Potassic fertilizers can be applied in one dose as basal application but for long duration crops the fertilizer application may be done in 2 to 3 splits.

Fertilizer calculations

Here is an example for the application of fertilizers based on the soil test recommendations. Suppose the recommendations are 120 kg N, 60 Kg P and 40 Kg K per ha. Calculate the quantity of urea, superphosphate and muriate of potash fertilizers needed to supply the recommended doses!

Urea content is 46%, so to supply 46 kg N/ha 100 kg urea is required. To supply 120 kg N/ha 100/46*120=260.9 or 261 kg urea is required. Similarly super phosphate content is 16% P2O5 and the recommendation is given in the form of **Phosphorous.**

- % P = %P2O5 X 0.44
- Kg P = kg P2O5 X 0.44
- Kg P = 16 X 0.44
- Kg P = 7.04

So super phosphate contains 7.04 P and the recommendation is given as 60 kg P.

To supply 60 kg P=100/7.04*60=852.27 or 852 kg single super phosphate is required. Similarly, we have to calculate the dose of potash through Muriate of Potash (MOP) as MOP contains 60% K2O.

So MOP contains 49.8 K and the recommendation given is 40 kg K/ha.

To supply 40 kg K=100/49.8*40=80.3 kg or 80 kg MOP is required. So based on fertilizer analysis and soil test information the fertilizer application rates are calculated as above.

The formula for calculating the fertilizer to be applied (kg/ha) =

$$\frac{100}{\text{Nutrient content in the fertilizer material (\%)}} \times \text{recommended dose (kg/ha)}$$

Nutrient cost comparison

Example - 1

1. Urea with 46% N costs Rs.562.20 per 100 kg.
2. Ammonium sulphate 20% N costs Rs.1029 per 100 kg.

Urea has 46% N i.e. 46 kg N in every 100 kg urea. Therefore unit value of N in urea: 562.2/46= Rs.12.22 per kg N. Ammonium sulphate has 20.6% N i.e. 20 kg N in every 100 kg fertilizer and 24% sulphur. Therefore unit value of N in ammonium sulphate: 1029/20.6= Rs.49.95 per kg N. Thus, the nitrogen is cheaper in urea. Yet, ammonium sulphate also has 24% sulphur in it.

Therefore, for soils with sulphur deficiency, ammonium sulphate is a better choice and for soils with normal sulphur levels, urea presents a better N source.

Example - 2

1. Single Super Phosphate (SSP) with 7 % P costs Rs.480 per 100 kg.
2. Di - Ammonium Phosphate (DAP) 20% P and 18% N costs Rs.1596 per 100 kg.

SSP has 7% P i.e. 7 kg P in every 100 kg SSP. Therefore unit value of P in SSP: 480/7= Rs.68.57 per kg P.

Whereas, DAP has 20% P and 18% N, i.e. 20 kg P and 18 kg of N in every 100 kg of DAP.

Cost of Nitrogen in 100 kg of DAP = (18 X 12.22) = Rs.219.96. Therefore unit value of P in DAP: (1596 – 219.96 = 1376.00); i.e. 1376/20= Rs.68.80 per kg P

Thus, the unit cost of P is the same in both the fertilisers. Yet, DAP also has 18% N in it. Therefore, for soils with nitrogen requirement, DAP is the better choice.

GUIDE FOR MIXING FERTILIZERS

1. Muriate of Potash	2. Sulphate of Potash	3. Sulphate of ammonia	4. Calcium ammonium nitrate	5. Sodium Nitrate	6. Calcium cynanamide	7. Urea	8. Superphosphate single or triple	9. Ammonium phosphate	10. Basic slag	11. Calcium carbonate	
☐	☐	☐	☒	☒	☒	☒	☐	☐	☐	☐	1. Muriate of Potash
☐	☐	☐	☒	☒	☐	☒	☐	☐	☐	☐	2. Sulphate of Potash
☐	☐	☐	☐	☒	■	☒	☐	☐	■	■	3. Sulphate of ammonia
☒	☒	☐	☐	☒	■	☒	☒	☒	■	☐	4. Calcium ammonium nitrate
☒	☐	☐	☐	☐	☒	☒	☐	☐	☐	☐	5. Sodium Nitrate
☒	☐	■	■	☒	☐	☒	■	■	☐	☐	6. Calcium cynanamide
☒	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒	7. Urea
☐	☐	☐	☒	☒	■	☒	☐	☐	■	■	8. Superphosphate single or triple
☐	☐	☐	☒	☒	■	☒	☐	☐	■	■	9. Ammonium phosphate
☐	☐	■	■	☐	☐	☒	■	■	☐	☐	10. Basic slag
☐	☐	■	☐	☐	☐	☒	■	■	☐	☐	11. Calcium carbonate

☐ Fertilizers which can be mixed
☒ Fertilizers which may be mixed shortly before use
■ Fertilizers which can not be mixed

Note: The crossing point of the required vertical column and horizontal colum indicates the possibility of mixing or otherwise of the fertilizer

Compatibility of Fertilizers

Organic fertilizers

Organic manures are natural products used to provide the nutrients to crops. Examples of organic manure are cow dung, farmyard manure, green manure, compost from crop residues, vermicompost and other biological waste.

Organic manures increase the organic matter in the soil. Organic matter in turn releases the plant food in available form for the use of crops. However, organic manures should not be seen only as carriers of plant food. These manures also enable a soil to hold more water and also help to improve the drainage in clay soils. They provide organic acids that help to dissolve soil nutrients and make them available for the plants.

Additionally, organic manures have low nutrient content and, therefore, need to be applied in large quantities. For example, to get a standard mixture of 25 kg NPK, almost 1000-2000 kg organic manure is required, whereas the same amount of NPK can be easily obtained with a 50 kg bag of NPK fertilizer.

Compost: Compost is well decomposed organic wastes like plant residues, animal dung and urine earth from cattle sheds, waste fodder, etc.

Tips for FYM preparation

- Provide shade to compost site
- Make compost pit in a well drained elevated place
- Smaller heaps of manageable shape are better
- Keep the pit free from weeds
- Pit should be near to cattle shed and water source
- Allowing for full maturity
- Also use urine
- If ash is added do not heap in the pit but spread uniformly

How to use

- Manure should not be kept exposed in the field before application
- Preferably apply in bands into the soil instead of broadcast. In case broadcast, work it into the soil immediately

Advantage

- Complete plant nutrient
- Improved soil structure increases soil aeration and drainability
- Organic matter acts as biological clay and increase nutrient and water holding capacity of soil
- It reduces the bulk density to desirable level (1.3g/cc)
- It provides enough food for micro organisms in the soil and helps to build up microorganisms in the soil
- Nutrients slow release nature is very useful for long duration and plantation crops

Disadvantage

- It is bulky
- Cost of handling, transportation and storage are high
- High labour input
- High cost per kg of nutrients

Vermi-Compost: Vermi-composting is a process by which earthworms convert organic waste into fertile manure. Important spp of earthworm used for vermicomposting in Indian conditions are Epigeic Eugeniae, Eisenia Foetida, and Perionyx Excavatus.

Earthworms

- Feed on soil and soil organic matter and convert it into compost
- Encourage growth of useful micro organisms
- Aerate and pulverize the soil
- Make soil porous and improving drainage
- Increasing water holding capacity of soil
- Give strength to plant immunity system

Advantages of vermi-compost

- Easy to use
- Low cost to produce
- Convert organic matter in to good manure in short time

How to make it

- Enrich vermi-compost with bio-agents
- Earthworms are our friends. Protect them and nurture in the field

Green manures

- Select important species
- Find a season to fit the green manuring crop in to cropping system
- Incorporate when 45 days old
- Allow 15-20 days before next planting for decomposition

Example: Sunhemp and Daincha

Green leaf manure

- Grow leguminous trees on road side, on bunds and waste land

- Loppings of trees can be incorporated 15 days before planting
- Provide approximately 5 to 10 tons

Example: Pongamia and Neem

Oilcakes

- Concentrated organic matter
- Mix with chemical fertilizer is useful as they make fertilizer slow release
- Use to enrich compost/organics
- Use preferably non edible oil cakes

Liquid Fertilisers

- Liquid form of fertilizers are applied with irrigation water or for direct application through foliar spray to augment yield and improve quality of a variety of crops like fruits, vegetables, oil seeds, pulses, cereals, cotton, tobacco, sugarcane, tea, etc.
- It will ease handling, less labour requirement as well as the possibility of mixing with herbicides have made the liquid fertilisers more acceptable to farmers

Fertigation

- Fertigation is the judicious application of fertilizers by combining it with irrigation water
- Fertigation can be achieved through fertilizer tank, venturi system, injector pump, Non-Electric Proportional Liquid Dispenser (NEPLD) and automated system

Advantages of Fertigation

- Ensures a regular flow of water as well as , resulting in increased growth rates for higher yields
- Offers greater versatility in the timing of the nutrient application to meet specific crop demands
- Improves availability of nutrients and their uptake by the roots
- Safer application method which eliminates the danger of burning the plant root system
- Offers simpler and more convenient application than soil application of fertilizer, thus, saving time, labour, equipment and energy
- Improves fertilizer use efficiency
- Reduction of soil compaction and mechanical damage to the crops
- Potential reduction of environmental contamination
- Convenient use of compound and ready-mix nutrient solutions which also contain a small concentration of micronutrients

Nutrient content of important organic manures

Nutrient content of important organic manures

Organic Manure	Percentage of Nutrients		
	Nitrogen	Phosphorus (P2O5)	Potassium (K2O)
Poultry Manure	1.2-1.5	-	-
Sheep Manure	0.8-1.6	-	-
Farmyard Manure	0.4	0. 3	0.2
Compost	0.5	0.25	0.5
Bone Meal	3.5	21.0	-

Bio-Fertilizers

These are products of microbial origin containing live cells of micro organisms multiplied in a laboratory and mixed with a carrier material like finely powdered coal, lignite or humus and supplied in a solid form.

Basically two types of bio fertilizer distinguished based on their capacity to supply nitrogen or phosphorus. Some bio-fertilizers have the capacity to supply nitrogen because they have the capacity to absorb nitrogen gas from the atmosphere in association with plants and use the nutrient for their cell synthesis. The nitrogen fixed becomes available to crop plants after the plant material is incorporated in to soil. This process is known as symbiotic nitrogen fixation. Other nitrogen fixing organisms can live freely in soil and fix nitrogen. There is one more type where the organism establishes a weak or associative symbiosis and fixes nitrogen.

In addition to nitrogen fixers there are some bacteria and fungi, which are used as bio-fertilizers to enhance phosphorus supply to crops. They solubilize insoluble or difficulty soluble phosphorus through their capacity to produce organic acids.

Plant Nutrient	Microorganism	Crops Benefited
NITROGEN	a) Symbiotic Rhizobium Azolla	All leguminous crops; Rice
	b) Associative Symbiosis Azospirillum	All cereal crops; Sugar cane
	c) Non Symbiotic Azotobacter Blue- green Algae	All crops; Rice

PHOSPHORUS	Microorganism	Crops Benefited
a) Solubilizing Effect	Bacteria: (Bacillus, Pseudomonas)	All Crops
	Fungi: (Aspergillus Penicillium)	All Crops
b) Absorbing Effect	Ecto Mycorrhizal	Tree crops
	Endomycorhizae	All Crops

All bio-fertilizers are microorganisms belonging to group of bacteria or fungi or blue-green algae.

The capacity of N-Fixers to supply N to crops varies from 10 kg – 100 kg/ha and symbiotic nitrogen fixers particularly rhizobium bacteria are very efficient. The P-Solubilizers and P-Absorbers can mobilize insoluble phosphorus to the extent of 10 Kg to 50 kg P/ha. Mycorrihizae are symbiotic fungi, which get associated with plant roots. Azolla is a water fern which establishes a very active and beneficial association with blue-green algae - Anabaena

Algal inoculants are cultivated on submerged soils in tanks, dried and supplied as soil based culture (Inoculants). Endomycorrhizal inoculants are supplied as pieces of roots of grass plants infected with mycorrhizae. Ectomycorrihizal inoculants are cultivated in the laboratory and mixed with carrier material for supply. Azolla are supplied as fresh fronds (leaf). The bio-fertilizers are inoculated to seeds or applied directly to soil after suspending them in water. Azolla is applied directly to soil, which on multiplication establishes a very active symbiotic association with Anabaena, a blue-green alga.

- The best nutrient management method is Integrated Nutrient Management.
- Use bulk quantity of organic matter to provide good physical and chemical properties to soil.
- Smaller quantities of chemical fertilizer to provide quick release, which matches the uptake pattern of crops.

I. Package of Practices – Banana

Banana is one of the most commonly grown fruit crop of the country. India produces about 26.217 MT of banana from an area of 0.709 Mha with an average productivity of 37.0 mt/ha. Major producing states are Tamil Nadu, Maharashtra, Karnataka, Gujarat, Andhra Pradesh, Assam and Madhya Pradesh. Tamil Nadu has 0.1244 mha under banana and the total production during 2008-09 is 6.667 Mmt with 53.6 mt/ha productivity. In Gujarat, banana crop is cultivated in 11 districts covering an area of about 60900 ha. Gujarat is ranking 3rd among the states of India with an average productivity of 58.7t/ha. Unripe fruits are used for making chips, vegetable flour, etc. Ripened fruits are used for preparing drink, jam, beer, salad, etc. Banana fruits are having numerous medicinal uses (as per Ayurveda). From psedostem threads are obtained and used for making rope and cloths.

Prevailing Varieties: Basarai, Lokhandi, Robasta, Shreemanti and Grand Naine.

Climate: Banana grows well in warm and humid climate with an average temperature of 27oC and rainfall of 2,000 to 2,500 mm. However, it requires assured irrigation facility.

Soil type suitability: Loamy and salty clay loam soils with good fertility status are best suited for banana cultivation. However, extremely clayey and sandy soils are not suitable for banana crop.

Land preparation: The land should be ploughed, harrowed and planked to achieve levelled fields. For planting banana, dugout a pit of 30 x 30 x 30 cm size at a spacing of 1.5 x 1.5 m.

Soil sterilization: After land preparation, the pits should be exposed to the sunlight for 10 to 15 days. In the case of heavy infestation of soil born pest and diseases, the soil solarization should be done using transparent plastic for a period of 15 to 20 days during summer season (preferably during May).

Time of plating: Optimum time for planting is 15th June to 15th July. Banana planting is done either by suckers or tissue culture plant.

Suckers: Suckers should be selected from a healthy field of banana. Sufficient care should be taken that suckers should not be damaged during digging and transport. Fresh sucker, weighing about 500 to 1,500 g, should be selected. For control of fungal diseases, suckers should be dipped in a solution of aurofugine[1] (10 g / 100 litres of water) for 1.5 hour prior to planting.

Tissue culture plant: Select healthy and uniform plants.

Irrigation and scheduling: The water requirement of banana crop by surface method of irrigation (25-30 irrigations) is 1,500-1,800 mm. By drip irrigation method, water requirement is 900-1,080 mm. The drip system should be operated on alternate day for a period of 1.05 to 2.25 hours during winter and 2.5 to 2.75 hrs during summer at a pressure of 1.2 kg/cm2.

[1] ***Aurofugine***- *To be used with caution.*

Details of Drip System

Lateral spacing	1.5 m
Dripper per plant	2
Spacing between two drippers	30 cm away on either side of stem
Dripper discharge rate	4.0 lph

Water requirement for banana under drip irrigation

Months after planting	Liters / day / plant
1-3	5
3-5	9
5-8	11
8-11	10

Application of Fertilizers

Basal

FYM should be applied @ 20 to 25 t/ha at the time of land preparation. Top dressing of fertilizer should be done as given below

For using water-soluble fertilizers, the following schedule of fertigation may be followed:

Nutrient	Fertilizer application schedule
N (180g/plant)	7 to 8 splits at an interval of 15 to 20 days.
P (72g/plant)	7 to 8 splits at an interval of 15 to 20 days.
K (180g/plant)	7 to 8 splits at an interval of 15 to 20 days.

Time of application: The total quantities of water-soluble fertilizers should be applied in 7 to 8 splits at an interval of 15 to 20 days. The first split should be applied at the time of planting before monsoon and the rest should be applied after the cessation of monsoon.

Weeding: Banana fields should be kept weed free either by hand weeding/interculturing or by weedicide (diuron @ 1.2 kg/ha as pre emergence) application. Mulching with black plastic (50 micron) or sugarcane trash (@ 10 t/ha) should be done. If required, hand weeding should be done prior to mulching.

Desuckering: First desuckering should be done manually. To minimize the regeneration of suckers, 3 ml of diesel or kerosene should be injected into the cut portion of the suckers.

Plant protection measures

Diseases	Control Measures
Bunchy top	Aphid should be controlled by applying systemic insecticide viz; Acetamiprid: (0.2gm/litre).
Premature fruit ripening	Sucker should be dipped in the solution of Aurofugin[1] 10 g in 100 litres of water for 1.5 hrs.
Pests	**Control Measures**
Rhizome weevil and nematode	Neem based products such as Neem oil, Neem cake, Neem seed Kernal extract (NSKE) can be applied.

Bunch coverage: **After the complete formation of the bunch, it should be covered by LLDP film bag (blue, white or black colour). This improves quality as well as yield of banana.**

Time of harvesting: **Maturity varies with variety but usually the crop takes about 12-14 months to mature.**

Yield: **By adopting the above practices, the banana yields about 70 to 80 t/ha. Generally, size and colour based grading is done.**

Post harvest handling and storage: **Bananas can be stored for up to a week in a cool place but unripe bananas should not be stored in the refrigerator as this may irreversibly interrupt the ripening process. If the banana is no longer green, then it is ripe and can be stored for a maximum of one week. For storage, banana should be stored at 13o to 140C. Bunches should be kept out of light after harvest since this hastens ripening and softening. For export, hands are cut into units of 4-16 fingers, graded for both length and girths and carefully placed in poly-lined boxes to hold 12 to 18 kg depending on export requirement. Prior to packaging fruits are cleaned in water or dilute sodium hypochlorite solution to remove the latex and treated with thiobendazole.**

Plastic packaging: **Keeping quality of banana can be increased when packed in 400 gauge LDPE (Low-Density Poly Ethylene) bags with or without ventilation either under ambient temperature or in a zero energy cool chamber (13.5°C). Storing of banana fruits in unvented polybags at low temperature could extend the shelf life of the fruits up to 19.33 days.**

Cost economics

Annual system cost (Rs /ha)	17,500
Cost of cultivation (Rs/ha)	1, 08,000
Total cost (Rs/ha)	1, 25,500
Total income (Rs/ha)	2, 10,000 to 2, 40,000
Net income (Rs/ha)	84,500 to 1, 14,500
C:B ratio	1:1.67

[1]***Aurofugin***- *To be used with caution.*

II. Packages of Practices – MANGO

Mango is the most important fruit crop of India. India produces about 12.750 Mt of Mango from an area of 2.309 Mha with an average productivity of 5.5 mt/ha. Major producing states are Andhra Pradesh, Uttar Pradesh, Bihar, Karnataka, Tamil Nadu, West Bengal, Orissa & Maharashtra. Uttar Pradesh has 0.2712 mha under mango and the total production during 2008-09 is 3.465 Mt with 12.8 mt/ha productivity. India ranks first in the world for mango production and area under cultivation. Mango is a rich source of vitamin A and has a fairly good content of vitamin C. Mango fruits are used for preparation of pickle, chatani, amchur, jam, squash, nectar and many other delicious products.

Climate and soil: Mango can be grown from alluvial to lateritic soils except black cotton soil which has poor drainage. The temperature between 24 and 270C is ideal for mango cultivation.

Variety: Dashehari, Langra, Chausa, Bombay Green, Lucknow Safeda, Mallika and Amrapali.

Multiplication of genuine planting material: Mango can be propagated by veneer, wedge and soft wood grafting. The protected nurseries in polyhouses and use of sprinkler and drip is becoming common for raising humidity level, which is required for higher grafting success rate.

Preparation of land: The land should be prepared one month before planting. The pits of 1m x 1m x 1m size are dug. The pits are exposed for 2 to 4 weeks to kill harmful soil organisms.

Soil sterilization: Soil sterilization can be achieved through both physical and chemical means. Physical control measures include steam and solar energy. Chemical control methods include herbicides and fumigants. Soil sterilization can also be achieved by using transparent plastic mulch film (25 micron thickness) termed as soil solarisation.

Planting: Square and rectangular systems are popular. Before planting, pits are filled with FYM at the rate of 15-20 kg/plant. The grafts should be planted during July to September.

Planting density: High density (3m x 6m or 5m x 5m) planting helps increase the yield/unit area. Normal planting distance of mango is 8m x 8m.

Canopy Management: Training should be done after 6th month of planting. It is essential to space the branches properly and to help in intercultural operation. At initial branching height between 60 to 70 cm is appropriate.

Water requirement of the crop

Age of the plant	Water requirement of the crop in litres/day/tree
Young plant (up to 3 years)	9-12 lts
3-6 years	30-35 lts
6-10 years	50-60 lts
9-12 years	80-90 lts
Fully grown trees	120 lts

A young tree requires 2 drippers at a distance of 1m on lateral lines, while fully-grown tees require 2 drippers with double lateral lines at 1-1.5 m distance.

Application of fertilizers: Mango should be manured with phosphorus twice in a year i.e. the beginning of the monsoon (June-July) and during the period of post-monsoon (September-October). Usually fertilizers (N and K) are applied in split doses in the month of June-July, September-October, January-February and March-April. For adult trees (10 years or above) 1,000g N, 75g P2O5, 75g K2O and 100 kg FYM per year should be applied. Application of micronutrients such as Zinc and Boron help in cell elongation process.

Malformation: Deblossoming at bud stage (1 cm long) alone or in combination with spray of 200 ppm NAA lowers the number of malformed panicle.

Alternate bearing management: Use of paclobutrazol (5-10g/m canopy diameter), 3 months before budburst applied through soil drenching can be used for obtaining regular bearing.

Weed management: Black plastic mulch (100 micron) restricts the germination of weed seeds and suppresses the weed growth. The size of the film requirement for young plant is 1 m x 1 m, and for 8 years onwards film requirement is 2.5 m x 2.5 m around the tree.

Intercropping: In the interspaces of mango orchard, certain vegetable can be intercropped viz. onion, tomato, radish, carrot, ginger, turmeric, methi, cabbage, etc. Moreover, fruit crops can also be grown viz. papaya, pineapple, etc. for the initial 4-5 years.

Mulching: Soil drenching with paclobutrazol (5g and 10g/tree) coupled with black polythene mulch (100 micron) results into minimum outbreak of September to October vegetative flushing, giving an early and profuse flowering and a higher annual yield.

Plant Protection Measures

Insect pests	Symptoms	Control measures
Mango hopper	Pest starts attacking during flowering season.	Spraying of acetamiprid or thiamethoxam (0.2gm/litre of water).
Mealy bug	Nymphs suck juice from young shoots, panicles and flower pedicels.	Raking of soil around the trunk and mixing with neem cake around tree trunk is effective.
Stem borer	Pest makes tunnel through the main trunk and branches.	Clearing tunnels with hard wire, pouring, Emamectin benzoate (50 gm / litre) and plugged with mud.
Fruit fly	Pest makes the fruits rot by laying its eggs in clusters, just before the ripening, under the peel of fruits.	Application of acetamiprid or thiamethoxam (0.2gm/litre of water).

Disease	Symptoms	Control measures
Anthracnose	It attacks leaves, flowering panicles and fruits.	Spraying of Copper Oxychloride (0.03%)can control this disease.
Powdery mildew	Whitish powdery growth on the leaves.	Wettable sulphur (0.02%) and Bayleton (0.05%).
Bacterial Canker	Unattractive fruits because affected parts of fruits show longitudinal crack and oozing of bacterial exudate and leading to fruit drop.	StreptocycIine(100-200ppm), agrimycin (17% streptomycin) -100 (100ppm) and copper oxychloride[1] (0.03%).

Yield and quality control:

From a well grown up tree one can expect an average yield of 50-225 marketable fruits (50 kg) per plant per year.

Harvesting and post harvest management:

From the 4th year onwards the mango fruits can be harvested at the mature green stage during morning hours. After harvesting fruits are graded according to their size, weight, colour and maturity. Packaging of fruits should be done in corrugated fibreboard (CFB) boxes. Tissue paper and polythene foam paper are used for wrapping high-value fresh mangoes. Polyethylene lining has been found beneficial as it maintains humidity, which results in lesser shrinkage during storage. Dashahari treated with calcium chloride solution (4%) at sub-atmospheric pressure of 500 mm Hg for 5 minutes can be stored at 12 oC for 27 days.

Cost economics of drip irrigated mango (one ha)

Rate of interest	10.5%
Life of system	7.5 years
Expected yield	19 t/ha
Planting distance	5m x 5m
Cost of cultivation	Rs.24, 000
Fixed cost	Rs.30, 298
Annual cost of drip system	Rs.8,713
Expected cost benefit ratio	1: 6.0

III. Package of Practices - Sugarcane

Time Schedule	Recommended Operations
Before planting	Plough the land up to 45 cm depth. Apply 25 tons per hectare of well-decomposed Farm Yard Manure (FYM) or decomposed molasses or compost and deep plough the field with tractor. Make ridges and furrows with 80 cm spacing having 20 cm height and up to 10 m length. Apply 375 kg super phosphate in the furrows.
Planting day	Select about 75,000 two-budded setts per hectare from 6-8 months old nursery or raise seedlings in poly bags with single-budded setts. Treat the setts for 10 minutes by dipping in a solution prepared with Thiamine, 2.5 kg urea and 2.5 kg lime in 250 litres of water. Plant the setts 2 cm deep with buds on the sides. For every 10 furrows, plant setts on 2 rows in one furrow for gap filing purpose.
3rd day after planting	To control weeds, spray atrataf @ 2.5 kg/ha in 500 litres of water with hand sprayer.
5th day after planting	Spread sugarcane trash up to a height of 15 cm on the ridge.
25th day after planting	Gap-fill with seedlings raised in polybags or the plants taken from the 2 rows planted- furrows in every 10th furrow.
30th day after planting	Mix 5 kg Azospirillum and 5 kg phosphobacterium per ha mixed with 250 kg powdered FYM. Apply at the bottom of the plants and irrigate immediately.
From 35th day after planting	From 35 to 100 days after planting Irrigate once in 7-10 days. To prevent attack by early shoot borer, apply Sulphur on the setts and cover with soil. If 25-30% of the shoots are affected, then for every 100 metre length of furrow, mix Sulphur and apply using a hand sprayer on the tips and bottom of the shoots.

45th day after planting	Do hand weeding. Apply in pits a mixture of 110 kg nitrogen[1], 60 kg of potash and 35 kg of neem cake per hectare.
60th, 90th and 120th days after planting	Spray a mixture of urea 2.5%, potassium chloride 2.5% during periods of drought. On the 60th day, apply a mixture of 5kg azospirillum, 5kg phosphobacterium and 250 kg of decomposed FYM in powder form at the bottom of the plants and irrigate immediately. On the 90th day, do hand weeding; apply in pits (after earthing up) a mixture of 110 kg of nitrogen[1], 60 kg of potassium and 35 kg of neem cake per hectare.
120th day after planting	Under drought conditions, apply 60 kg of potassium and irrigate immediately.
150 days to 225 days after planting	Carry out de-trashing at 150 days after planting. If inter-node borer exists, release parasites 6 times @ 5 cc per hectare once in 15 days. 101-210 days – irrigate once in 7 days. 210th day- detrash and tie lodged canes. 225th day- spray acetamiprid or thiamethoxam (2ml/litre of water) to control mealy bugs, white fly and scales.
260th day	Spray Emamectin benzoate (50 gm / litre) of water under the leaves (if required) to control pyrilla and all sucking pests.
270 to 360 days	Irrigate once in 15 days. Stop irrigation 15 days before harvesting.
Harvest	Cut canes at the bottom close to the ground with sickles or sharp knife. Remove trash, roots, water shoots and cane tops and send clean canes to the factory.

[1]***Nitrogen***- *To be used with caution and preventive measures (Gloves, etc).*

SCHEDULE OF OPERATIONS: RATOON CROP

Time Schedule	Recommended Operations
	Remove trash, stubble shave uniformly under correct moisture conditions with sharp spades.
	Mix 15 tons of FYM or 25 tons of compost or 25 tons of decayed molasses with 375 kg Superphosphate (75 kg of P2O5), 135 kg of nitrogen and 35 kg of neem cake per hectare in pits.
	Irrigate immediately, cutting the sides of the ridges and ensuring mixing of applied manures well with the soil.
	Control weeds by spraying atrataf @2.5 kg in 500 litres of water with hand spray.
9th -10th day	Mix 5kg azospirillum and 5 kg phosphobacteria per hectare with 250 kg of powdered FYM and apply at the bottom of the plants and irrigate immediately.
	Spread trash obtained from plant canes on the furrows.
25th – 30th day	Gap-fill with grown up plants.
	To prevent attack by early shoot borer, apply Sulphur on the setts and cover with soil.
35th day	Mix 5kg azospirillum per hectare with 250 kg of powdered FYM and apply at the bottom of the plants and irrigate immediately.
From 1-35 days	Irrigate once in 7 days.
From 35-90 days	Irrigate once in 10 days.
60th day	Do hand weeding.
	Apply in pits a mixture of 110 kg nitrogen, 60 kg of potash and 35 kg of neem cake per hectare and follow light earthing-up.
90th day	Apply 60 kg of potash additionally in drought situations.
91- 250 days	Irrigate once in 7 days.

30, 60 and 90 days	Spray a mixture of urea 2.5%, and potassium chloride 2.5%on the leaves in drought situations.
120th day	Detrash and earth up well.
121 to 210 days	Release Trichogramma parasites (when required) once in 15 days.
180 days:	Detrash a second time.
210 days	Spray acetamiprid or thiamethoxam (0.2gm/litre of water) to control mealy bugs, white fly and scale insects.
251-360 days	Irrigate once in 15 days and stop irrigation 15 days before harvest.
Harvest	Cut canes at the bottom close to the ground with sickles or sharp knife. Remove trash, roots, water shoots and cane tops and send clean canes to the factory.

Note: Water Management – Irrigation gap has to be adjusted depending up on quantity of rain fall Reduce gap between irrigation in sandy soils and increase it in block soils.

IV. Package of Practice for Nagpur Mandarin Cultivation

Selection of site

Orchard: Soil should be well drained and of shallow or medium depth. Deep heavy soils having more than 60% clay contents are not suitable for citrus plantation. **Nursery:** Nursery should be located at least 500 meters away preferably on western side of the orchard to minimize incidence of insect pests and diseases.

Raising of citrus nursery

Sowing of Rootstock Seeds

Potting mixture of soil, sand and FYM or compost should be used in equal proportion (1:1:1) for filling of trays in primary nursery and polythene bags in secondary nursery. Before it is used for filling the bags/trays, the potting mixture should be solarised. For solarisation, it is spread on the concrete platform in 4" thick layer in the month of April – May, sufficiently moistened with water, then covered fully with the white polythene sheet, sealing its edges with soil. Then it should be left undisturbed for 11/2 to 2 months in the hot sun for solarisation.

- Only certified seeds of rough lemon or Rangpur lime rootstocks should be used.
- Shade dried medium size bold seeds of rootstocks should be treated with vitavax or thiram (@ 3g/kg seed) and sown on the raised beds or in plastic trays during September – October.
- Stagnation of water in beds should be avoided otherwise roots of young plants may start rooting.
- Uniform seedlings of medium height only be selected discarding either vigorous or the weak and dwarf ones while transferring to secondary nursery. Plants having hooked or bent roots should be discarded.
- Disease free budgrafts of nagpur mandarin

Plant Protection Measures

- Phytophthora infected plants must be eliminated. In case of phytophthora infection drenching of plants with either metalaxyl MZ72 @ 2.75 g/l water or fosetyl Al @ 2.5 g/l water should be done. Second spray should be given after 40 days.
- To prevent infestation of insect pests like citrus leaf miner and thrips plants should be sprayed either with acetamiprid or thiamethoxam (0.2gm/litre of water) at 10 days intervals.
- The growth of plants is also affected adversely due to mite attack which can be controlled by spraying plans with Fenazaquin 10% EC @ 4ml/litre and wettable sulfur @ 3 g/l water, alternatively at 15-20 days interval.
- In containerized nursery irrigation, fertilizer application, weed control, insect pest and disease control as well as cultural operations can be performed at ease.

Budding

- Budsticks should be used from the authorized and certified source only.
- Budstick should be drawn from the last years flush. The stick should have pencil thickness, be roundish and have whitish longitudinal streaks.
- Budstick should not be drawn from rubbery wood or kikarpani plants.
- Budding should be performed at 10" – 12" height on the rootstock seedling.



Disease Free Bud Grafts of Nagpur Mandarin

Orchard Establishment

Pits size

- Pits for planting should be 2'6" x 2'6" x 2'6" (75 x 75 x 75 cm) size and spaced at 6 x 6 m distance.
- To avoid soil borne fungi or nematodes soil of roots should be removed.

Pre-planting treatment for budlings

- Roots of budling should be dipped in the solution of metalaxy1MZ72 2.75 g for 10-15 minutes before planting.

Planting of budlings

- While planting care should be taken that rootstock union remains at least 6" above ground.

Manure and fertilizer application

- Nitrogen containing fertilizers should be applied in three equal splits in January, July and November months; phosphorus containing fertilizers in two splits in January and July months and Potassium containing fertilizers may be applied as singly dose in January.
- Surveys conducted by NRCC in Kalmeshwar, Katol, Narkhed, Saoner, Hingna and Ramtek tehsils of Nagpur district have revealed N deficiency in most of the orchards and P deficiency in leaf and soil of 50% orchards. Similarly, leaf and soil K was either at desired levels or even more than it. Citrus trees are nitrogen loving plants. They respond well to the applied nitrogenous fertilizers. During fruit development K may also be applied as it may fell deficient. Supplementary doses of P and K at 200 and 100 g/tree, respectively, may be included in the fertilizer package recommended for bearing orchard.

Fertilizer doses

Fertilizers /Age of tree	I Year	II Year	III Year	IV Year and Above
Nitrogen	150	300	450	600
Phosphorus	50	100	150	200
Potassium	25	50	75	100

As far as possible 1/3rd of the dose of N may be given through farm yard manure/compost, oil cakes etc.

Leaf sampling

For correct diagnosis of nutritional status use of correct sampling technique is very important. For this, it is important to know as to how many leaves, when, from which part of the plant and from how many trees should be sampled. In case of ambia bahar, 5-6 month old leaves in August-October and for mrig bahar 6-8 months old leaves in December and February should be sampled. As far as possible, the 2nd, 3rd of 4th leaf should be picked from the tip of the non-bearing shoot, preferably at 1.5 – 2 m above the ground and sampled.

Drip irrigation

With the help of a drip system of irrigation, the required quantity of water can be provided right at the feeder root system. Similarly, water-soluble fertilizers and micronutrients also can be given through the drip system. Water requirement of irrigation depends upon age of the tree and season of the year (Table 1). Mulching with drip irrigation maintains moisture in soil for a longer period.

Water requirement of the nagpur mandarin (litres / day / tree)

Month	Age of the three (years)									
	1	2	3	4	5	6	7	8	9	>10
January	7	15	22	30	44	62	72	82	92	102
February	9	20	30	40	60	82	96	101	121	137
March	12	26	40	53	78	109	127	145	163	181
April	14	29	43	63	87	123	143	163	183	204
May	17	34	52	74	102	143	166	188	211	235
June	11	22	34	48	67	95	110	126	142	157
July	8	18	26	41	56	79	92	105	118	131
August	7	14	23	34	42	60	70	80	90	100
September	8	15	25	36	45	65	76	87	98	108
October	9	17	27	40	52	79	92	105	118	131
November	8	15	25	36	45	63	74	85	96	150
December	6	11	19	24	35	49	57	65	73	82

Weed control

For effective and economic control of mono and dicotyledonous weeds, pre-emergence weedicides, diuron 3 kg at the end of May and 120 days thereafter should be done. For post-emergence weed control, glyphosate @ 4 l/ha should be sprayed on weeds before flowering.

Fruit drop

Fruit drop in citrus is of serious nature which occurs at least twice; i.e. the first time when the fruits are little more than the marble size and the second time when the fruits are fully developed or at the time of colour break. This drop is very serious in the ambia bahar crop pre-harvest fruit drop in

nagpur mandarin, which is called as pre-harvest fruit drop and is important from economic point of view to the orchardists. To control the fruit drop that occurs after fruit set, two foliar sprays of either 2,4-D or GA3 at 15 ppm + urea 1% and copper oxychloride[2] (0.3%) at monthly intervals in April – May are recommended. Same spray concentration is recommended for controlling pre-harvest drop in the months of September and October. 2,4-D and GA3 may be dissolved earlier in little quantity (30 – 40 ml for 1g) of some organic solvent such as alcohol or acetone before making the spray solution.

Control of Insect Pests

Blackfly (Kolshi): Nymphs of citrus blackfly attack the young flush, suck the sap and excrete sweet and sticky liquid, which favours rapid development of black sooty mould that covers entire plant surface. The process of photosynthesis is hampered greatly resulting in stunted growth of plants, low intensity of flowering, scarce fruiting which are insipid in taste and decline of citrus orchard sets. For control of blackfly two sprays of insecticides viz. acetamiprid 0.2 g/litre water should be given at 50% egg hatching stage that normally occurs in the II weed of July and the I week of December and April. One additional spray which targets the adult blackfly population when it is as its peak helps tremendously in controlling the pest.

Method of spraying: Spraying should be directed at the underside of the leaf, ensuring the complete drenching of the tree. For proper coverage and penetration of canopy, use of power sprayers for spraying operation should be envisaged. Insecticides should be used alternatively for better results.

Citrus psylla: Numerous young brownish nymphs of psylla are seen crawling on the young flush. Several dirty gray colour adults can be seen sitting in line with tails upwards. Voluminous desapping by the nymphs results into the drop of flush, flowers and berries. Affected branches dry and die-back sets in. The nymphs also excrete white crystalline powder, which invites fungal infestation. Psylla can be controlled by spraying acetamiprid 0.2 g/litre water twice with 10 days interval during the initial days of flushing. Bark eating caterpillar (Larva) are incidence of mites on fruits (Lalya).

Leaf miner: Serpentine mines are seen on the new leaves and also young stems are mined when the incidence is severe. At times death of young shoots may occur. The problem is quite serious in nursery and in young orchard. To control citrus leaf miners, spray either imidacloprid 5 ml (To be used in proper doses under guidance of an agriculture expert) and thiamethoxam 0.2g/litre water.

Bark eating caterpillar: The pest is noticed predominantly in the older and ignored orchards. The hanging wooden frass and tunnel at the joint of two branches during October – April indicates the presence of the pest. Larva remains hidden inside the tunnel during daytime and becomes active in the night and feeds on the bark nearby the tunnel. This results into snapping of food supply, ultimately yellowing of leaves on the branch and its slow decline. For control of the pest the wooden frass should be cleaned and each tunnel should be administered with Emamectin benzoate/spinosad @0.4 ml/litre water.

Fruit sucking moth: The moth attacks the ripening fruits during late hours in the evening. The moth punctures it making a hole in the ripening fruit to suck the juice through which an infection may take place. Soon rottening starts, leading to fruit drop. Collection of dropped fruits and their

destruction followed by smoking of orchard late evening hours is suggested. For fruit fly control, hanging of methl eugenol (feromone) traps is an effective method to check the pest.

Mites: Citrus rust mites attack in mrig bahar fruits especially during September – November. The fruit surface, particularly the side exposed to sun, is brushed and develops a big patch of dark brown colour – called 'Lalya' – only after 1 to 11/2 months. To check this, two alternating prophylactic sprays at 15-20 days interval with Fenazaquin 10% EC @ 4ml/litre and wettable sulphur @ 3 g/l water are recommended in September – October. Similarly, a phytophthora caused gummosis spray of acetamiprid/thiamethoxam @0.2g/litre water at the berry stage of the ambai fruits is required to protect the fruits from unpleasant scars.

Precautionary measures: Avoid water stagnation in orchard by providing channels along the slope for proper drainage. Moreover, pruning of intermingling branches to allow aeration and sunlight to prevent dampness in orchard are suggested. Presence of guava, pomegranate, chiku, mango trees, etc. near the orchard act as the alternate hosts for blackfly, therefore, such trees should also be covered with insecticidal sprays.

Control of Diseases

Twig Blight: Drying of fruit bearing branches after harvest is a common phenomenon. Removal of such dried shoots along with the 2 cm lower live part, followed by a fungicidal spray i.e. copper oxychloride or bordeaux paste application is recommended to check twig blight.

Gummosis, root rot and collar rot

- Proper diagnosis of the disease is must.
- Affected trees should be treated with metalaxyl MZ 72 @ 2.75 g or fosetyl AL @ 2.5 g/l water till drenching once in May – June. The second spray should be given after 40 days. The tree trunk and soil of the tree basin should also be sprayed/drenched.
- Removal of the rotten roots, cleaning the wound on the gum-oozing trunk with sharp knife and then pasting with metalaxyl[3] MZ 72 should be done.
- Apply bordeaux paste (1 kg CuSO4 + 1 kg CaOH + 10 l water) on tree trunk upto 2 to 21/2 ft from ground twice i.e. before monsoon (May) and after monsoon (October).

Precautionary measures

- Avoid flood irrigation
- Follow the double ring system of irrigation so that water does not come in contact with the tree trunk.
- Avoid deep ploughing under the tree to prevent damage/injury of the root system. Declining nagpur mandarin orchard rejuvenated orchard.

Pre-harvest spray of fungicide: Three pre-harvest sprays of fungicides like difenoconazole @2ml/litre water at 15 day intervals till drenching prevent pre-harvest fruit drop by 54%. It also controls post harvest diseases upto 70%. This spray remains effective for 3 weeks at normal temperatures.

Post-harvest fungicidal treatment: Fruits dipped in fungicidal solution difenoconazole for 5 minutes are safe to minimize rotting upto 70% and can safely be stored for 3 weeks under normal condition.

Precautionary measures

- Follow plant protection measures to keep bearing tree disease free and healthy.
- Post-harvest handling of fruits must be followed carefully to avoid any sort of injury of fruits.
- Avoid the use of copper fungicides like b,ordeaux mixture, blitox, phytolan etc. at this stage.

Rejuvenation of declining citrus orchards: Declining citrus orchards can be rejuvenated with the use of developed technologies and brought into productive stage.

Harvesting

- Traditionally fruits are harvested by twisting and pulling forcefully which may lead to a hole in the neck of the fruit or injury to the stem end. The fruits should be selectively harvested when ¾th of the skin turns yellow. TSS: Acidity ratio should not be less than 14 in both ambia and mrig bahar fruits. TSS should be at least 10%. Once this stage is reached, harvesting should not be delayed for colour development since other fruits may turn loose.
- Fruit should not be allowed to come in contact with soil and straw and also should not be exposed to hot sun.
- Packing should be done immediately after harvesting.
- For waxing on a large scale packing line can be used. Washing and waxing (stay fresh high shine was 2.5 g + Tebuconazole) and grading of fruits is also done automatically at the end.
- Graded fruits are packed in corrugated fibre board boxes (50 x 30 x 30 cm) which are telescopic with holes on both the sides covering 5% of total side portion. To keep the fruits safe from moisture in the store house, boxes should be externally laminated with plastic.
- For small scale (1 – 1.5 tonne) and short duration (20 – 25 days) storage an evaporative cool chamber is recommended, costing Rs.10,000/- to 12,000/-.
- For delaying or postponing the harvest of the fruits, two sprays of GA3 (100 mg / 10 litre of water) in 15 day intervals at the point of the colour break are recommended.
- For long duration cold storage, a temperature of 6-7 0 C and humidity 90-95% is desirable. In such a situation, citrus fruits should not be stored for more than 45 days. To avoid chilling injury, care should be taken that the temperature does not go down upto 40 C.

2.6. Lessons Learnt

1. Dark coloured soils are more fertile compared to light coloured soils. Neutral pH is more suitable for majority of crops. Lime is used for amending acid soils and gypsum for alkaline soil.
2. Soil testing is essential for judicious application of fertilisers based on nutrients status in the soils.
3. Integrated Nutrient Management (INM) practices are useful for efficient, economic and sustainable production.
4. Package and practices recommended by research stations may be adopted in totality

3. Plant Protection

3.1. Objectives of the session

- To increase the awareness and understanding about the crop pest, diseases and weeds.
- To strengthen the farmer's knowledge on effective management of insects, diseases and weeds in crops through Integrated Pest Management.
- To sensitise farmers on safe handling of chemicals.

3.2. What we know at the end of the session

- Insects and their life cycles
- Methods of insect control including Integrated Pest Management (IPM)
- Plant protection equipments
- Symptoms of major diseases
- Integrated disease management
- Major weeds
- Methods of controlling weeds
- Safe handling of chemicals

I. Pest Management

3.3. Crop pest and their importance

Pest is any organism which is detrimental to crop production. Pest cause damage to the plant to the extent of 30 - 90 per cent, sometimes it even causes total loss. Pest includes insects, diseases and weeds. Non insect pest includes nematodes, snails and rodents.

Insect is any of many small invertebrate animals having a segmented body and three pairs of legs and usually two pairs of wings. Some insects are beneficial and some are harmful to agriculture.

What is the difference between complete and incomplete metamorphosis in insects?

Incomplete and complete metamorphosis differs in the number of life cycle stages that insects go through during their transformation from egg to adult. The complete metamorphosis has 4 life cycle stages and an incomplete metamorphosis has 3 life cycle stages.

Complete metamorphosis

Complete metamorphosis has four distinct life cycle stages: **egg, larva, pupa and adult.**

Examples of insects that go through complete metamorphosis are butterflies, silkworms, mealworms and ladybugs. The larva can be worm-like even though the six legs are still visible. The larva form moths and butterflies are called caterpillars. Maggots are the larval stage of flies. The larvae eat constantly and grow rapidly. A hard, protective case forms around the larva at pupa stage. The pupa stage for a butterfly is called a chrysalis. The pupa stage for moth is called cocoon.

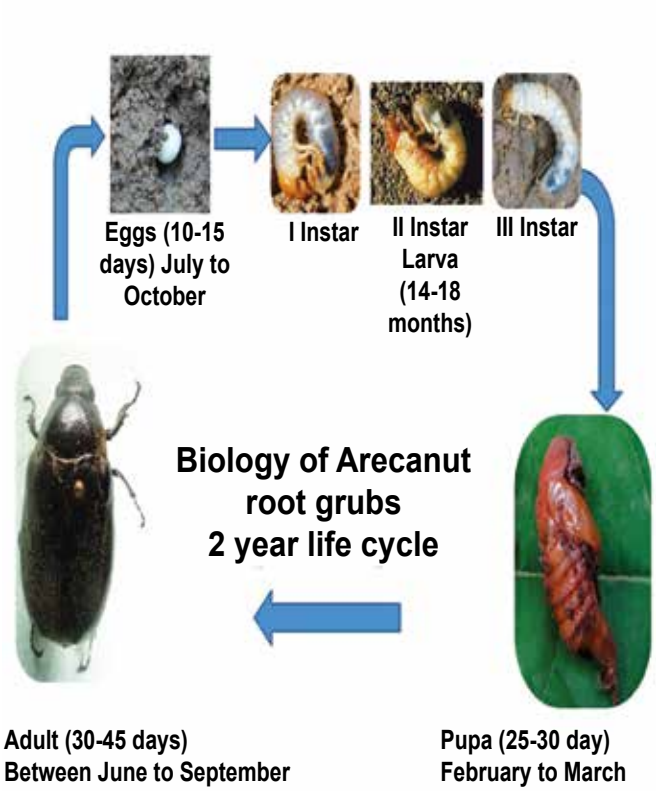
Incomplete metamorphosis

Incomplete metamorphosis has three life cycle stages: **egg, nymph and adult**. The nymph looks similar too but is a smaller version of the adult. The nymph is also wingless. Examples are grasshoppers, cockroaches, ants and praying mantids.

3.4. Life stages of insects

- **EGG** is the initial stages of the insect. Normally an insect lays at least 30 to 300.
- Egg hatch into **LARVA** or worms. Larva is the

damaging stage of insect to any crop. This stage is normally seen in the field.



Life stages of insects

- **PUPA** is an inactive stage preparing itself to develop to an adult.
- Mostly **ADULTS** insects are harmless but many bugs and beetles are harmful to plants.



Mite damage in coconut buttons



Larva damaging leaves



Bollworm damaging cotton



Semilooper feeding on caster

How the damage is visible?

- The larva eats the leaf, fruits or the whole plant parts. Hence the damage is visible.
- Some insects scrape the plant tissues. They also cut the growing parts. Beetles, bugs, thrips and hoppers usually suck the sap from the plants and growing parts, affecting the healthy plant.



Grub of green lace wing



Ladybird beetle

- Not at all. HONEY BEES are also insects which help in pollination (brings pollen from one plant to another), thereby, increasing the yield.

They also provide us valuable HONEY and other products. Another example of "friendly" insects are SILK WORMS.

- NATURAL ENEMIES are insects that are beneficial to man since they feed from the egg or larvae and pupae of crop pest. They are called biological control agents. Examples are the trichogramma parasite for sugarcane borers, crub of green lacewing and ladybird beetle.

3.5. Insect classification

Insects can be classified into 3 groups, depending on their behaviour in the farm such as

- Pests
- Beneficial insects
- Neutral insects

Pest: Whether an insect species is a pest or not depends on the situation. This means that a certain insect could be a pest in one situation but not in another situation. For example, the caterpillars of diamondback moth feed on cabbage and other plants of the cruciferae family. A farmer who grows cauliflower will therefore consider it a pest. Yet, for a farmer who grows potatoes or bananas, the diamondback moth is a neutral insect. When there is nothing to feed it will not even occur. In a paddy crop, black gram is a weed/pest since it is growing unnecessarily. In a black gram field, cow pea is a weed since it can grow. Pest will occur only if there is a host.

Beneficial insects

Some insects are beneficial to the farmer, because they are the natural enemies of harmful insects. Predators feed on other insects and hence control the pest. For example, the assassin bug kills caterpillars and ladybird beetles feed on aphids. Some other insects are beneficial as they help in pollination of plants, e.g. honey bees. There are commercially beneficial insects such as silkworm, which produces silk.

Neutral insects

A neutral insect is neither a pest nor beneficial. Yet again, it really depends on the context. A mosquito in the rice field can be considered as neutral insect.

3.6. How to control pest

Natural control

Understanding insect life cycle: Insects multiply in large numbers. However the survival rate is very less as the nature maintains insect population. Factors like extreme temperature, heavy rains, heavy wind, water stagnation, birds, lizards, spiders, animals, other insects and diseases control the pest. Natural barriers like sea, river, lake, mountains, etc. prevent movement and spread of pest.

Mechanical control: Some of the recommended practices are

- Removal of affected parts
- Collection and destruction of insects
- Drying of seeds
- Tar coating of trees to protect from termites
- Provision of barriers to prevent the entry of pests like green house / screen house, covering of pomegranate fruits with butter paper, etc.
- Clipping off the withered shoots

Agronomical methods: Recommended cultivation practices are as follows.

- Summer ploughing: Opens up the soil and exposes pest to hot sun and predators.
- Trap crop: Is growing the most favoured crop of the insect along with the main crop. The insect feeds from the trap crop and the main crop remains not unaffected. Growing of resistant varieties also prevents the pest attack.

Main Crop	Trap Crop
Tomato	Marigold
Cotton	Okra, Castor, Onion, Garlic
Maize	Sorghum

- **Mixed cropping:** Is growing more than one crop.
- **Inter cropping:** Growing another crop along with main crop, which increases the population of natural enemy.

Main Crop	Intercrop
Cabbage	Tomato, mustard
Cotton	Black gram, green gram
Maize	Sorghum

- **Crop Rotation:** Is growing of different crops in

sequence instead of one single crop.

- **Keeping the fields clean:** Managing the weeds which provides home for pests, treating the seeds with pesticides, growing the crops in seasons where pest incidence is less, application of correct dose of fertilizer at correct time, optimum use of water are some of the agronomical methods to control pests.



Ladybird beetle

Pest load and monitoring

Pest monitoring: Is the practice of examining the crop to know whether the pest has affected the crop and the extent of damage to decide whether to spray the pesticide or not.

Economic Threshold Level (ETL): ETL is the pest population density at which the control measure has to be taken up to prevent the pest from reaching economic injury level.

ETL for some of the crops are as follows

- Brown plant hopper in paddy - nymph or adult hopper 5-10 / hill
- Leaf miner in groundnut: 2 larvae / 10 plant or 20-30% plant infestation
- Whitefly in cotton: 5-10 adults / leaf or 20 nymphs / leaf

Biological control methods: There are several biological agents that controls pest such as

- The insect that kills another insect by living inside its body is called Parasite. There is egg, larval and pupal parasites. Trichogramma is an egg parasite used commercially on sugarcane borers.
- Predators are insects that eat other insects. For example, the Ladybird Beetle, dragonfly, damselfly, etc. Insects are also affected by many diseases. Some are fungal, bacterial and viral. Verticillium, a fungal agent, bacillus, a bacterial agent, and NPV (Virus) are commercially used.

Physical Control

- An age old practice is to mix pulses with red earth to protect pulses from pulse beetle.
- Drying of seeds in hot sun and using radiation are some of the physical control methods.

Traps

- **Light Trap:** Most of the insects are attracted towards light. This principle is used to monitor and control pests.
- **Pheromone Trap:** Insects are trapped using the scent of one sex. This is commonly used for the control of cotton bollworms.
- **Yellow Sticky Traps:** Some of the small sucking insects are attracted by yellow colour. Hence yellow coloured containers smeared with sticky materials are kept inside the field to attract sucking insect.

Chemical control

Manual pesticide spraying

- Pesticide application is the last resort to control pests.
- Pesticides to be selected carefully to control the pests.
- Dose and spraying equipments should be selected carefully.
- Get the correct recommendation from the extension worker.
- Understand the difference between chemical name and trade name. Trade name is the trader given name but the chemical name is originated based on the chemical ingredient.

Botanicals: are plant origin pesticides like neem based formulations. We can also use neem seed kernal extract or neem oil along with soap as pesticides. Commercial botanicals are available in the market and botanicals can be prepared, which cost less and are eco friendly.



Trap

For example, Neem Seed Kernal Extract (NSKE) 5%, Gronim, Achook and Neemazal.

3.7. Pesticides

Pesticides are the chemicals used to control pests. The pesticides can be broadly classified into three groups based on how they act on insects.

Contact and stomach poison: When insect comes into contact or when the insect eats the pesticide sprayed parts, it gets killed. Contact and stomach poison is used for controlling larvae that feed on leaves. Examples include Flubendamide. Some of the pesticides derived from plants also have contact action, for example pyrethrum, sabadilla, etc.



Manual Pesticide Spraying

Systemic poison: When the chemical is sprayed on the plant, it is absorbed by the plant and transferred to the entire plant system. Sucking type of pest like aphids, leafhoppers thrips suck the sap from plants. This kind of sucking pests can be controlled by systemic poison like Imidacloprid (To be used with caution), flubendamide, acetamiprid, thiamethoxam, etc.

1) **Fumigant:** Forms vapour and acts on breathing system of the insects. When insects breathe it, they get killed. Examples include, hydrogen cyanamide, sulfuryl fluoride, etc.

Insecticides are poisons which cannot be used directly. They need to be used as per directions given on the container or as per the recommendation of extension workers. Insecticides are available in different forms of which some are:

i) **Dust (D):** Poisons are mixed with gypsum, talc or clay so that it can be used in powder form. There is no need to mix with water.

ii) **Wettable Powders (WP/WDP):**Mixed with dry fillers with sticking agents but these dry powders can be mixed with water.

iii) **Granules (G):** Dry formulations in which poisons are mixed with calcium or gypsum in granulated form. Main merit of this formulation is handling will be easy, it is not carried away by the wind and it is less toxic to plants.

iv) **Liquid forms, Soluble Liquids (SL), Emulsifiable Concentrates (EC), and Soluble Concentrates (SC):** Liquid forms of pesticides. Since the pesticides are directly soluble in water, they are mixed with organic solvents, EC formulations disperse in water easily as they are mixed with emulsifiers.

v) **Fumigants:** are the poisons in gas form. Normally it is used to fumigate godowns, grains, storage rooms and ships even for rat control.

vi) **Poison baits:** Poisons mixed with food material. The latter acts as attractants. Baits are exclusively used to control rats. As the rats are more sensitive, pre baiting is necessary.

3.8. Types of chemicals

- Organochlorines, Organophosphates, Carbamates, Synthetic Pyrethroids are types of chemicals which were used before, but are not recommended now.
- New Molecules: Invention is a continuous process to identify more effective and less hazardous chemical search for better insect management. Examples include Nicotinamides, spinozids, triazoles (Hexaconozole, Propiconozole).

3.9. Sprayers/Dusters:

Types of applicators

Based on what to apply

- Sprayers are used to sprinkle the soluble chemicals.
- Dusters are used to spread the dust formulation of the pesticide.

Based on power source

- Manually operated sprayers/duster
- Power operated sprayers/duster

- Fuel operated
- Battery operated
- Solar panel powered operated
- Self propelled; i.e. they have their own power source for movement
- Normally 50 -100 litres of spray fluid are required for an acre. If we use manually operated sprayers 200-250 litres of spray fluid are required to cover an acre.
- There are a number of models with a variety of features available in the market. One can choose the model depending upon the requirement. Each instrument has got its own merits and demerits.
- Nozzle is the terminal part of the sprayer, which delivers the spray liquid to the plants. Different types of nozzles are available for different purpose.

Types of sprayer



Battery Operated Sprayer

Capacity 16 litres
Battery 12V/8AH (Fuse:6A)
Maintenance free Power Battery
Charger: Input: 220V/50HZ
Output: 12V/1000MA
Nozzle: Single, Double



Rocker Sprayer:

A minimum of 3 persons are necessary. One person to use the lever to create pressure and other persons to spray.



Motorized Power Sprayer is used for spraying of vegetable and ornamental crops in a large area.



Battery Operated Power Sprayer meant for use in the field.

3.10. Integrated Pest Management (IPM)

"IPM is a sustainable approach for managing pests by combining biological, cultural, physical and chemical tools in a way that minimizes economic, health, and environmental risks."



3.11. Some insecticidal materials for common household use

Kerosene emulsion

This is a contact insecticide useful against many sucking insects. Finely divide 500 g of ordinary bar soap and dissolve it in 4.5 litres of boiling water. Cool and add 9 litres of kerosene. The mixture is then vigorously agitated until the oil is completely emulsified. The stock solution can be diluted with 15-20 times of water before spraying.

Tobacco decoction

This is very effective for controlling aphids infesting vegetable crops. Tobacco decoction can be prepared by steaming 500 gm of tobacco in 4.5 litres of water for 24 hours. Then, 320 gm of ordinary sliced bar soap is dissolved separately in another vessel. The soap solution is added to tobacco decoction and the stock solution is diluted 6 - 7 times.

Neem seed suspension

This is very effective as a repellent against locusts and grasshoppers. Kernels of mature neem fruits should be crushed into a coarse powder. For obtaining 0.1% concentration, 1 g of powdered neem seed is required for a litre of water. The required quantity of the coarse powder should be put in a small bag of muslin cloth and dipped in water contained in a bucket and squeezed till the water becomes light brownish. This has to be sprayed on crops.

The latest registered chemicals are available in www.cibrc.nic.in or please verify with extension officer.

II. Disease Management

3.12. Disease

Disease is an impairment of the normal state of a plant that interrupts or modifies its vital functions. All species of plants, wild and cultivated alike are subject to disease. Although each species is susceptible to characteristic diseases, these are, in each case, relatively few in number. The occurrence and prevalence of plant diseases vary from season to season, depending on the presence of the pathogen, environmental conditions, and the crops and varieties grown. Some plant varieties are particularly subject to outbreaks of diseases.

Disease of crops and their importance

- Plant become diseased when it is continuously disturbed by some causal agent including an abnormal process that disrupts the plants.
- There are more than 80,000 plant diseases.
- Diseases reduce the yield of the crops and sometimes lead to disaster e.g. late blight of potato, Panama of banana, etc.
- Managing outbreak of diseases is challenge to the farmer.

Causes of plant disease

Infectious diseases: caused due to fungi, bacteria, viruses, nematodes, etc.

Non-infectious diseases: caused due to unfavourable extraneous condition such as scorching sunlight, high temperature, moisture stress or deficiency of micronutrients, pH, heavy metal toxicity, atmospheric pollution, etc.

3.13. The disease cycle

The main events of stages comprising the disease cycle include the following: production and dissemination of the primary inoculum, primary infection, growth and development of the pathogen, secondary infection and over wintering.

The primary inoculum is the part of the pathogen (that is, bacterial or fungal spores or fungal mycelium) that over winters (over-seasons) and causes the first infection of the season, known as primary infection. In general, the greater the amount of inoculum and the nearer it is to its host, the greater the potential for a disease epidemic.

Dissemination refers to the spread or dispersal of the pathogen from an inoculum source to a host. Dissemination can occur by wind, splashing rain, insects, infested pruning tools, infected or infested transplants, and other means. Spread can occur over short distances within the tree canopy or from distant sources.

Primary infection occurs when the pathogen comes into contact with a susceptible host under favourable environmental conditions. Pathogens penetrate the surface of a plant directly or enter through wounds or natural openings.

Growth and development of a pathogen usually occurs on or within infected plant tissue.

Secondary infection results from spores or cells produced following primary infection or from other secondary infections. The secondary infection cycle can be repeated many times during the growing season. The number of cycles is dependent on the biology of the pathogen and its host and the duration of environmental conditions needed for infection.

Over wintering or over seasoning is the ability of a pathogen to survive from one growing season to the next. Pathogens of apple survive the winter in a number of different ways.

For a plant disease to occur, a susceptible host, a pathogen (casual agent), and favourable environmental conditions must be present and interact with one another. If any one of these requirements is not met, a plant disease will not occur. At present, our ability to manipulate the environment is limited to only a few practices such as pruning to promote drying, bedding to improve soil drainage, and scheduling of irrigation. Severe disease outbreaks can be prevented by manipulating the host - through the use of resistant cultivars - and the pathogen -- through cultural practices and fungicidal or bactericidal sprays.

3.14. Disease control measures of important crops

Crops	Disease	Symptoms	Control Measures
Banana	1. Sigatoka leaf spot	On leaves small light yellow or brownish green narrow streaks appear. They enlarge in size becomes linear, oblong, brown to black spots with dark brown brand and yellow halo. Black specks of fungal fructification appear in the affected leaves. Rapid drying and defoliation of the leaves.	Remove affected leaves and burn. Spray any one of the following fungicides commencing from November at monthly interval-difenoconazole (2ml/litre), azoxystrobin (2ml/litre), Copper oxychloride 2.5 g/lit. Alternation of fungicides for every spray prevents fungicidal resistance. Always add 5 ml of wetting agent like Sandovit, Triton AE, Teepol etc. per 10 lit of spray fluid.
	2. Anthracnose	The skin at the distal ends of the fingers turn black, shrivels. The fungus produces masses of conidia which form a pinkish coat. The entire fruit and bunch is affected in severe cases. Sometimes main stalk of bunch diseased. The bunch becomes black and rotten. Acervuli produces cylindrical conidiophores, hyaline, septate, branched. Conidia hyaline, non-septate, oval to elliptical.	Spray copper oxychloride 0.25% or Bordeaux mixture 1%. Post harvest dipping of fruits in difenoconazole (2ml/litre).
	3. Bunchy-top	Dark broken bands of green tissues on the veins, leaves and petioles. Plants are extremely stunted. Leaves are reduced in size marginal chlorosis and curling. Leaves upright and become brittle. Many leaves are crowded at the top. Branches size will very small. If infected earlier no bunch will be produced. The disease is transmitted primarily by infected suckers. Secondary spread is through the aphid vector.	Spray Methyl Demeton 2 ml/lit to control it. The sprays may be directed towards crown and pseudostem base upto ground level at 21 days interval atleast thrice.

Crops	Disease	Symptoms	Control Measures
Banana cont...	4. Panama disease	Yellowing of the lower most leaves starting from margin to midrib of the leaves. Yellowing extends upwards and finally heartleaf alone remains green for some time and it is also affected. The leaves break near the base and hang down around pseudostem. Longitudinal splitting of pseudostem. Discolouration of vascular vessels as red or brown streaks. The fungus spreads through use of infected rhizomes. Continuous cultivation results in build up of inoculum.	Uproot and destroy severely affected plants. Apply lime at 1 – 2 kg in the pits after removal of the affected plants. In the field, Panama wilt disease can be prevented by corm injection methods. A small portion of soil is removed to expose the upper portion of the corm. An oblique hole at 45° angle is made to a depth of 10 cm. Capsule application for 50 mg of Pseudomonas fluoresces is injected into the hole with the help of 'corm injector' on 2nd, 4th and 6th month after planting.
Mango	1. Powdery mildew	It attacks the leaves, flowers, stalks of panicle and fruits. Shedding of infected leaves occurs when the disease is severe. The affected fruits do not grow in size & may drop before attaining pea size.	Application of Sulphur dust (350 mesh) in the early morning will protect new flush or spraying Wettable sulphur 0.2% will control powdery mildew.
	2. Anthracnose and stalk end-rot	Produces leaf spots, blossom blight, wither tip, twigs blight and fruit rot. Small blister like spots develop on the leaves and twigs. Young leaves wither and dry Tender twigs wither and die back symptom appears. Affected branches ultimately dry up. Black spots appear on fruits. The fruit pulp becomes hard, crack and decay at ripening. Infected fruits drop.	Pre-harvest spraying of Thiophanate methyl 1g/lit 3 times at 15 days interval will control anthracnose and stalk end-rot.
	3. Sooty mould	The fungi produce mycelium which is superficial and dark. They row on sugary secretions of the plant hoppers. Black encrustation is formed which affect the photosynthetic activity. The fungus grows on the leaf surface on the sugary substances secreted by jassids, aphids and scale insects.	Spraying thiamethoxam @ 2 ml/ litre + Maida 5% (1 kg Maida or starch) boiled with 1 lit of water and diluted to 20 litres will control the incidence of sooty mould. Avoid spraying during cloudy weather.

Crops	Disease	Symptoms	Control Measures
Mango cont...	4. Mango malformation	The dark epicarp around the base of the pedicel. In the initial stage the affected area enlarges to form a circular, black patch. Under humid atmosphere extends rapidly and turns the whole fruit completely black within two or three days. The pulp becomes brown and somewhat softer. Dead twigs and bark of the trees, spread by rains.	Apply plant growth regulators (NAA/ GA/Ethephon @ 50-200ppm) at Bud Inception stage. Harvest mangoes on clear dry day. Injury should be avoided to fruits at all stages of handling.
Cotton	1. Seedling diseases (seed-rot, rootrot, and damping off	Seed-rot, root-rot, pre emergence and-post emergence damping-off.	Fungicide seed treatments help control seed rots and some pre emergence damping off. However, an additional soil treatment of fungicide must be used to control root-rots and most damping-off. In addition, producers must follow all other recommended cotton production practices to decrease seedling diseases. Some of these practices include use of correct planting equipment and date of planting, good seed bed preparation, correct use of herbicides and insecticides and use of high germinating seed.
	2. Fusarium wilt	Plants become stunted, yellowed, followed by defoliation. Yellowing first occurs around leaf edges and advances inward. Cross sections of infected stems usually reveals a brown Discoloration which is more intense in outer layers of tissue. Infected plants fruit earlier and produce smaller boll.	Reduce nematode population. Crop rotations. Use resistant varieties.
	3. Boll rot	Boll rots usually first appear as water soaked spots. Later, as infection-spreads, bolls turn black and may be-covered with a moldy fungus growth. Badly infected bolls may drop from-plant.	Avoid excessive rates of nitrogen. Practice skip-row planting. Timely defoliation will reduce boll rots. Reduce insects which injure bolls. Growth regulators such as Pix can be used effectively to reduce boll rots.

Crops	Disease	Symptoms	Control Measures
Cotton cont...	4. Leaf spot	Various types of leaf spots and blights. Many spots occur on leaves toward maturity, but these are not usually damaging to the plant at this stage of growth.	Use fungicide seed treatments. Destroy crop residues. Use crop rotations and plant resistant varieties when available (esp. when Bacterial Blight is severe). Keep potash levels at least medium to high.
	5. Verticillium wilt	Seedlings may become infected and turn yellow, dry out and die. Plants that become infected later in the season are stunted and exhibit a yellow condition along leaf margins and between the major vein. Severely affected plants will shed their leaves. A brown Discoloration of the interior of the stem can usually be found later in the season. This discoloration is. Distributed evenly across the inside of the stem.	Plant resistant varieties when Verticillium Wilt is severe. A variety that matures very early may in some years escape injury from Verticillium Wilt.
Citrus	1. Root rot, foot rot and gummosis	Rotted roots, cracked bark, accompanied by gumming Water-soaked, reddish-brown to black bark at the soil line Discoloured tissue in the lower trunk; yellowing, sparse foliage and death of the tree.	Two sprays with drenching either by Fosetyl-Al (2.5g/L)or Metalaxyl MZ-72(2.75g/1 water covering the whole plant canopy and basin of affected plant at 40 days interval after onset of monsoon provided significant control .For the control of gummosis, scraping of the affected parts followed by application of Metalaxyl[2] MZ-72 paste.
	2. Citrus canker	Disease affecting citrus species that is caused by the bacterial Infection causes lesions on the leaves, stems, and fruit of citrus trees. While not harmful to humans, canker significantly affects the vitality of citrus trees, causing leaves and fruit to drop prematurely.	Pruning and destruction of infected twigs followed by three to four sprays with copper oxychloride (COC) 0.3% + streptocycline 100 ppm at monthly intervals after the onset of monsoon.

Crops	Disease	Symptoms	Control Measures
Citrus cont...	3. Citrus decline	Symptoms vary with the cause of the malady. The affected trees do not always die completely, but remain in a state for decandance and unproductive for a number of years. Some-times they may suddenly wilt and die in a day or two. In early stages, Symptoms are restricted to a few limbs, but eventually the whole tree is involved. Trees show sparse mottling leaves, stunted growth, sickly appear-ance. Midrib and lateral veins of old, mature leaves turn yellow with in-terveinal areas along the veins showing diffuse yellowing. Leaves may turn yellow and are shed with the onset of summer or autumn and the die-back of twigs starts. Dead shoots stand out prominently and may be found dead right down to the main trunk. The entire tree bears short twigs carrying narrow small leaves on their lower portion. Subsequent secondary growth consists of short, upright small, weak shoots showing a variety of discolouration of leaves. Often these leaves have green veins of green blotches. Occasionally, small, circular, green spots appear on yel-low tissue on leaves. The die-back of weak shoots continues. There is exces-sive flowering, but the fruits are not carried to maturity. The fruits show distinct sun-blotching. The feeder root system becomes depleted, roots turn black and sometimes are covered with rotting bark. Either only a few trees or entire orchard may be affected.	Good cultural practices, improvement in soil fertility and drainage, control of insect pests, nematodes, etc. may be useful to minimize the incidence of decline. Use of resistant rootstocks and certified budwood for propagation is also useful.

3.15 Concept of Seed Treatment

The concept of seed treatment is the use and application of biological and chemical agents that control or contain primary soil and seed borne infestation of insects and diseases which pose devastating consequences to crop production and improving crop safety leading to good establishment of healthy and vigorous plants resulting better yields.

The benefits of seed treatment are as follows:

- Increased germination
- Ensures uniform seedling emergence.
- Protect seeds or seedlings from early season diseases and insect pests improving crop emergence and its growth.
- Use of plant growth hormones may enhance crop performance during the growing season.
- Rhizobium inoculation enhances the nitrogen fixing capability of legume crops, and their productivity.
- Improved plant population and thus higher productivity.

Pest/Disease	Seed Treatment	Remarks
Root rot, wilt	Trichoderma spp. 4-6 gm/kg. seed.	For seed dressing metalseed dresser / earthern pots or polythene bags are used.
Root rot disease	Trichoderma 5-10 gm/kg. seed (before transplanting).	-do-
Bacterial sheath blight	Pseudomonas flourescens 0.5% W.P. 10 gm/kg.	
White tip nematode	Seed soaking in 0.2% solution.	-do-
Anthracnose spp. **Damping off**	Seed treatment with Trichoderma viride4g/kg.	-do-
Soil borne infection of fungal disease	Trichoderma viride @ 2 gm/kg. seed and Pseudomonas flourescens,@10gm/kg. Captan 75 WS @ 1.5 to 2.5 gm a.i./litre for soil drenching.	-do-
Jassid, aphid, thrips	Imidacloprid 70 WS @ 10-15 gm a.i./kg seed (To be used in proper doses under guidance of an agriculture expert).	
Wilt, blight and root brot	Trichoderma spp. @ 4 gm/kg. seed.	For seed dressing metal seed dresser/earthern pots or polythene bags are used.
Root knot nematode	Paecilomyces lilacinusand Pseudomonas fluorescens @ 10 gm/kg as seed dresser.	- do-
Soil borne infection of fungal disease **Early blight** **Damping off** **Wilt**	T. viride @ 2 gm/100gmseed. Captan 75 WS @ 1.5 to 2.0 gm a.i./litre for soil drenching. Pseudomonas fluorescens and V. clamydosporium @ 10gm/kg as seeddresser	For seed dressing metal seed dresser/earthern pots or polythene bags are used.

Bacterial wilt	Pseudomonas fluorescens @ 10gm/kg.	-do-
Pest/Disease	**Seed Treatment**	**Remarks**
Seed rot	Trichoderma viride @ 6 gm/kg seed.	-do-
Jassids, whitefly	Thiamethoxam (0.2gm/litre of water)	
Termite	Treat the seed before sowing with any one of the following insecticides. i) Chlorpyriphos @ 4 ml/kg seed (Best available option and use with caution).	For seed dressing metal seed dresser / earthern pots or polythene bags are used.
Bunt/false smut/ loose smut/covered smut	Carboxin 75 % WP. Tebuconazole 2 DS @ 1.5 to 1.87 gm a.i. per kg seed. T. viride 1.15 % WP @ 4 gm/kg.	
Wilt and damping off	Seed treatment with Trichoderma viridi 1% WP @ 9 gm/kg seeds.	
Soil and tuber borne diseases	Seed treatment with boric acid 3% for 20 minuts before storage.	

3.16. Nematode management

Nematodes are thread-like roundworms invisible to the naked eye. Species parasitic on plants attack roots and other plant parts, causing stunting and yield reduction. Nematode-infected plants are not only weakened but their root systems are more susceptible to secondary infections by fungi or bacteria.

Correct identification is the first step when a nematode problem is suspected. The second step is to determine whether populations are high enough to threaten the crop. Root knot nematodes, the most common pathogenic nematodes in vegetables, cannot penetrate roots when soil temperatures are below 50 degrees F, and will not reproduce when soil temperatures are below 58 degrees F. Their reproductive rate is slower at cooler temperatures, so populations build up more slowly. Thus, cool season crops are less likely to be damaged. E.g., early spring potatoes, are rarely damaged by nematodes.

Nematode management practices

Isolation: Once a nematode problem is confirmed, affected areas and plants should be isolated because transplants, machinery and irrigation water can all spread nematode infections. From initially small-infested areas, nematodes can spread across a field at a rate of 3 feet per year.

Crop rotation and cover crops: Crops susceptible to root knot nematodes include all cole crop species, beans, cucumber, muskmelon, watermelon, bendi, potato, sweet potato and tomato. All potatoes are susceptible to nematodes except for a few cultivars resistant to the golden nematode.

Rotation to non-host crops such as corn, cucurbits, potatoes and tomatoes is an effective control for the cyst nematode. But is less likely to control the root knot nematode because of its wider host range. All species of Meloidogyne are called 'root knot' nematode but each species has a different host range, causing confusion over which crops or cultivars are resistant or tolerant to which species of root knot nematode. Rotations to non-host crops for more than a year reduce populations below damaging levels but will not eliminate them.

Cover crops: significantly reduce subsequent damage to crop.

Increasing Soil Organic Matter: Higher soil organic matter content protects plants against nematodes by increasing soil water-holding capacity and enhancing the activity of naturally-occurring biological organisms that compete with nematodes in the soil.

Fallow Period: A fallow period of two years with no susceptible plants in the field decreases nematode populations. *Marigold* as a rotation crop suppress nematodes.

Plant resistance: Nematode resistant cultivars may be used to reduce the incidence of nematodes

Symptoms

- Understanding symptom and description of disease will help in identification at field level.



Blast infected leaf of paddy



Bacterial blight of paddy



Altenaria leaf spot of redgram



Wilt affected redgram plant



Red rot in sugarcane



Ratoon stunting in sugarcane



Anthracnose in cotton



Gray mould in cotton

3.17. Control measures - Tips for the farmers

- Correct identification of disease in your farm is essential for effective control of disease.
- With little experience, you can identify the disease. However you can contact agricultural officers of your area along with disease specimen and seek their help in identifying the disease.
- You can also give disease-affected plant parts to the Agri clinics for clinical test before undertaking control measures.
- Follow Integrated Diseases Management such as host plant resistance, agronomic practices, judicious use of fungicides, pesticides for vector control, bio-pesticides for pathogen control etc., as indicated below.

3.18. Integrated disease management practices in the field

- Select varieties and hybrids resistant to the most common or economically important diseases in consultation with agricultural officers of your area.

For example:

Diseases of Cotton	Resistant Variety/Hybrid
Verticillium Wilt	MCU 5 VT, Surabhi, Savitha (Hybrid)
Bacterial Leaf Blight	MCU 10, L 604, L 389

- Plant only good quality, disease-free seed having good germination. E.g. Cotton seed having above 80% germination will have vigorous growth and will not suffer from infection of soil borne diseases
- Use seed-treatment with fungicides to control diseases.
- Plant when soil temperature and moisture are most favourable for specific crop. E.g: If the farmers take up sowing of cotton during the warmer temperature (>65 F), there will be better germination and growth.
- Avoid planting the same crop in a field year after year. E.g.:
  I) Growing paddy in the Veriticillium wilt infested field will reduce the incidence of microbial population in the soil.
  II) Growing Chrysanthemum will inhibit the growth of the Veriticillium in the soil.
- Incorporate the crop residues of the previous crop by tilling well before planting season. E.g. Disease affected plants should be burnt immediately.
- Choose right sowing time and maintain appropriate plant population by adopting recommended spacing.
- Apply a balanced fertilizers based on a soil test. E.g.: Potassium deficiency leads to susceptibility of Alternaria leaf spot.
- Apply recommended amount of Farmyard Manures/compost at regular intervals and maintain soil-organic matter content. E.g.: Over dose of chemical fertilisers lead to more vegetative growth and more disease.
- Enrich soil with beneficial micro-organisms like Trichoderma.
- Keep the land weed free. Weeds can serve as alternate hosts for pathogens and helps disease development.
- Timing and duration of irrigation should match the crop and water requirement without allowing for excess water. E.g.: Excessive irrigation favour soil borne pathogen
- Maintain insect population below ETL to reduce the incident of disease transfer by insects.
- Regular crop monitoring is essential for effective disease management.
- Use bio-pesticide as far as possible to control disease. Chemical may be used only as last resort.

III. Weed Management

3.19. Weed and its relevance in crop production

- Weeds are the plants, which grow where they are not wanted
- Weeds compete with crops for water, soil nutrients, light and space
- Weeds reduce crop yields to the extent of up to 50 percent
- Critical period of weed competition is approximately 1/3rd of the duration of the crop

Characteristics of weeds: "One year seeding, seven years weeding"

- Produces larger number of seeds compared to crops. E.g.: Amaranthus retroflexus produces 1,96,405 seeds/plant, whereas wheat & rice produces only 90 to 100 seeds/plant
- Most of the weed seeds are small in size
- Easy and diverse means of seed dispersion
- Seeds germinate earlier and grow faster
- Flower earlier and mature ahead of the crop
- Germinate under tough conditions, season bound
- Seeds are dormant for long period and germinate during suitable season
- Good viability for years
- Tolerate moisture stress
- Possess stronger and deeper root system

Effect of weed competition on crop growth and yield

- Crop suffers from nutritional deficiency
- Growth is reduced
- Water requirement will be more
- Lowers the input response
- Pest and disease incidence will be more
- Yield is affected
- Cost of production will increase

3.20. Critical period of weed competition for important crops

Crops	Days from sowing
Rice (lowland)	35
Rice (upland)	60
Sorghum	30
Maize	30
Cotton	35
Sugarcane	90
Groundnut	45
Soybean	45
Onion	60
Tomato	30

3.21. Different types of common weeds

GRASSY WEEDS


Acrachne racemosa


Aristida funiculata


Avena ludoviciana


Brachiaria reptans


Chloris inflata
Syn: Chloris barbata


Cynodon dactylon


Dactyloctenium aegyptium


Digitaria bicornis


Digitaria longiflora


Dinebra arabica


Dinebra retroflexa


Dinebra


Echinochloa colona


E. crus-galli


Echinocloa glabrescens

BROAD LEAVED WEEDS


Abutilon hirtum


Abutilon indicum


Acalypha ciliata


Acalypha indica


Acanthospermum hispidum


Achyranthes aspera


Aerva lanata


Aerva tomentosa


Aeschynomene indica


Ageratum conyzoides


Alternanthera

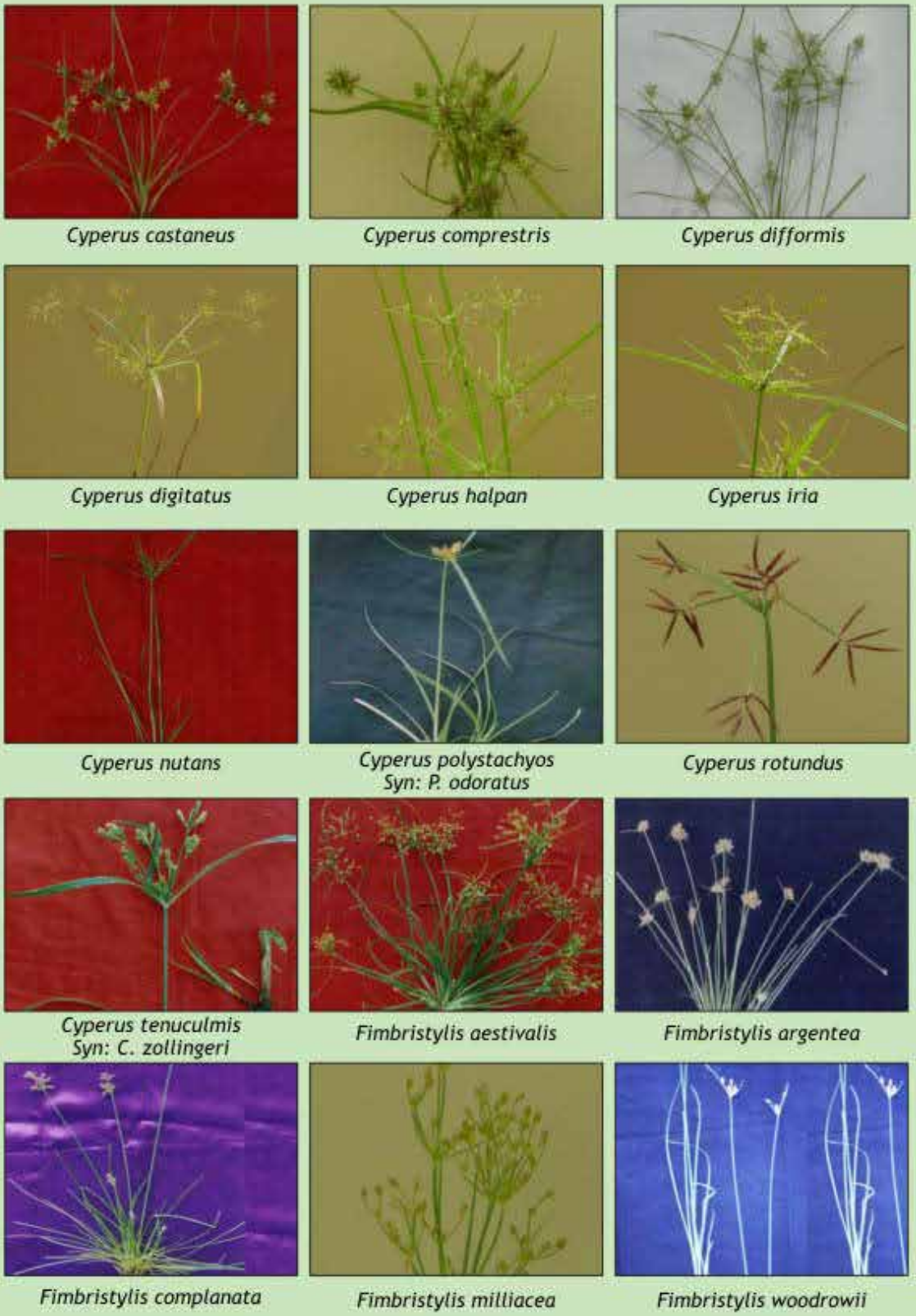

Alternanthera paranychioides


Alternanthera sessilis


Alternanthera pungens


Allmania nodiflora

SEDGES



Cyperus castaneus
Cyperus comprestris
Cyperus difformis
Cyperus digitatus
Cyperus halpan
Cyperus iria
Cyperus nutans
Cyperus polystachyos
Syn: *P. odoratus*
Cyperus rotundus
Cyperus tenuculmis
Syn: *C. zollingeri*
Fimbristylis aestivalis
Fimbristylis argentea
Fimbristylis complanata
Fimbristylis milliacea
Fimbristylis woodrowii

Source: Directorate of Weed Science Research, Jabalpur

3.22. Control measures of weeds

Principles of weed control

- Prevention
- Eradication
- Control
- Management

a) Preventive weed control

- Avoid using crop that are infested with weed seeds for sowing
- Avoid adding weeds to the manure pits
- Nursery or planting material should be free from weeds
- Keep irrigation channels, fence-lines and uncropped areas clean
- Constantly look for weed, destroy the weeds then and there
- Use good quality certified seeds, which are free from weed seeds
- Use pre-emergence herbicides to prevent germination of weeds

b) Eradication (complete removal): Weeds are killed or completely removed from a given area, will not reappear unless it is introduced again. However,

- It is very difficult
- Involves high cost
- Can be used in green houses and nurseries

c) Control: Weeds growth is restricted and killed when necessary so that it does not affect crop growth.

d) Weed management: Managing the population of weeds using all possible methods.

(i) Mechanical method:

- **Tillage:** Using plough or disc, weeds are removed from soil and exposed to sunlight.
- **Hoeing:** Using hand hoe, annual and biennials are completely removed.
- **Hand weeding:** Either by physical removal or pulling out of weeds by hand or using some implements.
- **Digging:** Advisable in the case of perennial weeds.
- **Using sickle**: Top portion of weeds are removed using sickle, thereby weeds seed production is controlled.
- **Burning:** Burning is often an economical and practical means of controlling weeds but not always possible in crop production field.
- **Flooding:** Kills weeds by reducing oxygen to plant growth. This is possible only under garden land or wetland condition.

Merits of mechanical method

- Oldest and effective method
- Safe method for environment
- High skill is not necessary
- Weeding is possible in between plants
- Deep rooted weeds can be controlled effectively

Demerits of mechanical method

- Labour and time consuming
- Possibility of crop damage
- Requires ideal moisture
- Costly

ii) Cultural weed control

- **Summer ploughing:** Is done immediately after summer showers. This exposes weeds to hot sun.
- **Field preparation:** Makes the field weed free by constant removal.
- **Select crop** that can compete better with weeds like cowpea, sudan gross, sorghum are good competitors. Fast growing crops suppresses the weed effectively.
- **Maintenance of optimum plant population:** Adequate plant population covers the land and, hence, growth of the weed will be difficult. Close row crops are better than wide row.
- **Crop rotation:** Minimize the dominance of particular weed in the cropping system.
- **Growing of intercrops:** Inter cropping covers the land quickly and reduce growth of the weeds. E.g,: Growing crop such as cowpea/ soybean, etc. in wide spaced crops like maize/pigeon pea/sugar cane, etc.
- **Mulching:** Mulch is a protective covering of material maintained on soil surface. It has smothering effect and reduces the weed growth. Mulching can be done through degradable farm waste or through plastic sheets. In case of plastic, black is the most popular colour used in commercial horticulture crop production, especially for weed control.
- **Solarisation:** Done by covering the pre soaked field with transparent polythene cover, which increases the temperature by 5 – 10 °C.

- **Stale seedbed:** Weeds are allowed to germinate and non-residual herbicide is sprayed to kill the young weed seedlings.
- **Blind tillage:** Ploughing after sowing of the crop and before plants emerge. Normally seed drills are used under rainfed condition for this purpose.
- **Crop management practices:** Vigorous and fast growing crop varieties for better competition with weeds.

Merits of cultural method

- Low cost
- Easy to adopt
- Less technical skill is sufficient
- No damage to crops
- Effective weed control

Demerits of cultural method

- Time taking and difficult
- Perennial and problematic weeds can not be controlled

iii) Biological weed control: Natural enemy of a weed plant is used to control the weed

Examples:

- Zygogramma biolorata for control of Parthenium
- Hirsch – Manniella spinicaudata is a rice root nematode to control most upland rice weeds
- Azolla in rice

Advantages

- Eco friendly
- Easy
- Low cost

Disadvantages

- They may have alternate host or switching over to alternate host
- Multiplication of bio agent in many cases is difficult

iv) Chemical Weed Control: Herbicides are chemicals used to control weeds.

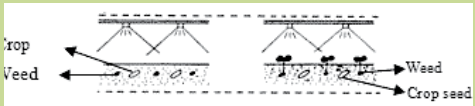
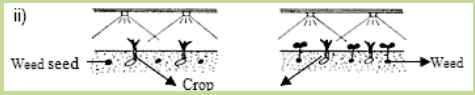
Merits

- Recommended for adverse soil and climatic conditions
- Applied even before weeds emerge and make the environment weed free
- Suitable for all types of crops
- Controls the targeted weeds only
- Controls many perennial weed species
- **Cost effective** compared to labour

Demerits

- Pollutes the environment
- Affects the soil
- Herbicide drift affects adjoining field
- Requires minimum technical knowledge
- Leaves residual effects
- Some herbicides are costly
- No suitable herbicides are available for mixed and inter-cropping system

3.23. Classification of herbicides

Method of application	**Soil herbicides**; e.g: Fluchloralin **Foliar herbicide:** e.g. Glyphosate
Mode of action	**Selective herbicide:** Kills only weeds. **Non selective herbicide:** Kills the entire vegetation.
Mobility	**Contact herbicide;** Kills when comes in contact with plant. **Translocated herbicide:** poison moves from treated parts to untreated part: Eg. Glyphosate
Time of application	**Pre-plant:** Before sowing or along sowing Eg. Glyphosate for Hariyali, Basalin for groundnut  **Pre emergence:** Before weeds germinate;Eg. Thiobencarb  **Post emergence**: applied after weeds germinate i) Eg. Bispyribac Sodium. 

Formulations	**Emulsifiable Concentrate (EC):** Liquid form. **Wettable powders:** Poisons mixed with inert carrier. **Granules (G)** poisons are mixed with granular forms. **Water Soluble Concentrates (WSC)** forms are also available

3.24. Methods of application

- Spraying
- Broadcasting

Foliar application

Blanket spray: Both crop and weed is sprayed with weedicide.



Blanket Spray

Directed spray: Application of herbicides on weeds only avoiding crop using hood.

Eg: Spraying glyphosate[1] in between rows of tapioca using hood to control Hariyali.

Protected spray: Crops are covered and herbicides are sprayed on weeds. This method is expensive.

Spot treatment: Applied only where weeds are present.

3.25. Control of parthenium (perennial weed)

- Manual removal and destruction of Parthenium plants before flowering using hand glouse/ machineries (or)
- Uniform spraying of sodium chloride 200g + 2 ml soap oil/litre of water (or)
- Spraying of 2,4-D sodium salt 8 g or glyphosate10 ml + 20g ammonium sulphate + 2 ml soap solution/litre of water before flowering (or)
- Post-emergence application of metribuzin 3 g / litre of water under non-crop situation.
- Raising competitive plants like Cassia serecea and Abutilon indicum on fallow lands to replace Parthenium (or)
- Biological control by Mexican beetle, fungal pathogen and nematodes

3.26. Control of perennial weeds in orchards

Perennial weeds like Cyperus rotundus, cynodon dactylon, etc. in orchards can be controlled effectively by spraying glyphosate[1] at 2.5 to 5.0 L ha-1 dissolved in 500 liters of water. Falling of the spray fluid on young fruit plant foliage should be avoided. Second spray is required when there is re-growth of weed. (Cost Rs.700/- to 1400 ha).

3.27. Precautions while spraying the herbicides

- Select right kind of herbicide for right kind of crop and spray. Any mistake in choosing the herbicide may result in loss of total crop.
- Dosage should be accurate and good quality of water should be used.
- Use always correct nozzle for spraying. Spraying should be done from front to backwards. (We should not step into the sprayed field for a minimum of 3 days).
- The soil should have sufficient moisture for effective control.
- For paddy, a thin film of water should be maintained for 3 days and it should not be drained.

3.28. List of Herbicides for different crops

✶ There may be slight variation in doses. Please consult local Agricultural Extension Officer/Scientists from KVK or read carefully the leaflet attached along with the herbicide:

Crop	Herbicide	Dose (kg ai/ha)	Trade name and formulation	Time of application
Rice	Thiobencarb	1.25	Machete 50% EC Delchlor 50% EC	Pre-emergence
	Anilophos	0.40	Thunder 50% EC Saturn 50% EC	Pre-emergence
	Pendimethalin	0.90	Arozin 30% EC Aniloguard 30% EC	Pre-emergence
Finger millet	Pendimethalin	0.90	Stomp 30% EC	Pre-emergence
	2,4-D Na salt	1.00	Fernoxone 80% SS	Post-emergence
Maize	Pendimethalin	0.75	Stomp 30% EC	Pre-emergence
Cotton	Metolachlor	1.00	Dual 50% EC	Pre-emergence
	Pendimethalin	1.00	Stomp 30% EC	Pre-emergence
Groundnut	Metolachlor	1.00	Dual 50% EC	Pre-emergence
	Pendimethalin	0.90	Stomp 30% EC	Pre-emergence
Vegetables	Pendimethalin	1.00	Stomp 30% EC	Pre-emergence
Pulses	Pendimethalin	0.60	Stomp 30% EC	Pre-emergence
Wheat	Isoproturon	0.60	Arelon 75% WP	Pre-emergence
Citrus	Glyphosate 4	4 Kg/ha		Post-emergence

3.29. Herbicide mixtures

Sometimes involves mixing of two or more herbicides used for effective and economical weed control at reduced dosage.

Two types of mixtures available are:

1. Tank mixtures made with the desired proportion of herbicides before application. eg: Anilophos + Bispyribac Sodium – Rice
2. Ready mix – formulated by the manufacturer. Ready mix available in the world market eg: Bispyribac Sodium+Glyphosate.

Mixing Ammonium sulphate with glyphosate increases the efficiency where nitrogen increases the translocation.

3.30. Integrated Weed Management (IWM)

Combination of two or more weed control methods at low input levels to reduce weed composition in a given cropping system below the economic threshold level. It:

- Aims to minimize the residue problems
- Minimize the effect on the ecosystem

3.31. Beneficial Effects of Weeds or Economic Uses of Weeds

Several weeds have been put to certain economic uses since ages. Some of the examples are:

- Typha and Saccharum sp is used for making ropes and thatch boards.
- Chicory Cichorium intybus roots are used for adding flavor to coffee powder.
- Amaranthus viridis, Chenopodium album and Portulaca sp. are used as leafy vegetable.
- Hariyali grass (Cynodon dactylon) and Cenchrus Ciliaris, Dichanthium Annulatum andEclipta alba weeds of grass land serve as food for animals.
- Weeds act as alternate host for predators and parasites of insect pests, which feed on the weeds. For example, Trichogramma chilonis feed upon eggs of caster semi looper, which damage the castor plants. E.g. Commelina sp (Copper), Eichornia crassipes (Copper Zinc, lead and cadmium in water bodies.
- Several species of weeds like Tephorsia purpurea and Croton sparsiflora in South India are used as green manures, whereas Eichornia crassipes and Pistia stratiotes are used for composting.
- Argemone Mexicana is used for reclamation of alkali soils.
- Some weeds have medicinal properties and used to cure snake bite (Leucas aspera), gastric troubles (Calotropis procera), skin disorders (Argemone mexicana) and jaundice (Phyllanthus nirur) and Striga Orobanchioides to control diabetes.
- Agarbathis (Cyperus rotundus), aromatic oils, (Andropogan sp & Simbopogon sp) are prepared from weeds.
- Air pollution determined by wild mustard and chickweed respectively.
- Aquatic weeds are useful in paper, pulp and fiber industry.
- Chenopodium album is used as mulch to reduce evaporation losses, whereas Agropyron repens (quack grass) is used to control soil erosion because of its prolific root system.
- Weeds like Lantana camara, Amaranthus viridis, Chenopodium Albu and Eichhornia crassipes are used for beautification.
- Agropyron repense are used for soil conservation, whereas Dicanthium Annulatum are used as stabilizing field bunds.
- Opuntia Dellini is used as biological fence.

3.34. Lessons Learnt

1. Not all insects are harmful. Farmers need to identify beneficial and harmful insects.
2. Summer ploughing, growing trap crops, adopting mixed cropping, intercropping, crop rotation and keeping the field clean are the important agronomical practices farmers needs to follow for effective pest management.
3. Farmers need to understand the Economic Threshold level of major pest of the crops grown.
4. Biological control methods results in sustainable pest control.
5. Farmers are advised to harvest the fruits and vegetables after the waiting period to minimize the residual effect.
6. Seed treatment prevents soil and seed borne infestation of insects and diseases
7. Select varieties and hybrids resistant to most common and economically important diseases.
8. One year seeding, seven years weeding. Hence, prevent spreading of seeds of weeds.

Categories of Pesticides and Precautions

Insecticide	Group	Label of Toxicity	Precaution
Flubendiamide	Diamide	Category D	To be used with Caution.
Emamectin Benzoate	Macrocyclic Lactone - Avermectin	Category D	To be used with Caution.
Spinosad	Macrocyclic Lactone – Spinosyn	Category B	To be used with great caution (since toxic to bees etc.) and under the guidance of an Agriculture Expert.
Acetamiprid	Neonicotinoids	Category D	To be used with Caution.
Imidacloprid	Neonicotinoids	Category B	To be used with great caution (since toxic to bees etc.) and under the guidance of an Agriculture Expert.
Thiamethoxam	Neonicotinoids	Category B	To be used with great caution (since toxic to bees etc.) and under the guidance of an Agriculture Expert.
Chlorpyriphos	Organothiophosphate	Category B	To be used with great caution (since toxic to bees etc.) and under the guidance of an Agriculture Expert.
Profenophos	Organothiophosphate	Category B	To be used with great caution (since toxic to bees etc.) and under the guidance of an Agriculture Expert.
Fenazaquin	Unclassified	Category B	To be used with great caution (since toxic to bees etc.) and under the guidance of an Agriculture Expert.

Fungicide	Group	Label of Toxicity	Precaution
Thiophanate methyl	Benzimidazole precursor	Category B	To be used with great caution (since highly hazardous for human health and/or environment) and under the guidance of an Agriculture Expert.
Carboxin	Carboxamide/ oxathin	Category D	To be used with Caution.
Copper oxychloride	Inorganic copper	Category C	To be used with caution and under the guidance of an Agriculture Expert.
Fosetyl	organophosphorus	Category D	To be used with Caution.
Captan	Phthalimide	Category B	To be used with great caution (since toxic to bees etc.) and under the guidance of an Agriculture Expert.
Metalaxyl	phenylamide	Category C	To be used with caution and under the guidance of an Agriculture Expert.
Azoxystrobin	Strobilurins	Category D	To be used with Caution.

Difenoconazole	Triazoles	Category C	To be used with caution and under the guidance of an Agriculture Expert.
Triadimefon	Triazoles	Category B	To be used with great caution and under the guidance of an Agriculture Expert.
Tebuconazole	Triazoles	Category C	To be used with caution and under the guidance of an Agriculture Expert.

Herbicide	Group	Label of Toxicity	Precaution
2,4-D	Chlorophenoxy acid or ester	Category C	To be used with caution and under the guidance of an Agriculture Expert.
Butachlor	Chloroacetanilide	Category B	To be used with great caution and under the guidance of an Agriculture Expert.
Metolachlor	Chloroacetanilide	Category B	To be used with great caution (since toxic to bees etc.) and under the guidance of an Agriculture Expert.
Fluchloralin	Dinitroaniline	Category C	To be used with caution and under the guidance of an Agriculture Expert.
Pendimethalin	Dinitroaniline	Category C	To be used with caution and under the guidance of an Agriculture Expert.
Anilophos	Organophosphorus	Category B	To be used with great caution (since toxic to bees etc.) and under the guidance of an Agriculture Expert.
Diuron	Phenylurea	Category B	To be used with great caution, since probably carcinogenic, and under the guidance of an Agriculture Expert.
Isoproturon	Phenylurea	Category C	To be used with caution and under the guidance of an Agriculture Expert.
Glyphosate	Phosphonoglycine	Category B	To be used with great caution, since probably carcinogenic, and under the guidance of an Agriculture Expert.
Atrazine	Triazines	Category B	To be used with great caution and under the guidance of an Agriculture Expert.
Thiobencarb	Thiocarbamate	Category B	To be used with great caution (since toxic to bees etc.) and under the guidance of an Agriculture Expert.
Metribuzin	Triazinone	Category B	To be used with great caution and under the guidance of an Agriculture Expert.

4. Farm Management

4.1. Objectives of the session

- To equip the farmer to take advantage of improved technologies and market opportunities to increase income.
- To assist the farmer to make proper plan and adopts his production to assure food security for the family.
- To empower the farmer for professional negotiations with buyers, input dealers and credit institutions.
- To educate the farmer to make profitable decisions considering available resources and anticipate risks including market fluctuations.

4.2. What we know at the end of the session

- Importance of farm management
- Basic information which support better farm management decision
- Market driven enterprises
- Matching resources with calendar of activities
- Selection of cropping pattern
- Understanding of cost benefit analysis
- Risk analysis in agriculture

Module 1: Is farming a business?

What is a farm?

Farm is a socio economic unit. It is composed of farm family, farm enterprises and structures. Farmer is a grower cum manager. The farmer has to decide how much land, labour, capital and type of technology to use to produce in a given season. Ultimately, the farmer has to earn a profit to support his livelihood.

What will the farmer do in farm management?

The farmer would efficiently use the available resources to increase profits through deciding among the best alternatives available.

Some basic functions of farm management

The farmer performs following basic functions to effectively manage the farm:

Diagnosis: Analysis of past performance of farm, its weakness/strengths.

Planning: Planning for the future crops/animals considering the opportunities and threats.

Implementation: Efficient implementation with least cost.

Monitoring: Reduce the losses and increase the profits by reducing the costs and choosing better technologies based on the observed opportunities.

Evaluation: Evaluating the actions for repeating the succeses in future.

What will happen if the farmer does not do farm management?

In the absence of good farm management, the farmer may experience losses in farming for the following the reasons:

- There is continuous changes in supply and price of agri inputs like seeds, fertilizers, irrigation, power etc.
- There is continuous change in prices of the produce (outputs) in the market due to demand and supply changes.
- There is continuous change in the farm technologies.

Therefore the objectives of farming as a business are:

- How to choose best variety/crop/cropping pattern
- How to minimise input cost by judicious use
- How to increase the production and productivity
- How to enhance the quality
- How to plan market driven production
- Choosing the better source of finance and better avenues for investment
- Efficient risk management

For better farm management, the farmer should have thorough knowledge of the following aspects:

- Farm map
- Soil slope and topography
- Soil type (physical and chemical properties)
- Soil colour such as red soil or black soil
- Weather parameter such as rainfall, temperature, relative humidity, etc.
- Vegetative cover such as trees, weeds, etc.
- Irrigation potential from borewell/tubewell/nala/channels
- Drainage facilities - whether water gets logged or not
- Technology available and whether the farmer can access them easily
- Risk factors like hand loans and high rate of interest
- Market facilities - whether they are near to his farm or far off
- Communication facilities like cell phone and internet connectivity
- Physical and infrastructure facilities such as godowns, roads for transport, vehicles, custom hiring centers, etc.
- Whether a farmer can afford the crop/animal he/she wishes to grow or rear considering the above conditions
- Supporting programmes and schemes/subsidies

Farmers should maintain farm records to have a holistic knowledge of their production system

For example: If a farmer maintains a record for all the cost of production such as inputs, labour, etc. for the entire crop cycle along with yield and income obtained from selling the produce, he/she can compare with the next crop cycle to understand whether his/her profit increased or decreased. The records also provides information on activities which contributed for his/her profit or loss so that, the farmer can take alternative decisions to enhance his/her net income.

Farm Resources: To make good farm management decisions, farmers need some basic knowledge on farm resources such as the extent of land available for cultivation, source of irrigation, family labour, availability of labour, skill level of labours, livestock, availability of fodder, availability of farm machinery, availability of inputs such as seeds and fertilizers, credit requirement and availability, source of credit, market demand for produce, infrastructure such as cold storage and godowns. etc. For example, regarding manpower and livestock the farmers have to understand following issue:

Man Power

- **Skill:** Is the labour employed is skilled, e.g. cotton picking skill?
- **Knowledge:** Does the farmer/labour have a thorough knowledge of the package of practices of the crop?
- **Attitude:** Does he/she have a positive attitude towards the technology?

Live Stock

- **Breed:** Selection of a suitable breed
- **Production capacity:** For instance, milk production, meat production and egg laying capacity.
- **Adaptability:** Does the selected breed adapt to the local situation?
- **Drafting capacity:** Knowledge on the draught capacity of animal.
- **Resistance:** Is the foreign breeds resistant to local Indian conditions. E.g. the Holstein Fresian is highly sensitive to high temperatures.
- **Feeding habit:** Are the upgrade breeds or imported breeds capable of feeding on locally available feed materials

Module 2: Know your farm resources

Inputs	Tools and equipment	Labour	Money	Land
Seeds **Fertiliser** **Insecticide** **Fungicide**	Plough, hoe, sprayer, thresher, Tractors	Family and Paid workers	Self finance and credit	Own / Rented land Share-cropping

What does one need to know about the market if one wants to do good business?

The market for agricultural produce	The market for inputs and equipment
The location of the market	The locations of sale
Who is the buyer?	Who sells the inputs and equipments?
The quality of product that is demanded by the market	The quality of inputs and equipment
The price of the product compared to other markets	The price of sale of the inputs and equipments
When to sell	When to buy

How does the price of agricultural products change?

The price of agriculture products change according to the season of the year	The price of agriculture products change between years
At times of abundance, the prices are lowest	The price of a product that is needed by more and more people will rise from one year to the next
At times of scarcity, the prices are highest	The price a product that is produced in greater abundance will fall from one year to the next
The quality of inputs and equipment	
The price of sale of the inputs and equipments	
When to buy	

Important Lesson

To DO successful farm business, the farmer must be well aware of prices (of inputs and produce) at different markets. This allows the farmer to plan production as well as make decisions on the purchase of inputs and the sale of produce.

Module 3: Manage your farm for enough income to sustain yourself

FARM PLANNING: Farm planning is to help the farmers to move to a higher level of production and income, starting from where he/she is now with the resource available to him/her. In this process, the farmer has to consider different types of enterprises like:

- Land based (agril. production activities, pisciculture, plantation, seed production, etc.)
- Animal component based (diary, poultry, goatery, piggery, duckery, etc.)
- Nursery/orchard
- Non-land based (mushroom, apiculture, vermiculture, etc.)

In the present example, three crops namely paddy, cotton and maize have been taken into comparison

for one season. With these three crops, farmers can grow the following combinations in a year:

- Paddy – Paddy (Kharif followed by Rabi)
- Paddy – Maize (Kharif followed by Rabi)
- Cotton – Maize (Kharif followed by Rabi)
- Maize – Maize (Kharif followed by Rabi)

Exercise 1: Agricultural Calendar of Operations to Plan the Production of Paddy.

The times of work…													
Of the main season are shown by a square													
Of the off-season are shown by a circle													
The tasks of the farmer		**January**	**Feb.**	**March**	**April**	**May**	**June**	**July**	**August**	**Sept.**	**October**	**Nov.**	**Dec.**
	Prepare the field												
	Plough the field												
	Purchase seeds												
	Sow												
	Fertilizer application												
	Weeding												
	Apply insecticide												
	Harvest and store												

Important Lesson:

For a good yield, the farmer plans to do the necessary work in the field and apply the inputs at the right time based on the calendar of operations throughout the year

Here we will see how to determine if farm business was good or bad. We will calculate the "Income" and "Expenditure" from different produce. Master Trainers may give the following exercise sheet to the farmers to work out the details for the respective crops by changing the relevant package of practices as applicable to local situations.

Exercise Sheet 2: Paddy

Blank sheet to be filled by Farmer based on worked solution given below.

Steps:

- Multiply the quantity with the price in each line
- Add the money spent ("Expenditure") on inputs and labour
- Multiply the yield by the price of sale ("Income")
- Subtract the sum of "money-out" from the "Income"
- Determine if there was a gain or a loss

Activity	Unit	Quantity	Price	Total (Rs.)
Preparatory cultivation				
a) Machine / labour	No of hours			
b) Animal / labour	Days			
Sub Total				
Seeds and sowing				
a) Cost of seed	Kgs			
b) Cost of seed treatment				
c) Cost of sowing (Human Labour)	Days			
d) Cost of thinning/gap filling	Days			
Sub-Total				
Manures and Fertilizers				
a) Cost of organic & Green Manuring (In-situ plouging)				
b) Application cost				
c) Cost of fertilizer	Kgs	N		
		P		
		K		
d) Application cost (Human Labour Male)	Days			
Sub-Total				
Weed control				
a) Cost of Manual weeding	Labour			
b) Cost of herbicide if any (butachlor)	Litre			
Sub-Total				
Plant Protection				
a) Cost of bio-agents				
b) Cost of pesticides (Thiamethoxam/pro-fenophos)	Litres			

Activity	Unit	Quantity	Price	Total (Rs.)
Furadon-3G	Kg			
c) Cost of Application	Labour			
Sub-Total				
Irrigation cost if any	Power	month		
Sub-Total				
Cost of harvest				
a) Combined harvester	Hours			
Post harvest charges				
b) Cleaning and bagging (Human Labor)	days			
Sub total				
Total cost of cultivation				
Yield Kgs/Ha. and returns				
a) Qty. produced Qtls. per ha	qtls			
b) Gross returns received per ha (Rs.)				
c) Total cost involved per Ha (Rs.)				
d) Net returns per Ha (Rs.)				
e) Cost benefit ratio (Gross Returns divided by Total Cost)				

Solution to exercise 2 – Paddy

Cost of Cultivation of Paddy (per hectare)

Activity	Unit	Quantity	Price	Total (Rs.)
Preparatory cultivation				
a) Machine / labour	No of hours	8	800	6400
b) Animal / labour	Days	6	600	3600
Sub Total				**10000**
Seeds and sowing				
a) Cost of seed	Kgs	50 kgs	20	1000
b) Cost of seed treatment				50
c) Cost of sowing (Human Labour)	Days	25	200	5000
d) Cost of thinning/gap filling	Days	5	200	1000
Sub-Total				**7050**
Manures and Fertilizers				
a) Cost of organic & Green Manuring (In-situ plouging)				350
b) Application cost				100

Activity	Unit	Quantity	Price	Total (Rs.)
c) Cost of fertilizer	Kgs	120 N	12	1440
		60 P	50	3000
		40 K	28	1120
d) Application cost (Human Labour Male)	Days	3	200	600
Sub-Total				**6610**
Weed control				
a) Cost of Manual weeding	Labour	30	200	6000
b) Cost of herbicide if any (butachlor)	Litre	2.5	200	500
Sub-Total				**6500**
Plant Protection				
a) Cost of bio-agents				
b) Cost of pesticides (Thiamethoxam/pro-fenophos)	Litres	5	350	1750
Furadon-3G	Kg	15	60	900
c) Cost of Application	Labour	6	200	1200
Sub-Total				**3850**
Irrigation cost if any	Power	5 months	500	2500
Sub-Total				**2500**
Cost of harvest				
a) Combined harvester	Hours	4.5	1700	7650
Post harvest charges				
b) Cleaning and bagging (Human Labor)	days	10	200	2000
Sub total				**9650**
Total cost of cultivation				**46160**
Yield Kgs/Ha. and returns				
a) Qty. produced Qtls. per ha	qtls	50	1500	75000
b) Gross returns received per ha (Rs.)	Cart load	8	800	6400
c) Total cost involved per Ha (Rs.)				**81400**
d) Net returns per Ha (Rs.)				**46160**
e) Cost benefit ratio (Gross Returns divided by Total Cost)				**35240**
e) Cost benefit ratio				**1:1.76**

Exercise 1: Agricultural Calendar of Operations to Plan the Production of Maize

The times of work…													
Of the main season are shown by a square													
Of the off-season are shown by a circle													
The tasks of the farmer		**January**	**Feb.**	**March**	**April**	**May**	**June**	**July**	**August**	**Sept.**	**October**	**Nov.**	**Dec.**
	Prepare the field												
	Plough the field												
	Purchase seeds												
	Sow												
	Fertilizer application												
	Weeding												
	Apply insecticide												
	Harvest and store												

Important Lesson:

For a good yield, the farmer plans to do the necessary work in the field and apply the inputs at the right time based on the calender of operations throughout the year.

Here we will see how to determine if farm business was good or bad. We will calculate the "Income" and "Expenditure" from different produce. Master Trainer may give the following exercise sheet to the farmers to work out the details for the respective crops by changing the relevant package of practices as applicable to local situations.

Exercise Sheet - 2 (Maize – Blank sheet to be filled by Farmer based on worked solution given below.

Steps:

- Multiply the quantity with the price in each line
- Add the money spent ("Expenditure") on inputs and labour
- Multiply the yield by the price of sale ("Income")
- Subtract the sum of "money-out" from the "Income"
- Determine if there was a gain or a loss

Activity	Unit	Quantity	Price	Total (Rs.)
Preparatory cultivation				
a) Machine / labour	No of hours			
b) Animal / labour	Days			
Sub Total				
Seeds and sowing				
a) Cost of seed	Kgs			
b) Cost of seed treatment				
c) Cost of sowing (Human Labour)	Days			
d) Cost of thinning/gap filling	Days			
Sub-Total				
Manures and Fertilizers				
c) Cost of fertilizer	Kgs	N		
		P		
		K		
d) Application cost (Human Labour Male)	Days			
Sub-Total				
Weed control				
a) Cost of Manual weeding	Labour			
b) Cost of herbicide if any (butachlor)	Days			
Sub-Total				
Plant Protection				
a) Cost of bio-agents				
b) Cost of pesticides (Thiamethoxam/profenophos)	Litres			
Furadon-3G	Kg			
c) Cost of Application	Labour			
Sub-Total				

Activity	Unit	Quantity	Price	Total (Rs.)
Irrigation cost if any	Power	month		
Sub-Total				
Cost of harvest				
a) Combined harvester	Hours			
Post harvest charges				
b) Cleaning and bagging (Human Labor)	days			
Sub total				
Total cost of cultivation				
Yield Kgs/Ha. and returns				
a) Qty. produced Qtls. per ha	qtls			
b) Gross returns received per ha (Rs.)				
c) Total cost involved per Ha (Rs.)				
d) Net returns per Ha (Rs.)				
e) Cost benefit ratio (Gross Returns divided by Total Cost)				

Solution to exercise 2 - Maize

Cost of Cultivation of Maize (per hectare)

Activity	Unit	Quantity	Price	Total (Rs.)
Preparatory cultivation				
a) Machine / labour	No of hours	5	650	3250
b) Animal / labour	Days	5	500	2500
Sub Total				**3750**
Seeds and sowing				
a) Cost of seed	Kgs	20	81.25	1625
b) Cost of seed treatment				
c) Cost of sowing (Human Labour)	Days	5	200	1000
d) Cost of thinning/gap filling	Days	5	200	1000
Sub-Total				**2625**
Manures and Fertilizers				
c) Cost of fertilizer	Kgs	150 N	12	1800
		60 P	50	3000
		50 K	28	1400
d) Application cost (Human Labour Male)	Days	10	150	1500
Sub-Total				**7700**
Weed control				

Activity	Unit	Quantity	Price	Total (Rs.)
a) Cost of Manual weeding	Labour	20	150	3000
b) Cost of herbicide if any (butachlor)	Days	4	500	2000
Sub-Total				**5000**
Plant Protection				
a) Cost of bio-agents				
b) Cost of pesticides	Litres	5	300	1500
Furadon-3G	Kg	10	60	600
c) Cost of Application	Labour	4	200	8
Sub-Total				**2900**
Irrigation cost if any	Power	4	500	2000
Sub-Total				**2000**
Cost of harvest				
a) human labour	days	20	200	4000
b) Threshing (machine)	quintals	40	60	2400
Post harvest charges				
b) Cleaning and bagging (Human Labor)	days	15	200	**3000**
Sub total				**7400**
Total cost of cultivation				**31375**
Yield Kgs/Ha. and returns				
a) Qty. produced Qtls. per ha	qtls	40	1000	40000
b) Gross returns received per ha (Rs.)	Cart load	4	500	2000
c) Total cost involved per Ha (Rs.)				42000
d) Net returns per Ha (Rs.)				**31375**
e) Cost benefit ratio (Gross Returns divided by Total Cost)				**10625**
e) Cost benefit ratio				**1:1.34**

Exercise 1: Agricultural Calendar - Operations to Plan the Production of Cotton

The times of work…													
Of the main season are shown by a square													
Of the off-season are shown by a circle													
The tasks of the farmer		**January**	**Feb.**	**March**	**April**	**May**	**June**	**July**	**August**	**Sept.**	**October**	**Nov.**	**Dec.**
	Prepare the field												
	Plough the field												
	Purchase seeds												
	Sow												
	Fertilizer application												
	Weeding												
	Apply insecticide												
	Harvest and store												

Important Lesson:

For a good yield, the farmer plans to do the necessary work in the field and apply the inputs at the right time based on the calender of operations through out the year.

Here we will see how to determine if firm business was good or bad. We will calculate the "Income" and "Expenditure" from different produce. Master Trainer may give the following exercise sheet to the farmers to work out the details for the respective crops by changing the relevant package of practices as applicable to local situations.

Exercise Sheet - 2 (Cotton - Bt Cotton) – Blank sheet to be filled by Farmer based on worked solution given below.

Steps:

- Multiply the quantity with the price in each line.
- Add the money spent ("Expenditure") on inputs and labour
- Multiply the yield by the price of sale ("Income")
- Subtract the sum of "money-out" from the "Income"
- Determine if there was a gain or a loss

Cost of Cultivation Cotton (per hectare)

Activity	Unit	Quantity	Price	Total (Rs.)
Preparatory cultivation				
a) Machine / labour	No of hours			
b) Animal / labour	Days			
Sub Total				
Seeds and sowing				
a) Cost of seed	Kgs			
b) Cost of seed treatment				
c) Cost of sowing (Human Labour)	Days			
d) Cost of thinning/gap filling	Days			
Sub-Total				
Manures and Fertilizers				
c) Cost of fertilizer	Kgs	N		
		P		
		K		
d) Application cost (Human Labour Male)	Days			
Sub-Total				
Weed control				
a) Cost of Manual weeding	Labour			
b) Cost of herbicide if any (butachlor)	Days			
Sub-Total				
Plant Protection				
a) Cost of bio-agents				
b) Cost of pesticides (Imidacloprid/Thiamethoxam/profenophos)	Litres			
Furadon-3G	Kg			
c) Cost of Application	Labour			
Sub-Total				

Activity	Unit	Quantity	Price	Total (Rs.)
Irrigation cost if any	Power			
Sub-Total				
Cost of harvest				
a) Picking	Kgs			
Sub total				
Total cost of cultivation				
Yield Kgs/Ha. and returns				
a) Qty. produced Qtls. per ha	qtls			
b) Gross returns received per ha (Rs.)				
c) Total cost involved per Ha (Rs.)				
d) Net returns per Ha (Rs.)				
e) Cost benefit ratio (Gross Returns divided by Total Cost)				

Solution to exercise 3 – Cotton

Cost of Cultivation of Cotton (per hectare)

Activity	Unit	Quantity	Price	Total (Rs.)
Preparatory cultivation				
a) Machine / labour	No of hours	5	650	3250
b) Animal / labour	Days	5	500	2500
Sub Total				5750
Seeds and sowing				
a) Cost of seed	Kgs	0.9	1860	1674
b) Cost of seed treatment				
c) Cost of sowing (Human Labour)	Days	8	200	1600
d) Cost of thinning/gap filling	Days	2	200	400
Sub-Total				**3674**
Manures and Fertilizers				
c) Cost of fertilizer	Kgs	150 N	12	1800
		60 P	50	3000
		60 K	28	1680
d) Application cost (Human Labour Male)	Days	30	150	4500
Sub-Total				10,980
Weed control				
a) Cost of Manual weeding	Labour	75	150	11250

Activity	Unit	Quantity	Price	Total (Rs.)
b) Cost of herbicide if any (butachlor)	Days			
Sub-Total				11250
Plant Protection				
a) Cost of bio-agents				
b) Cost of pesticides (Imidacloprid/ Thiamethoxam/profenophos)	Litres	10	350	3500
Furadon-3G	Kg			
c) Cost of Application	Labour	20	200	4000
Sub-Total				**7500**
Irrigation cost if any	Power	**6**	**500**	**3000**
Sub-Total				**3000**
Cost of harvest				
a) Picking	Kgs	2500	6	15000
Sub total				**15000**
Total cost of cultivation				**57154**
Yield Kgs/Ha. and returns				
a) Qty. produced Qtls. per ha	qtls	25		
b) Gross returns received per ha (Rs.)			3600	90000
c) Total cost involved per Ha (Rs.)				57154
d) Net returns per Ha (Rs.)				**32846**
e) Cost benefit ratio (Gross Returns divided by Total Cost)				**1:1.57**

Note: The fixed cost and recurring cost like the interest, depreciation, opportunity cost, etc have not been taken into calculation in the three crops above. Therefore, other things remaining constant, Only the variable cost and returns have been used for the exercise.

Module 4: Income and expenditure statement

Comparing results to know whether you are doing successful farm business

Please tell what is good and what is bad business and indicate the reasons.

Crop	Unit	1 ha of Paddy	1 ha maize	1 ha Cotton
Production	Quintal	50 q	Main product = 40 q By Product (stover/ Straw) = 4 cart load	25 q
Income	Rs/ha	81400	42000	90000
Expenditure	Rs/ha	46160	31375	57154
Profit or Loss?	Rs/ha	+ 35240	+ 10625	+ 32846
		Profit	Profit	Profit
Rank		I	III	II

* It proves that Paddy after Paddy realises the highest profit throughout the year followed by Cotton – Maize. Paddy – Cotton is not feasible.
Remember: Even if Paddy after Paddy promises the highest profit throughout the year, farmer should avoid continuous monocropping to ensure soil fertility through crop diversification and rotation to realize sustained profitability of their farm.

Main Lessons

- To know if you are doing successful business with a crop, you need to know the "Income" and "Expenditure" accurately.
- The farmer records the inputs & labour used in a field, and calculates the "Income" and "Expenditure"
- From the "Income" the farmer subtracts the Expenditure. The result indicates whether farm is making profit or loss.
- The farmer makes a **PROFIT** or **GOOD BUSINESS** if the "Income" is greater than the "Expenditure."
- It is a **LOSS**, if the "Expenditure" is greater than the "Income." In that case it is **BAD BUSINESS.**
- A loss is illustrated by the the minus (dash) and a profit by the plus in front of the number.
- A good farmer will abandon loss making crop or use a better technique to make a profit.
- To ensure a profit, the farmer needs to visualise Income and Expenditure before production.
- The Difference between Income and Expenditure indicates whether we are making a loss or profit from the use of the land.
- The Unit Cost of a crop indicates if it can compete with the same crop produced elsewhere. In the case of food crops, the Unit Cost indicates if it is preferable to buy the produce in the market.
- The good farmer calculates well ahead of the season to decide what to produce and which techniques to use.
- During the production season the good farmer keeps records on money spent for farm operations and inputs.
- After the harvest, the good farmer evaluates the profit and identifies what changes are needed to improve the planning and profit for the next production season

What are Fixed Costs?

Certain costs are called fixed costs. These are costs for equipment and tools that the farmer owns and are used for various crops over several years, such as sprayers, irrigation pumps, buildings etc. The Fixed Costs do not vary with the size of the field.

Blank Exercise sheet of Income and Expenditure Statement: Comparing Results to know whether you are doing successful Farm Business

After, all the calculations farmers will determine the opportunities to increase revenues. By looking at the money on this page farmers will learn

- How to make investment decisions and determine the best opportunities by using Gross Margin, Labour Productivity and Capital Productivity.
- Rank crops based on Profit or Loss
- What crops and techniques to choose?
- Make a choice based on this ranking

	Unit	Paddy	1 ha maize	1 ha Cotton
Surface Area	Ha	1	1	1
1. Money-Out (Variable Costs) Rs/ ha	Rs/ha			
Cost of Inputs	Rs/ha			
Labour Costs	Rs/ha			
2. Income (Gross revenue)	Rs/ha			
Production	Kg			
Price	Rs/kg			
Yield x Price of Sale	Rs/ha			
Value of Produce (Gross Income)	Rs/ha			
Expenditure	Rs/ha			
Profit or Loss?				
Rank				

Module 5: Manage your Money throughout the Year

Bad management of money

- How does one know if the money is managed badly?
- What are the causes?
- What must one do to manage money well during the year?

One must plan! The person, who fails to plan, plans to fail!

First Step: Please look at Income and Expenditure for different crops on the farm and also look at household expenditures. Below are the expenditures of a Household of 5 persons (1 child not yet in school, 1 child in primary school, 1 old parent along with husband and wife). We discuss if we can predict all these expenditures.

Money Needs	Expenditures (Rs)	Period	Can be foreseen
Provisions (food, fuel and household items for an average family size of 5 members)	36,000 (3000 X12)	Each month	Yes
School fees **Free education for primary school children**			
Clothing			
School uniforms for 1 child	1,000	June	Yes
Clothing /year	5,000	April	Yes
Happy events			
Diwali /dussehra	1000	November	Yes
Baisakhi/Ugadi	1000	January	Yes
Unexpected events	3600	Each month @ Rs 300	No
Health expenditure	2400	Per year	No
Total expenditure	50000		

Second Step:

- Let us put these numbers into a financial calendar. On the next page you will see the numbers calculated in Module 5.
- How much money is left at the end of each month?
- How much money is left at the end of the year?

The trainer explains how to do it.

Case 1: Paddy – Paddy financial calendar based on a farm using current practices (Rs) – Exercise

Expenditure	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sept	Oct	Nov	Dec
Paddy (1 ha) - Kharif												
Cost of Inputs including labour						6550	18610	7400	3450	8150	2000	
Paddy(1 ha) - Summer												
Cost of Inputs including labour	18610	7400	3450	8150	2000							6550
Household	3000	3000	3000	3000	3000	3000	3000	3000	3000	3000	3000	3000
School fees and material												
Happy events				1000							1000	
Clothing				5000		1000						
Unexpected events	300	300	300	300	300	300	300	300	300	300	300	300
Total per month	**21910**	**10700**	**6750**	**17450**	**5300**	**14850**	**21910**	**10700**	**6750**	**11450**	**6300**	**9850**

Expenditure	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sept	Oct	Nov	Dec
Paddy (81 Paddy (81400) Summer					81400							
Paddy (81400) - Kharif											81400	
Total per month					81400						81400	

Expenditure	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sept	Oct	Nov	Dec
Expenditure per Month	21910	10700	6750	17450	5300	14850	21910	10700	6750	11450	6300	9850
Total Expenditure for the Year	142320 (Total expenditure per year = 46160 (Cost of Cultivation for Paddy per season) X2 seasons + 50000 (Family Expenditure per year											
Total balance per year	Income = (81400 X 2 = 162800 -142320 = 20480 (Net balance)											

* As the income is realised only during May and November, the monthly balance is shown as negative which indicates that the farmers raises a hand loan/personal loan or shall keep the gross income (Rs 81,400 X 2 = 162,800) and spend through out the year based on the need.

Note: In this example all produce from the farm is sold as per the prevailing market rates. The profit of the farmer depends on the price for the produce i.e., whether the Minimum Support Price (MSP) or market rates are prevailing - which are both fluctuating.

Third Step:

Fill the second financial calendar. The expenditures for inputs and labour are those from the exercise Sheets in Module 5 – using improved practices.

Fourth Step:

Discuss the differences and which situation is preferable. What changes are necessary?

Main Lessons

- In the agricultural enterprise, expenditure for the farm and the household are incurred every month. But the revenue comes only after sale of produce. Therefore there are months of the year where the expenditures are greater than the revenues. These months are called "deficit months."
- For this reason, the good farmer makes a financial calendar. The farmer plans expenditures for production and household needs in consultation with his spouse.
- To be able to cover the expenditures in deficit months, the good farmer saves money from the sales of produce ("surplus months").
- Improved techniques may contribute to improve the revenues of the farmer.
- The needs for inputs can be identified with calculations of gross margin and the financial calendar. This information can be used to make savings in a targeted way or to solicit credit for production.
- The above calculations are done for a farmer with one hectare for Paddy – Paddy combination in Kharif and Rabi for one year. The same exercise should be repeated for taking farm management decisions for the combinations such as
  - Cotton – Maize (Kharif – Rabi)
  - Paddy – Maize (Kharif – Rabi)
  - Maize – Maize (Kharif – Rabi)

Risk in Agriculture

The farmer does not like risks because they are difficult to predict. However, he/she can determine the risk during the planning so that the impact on revenues could be minimised. For example, Module 4 indicates income and expenditure of paddy, maize and cotton. With available data, Paddy - Paddy is the best preferred cropping pattern. However, the farmer can make a judicious decision taking prevailing and anticipated risks into considerations. The anticipated risks and possible decisions are as follows:

Anticipated Risks	Remarks
Reduction in Paddy Market price by 10%	Cotton – Maize is preferred
Inadequate water due to drought for Paddy - Paddy	Cotton – Maize is preferred followed by Maize - Maize. Quantity of water available decides the cropping pattern.

Anticipated Risks	Remarks
Lack of capital for investment for Paddy – Paddy or Cotton - Maize	Paddy and Maize is preferred followed by Maize - Maize
Serious pest problems in Cotton	Paddy – Paddy is preferred

Main Lessons

- Comparing the gross margins of different crops and the production techniques, helps to make decisions on using the land to maximise revenue. This comparison is important to all agricultural entrepreneurs.
- Comparing the labour productivity helps to identify the crops and techniques that make best use of labour (family or wage labour). It also indicates if it is profitable to work on your own farm.
- Comparing the capital productivity indicates which crops or production techniques make best use of money invested.
- Production decisions are based on these comparisons.
- The good agricultural entrepreneur knows that a fluctuation in prices constitutes a risk and revenues. Risks are concern for traditional as well as improved varieties and techniques.
- To evaluate the impact of this market risk, the entrepreneur calculates estimated the gross margin with a much lower price (pessimistic) than the current price (or last season's price). If the pessimistic gross margin estimate can still satisfy the revenue objectives, then the risk is acceptable.
- To evaluate the impact of production risks, the agricultural entrepreneurs calculates a gross margin using a yield lower (pessimistic) than expected. If the pessimistic gross margin estimate can still satisfy the revenue objectives, then the risk is acceptable.

5. Occupational Health and Safety of Farmers

5.1. Objectives of the session

- To create an awareness about causes of health hazards, risks and fatalities in agriculture.
- To impart the knowledge on preventive measures of health hazards in agriculture.
- To enlighten the farmers on use of first aid in emergencies.

5.2. What we know at the end of the session

- Important occupational health hazards in agriculture
- Factors responsible for increasing risk of injury or illness for farmers
- Safe handling of agro chemicals
- Colour coding of pesticides
- First aid measures for pesticide poisoning
- Care in use of pesticides by farmers
- Safety tips to reduce the risk of injuries and fatalities while handling machineries
- First aid for accidents

In agriculture, farmers work under open condition in natural environment, which expose them to various occupational hazards, especially more caused due to handling of chemicals and machineries. Besides, natural hazards are caused due to snakebite, wild animal attack, etc. Knowledge on preventive and curative aspects of these occupational hazards would reduce risks and ensure the safety to the farmers.

5.3. Important occupational health hazards in agriculture

Exposure	Health Effect	Specificity to Agriculture
Hot Weather	Dehydration, heat cramps, heat exhaustion, heat stroke, skin cancer	Most agricultural operations are performed outdoors
Snakes, insects	Fatal or injurious bites and stings	Close proximity results in high incidence
Sharp tools	Injuries ranging from cuts to fatalities	Most farm situations require a wide variety of skill levels for which workers have little knowledge
Physical labour, carrying loads	Back pain and body pain	Agricultural work involves uncomfortable conditions and sustained carrying of excessive loads
Pesticides	Acute poisoning or chronic poisoning	Pesticides can be hazardous and must be used with Personal Protective Equipment (PPE)
Dusts, fumes	Irritation of the eyes and respiratory tract	Agricultural workers are exposed to a wide range of dusts and gases during plant protection with few exposure controls and limited use of PPE

Gases, pathogen	• Skin diseases such as fungal infections and allergic reactions • Parasitic diseases such as malaria, sleeping sickness and hookworm • Animal related diseases such as anthrax, bovine tuberculosis and rabies (at least 40 of the 250 animal related diseases are occupational diseases in agriculture) • Cancers	• Workers are in direct contact with environmental pathogens, fungi, infected animals, and allergenic plants • Workers have intimate contact with parasites in soil, waste water/sewage, dirty tools and unhygienic housing • Workers have ongoing, close contact with animals through raising and sheltering • Agricultural workers are exposed to a mix of biological agents, pesticides, and diesel fumes, all linked with cancer
Others	• Electricity shocks, fire, road accidents, livestock and wild animal attacks, falling into wells, lightening, psychological depression, suicides, etc.	• Loss of life and injuries and suffering to the dependents

Adapted from IFPRI, 2006

5.5. Factors that may increase risk of injury or illness for farm workers

Age: Injury rates are highest under 15 and over 65 years of age.

Equipment and machinery: Most farm accidents and fatalities involve machinery. Proper machine guarding and regular equipment maintenance according to manufacturers' recommendations can help prevent accidents.

Non-availability of protective equipments: Using protective equipment, such as seat belts on tractors and personal protective equipment (such as safety gloves, cover alls, boots, hats, aprons, goggles and face shields) could significantly reduce farming injuries.

Lack of medical care: Hospitals and emergency medical care are typically not readily accessible in rural areas near farms. However, the farmers may utilize for example 108 mobile medical emergency services during emergencies.

Socio - economic risks

- There are dangers due to indiscriminate pesticide/weedicide application to soil, water, air, food chain, natural enemies of pest, resistance of pest, etc.
- Pesticides are hazardous, so there will be associated risk and impact of risk on social system including economic damage. So there should be a proper understanding on hazards of pesticides.

Effects on human beings

- Pesticides accumulate in fatty tissues and reproductive cells lead to birth defects, abnormalities, abortions, premature deliveries, etc.
- Farm workers who regularly spray pesticides are susceptible to impaired eyesight.
- The liver is particularly susceptible to damage by chlorinated hydrocarbons, which can lead to higher risk of serious infection.

Environmental exposures lead to

- Air pollution
- Soil and water pollution
- Food contamination

Looking into the implications of pesticides on socio-economic dimensions, use of pesticides should be judiciously planned only in critical situations with utmost care. Management of pest below Economic Threshold Level (ETL) is most desirable.

5.6. Safe handling of agro chemicals

Agro chemicals are widely used in agriculture, which are major source of health hazards to farmers. Important tips to farmers for safe use of agro chemicals are as follows:

Buying pesticides

Before buying think over this

- Which pest is to be controlled?
- Is damage crossing threshold level requiring pesticide use?
- Have you observed any predators that check pest attack?
- What is the recommendation for the observed problem?
- Which is the least toxic and low persistent chemical among recommendations?

While buying

- Buy from a reputed and licensed store.
- Buy only required quantity and not to go for bulk purchase.
- Do not buy leaky containers.
- See the expiry date.
- Buy only ISI / BIS marked products.
- Don't buy banned and restricted chemicals.

Transportation

- Do not transport/carry pesticides along with food products.
- Do not spill or allow leakage while transporting.
- If leakage is observed clean the vehicle.

Storage

- Don't keep chemicals in kitchen or with animal feed.
- Keep out of reach of children.
- Keep under lock and key.
- Keep in well ventilated separate room.
- Avoid cross contamination of different chemicals.
- Do not keep with human or animal medicine.
- Properly reseal before storing again.

Application

- Read the label and follow the instructions.
- Don't work alone while handling and applying.
- Don't allow children and animals near mixing and application site.
- Use long wooden stick for mixing.
- Mix only the quantity that you need for the next application.
- Avoid excessive spraying and unintended site application.
- Read the label and instructions carefully before opening the pack.
- Never eat and drink while applying.
- Use Personal Protective Equipment (PPE), particularly long shirt and pants, closed shoes as well as protection mask and gloves.
- Avoid application during rainy period.
- Never blow out clogged nozzles or hoses with your mouth, use pin or fine wire for cleaning.
- Avoid windy conditions and do not spray against wind direction.

After application

- Immediately after application take bath and change cloths.
- All clothes must be washed after spraying/dusting and wash them separately.
- Never leave residues of pesticide in sprayers and dusters.
- While cleaning see that water used for cleaning should not enter the drinking water stream/wells.
- Do not go into the treated field until the recommended safety period has passed.
- Do not harvest produce before safe period.

Safe container disposal

- Burry the containers in the field after use.
- Don't use for food and feed storage.
- Do not sell empty container.
- Do not wash the used containers in community water sources.

Colour Coding of pesticides:

The colour band on the pesticide container indicates the hazard level of pesticides. While handling pesticides farmers should take colour band into consideration.

- Red – Very toxic to toxic
- Yellow – Harmful
- Blue - Moderately hazardous
- Green - Acute hazard unlikely in normal use

5.7. First aid measures for pesticide poisoning

- In case of skin contact, remove contaminant contacts and wash with clean water.
- In case of inhalation, remove from site and provide good clean air site, keep the head and shoulder upright.
- In case of unconscious and breathing stops, provide artificial respiration.
- If pesticide is swallowed, induce vomiting by giving 2-3 liters salt water. Give milk after that.

- Take the patient to doctor at the earliest.
- Take the container along with patient to consult doctor.

Snake and other animal bites or attacks and precautions

Snakebite is a routinely occurring life threatening emergency in India. The mortality and morbidity associated with the diverse presentation of snakebites can be decreased if a proper history of the patient's background and habits combined with a thorough knowledge of the specific features of the regional snakes are kept in mind.

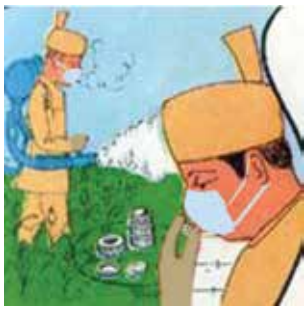
Physical and mental drudgery

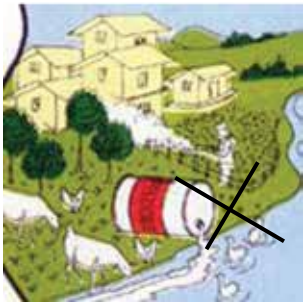
To reduce drudgery related to hard work in different agricultural operations, several technologies for land preparation, weeding, pesticide application and various other farm works have been developed including farm machinery which need to be used by the farmers to keep themselves fit and healthy. Farmers are advised to contact local extension functionaries and scientists to obtain information on such drudgery reducing farm machineries.

5.8. Care in use of pesticides by farmers

Useful tips for safe use of pesticides by farmers are:

	• Identify the pest and ascertain the damage done. • Use pesticide only if it has exceeded the economical injury level. • Use only the recommended pesticide, which is the least toxic.
	• Read instructions manual of the pesticide and equipment. • Check the spraying equipment and accessories, which are to be used.
	• Ascertain that all components are clean, especially filling and suction, strainer, sprayer tank, cut off device and nozzle • Replace worn out parts such as 'O' ring, seal, gasket, worn out nozzle tip, hose clamps and valves. • Test the sprayer and ascertain whether it pumps the required output at rated pressure. Check the nozzle spray pattern and discharge rate.
	• Calibrate the sprayer. Set sprayer speed and nozzle swath by adjusting spray height and nozzle spacing.

	• Make sure that appropriate protective clothing is available and is used • Train all concerned with the application and also understand the recommendations. Ensure that soap, towel and plenty of water is available.
	• Pesticide should be kept in dry and locked store.
During spraying	
	• Take only sufficient pesticide for the day's application from the store to the site. • DO NOT transfer pesticides from original container and packing into the containers. • Recheck the use instructions of pesticide and equipment. • Make sure pesticides are mixed in the correct quantities.
	• Wear appropriate clothing. • Avoid contamination of the skin especially eyes and mouth. • Liquid formulation should be poured carefully to avoid splashing.
	• Never eat, drink or smoke when mixing or applying pesticides. NEVER blow out clogged nozzles or hoses with your mouth.
	• Follow correct spray technique. Spray plant crop thoroughly by operating sprayer at correct speed and correct pressure.

	• Follow correct spray technique. Spray plant crop thoroughly by operating sprayer at correct speed and correct pressure.
	• Never allow children or their unauthorized persons to be nearby during mixing. Never leave pesticides unattended in the field. • Never spray if the wind is blowing towards grazing livestock or pastures regularly used. • Spraying should be done in the direction of Wind.
After spraying	
	• Remaining pesticides left in the tank after spraying should be emptied and disposed off in pits dug on wasteland. • Never empty the tank into irrigation canals or ponds.
	• Never leave unused pesticides in sprayers. Always clean equipment properly. After use, oil it and then keep away in storeroom.
	• Do not use empty pesticide containers for any purpose. • Crush and bury the containers preferably in a land filled dump.
	• Clean buckets, sticks, measuring jars etc used in preparing the spray solution.

	• Remove and wash protective clothing and footwear. • Wash yourself well and put on clean clothing.
	• Keep an accurate record of pesticide usage. • Prevent persons from entering treated areas until it is safe to do so. • Mark the sprayed plots with a flag.

Source : tnauagriportal

5.9 Basic Measures to Manage Weeds, Diseases and Pest Problems

- Prevention is better than cure, hence follow the preventive measures discussed above
- Keep pest below ETL using preventive techniques
- Give more attention to pest warnings by Agriculture Department officials and experts
- Be watch full on weather forecast by ICAR and IMD
- Consult KVK or Scientist in early stages
- Discuss the problems with fellow farmers, you may get many traditional tips
- Understand the scientific rationale behind each pest management ITKs
- Have regular contact with local Research and Development wings of agriculture and allied sectors to get a recent development.

5.10. Safety tips to reduce the risk of injuries and fatalities while handling machineries

Contrary to the popular image of fresh air and peaceful surroundings, a farm is not a hazard free work setting. Every year, thousands of farm workers are injured and hundreds die in farming accidents. Safety in agriculture is one of the main concern, especially when handling with farm tools and machinery. Many accidents in agriculture go unnoticed because they will not be reported. So learning from mistakes will be less. Safety tips to reduce the risk of injuries and fatalities while handling machineries are as follows:

- Safety can be improved on farm by increasing awareness of farming hazards and making a conscious effort to prepare for emergency situations including fires, vehicle accidents, electrical shocks from equipment and wires and chemical exposures.
- Be especially alert to hazards that may affect children and the elderly.
- Minimize hazards by carefully selecting the products to ensure safety.
- Always use seat belts when operating tractors.
- Read and follow instructions in equipment operator's manuals and on product labels.
- Inspect equipment routinely for problems that may cause accidents.
- Discuss safety hazards and emergency procedures with all concerned.
- Take precautions to prevent entrapment and suffocation caused by unstable surfaces of grain storage bins and silos. Never "walk on the grain."
- Be aware that methane gas, carbon dioxide, ammonia and hydrogen sulfide can form in unventilated grain silos and manure pits and can suffocate or poison farmers or explode.
- Wear clothing that fits well and is not loose fitting to avoid being caught in pinch points.

- Never reach over or work near unguarded rotating parts.
- Turn off machinery to attend to repairs.
- Always replace shields that were removed for maintenance.
- Plug unused or failed borewell pits
- Check all equipment for potential w.rap points (e.g. where clothing or hair could be wrapped around a shaft) and if possible, shield those points.
- Replace any damaged manufacturer installed warning labels and place warnings on equipment parts not previously labeled (consider painting them with a bright color, perhaps with a wide stripe).
- Stay alert and warn others when working with shear and cutting points (e.g. objects with blades or hard edges used to cut). Some shear and cutting points cannot be guarded, which can result in severe cuts, lost limbs or injuries from objects thrown by the cutting type equipment.
- Wait until a tractor has stopped completely before stepping into the hitching position.
- Never touch free-wheeling parts (i.e. parts that continue to spin after the power is shut off) until they have stopped moving. This could take 2-2½ minutes.
- Be aware of burn points: mufflers, manifolds and even gear cases.
- Hydraulic systems contain fluid under extreme pressure. Before loosening, tightening, removing or otherwise working with any fittings or parts, relieve this pressure (shut off the hydraulic pump, lower implements and follow instructions in the operator's manual).
- Keep equipment in good repair and safety features up to date.
- "Proper machine inspection and maintenance can help prevent accidents".
- When it comes to machinery maintenance, a shield and guard to cover spinning parts or blades should be kept in place.
- Follow the one seat, one rider rule. If there is only one seat on the equipment, there should only be one rider – an adult.
- Don't allow children to play or ride on equipment or in areas where machinery is used or stored.
- Under aged children should not operate 2, 3, and 4 wheeled vehicles.
- Do not allow riders or passengers in the back of pickup trucks.
- Before starting machinery, all operators should know where kids are located. You may be unable to hear or see children, especially behind large wheels or in blind spots.
- All equipment should be parked and locked with the keys removed when not in use.
- Keep hand tools out of reach of children, especially those with sharp or hot parts.

Better safety and health practices reduce farmer fatalities, injuries and illnesses as well as associated costs such as workers' compensation insurance premiums, lost production and medical expenses. A safer and more healthy workplace improves morale and productivity.

First aid for accidents: The farmers injured should be given first aid at the earliest in order to reduce the damage caused by the injuries. Important tips are as follows:

- The treatment should be given from a person trained in basic first aid, using supplies from a first-aid kit.
- Medical treatment and care given at the site of any medical emergency or while transporting any victim to a medical facility.
- Make sure that first-aid trained personnel are available to provide quick and effective first-aid. Alternatively, one of the family members should be trained.
- Make sure first-aid supplies at your workplace are appropriate to your occupational setting. The response time of your emergency medical services is very important.
- Good knowledge about locally available antidotes or medicinal plants, which can be used.
- Keep the first aid kit(s) in the work place.
- Keep clean water in the work place.
- Keep mobile numbers of trained person on first aid, ambulances, local hospital, local doctors and vehicle owners in the near by vicinity.

5.11. Suggested items for your first-aid kit

- Sterile adhesive bandages
- Small roll of absorbent cotton pads of different sizes
- Adhesive tape

- Triangular and roller bandages
- Cotton (1 roll)
- Band-aids (Plasters)
- Scissors
- Pen torch
- Latex gloves (2 pair)
- Tweezers
- Needle
- Moistened towels and clean dry cloth pieces.
- Antiseptic (Savlon or dettol)



First Aid Kit

- Thermometer
- Tube of petroleum jelly or other lubricant
- Assorted sizes of safety pins
- Cleansing agent/soap

Non-prescription drugs

- Aspirin or paracetamol pain relievers
- Antidiarrhea medication
- Antihistamine cream for Bee Stings.
- Antacid (for stomach upset)
- Laxative

Kits should be checked at least weekly to ensure adequate number of needed items is available. Kits may be kept in the work place.

Make sure that first-aid supplies are:

- Easily accessible to all farmers.
- Stored in containers that protect them from damage, deterioration or contamination.
- Containers must be clearly marked, not locked, and may be sealed.
- Able to be moved to the location of an injured or acutely ill worker.
- Make sure emergency washing facilities are functional and readily accessible.

5.12. Lessons Learnt

1. Awareness on occupational health and safety issues is must for every farmer.
2. Majority of the health hazards in Agriculture are preventive in nature provided farmers are aware about.
3. Farmers should handle the agro chemicals safely. Little negligence may cost the life of human beings and livestock.
4. Dispose the pesticide containers safely.
5. First aid knowledge and skills saves the lives.
6. Keep away children from farm machineries.

6. Farmers' Access to Services

6.1. Objectives of the session

- To enhance awareness about source of extension, information and services among farmers.
- To expose farmers to public and private extension services.
- To encourage farmers to avail extension services through ICT means.
- To enhance farmers knowledge on agricultural credit, insurance and legal aspects.

6.2. What we know at the end of the session

Sources of extension and nature of services provided by following extension service providers:

- Public extension services
- Private extension services
- Institutional sources
- ICT sources
- Agricultural credit
- Agricultural insurance
- Legal aspects

Farmers' Access to Services

6.3. Important services required for farmers

- Information
- Inputs (seed, fertilizer, pesticide, machinery, etc.)
- Infrastructure (cold storage, godown, feed mixing unit, etc.)
- Market (market yard, market intelligence, transport, etc.)
- Developmental schemes/programmes
- Credit and insurance, etc.

The above services are needed with dimensions of accessibility, quality, cost effectiveness and timeliness.

Information is the critical input required for the farmers to bring about changes starting from selection of crops till the marketing.

Past experience: The most important source of information to the farmers is his/her past experience itself. However before acting upon information based on past experience, he/she needs to cross check the relevance to the present context.

Progressive farmers: are the small segment of rural life who are socially, economically and technologically advanced compared to other farmers. They go in search of advanced technologies proactively, adopt and harvest the benefit of technologies. They are the nearest and easiest source of agricultural information to other farmers. For example, contact farmers, award winning farmers, Block Farmers Advisory Committee (BFAC) members, District Farmers Advisory Committee (DFAC) members, State Farmers Advisory Committee (SFAC) members, farmers running farm school, etc.

Input Dealers: are mostly village level businessmen who sell seeds, fertilizers, pesticides and machineries to the farmers. Also provides extension advisories by their strength of proximity to the farmers. However maximum precautions has to be exercised for accessing the advisory services as they are not professionally qualified extension functionaries. However, some input dealers are trained through various programme like Diploma in Agricultural Extension Service for Input Dealers (DAESI) who can provide quality agricultural information. In other cases, advisory from local public extension functionaries and inputs based on the advisory from input dealers can be accessed.

Cooperative Societies: District Central Cooperative

banks and Primary Societies

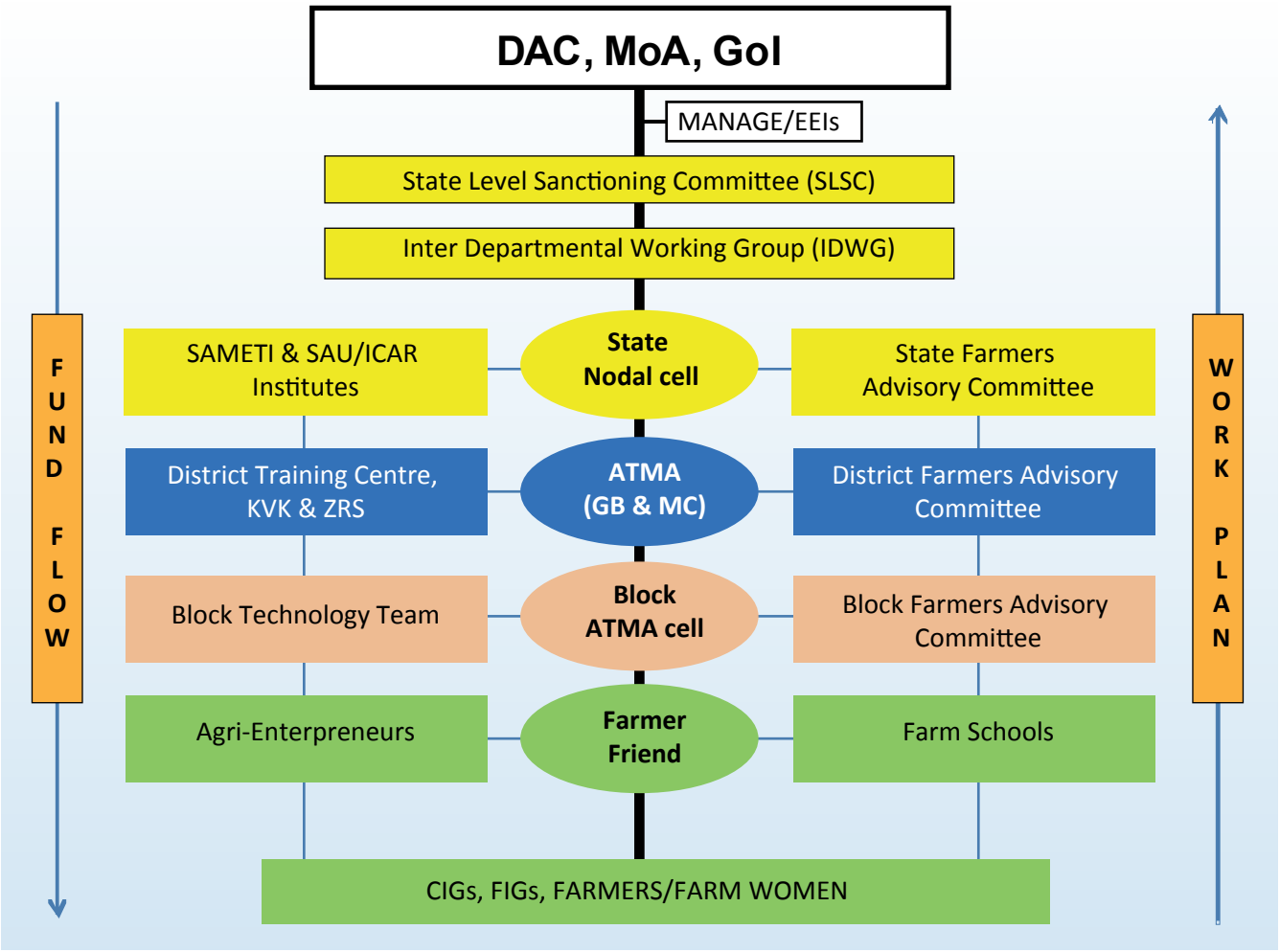
- Extend agriculture credits under priority sector
- Implement NABARD schemes
- Implement Government schemes
- Implement social security schemes
- Extends crop loans

Land (Agriculture & Rural) Development Banks

- Extend medium and long term credit to agriculture
- Implement government schemes

Public sector extension: Represented mainly by the State Agriculture and allied departments continues to be the most important source of information for the majority of farmers. Each department such as Agriculture, Horticulture, Sericulture, Animal Husbandry, Marketing, etc. have their own extension manpower and delivery mechanism to reach the farmers. The focus is on transfer of technology and undertaking agricultural development programmes of state and central Governments.

Agricultural Technology Management Agency (ATMA): An agency undertaking agricultural development programmes at district level in which agriculture and allied departments along with private sectors work together. At village level farmer friend, at block level Block Technology Manager and Subject Matter Specialists provide agricultural information and benefits of schemes/ programmes to the farmers. ATMA organize various extension activities like field visits, trainings, demonstrations, field days, farmers – Scientists interactions, exhibitions, exposure visits, campaign, etc. Publish extension information through print and electronic media, Provide alerts to farmers on agriculture operations and precautions.



Commodity Interest Groups of Farmers (CIGs): A group of farmers growing same crop/ enterprise, share common problems and benefits. Matured CIGs can collectively access information, inputs, infrastructure, credit and market linkages. Some CIGs have independent setup for research, extension, credit and marketing purpose. E.g. Amul for dairy farmers, Maha Grapes for Grape farmers.

Kisan Call Center (KCC): An online agricultural advisory service provided by Government to farmers. Farmers can access information on crops,

livestock, fisheries, inputs, credit, government scheme benefits, through toll free number **1800 180 1551 OR 1551** from 6 A.M. to 10 P.M. except on Sundays and Gazetted holidays. Beyond these hours the calls are attended in the IVRS mode. Qualified professional in local language provides the advisory.

Krishi Vigyan Kendra (KVK): A district scientific organizations that works on technology generation, refinement and dissemination. KVK consists of qualified multidisciplinary experts focusing mainly on locally relevant agricultural issues. KVKs organize front line demonstrations, exposure visits, training programmes, exhibitions, field days and provide agricultural literature to the farmers. Some of the KVKs do provide input support for farmers.

Agriculture university extension system

- Maintains supportive extension service to line departments.
- Supplements efforts of line departments for service through their extension units, research stations and through teaching campuses.
- Develop innovative extension strategies.
- Provide technical knowledge to line departments.
- Organize front line demonstration of their technologies.
- Disseminate technologies through public- private partnerships.

Indian Council of Agriculture Research (ICAR)

- Generates agricultural technologies.
- Provides extension support through its research institutes and KVKs.
- Develops innovative extension strategies.
- Provides technical knowledge to line departments.
- Organizes front line demonstration of their technologies.
- Disseminate technologies through public-private partnerships.

Commodity Boards and National Institutes: Commodity Boards like Coffee Board, Rubber Board, Spice Board, Tea Board, Coconut Development Board, Tobacco Board, Silk Board, Cotton Corporation of India, National institutes like National Institute of Agricultural Extension Management (MANAGE), Central Food and Technology Research Institute (CFTRI), Defence Research and Development Organisation (DRDO), National Institute of Plant Health Management (NIPHM), National Institute of Agricultural Marketing (NIAM), National Horticulture Mission (NHM), National Horticulture Board (NHB). etc., provide extension advisory services to their respective clientele group.

International institution: International institution namely International Center for Research in Semi Arid and Tropics (ICRISAT) in Hyderabad is also serving farmers on crops in semiarid and tropics.

Agriclinics and agribusiness centers

- Agriclinics and Agribusiness Centers are advisory and business centers managed by agricultural professionals in rural areas.
- They provide client specific advisory services free/payment basis.
- Details on Agripreneurs are available at www.agriclinics.net.

Agribusiness companies

- Almost all the Agribusiness Companies provide extension advisory services to farmers in their specialized crops/inputs.
- Contract farming assures extension, input, management and market access to farmers by agribusiness companies. Credit is also provided in few cases.
- Few agribusiness companies are providing different services to farmers through farmers one stop shop concept.

Non-Governmental Organisation (NGOs)

Many NGOs are working in various aspects of agricultural development such as farm advisory, input supply, infrastructures, processing, marketing, community mobilization, micro finance, livelihood development, etc. Farmers can take the advantages of such NGOs wherever available.

Mass Media

- Rapidly expanding mass media ensures easy access to information to farmers on real time basis.
- Agricultural magazines, newspapers provide updated information on agriculture to farmers regularly.
- Community radio, radio, and television provides updated information to the farmers.
- Mobile is also widely used as channel for reaching farmers by many organizations.

Internet opens worldwide information to the doorstep of farmers. Some of the important agri portals useful to the farmers along with key information available are as follows:

- www.icar.org: Research institutes and major technologies
- www.indiaagristat.com: Agricultural related statistics
- www.isapindia.org: Query Redress Services (QRS)
- www.indiaagronet.com: Agricultural jobs, buy and sell and exhibitions
- www.agriwatch.com: Market prices of agricultural commodities
- www.indiancommodities.com: Forecasts of prices of commodities, online trade and warehousing
- \www.krishiworld.com: Multilingual portal, organic farming, crop specific information, disease and pest, market watch and home gardening
- www.agriculture-industry-india.com: Export – import directory, agro trade leads, and agro Trade events
- www.agricoop.nic.in: central sector schemes and policies
- www.apeda.com: Export and import procedure and schemes
- www.fert.nic.in: Fertiliser details
- www.mofpi.nic.in: Food processing technologies and schemes
- www.agmarknet.nic.in: Prices of commodities and trends
- www.icrisat.org: Technologies for semi arid tropics
- www.ikisan.com: Agro informatics, soft ware services and education
- www.uttamkrishi.com: Hindi website and toll free help line
- www.nafed-india.com: Cooperative marketing of agricultural produce
- www.agritech.tnau.ac.in: Technologies, special technologies, schemes and services
- www.nhm.nic.in: Horticultural technologies and schemes

6.4. Accessing financial services – sources

- Self Help Groups (Micro Finance Institutions)
- Nationalized and other private banks
- Cooperative banks and societies
- Subsidy schemes of State/Central Governments

Important types of credits and savings

Kisan Credit Card Scheme (KCC) aims at providing adequate and timely support from the banking system to the farmers for their short-term credit needs for cultivation of crops. This mainly helps farmers for purchase of inputs during the cropping season. Credit card scheme proposed to introduce flexibility to the system and improve cost efficiency.

Benefits of KCC

- Simplifies disbursement procedures.
- Removes rigidity regarding cash and kind.
- No need to apply for a loan for every crop and every season.
- Assured availability of credit at any time enabling reduced interest burden for the farmer.
- Helps to buy seeds, fertilizers at farmer's convenience and choice.
- Helps to buy on cash-avail discount from dealers.
- Credit facility for 3 years – no need for seasonal appraisal.
- Maximum credit limit based on agriculture income.
- Any number of withdrawals permitted subject to credit limit.
- Repayment only after harvest.
- Rate of interest as applicable to agriculture advance.
- Security, margin and documentation norms as applicable to agricultural advance.

How to get Kisan credit cards

- Approach nearest public sector bank and get the details.
- Eligible farmers will get a Kisan Credit Card and a passbook. It contains details like name,

address, particulars of land holding, borrowing limit, validity period, a passport size photograph of holder which may serve both as an identity card and facilitate recording of transactions on an ongoing basis.
- Borrower is required to produce the card cum pass book whenever he/she operates the account.

Banks implementing KCC

- Allahabad Bank - Kisan Credit Card (KCC)
- Andhra Bank - AB Kisan Green Card
- Bank of Baroda – BKCC
- Bank of India - Kisan Samadhan Card
- Canara Bank – KCC
- Corporation Bank – KCC
- Dena Bank - Kisan Gold Credit Card
- Oriental Bank of Commerce -Oriental Green Card (OGC)
- Punjab National bank - PNB Krishi Card
- State Bank of Hyderabad –KCC
- State Bank of India –KCC
- Syndicate Bank –SKCC
- Vijaya Bank -Vijaya Kisan Card
- Personal Accident Insurance Package" is provided to the Kisan Credit Card (KCC) holders.

Salient features of the scheme

- This scheme covers all the Kisan Credit Card Holders against death or permanent disability within the country.
- All KCC holders up to the age of 70 years are eligible.

The benefits under the scheme are as under

- Death due to accident caused by outward, violent and visible means: Rs.50,000/-
- Permanent total disability: Rs.50,000/-
- Loss of two limbs or two eyes or one limb and one eye: Rs.50,000/-
- Loss of one limb or one eye: Rs.25,000/-
- Period of Master Policy - Valid for a period of 3 years.
- **Period of Insurance** - Insurance cover will be in force for a period of one year from the date of receipt of premium from the participating Banks in cases where annual premium is paid. In case of three year cover, the period of insurance would be for three years from the date of receipt of premium.
- **Premium** - Out of the Annual premium of Rs.15/- per KCC holder, Bank has to pay Rs.10/- and Rs.5/- has to be recovered from KCC holder.
- **Claims Procedure** - In case of death, disablement claims & death due to drowning: Claim administration will be done by the designated office of the Insurance Companies. Separate procedure is to be followed.

Credit support from nationalized banks

- Extend agriculture credits under priority sector, implement NABARD Schemes, Government schemes and social security schemes
- Extend crop and agriculture investment/term loans

Extension of Bank-SHG Linkage Programme to Agricultural Purposes

- The loans extended under both the SHG linkage programme and Primary Agriculture Co - operative Societies (PACS) originate from joint efforts of people and financial institutions.
- The microfinance (through SHGs- Bank linkage) meets both consumption and production needs of the people compared to the conventional system that caters only to the production needs.

Commodity Boards Finance

- Give crop loans and subsidies
- Crop pledge loan
- Market loan

6.5. Agricultural insurance

Insurance coverage is given to farmers mainly by Agricultural Insurance Corporation Ltd (AIC) and many other private agencies. All loanee farmers automatically eligible for agricultural insurance coverage. Non loanee farmers can also avail this benefit by payment of nominal premium. Some of the important agricultural insurance schemes available at present are as follows.

- National Agricultural Insurance Scheme
- WBCIS - Weather Based Crop Insurance Scheme
- MNAIS - Modified National Agricultural Insurance Scheme
- RISC - Rainfall Insurance Scheme for Coffee

(Coffee Insurance)
- Rubber Insurance
- Coconut Insurance
- Varsha Bima/Rainfall Insurance
- Rabi Weather Insurance
- Wheat Insurance (Weather & Biomass)
- Potato Insurance
- Bio-Fuel Tree/Plant Insurance
- Pulpwood Tree Insurance
- Cardamom Plant & Yield Insurance

6.6 . Legal aspects

It is important for farmers to know the legal aspects of important inputs namely seeds, fertilizers, pesticides, etc.

The Seeds Act 1966 deals with regulations related to production, certification, quality control, sales, seed analysis, seed inspection, export and import, penalty, exemptions and amendment. The details are available under **http://agricoop.nic.in/seedsact.htm**

The Fertiliser (Control) order 1985 deals with regulations related to fertilizer Price control, control on distribution, registration of dealers, manufacture of fertilizers, mixtures, restrictions on manufacturing, import, sale, enforcement authorities, analysis of samples, specifications and penalty. The details are available under **www.agricoop.nic.in/sublegi/FertilizerControlOrder.htm**

Insecticides Act, 1968: An act to regulate the import, manufacture, sale, transport, distribution and use of insecticides with a view to prevent risk to human beings or animals and for matters connected. The details are available under **http://cibrc.nic.in/insecticides_act.htm**

For more information, farmers may contact nearest banks, agricultural department or Kisan Call Centers.

5.12. Lessons Learnt

1. Important extension sources to farmers are progressive farmers, input dealers, Cooperative Society, ATMA, Kissan Call Center (Toll Free No. 1800 180 1551), Krishi Vigyan Kendra, Agricultural Universities, ICAR Institutions, Commodity Boards, National Institutes, Interantional Institutes, Agriclinics and Agribusiness Centers, NGOs, Radio, TV, Newspapers, Agricultural Magazines and agricultural related websites.
2. Important sources of finance to farmers are banks, cooperatives and SHGs.
3. Kissan Credit Cards provides adequate and timely financial support from the banking system to the farmers.
4. Non loanee farmers are advised to take the benefits of agricultural insurance coverage by paying nominal premium, maintain close liaison with extension, credit and insurance providers regularly.

WORKSHEET

WORKSHEET

WORKSHEET

WORKSHEET

WORKSHEET

WORKSHEET

WORKSHEET

WORKSHEET

For further information, contact:

Dr. P. Chandra Shekara
National Institute of Agricultural Extension Management (MANAGE)
Ministry of Agriculture
Government of India
Hyderabad, Andhra Pradesh
India

Mrs. Sabine Preuss
Programme Director
Natural Resource Management
Deutsche Gesellschaft für Internationale Zusammenarbeit (GIZ) GmbH
A2/18, Safdurjung Enclave
2nd floor, New Delhi - 110029
India

Mr. Rajiv Ahuja
Technical Expert
Natural Resource Management Programme
Deutsche Gesellschaft für Internationale Zusammenarbeit (GIZ) GmbH
A2/18 Safdurjung Enclave
2nd floor, New Delhi, 110029
India

german cooperation
DEUTSCHE ZUSAMMENARBEIT
giz
DESAI
FRUITS & VEGETABLES
MANAGE
develoPPP.de

CIFOR

World Agroforestry

AGROFORESTRY: A PRIMER

Design and management principles for people and the environment



Editors
Anja Gassner and Philip Dobie

AGROFORESTRY:
A PRIMER

Design and management principles for people and the environment

Editors
Anja Gassner and Philip Dobie

UFZ HELMHOLTZ Zentrum für Umweltforschung
Leibniz Universität Hannover
IUCN
CATIE Solutions for environment and development Soluciones para el ambiente y desarrollo
MUNI UNIVERSITY

AGROFORESTRY: A PRIMER

© 2022 CIFOR-ICRAF
All rights reserved.



The content of this publication is under license Creative Commons Attribution 4.0 International (CC BY 4.0), http://creativecommons.org/licenses/by/4.0/

ISBN 978-9-96-610861-6
https://doi.org/10.5716/cifor-icraf/BK.25114

Gassner A and Dobie P. eds. 2022. *Agroforestry: A primer*. Design and management principles for people and the environment. Bogor, Indonesia: Center for International Forestry Research (CIFOR) and Nairobi: World Agroforestry (ICRAF).

Illustrations by Roy Oliver Corvera and María Angela Espinosa
Designed by Dharmi Bradley
Copy editing by Monica Evans and Jonathan P. Cornelius
Content by Jonathan P. Cornelius

CIFOR
Jl. CIFOR, Situ Gede, Bogor Barat 16115, Indonesia
E cifor@cgiar.org

ICRAF
United Nations Avenue, Gigiri, PO Box 30677, Nairobi, 00100, Kenya
E worldagroforestry@cgiar.org

CIFOR-ICRAF
The Center for International Forestry Research (CIFOR) and World Agroforestry (ICRAF) envision a more equitable world where trees in all landscapes, from drylands to the humid tropics, enhance the environment and well-being for all. CIFOR and ICRAF are CGIAR Research Centers.

cifor-icraf.org

The designations employed and the presentation of material in this publication do not imply the expression of any opinion on the part of CIFOR-ICRAF, its partners and donor agencies concerning the legal status of any country, territory, city or area or of its authorities, or concerning the delimitation of its frontiers or boundaries.

CONTENTS

List of authors **7**
Foreword **10**
Acknowledgements **13**

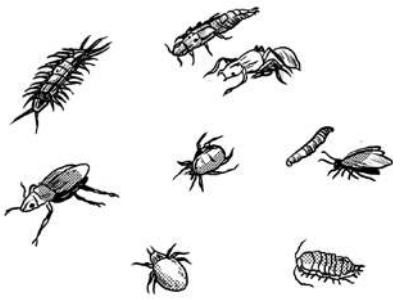
1. Introduction **14**

Why agroforestry? **15**
Why another agroforestry publication? **18**
Who will find this publication useful? **20**
How this publication is organized **21**
Key resources **23**

2. Components of agroforestry systems **24**

Attributes of crops **26**
Lifespan 27
Domestication 27
Light demand and shade tolerance 28
Water needs 30
Root types 31
Agroecological role 32
Attributes of livestock **33**
High value 34
High needs for care and maintenance 35
Mobility 36
Agroecological and environmental benefits 36
Attributes of trees **38**
The role of trees 38
Domestication 40





Lifespan 40
Crown characteristics 41
Deciduousness and leaf size 42
Root characteristics 43
Growth rate and invasiveness 44
Agroecological contribution 45
Soil life 45
Microsymbionts 48
Soil engineers 49
Decomposers and elemental transformers 49

3. Agroforestry systems as circular systems 50

Nutrients and water 52
Soil conservation and erosion control 56

4. Principles of agroforestry design 58

The principle of farmer-centredness 60
Farmers' goals and aspirations 61
Applying the principle of farmer-centredness 62
The principle of aptness to place, people and purpose 64
Aptness to place 65
Aptness to people 65
Aptness to purpose 66
Feasibility 66
The principle of synergy 67

5. Co-design and establishment of agroforestry systems 70

What kind of knowledge is needed for co-design of agroforestry systems? 72
Design process 76
Species selection 79
From design to action: The planting plan and its implementation 83
Importance of the planting plan 83
What needs to be in the planting plan? 84
Incentives 86



6. Planting material in agroforestry **88**

Quality of planting material and aptness for purpose **90**
Nursery plants 90
Seed 92
Genetics and aptness for purpose 93
Access to planting material **94**
Best practices 94
Planting material supply systems 100

7. Management of trees in agroforestry systems **104**

Agroforestry systems differ, but broad management goals are similar **106**
Thinning **107**
Crown management **108**
Pruning 108
Lopping 110
Coppicing 111
Pollarding 111
Weeding **112**
Fertilization **113**

8. From principles to practice: Key systems **114**

Annual crops with trees (ACT) **116**
Design guidelines 117
Management guidelines 121
Livestock with trees **122**
Description of the system 122
Design guidelines 127
Management guidelines 130
Multistrata perennial agroforestry **132**
Description of the system 132
Design guidelines 134
Management guidelines 135
Multistrata cacao agroforestry systems in Central America and the Philippines **137**
Description of the system 137





Design guidelines 138
Management guidelines 138
Case studies 141
Oil palm agroforestry 144
Description of the system 144
Design guidelines 145
Management guidelines 147
Rainforestation farming 148
System description 148
Design guidelines 150
Management guidelines 152

9. Stories from the front line 154

Hidden hunger and land degradation: An NGO responds to a village's request for assistance 156
The farmers' needs and the response 156
How the design principles were used and applied 158
Land restoration for livelihoods and biodiversity conservation 160
The farmers' needs and the response 160
How the design principles were used and applied 162
Forest restoration through "rainforestation" 166
The farmers' needs and the response 166
How the design principles were used and applied 168
When things haven't gone right 170
The farmers' needs and the response 170
How the principles were misapplied or ignored: What went wrong? 173

Scientific names of species and genera 174
Glossary 175



LIST OF AUTHORS

Marlito Bande
Institute of Tropical Ecology and Environmental Management, Visayas State University, Baybay City, Leyte 6521, Philippines.

Brian Chiputwa
World Agroforestry (ICRAF), United Nations Avenue, Gigiri, PO Box 30677-00100 Nairobi, Kenya.

Richard Coe
Statistics for Sustainable Development, Reading, UK; World Agroforestry (ICRAF), United Nations Avenue, Gigiri, PO Box 30677-00100 Nairobi, Kenya.

Jonathan P. Cornelius
James Cook University, Queensland, Australia; World Agroforestry (ICRAF), United Nations Avenue, Gigiri, PO Box 30677-00100 Nairobi, Kenya

Philip Dobie
World Agroforestry (ICRAF), United Nations Avenue, Gigiri, PO Box 30677-00100 Nairobi, Kenya.

Anja Gassner
Global Landscapes Forum Charles-de-Gaulle-Straße 5, 53113 Bonn, Germany; World Agroforestry (ICRAF), United Nations Avenue, Gigiri, PO Box 30677-00100 Nairobi, Kenya.

Rhett D. Harrison
World Agroforestry (ICRAF), East and Southern Africa Regional Office, Lusaka, Zambia.

Hanna J. Ihli
Chair of Economic and Agricultural Policy, Institute for Food and Resource Economics (ILR), University of Bonn, Nussallee 21, 53115 Bonn, Germany.

Clement A. Okia
Muni University, PO Box 725, Arua, Uganda; World Agroforestry (ICRAF), Uganda Country Office, PO Box 26416, Kampala, Uganda.

Andrew Miccolis
World Agroforestry (ICRAF), Brazil Country Office, Travessa Enéas Pinheiro, Marco, CEP 66.095-100, Belém, Pará, Brazil.

J. David Neidel
World Agroforestry (ICRAF), Philippines Country Office/Yale Environmental Leadership & Training Initiative, 2nd Floor, Khush Hall Building, International Rice Research Institute (IRRI) Los Baños, 4031, Laguna, Philippines.

Stepha McMullin
World Agroforestry (ICRAF), United Nations Avenue, Gigiri, PO Box 30677-00100 Nairobi, Kenya.

Agustin Mercado
World Agroforestry (ICRAF), Philippines Country Office, Rm. 15 Khush Hall Building, IRRI, College, Los Baños, Laguna 4031, Philippines.

Athanase Mukuralinda
World Agroforestry (ICRAF), Rwanda Country Office, Headquarters, PO Box 5016, Kigali, Rwanda.

Caroline Pinon
University of the Philippines Los Baños. Department of Social Development Services; Kanluran Road, Los Baños, 4031 Laguna, Philippines;
World Agroforestry (ICRAF), Philippines Country Office, Rm. 15 Khush Hall Building, IRRI College, Los Baños, Laguna 4031, Philippines.

Eduardo Somarriba
Agroforestry Program, Centro Agronómico Tropical de Investigación y Enseñanza (CATIE), Turrialba 30501, Costa Rica.

Peter Thorne
International Livestock Research Institute (ILRI), Addis Ababa, Ethiopia.

Etti Maria Winter
Institute for Environmental Economics and World Trade, Leibniz University of Hannover, Königsworther Platz 1, 30167 Hannover, Germany.

How to cite a chapter of this book

Chapter 1
Gassner A and Dobie P 2022. Introduction. *In* Gassner A and Dobie P. eds. *Agroforestry: A primer*. Design and management principles for people and the environment. Bogor, Indonesia: Center for International Forestry Research (CIFOR) and Nairobi: World Agroforestry (ICRAF) 14–23.
https://doi.org/10.5716/cifor-icraf/BK.25114

Chapter 2
Gassner A, Cornelius JP, Dobie P, Mercado A, Mukuralinda A, Okia CA, Pinon C, Somarriba E, Thorne P. 2022. Components of agroforestry systems. *In* Gassner A and Dobie P. eds. *Agroforestry: A primer*. Design and management principles for people and the environment. Bogor, Indonesia: Center for International Forestry Research (CIFOR) and Nairobi: World Agroforestry (ICRAF) 24–49.
https://doi.org/10.5716/cifor-icraf/BK.25114

Chapter 3
Dobie P, Gassner A, Somarriba E, Miccolis A. 2022 Agroforestry systems as circular systems. *In* Gassner A and Dobie P. eds. *Agroforestry: A primer*. Design and management principles for people and the environment. Bogor, Indonesia: Center for International Forestry Research (CIFOR) and Nairobi: World Agroforestry (ICRAF) 50–57. https://doi.org/10.5716/cifor-icraf/BK.25114

Chapter 4
Gassner A, Coe R, Cornelius JP, Dobie P, Miccolis A, Mukuralinda A, Okia CA, Somarriba E 2022. Principles of agroforestry design. *In* Gassner A and Dobie P. eds. *Agroforestry: A primer*. Design and management principles for people and the environment. Bogor, Indonesia: Center for International Forestry Research (CIFOR) and Nairobi: World Agroforestry (ICRAF) 58–69.
https://doi.org/10.5716/cifor-icraf/BK.25114

Chapter 5
Gassner A, Chiputwa B, Coe R, Dobie P, Ihli HJ, Somarriba E, Miccolis A, McMullin S, Pinon C, Winter EM. 2022. Co-design and establishment of agroforestry systems. *In* Gassner A and Dobie P. eds. *Agroforestry: A primer*. Design and management principles for people and the environment. Bogor, Indonesia: Center for International Forestry Research (CIFOR) and Nairobi: World Agroforestry (ICRAF) 70–87.
https://doi.org/10.5716/cifor-icraf/BK.25114

Chapter 6
Cornelius JP 2022. Planting material in agroforestry. *In* Gassner A and Dobie P. eds. *Agroforestry: A primer*. Design and management principles for people and the environment. Bogor, Indonesia: Center for International Forestry Research (CIFOR) and Nairobi: World Agroforestry (ICRAF) 88–103.
https://doi.org/10.5716/cifor-icraf/BK.25114

Chapter 7
Gassner A, Mercado A, Miccolis A, Mukuralinda A, Okia CA, Somarriba E. 2022. Management of trees in agroforestry systems. *In* Gassner A and Dobie P. eds. *Agroforestry: A primer*. Design and management principles for people and the environment. Bogor, Indonesia: Center for International Forestry Research (CIFOR) and Nairobi: World Agroforestry (ICRAF) 104–113.
https://doi.org/10.5716/cifor-icraf/BK.25114

Chapter 8
Gassner A, Bande M, Harrison RD, Mercado A, Miccolis A, Mukuralinda A, Neidel D, Okia CA, Somarriba E, Thorne P. 2022. From principles to practice: Key systems. *In* Gassner A and Dobie P. eds. *Agroforestry: A primer*. Design and management principles for people and the environment. Bogor, Indonesia: Center for International Forestry Research (CIFOR) and Nairobi: World Agroforestry (ICRAF) 114–153.
https://doi.org/10.5716/cifor-icraf/BK.25114

Chapter 9
Cornelius JP. 2022. Stories from the front line. *In* Gassner A and Dobie P. eds. *Agroforestry: A primer.* Design and management principles for people and the environment. Bogor, Indonesia: Center for International Forestry Research (CIFOR) and Nairobi: World Agroforestry (ICRAF) 154–173.
https://doi.org/10.5716/cifor-icraf/BK.25114

FOREWORD

At a time when the world faces interrelated global challenges of climate change, broken food systems, widespread degradation of land and water resources, and alarming biodiversity loss, it is easy to feel overwhelmed. This book is here to help by offering practical guidance on how we can tackle these seemingly unsurmountable problems, by working with farmers to make more and better use of trees in fields, on farms and across landscapes – that is, through agroforestry.

Agroforestry tackles global challenges systemically rather than one at a time because trees can help farmers adapt to climate change while sucking carbon dioxide out of the atmosphere, provide habitat for biodiversity, regenerate soil and water resources and provide nutritionally important food and other high value products that boost farm income. Put simply, trees on farms are an investment in ecological infrastructure. Their deep root systems, elevated leafy canopies and longevity add functional diversity to farming systems that make them more resilient to climate change effects, like increasingly frequent and severe floods and droughts.

The beauty of this book is that rather than offering a few prescriptive technologies for farmers to adopt, it sets out principles, that when followed, can support farmers and communities in driving their own local innovation.

The first principle is to start with farmers. To recognise their knowledge, their needs and wants, and their capacities and then to codesign agroforestry practices that match local circumstances. This is important because farmers differ from one another and so need different solutions that suit their context. This leads quickly onto the importance of the second principle of agroforestry co-design: aptness to people, place and purpose.

Agroforestry is not 'a' solution to local and global challenges, it is a basket of options with different tree species, cropping and livestock management practices that can deliver production and environmental benefits tailored to local conditions and changing priorities. It is not always easy to select appropriate tree species and management practices, but it is worth putting a lot of effort into doing this, precisely because trees are long lived components of farming systems. This highlights the importance of the third principle: synergy.

It is synergistic effects that confound the impact of global challenges. For example, the combined effect of climate change and agricultural intensification is reducing insect abundance and diversity, including pollinators that are, in turn, important for crop production, precipitating a downward spiral.

Farming and food systems are complex, involving interactions amongst many ecological and socio-economic components. Because trees add functional diversity to farms, they increase the potential to manage interactions to achieve positive outcomes. Their extensive root systems may enhance nutrient recycling, associations with soil microbes lead to biological nitrogen fixation reducing the need to purchase inorganic fertiliser, they shade crops and livestock reducing heat stress, promote groundwater recharge by increasing infiltration and penetration of water, as well as protecting against soil erosion. Well-managed trees in agricultural landscapes may enhance natural pest control, reducing the need for use of environmentally damaging pesticides as well as storing carbon in soil, vegetation and durable wood products. Trees are also important social and cultural components of landscapes, interacting with land tenure and having variable access and uses, depending on people's gender and ethnicity.

It is clear that integrating trees within farming practice can have positive outcomes. The difficulty is that with so many different tree species and management options, agroforestry is knowledge intensive. And that is why this book is so important. It sets out in simple language, concrete steps for applying the principles of codesign of agroforestry solutions and provides well-illustrated case studies of how this has played out in various crop and livestock systems across different regions of the world. It covers both successes and failures, so that we can all learn from the experiences that these expert authors bring together in this landmark publication.

I encourage people to not only read this book but to take it into the field and use it to guide action. Whether you are a farmer, an extension worker, a policy maker or someone interested in promoting equitable and sustainable agriculture, there will be much within these pages to enhance your understanding of agroforestry and how to put it into practice.

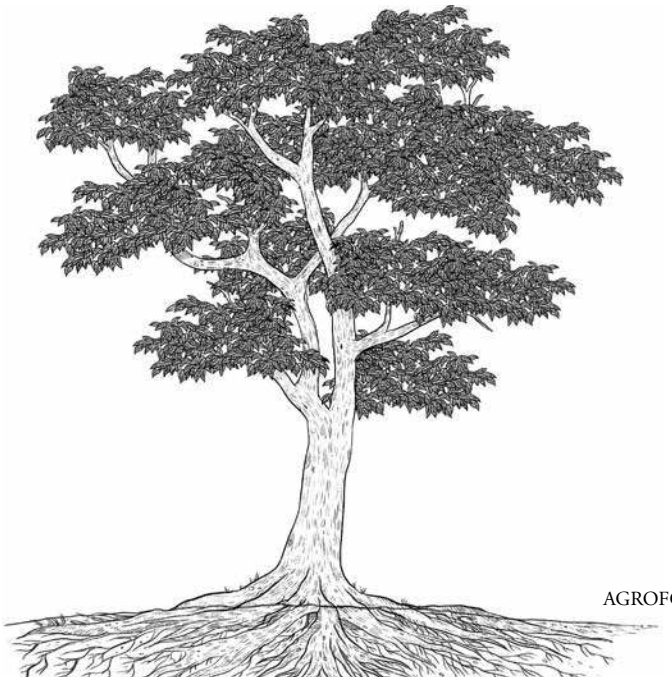
Fergus Sinclair
Chief Scientist, CIFOR-ICRAF

ACKNOWLEDGEMENTS

This book is a joint output of the Trees on Farms for Biodiversity and the Integrated Natural Resource and Environmental Management Project. We would like to thank both the German Federal Ministry for the Environment, Nature Conservation and Nuclear Safety and Consumer Protection (BMUV) and the Asian Development Bank for their support.

This work is linked to the CGIAR Research Program on Forests, Trees and Agroforestry. CIFOR-ICRAF are members of the CGIAR, a global research partnership for a food-secure future. We thank all donors who support research in development through their contributions to the CGIAR Fund. For a full list of the 'CGIAR Fund' funding partners please see: http://www.cgiar.org/our-funders.

We thank Jacklyn Care Linayao for helping to source key resources and content, Robert Frederick Finlayson for his input to the concept of the book, and Zarrel Gel Noza for coordinating and briefing the illustrators.



1

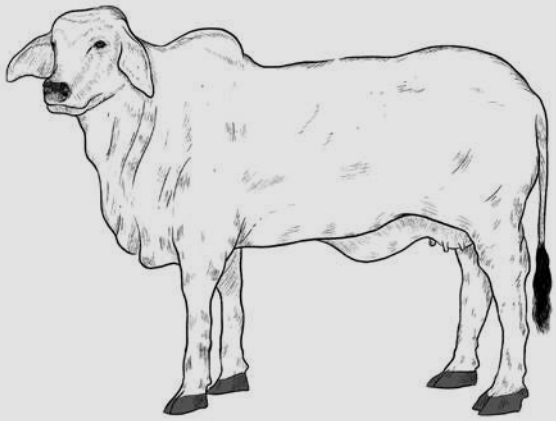
INTRODUCTION











Why agroforestry?

Conventional agriculture is very productive. But high productivity comes at a cost: soil that is depleted or eroded, watercourses that are polluted or drying up, and a food system that produces 20–40% of greenhouse gas emissions. Many people now agree that we urgently need to transform the food system, including agriculture. Agroforestry, as a nature-based approach to production and land use, will play an important role in this transformation.

Agroforestry is land use that combines trees with crops, trees with livestock, or trees with both crops and livestock. This mix of components creates an **agroforestry system** in which the components interact in a beneficial manner, improving agriculture in many ways; for example, by improving farm yields, increasing farm incomes, and contributing to soil and water conservation. Agroforestry is a form of 'trees-on-farms' (see Box 1).

Box 1. Agroforestry and trees-on-farms



We use the term 'agroforestry' to mean the use of trees in combination with crops, livestock, or both crops and livestock on the same area of land. Other ways of using trees-on-farms are also common: for example, woodlots; fruit trees planted next to a homestead; riparian buffer strips; and patches of natural forest. Some authors include these and other forms of trees-on-farms within a wider, '**landscape**' concept of agroforestry.

Trees on private farmland are an important part of global tree cover: 45% of the world's farms have more than 10% tree cover. Global carbon storage on agricultural land has been estimated at 45.3 billion metric tons, of which trees contribute more than 75%.[1] These figures exclude large areas of agroforestry on land usually classified as forest.

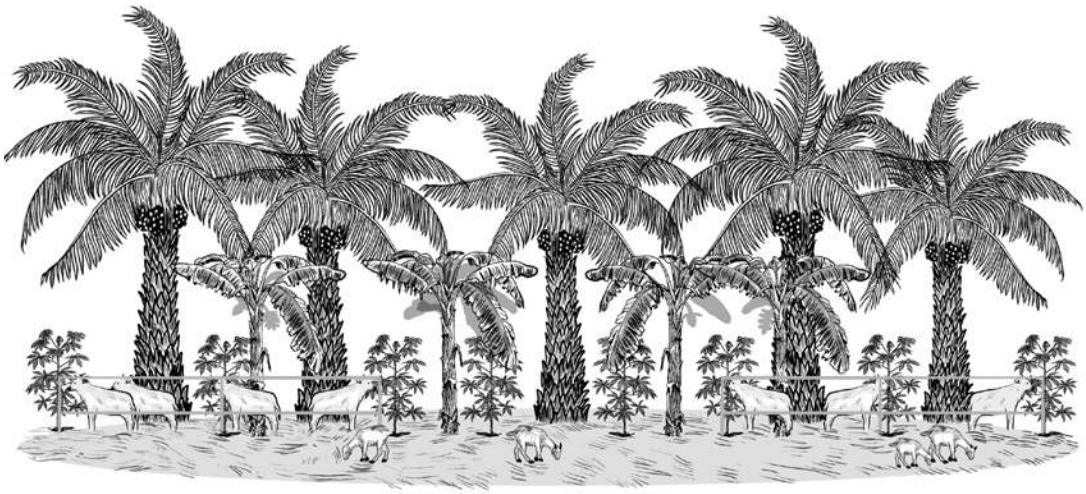
Trees-on-farms, including all types of agroforestry, have many different functions. They provide habitat that increases the biological diversity of agricultural land, including soil biodiversity and **agrobiodiversity**. They enhance the viability of protected areas by making it easier for animals, pollen and seeds to move between them. They contribute to nature-based solutions to conservation and food production challenges, despite being invisible in most countries' current 'green growth' strategies. Crucially, they supply useful and profitable goods and services to farmers and the community-at-large, including timber and fuelwood; improvement of soil fertility; water regulation; fruit, nuts, and edible leaves; and livestock fodder.





[1] Zomer RJ, Bossio DA, Trabucco A, Noordwijk M, Xu J. 2022. Global carbon sequestration potential of agroforestry and increased tree cover on agricultural land. *Circular Agricultural Systems* 2:3. https://doi.org/10.48130/CAS-2022-0003.

Conventional farming is often based on the production of a single crop. Agroforestry also often focuses on a single **flagship species** (which may be a tree species), but adds (other) trees to the system too. There are many ways in which trees can be incorporated. For example, a farmer may grow a cereal or vegetable crop combined with widely spaced trees. Or, the farmer might plant timber trees around the farm perimeter or along a watercourse. Agroforestry systems range from simple – such as one crop species and one tree species – to complex, with various crops species, tree species and livestock all meeting different needs.



Agroforestry is not new. Farmers have practised it for thousands of years, and scientists have recognized it since the 1970s as a productive and ecologically sustainable form of agriculture and land use. But now agroforestry is suddenly at centre stage. It is promoted as a land-use strategy to support **climate change mitigation** and **climate change adaptation**, biodiversity conservation, sustainable agriculture and other goals. Many organizations recommend or use it as a tool for restoring ecosystems, not only agricultural ones, but also forest landscapes.

Why another agroforestry publication?

Although not a cure-all, agroforestry has great potential to contribute to all the goals mentioned above. However, agroforestry is not just a matter of adding trees to farms. To realize its potential, practitioners need to understand its principles. *Agroforestry: A primer* is a guide to agroforestry principles and concepts, and how to use them effectively.

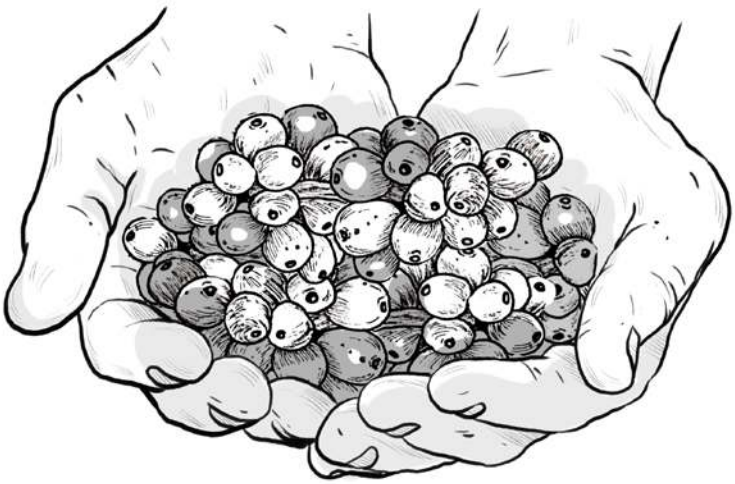
We took this approach partly because high-quality 'how-to' manuals and guides are already available – not just those with 'agroforestry' in their titles, but many too from the fields of forestry, horticulture and social sciences. More importantly, though, our approach reflects two basic needs.

1 First, it is important to avoid the preconceived '**technological package**' approach that can come from too much reliance on a specific manual. Instead, agroforestry practitioners and those supporting them need to understand how to use principles and concepts to design and adapt practices to their local conditions.

2 Second, it is critical to ensure that the aspirations, capabilities and dreams of farming families are central to the design and promotion of agroforestry. Agroforestry must support not only environmental goals, but also – and most importantly – rural people's goals and aspirations for better lives. Farmers rarely have the luxury of investing their time or their land in activities that do not give them direct, tangible benefits. The trees must contribute to food or nutritional security, or generate income.

Payments for **ecosystem services**, such as carbon credits, are usually of low value and are insufficient to persuade smallholders to plant and manage trees. Trees can be an important instrument to restore and maintain biodiversity and ecosystem services on farmland, but they need to be planted as part of a long-term development strategy. This should include the development of tree-based enterprises and markets for tree products.

Our aim is to support practitioners as they work to realize agroforestry's full potential to contribute to the local and global challenges of the twenty-first century.



Who will find this publication useful?

This book aims to guide professionals who are supporting farmers in implementing agroforestry systems, such as:

extension workers

planners and managers

researchers

trainers

teachers and students of agroforestry

and professionals from a range of disciplines working on projects and programmes that use agroforestry. These disciplines may include agroecology, biodiversity conservation, land restoration, rural development and others, as well as agroforestry itself.





How this publication is organized

Chapters 2–7 detailing different aspects of agroforestry systems follow this introduction:

2 **Components of agroforestry systems** describes the most important attributes of four key elements of agroforestry systems: crops, livestock, trees and soil life.

3 **Agroforestry systems as circular systems** describes how agroforestry promotes soil health and conservation.

4 **Principles of agroforestry design** outlines three guiding principles that are fundamental to successful agroforestry **interventions**.

5 **Co-design and establishment of agroforestry systems** explains some important considerations in implementing design principles and translating design into practical action.

6 **Planting material in agroforestry** presents guidance on ensuring adequate quality and quantity of seed and planting stock.

7 **Management of trees in agroforestry systems** describes the main activities needed to ensure that trees interact favourably with the other components of a system.

These are followed by **chapters 8 and 9** that provide different sorts of overviews of several agroforestry systems:

8 **From principles to practice: Key systems** presents generic characteristics of several widely used agroforestry systems.

9 **Stories from the front line** uses a series of synthetic case studies to illustrate how farmers, and those that support them, can apply the principles and concepts described in the previous sections.

We recognize that farmers are diverse and include people across genders, age groups, cultures and resource levels. The messages in this book are relevant to all, and we use the term 'farmer' to cover this diversity.

The appendices include a glossary and a list of scientific and common names used in the text. Words and terms included in the glossary are written in **bold blue type**.



Key resources

We have used many sources in preparing this publication. The following have been particularly useful:

1. Tengnäs B. 1994. *Agroforestry extension manual for Kenya*. Kenya: World Agroforestry (ICRAF). http://apps.worldagroforestry.org/downloads/Publications/PDFS/B08037.pdf.
2. Somarriba E and Quesada F. 2009. *Agroforestry farm planning: Manual for farming families*. 1st Edition. Turrialba, Costa Rica: Tropical Agricultural Research and Higher Education Center (CATIE). http://canacacao.org/wp-content/uploads/Manual-Agroforestry-Farm-Planning.pdf.
3. Miccolis A, Pereira AVB, Peneireiro FM, Marques HR, Vieira DLM, Arco-Verde MF, Hoffman MR, Rehder T, Pereira AVB. 2016. *Agroforestry systems for ecological restoration: How to reconcile conservation and production*. Options for Brazil's Cerrado and Caatinga biomes. Brasília: Instituto Sociedade, População e Natureza (ISPN) and World Agroforestry (ICRAF). http://apps.worldagroforestry.org/downloads/Publications/PDFS/B19034.pdf.
4. Somarriba EJ, Orozco-Aguilar L, Cerda R, López-Sampson A. 2018. *Analysis and design of the shade canopy of cocoa-based agroforestry systems*. *In* Umaharan P. ed. *Achieving sustainable cultivation of cocoa*. London: Burleigh Dodds Science Publishing,1–31. https://doi.org/10.1201/9781351114547.
5. Raintree JB. 1987. *The state of the art of agroforestry diagnosis and design*. Agroforestry Systems 5(3):219–250. https://doi.org/10.1007/BF00119124.
6. Kitalyi A, Miano DM, Mwebaze S, Wambugu C. 2005. *More forage, more milk: Forage production for small-scale zero grazing systems.* Nairobi, Kenya: Regional Land Management Unit (RELMA) of World Agroforestry (ICRAF). http://apps.worldagroforestry.org/downloads/Publications/PDFS/MN13558.pdf.

2

COMPONENTS OF AGROFORESTRY SYSTEMS





Agroforestry systems are composed of trees and crops, trees and livestock, or trees with both crops and livestock. In this section, we explore the attributes of these components in more detail.

A given agroforestry system is often centred on one species, which we call the 'flagship species'. This is the species that the farmer considers to be the most important one – often because it contributes most strongly to their livelihood. Other components, which we call **'flotilla species'**, are added to provide **agroecological services**, such as shade, that support the flagship species. The needs of the flagship species (for example, for light or water) determine the types of flotilla species needed, even when these also produce useful goods. For example, when growing shade-intolerant crops like cereals, farmers need to select and plant trees that do not compete with the cereals for light. In many agroforestry systems, the flagship species is an annual crop, a perennial (long-lived) crop (including tree crops), or a livestock species; it can also be a timber species. Some agroforestry systems may have more than one flagship species. Flotilla species can be trees or crops.

Attributes of crops



Rice



Maize



Potato



Cabbage



Soybean

Crops grown in agroforestry systems include well-known staples such as maize and rice, as well as longer-lived crops like cassava; cash crops like cacao, coffee and soybean; and crops used to feed livestock.

Lifespan

The lifespans of crops vary greatly. Many staple food crops, such as beans, maize, rice and wheat, are annuals: they germinate, grow, flower and die within a single season or year. All other plants are perennials, with lifespans that range from a few years to decades, or even centuries. Farmers shorten the lifespan of some perennial crops, such as sugarcane and cassava, when they harvest them. Lifespans are also shortened when farmers replace older, unproductive plants with new ones, such as in the replanting of coffee and cacao plantations.



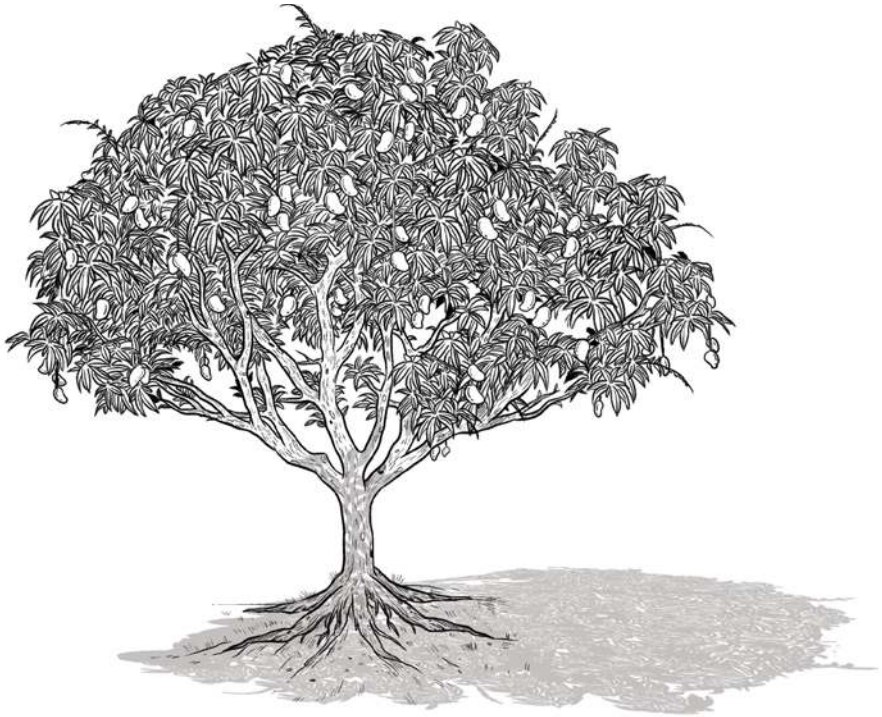
Domestication



Almost all crop species have been **domesticated** to some extent: that is, farmers and plant breeders have altered their genetic characteristics by selection and breeding. In many cases, domestication has altered them so much that they no longer look like their closest wild relatives. Many **modern varieties** of staple crops need more nutrients and water than traditional varieties. Similarly, many improved varieties of coffee and cacao, which are still often grown under shade in agroforestry systems, have been selected in unshaded conditions. For both staple and cash crops, thriftier and more resilient **traditional varieties** may be more suitable to agroforestry systems than alternative, highly improved varieties.



Light demand and shade tolerance



Light is one of the most important resources for plants, and limitations in its supply reduce survival and growth. Different kinds of plants have different levels of tolerance to shade, and some plants do better in shady conditions than under full sun.



Almost all major cereal crops are light demanding, so shade may reduce crop yields.



Fruit vegetables – those whose edible part is a fruit, such as cucumber, bell and chilli peppers, pumpkin and tomato – are some of the least shade-tolerant crops. When shaded, they will often fail to flower, which means they won't produce fruit.



Root vegetables – such as taro, arrowroot, yam, beet (beetroot), carrot and potato – will grow in partially shaded areas that have less direct sunlight, but usually benefit from at least a half-day of full sun.



Leafy vegetables – such as chard, spinach and salad greens – are the most shade tolerant of all vegetables.



Cacao and coffee can be grown successfully under shade in most climatic conditions, and their quality is often improved by doing so (see **Multistrata cacao agroforestry systems in Central America and the Philippines** in Chapter 8).



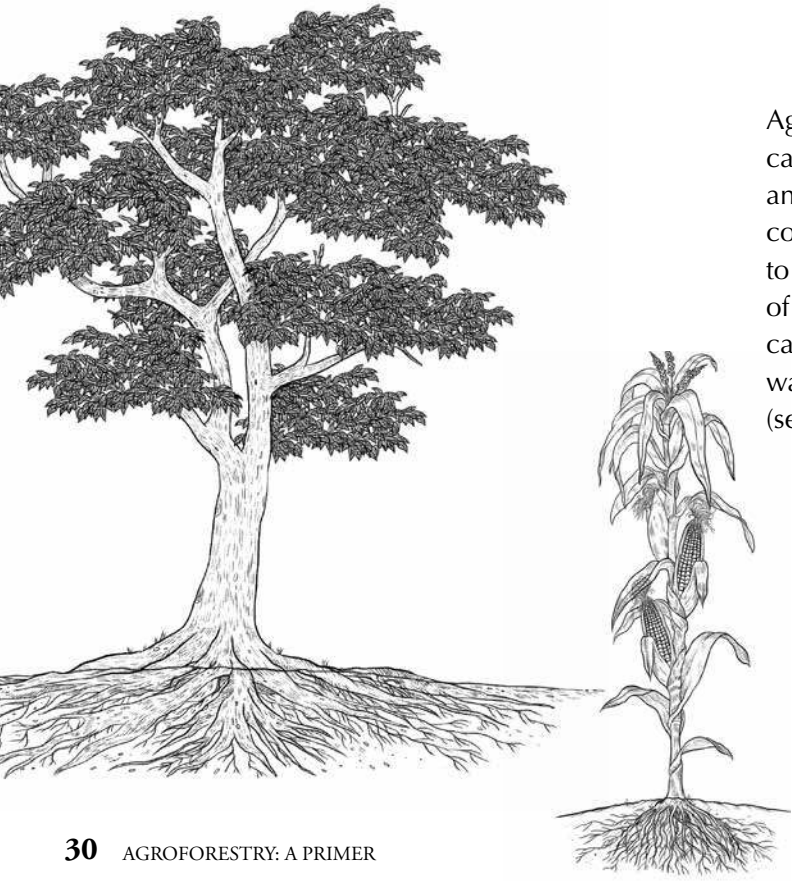
Many other crops – such as ginger, mint, parsley and turmeric – are also shade tolerant. Grass species vary in their shade tolerance, and several tropical forage legumes do well in shade.

Agroforestry systems carefully combine trees and crops to maximize their use of light. Care is taken not to grow light-demanding species in shade, while shade-tolerant species can exploit areas with lower light. Some crops, such as coffee and cacao, are often deliberately grown in the shade of trees, and thrive under those conditions.

Water needs

Crops differ greatly in their water requirements. When combined with other crops or trees, the various components may compete for water. Highly productive species that consume large amounts of water and nutrients, such as maize and other members of the grass family, can severely reduce the growth of tree seedlings. On the other hand, tree roots close to the soil surface can easily outcompete the roots of leafy vegetables.





Agroforestry systems carefully combine trees and crops to avoid harmful competition for water, and to maximize the benefits of available water and the capacity of trees to access water from deep in the soil (see Chapter 3).

Root types

Crops demand a lot of nutrients, and their root systems are adapted to absorb readily available nutrients from the topsoil and from water. Roots can form immense structures of millions of branch roots reaching tens of metres in length. They are both the feet and the hands of the plants; anchoring plants in the soil, but also extracting the nutrients the plant needs from the soil (Figure 1).

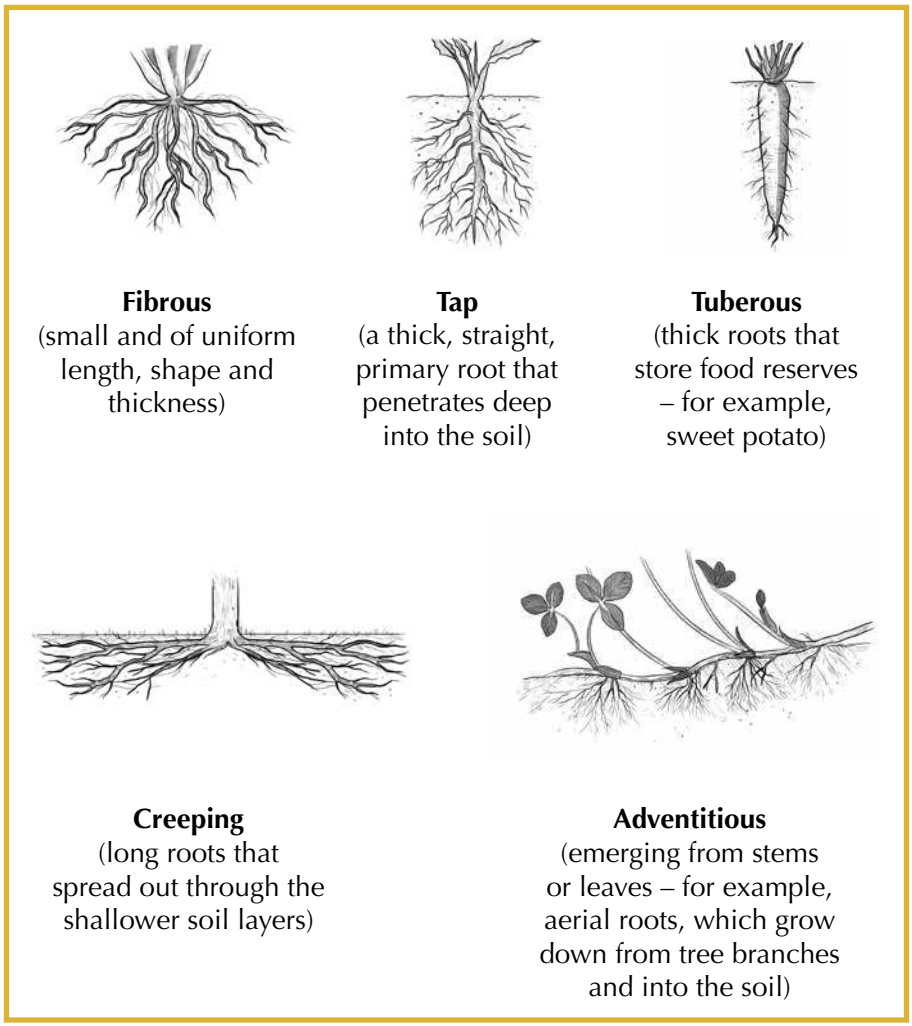


Figure 1. Some root types

Agroecological role

In agriculture, large quantities of **biomass** and nutrients are taken out of the landscape when the crop is harvested. This depletes the soil, and yields will fall unless these are replaced. Agroforestry can return important biomass and nutrients to the system (see Chapter 3, **Agroforestry systems as circular systems**).

Crops also make positive contributions to system functioning. The following are particularly important:



Crop residues are an important source of **organic matter**.



Leguminous crops increase availability of nitrogen to other system components. Other fertilizer species, such as bananas and Mexican sunflowers, can also play important roles in nutrient capture and flows.



Cover crops help to control weed competition.



Fodder crops provide nutrients and water to livestock.



The **high value** of crops often justifies weeding and cultivation activities that also benefit other system components.

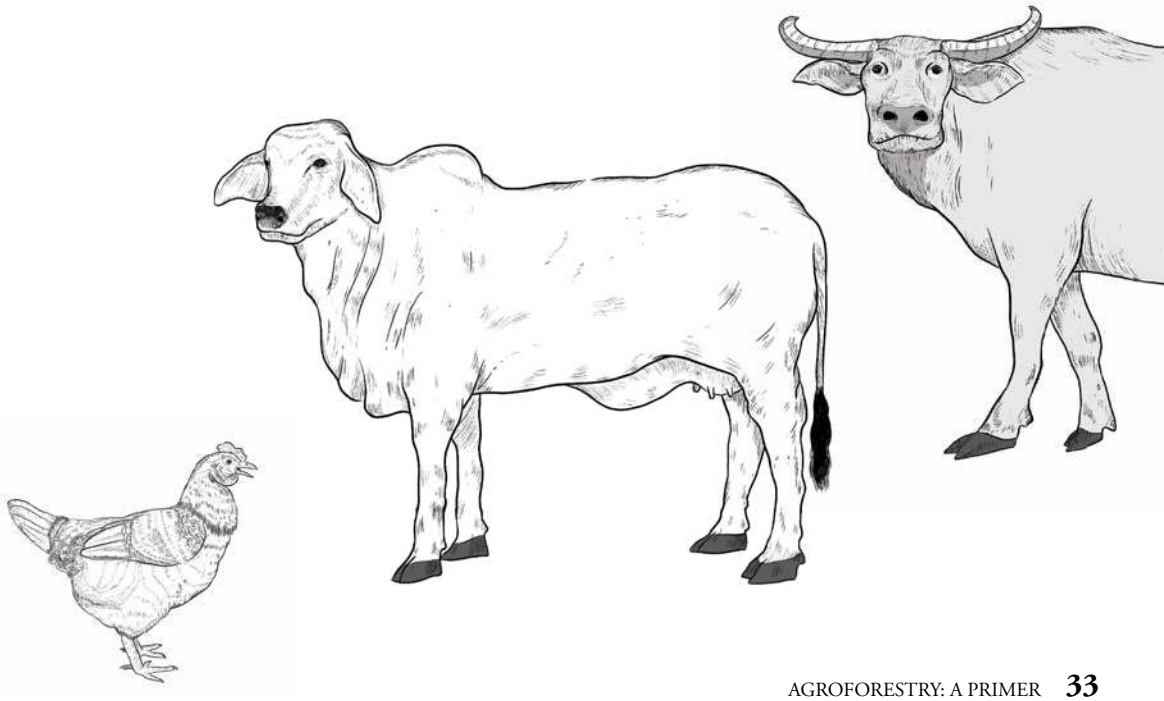
Tree crops play similar agroecological roles to those of other trees (see **Attributes of trees** in Chapter 2).

Attributes of livestock

The term 'livestock' means all domesticated animals that are kept for food and other agricultural products.

Associations between livestock and trees occur in a wide range of agroecosystems. The public often views these associations negatively – for example, when demand for cheap beefburgers leads to expansion of pastureland and to deforestation. However, livestock also interact synergistically with crops, trees and soils, and it is difficult to see how some mixed farming systems could be sustained without the contribution of livestock to household income and maintenance of soil health.

Livestock differ from other system components in several important ways. These affect both the commitment needed to manage them and the potential benefits they can bring. The most relevant characteristics are outlined below.



High value

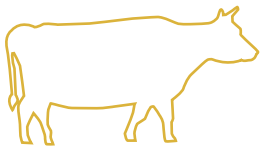
Almost all farming families in the tropics and subtropics keep livestock, ranging from a few chickens to herds of cattle. They do so because livestock offer a range of important benefits. Some of these are listed below:



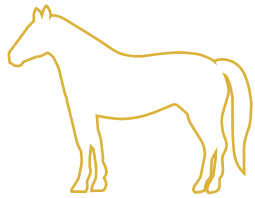
Nutritional benefits: dairy products, eggs and meat are important sources of proteins, fats and commonly deficient micronutrients such as iron.



Regular income: from sale of eggs and dairy products is very important for covering recurring expenses, such as food, transport, ongoing medical costs or school fees.



'Banks on legs': livestock, particularly cattle, can readily be sold to cover major expenses, such as university fees, weddings or unexpected medical costs.



Draught power: Oxen, horses and other large ruminants can be used on farm or rented out to pull ploughs, to power low-tech machinery such as sugarcane presses, or for transport.

High needs for care and maintenance

Each animal is valuable, and a large proportion of farmers' wealth may be invested in them. Their loss or devaluation through disease, accident and death – none of which are rare – can have catastrophic consequences: livestock farming is not risk free. This risk, coupled with the animals' need for access to feed and water, means that they require regular or even constant attention. If kept in pens, or confined in other ways, their keepers must feed and water them at least once a day, while free-ranging livestock must either be guarded (for example, by shepherds) or kept within regularly checked fencing. All these activities cost time and labour. On the positive side, the regular contact allows for close monitoring of health and reproduction.

Livestock need large amounts of food and water. For example, cattle need the dry feed equivalent of 3% of their bodyweight, plus 18 litres of water for each 100 kilograms of bodyweight, every day, so a cow weighing 400 kilograms has a daily requirement of 12 kilograms of dry feed and 72 litres of water.



When – as is common – resources are limited, either seasonally or for longer periods, consumption of feed and water is likely to be determined not by such guidelines, but by what is available. This limits productivity. In such 'lean periods', livestock keepers will often concentrate on keeping animals alive and as healthy as possible until the scarcity has passed. Long periods during which feed and water availability falls significantly below these levels will pose a risk to the health and well-being of the animal.

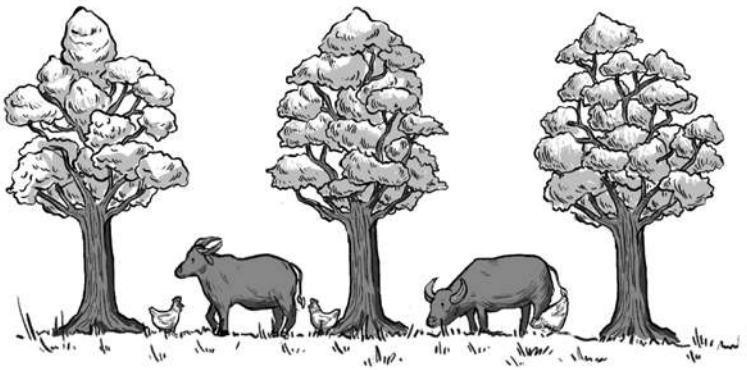
If more feed is available than what is needed for maintenance or survival, then livestock production will be proportional to the intake of balanced nutrients. Dairy livestock are a special case: they require sufficient water to support their milk output volume, irrespective of how much feed is supplied.

Mobility

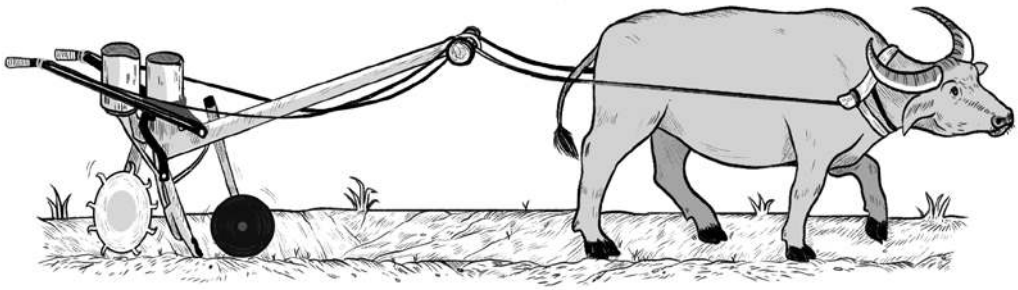
Unlike trees and crops, livestock have brains and often choose to pursue their own objectives, including going where they're not supposed to! This is one reason why they need careful oversight. But mobility also has its advantages. Unlike crops and trees, livestock are able to move to where food and water are available. This ability is useful even within a farm.

Agroecological and environmental benefits

Livestock offer both agroecological and wider environmental benefits. As part of balanced, well-managed agroecosystems, they can be highly beneficial to long-term sustainability. Some of these benefits are listed below.

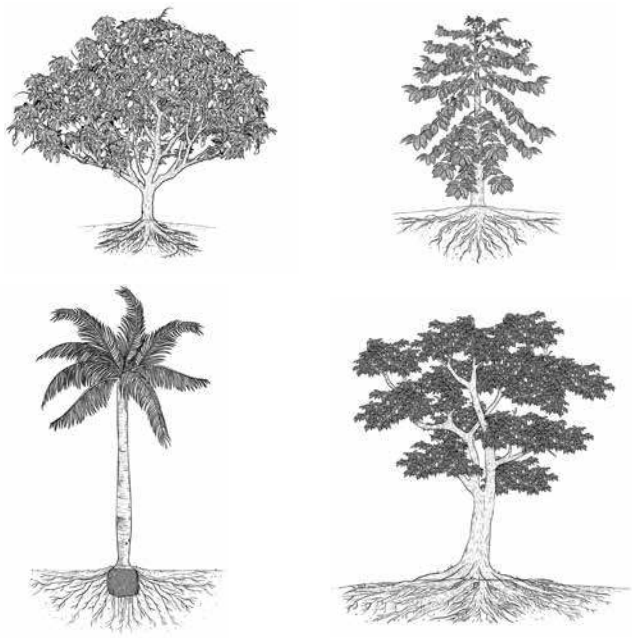


- **Livestock can make farming more efficient** by feeding on weeds or other plants that are planted as forage. Much of the nutrient content from these resources can then be recycled as manure to support other parts of the agroecosystem. Livestock waste (urine as well as manure) is a valuable resource for soil amendment, fuel and other uses.



- **Livestock can contribute to land restoration.** For example, they can provide draught power for operations that would otherwise require heavy and debilitating manual work or the use of expensive machinery. Their livelihood benefits can also act as drivers of adoption of environmental protection measures. For example, fodder plants – including trees – can be grown on contour banks, water diversion banks and other soil and water conservation structures to provide livestock with feed.

Attributes of trees



The role of trees

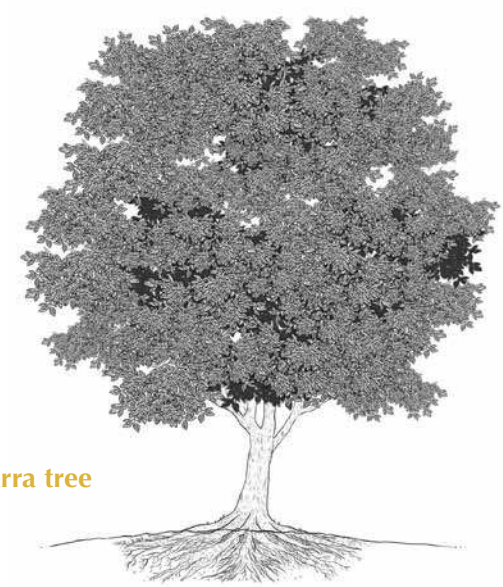
All agroforestry systems include trees. Flotilla tree species have two main functions:

1. They provide agroecological services to the flagship species or to the system as a whole.

2. Their products diversify the main income stream generated by the flagship species.

In selecting flotilla species, we need to consider not simply their own characteristics, including the goods and services they produce, but how well they will combine with the flagship species and other components of the agroforestry system.

In some traditional agroforestry systems, such as the damar and rubber agroforestry systems, the trees themselves are the flagship species. Some large-scale, mechanized agroforestry systems are also designed around timber trees.



Narra tree



Falcata tree



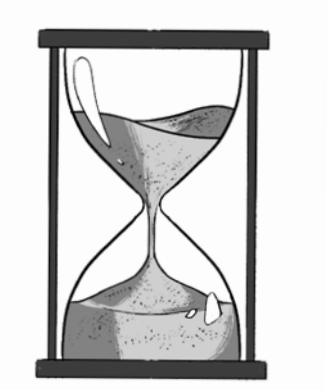
Musizi tree

Domestication

Unlike almost all crops, most trees in agroforestry systems are undomesticated. That is, they are very similar to their wild relatives.[2] Tree species are among the most genetically variable organisms on the planet. Because of this, seed sources for agroforestry trees must be chosen with care; the option of simply buying certified seed does not usually exist (see Chapter 6, **Planting material in agroforestry**).

By contrast, many fruit tree species have been domesticated and, as a result, have less genetic diversity. Clonal varieties of fruit species such as avocado and mango are an extreme, but common, case. They have no genetic variation, unless more than one variety is used.

Lifespan



The lifespans of trees vary from less than two decades to hundreds of years, but most timber or fuelwood species will be harvested long before the end of their natural lifespan. Lifespan and **rotation length** are important for two main reasons:

1. Farmers' choice of tree species is often strongly dependent on when they can expect a return.

2. Timber trees are difficult to harvest without damaging understorey crops. It is therefore better to synchronize felling with replacement (replanting) of flagship understorey species.

[2] Tree breeders have developed improved seed for some tree species. Trees of these genetically improved populations look very similar to those from unimproved or wild populations, although they may grow faster or have better form.

Crown characteristics

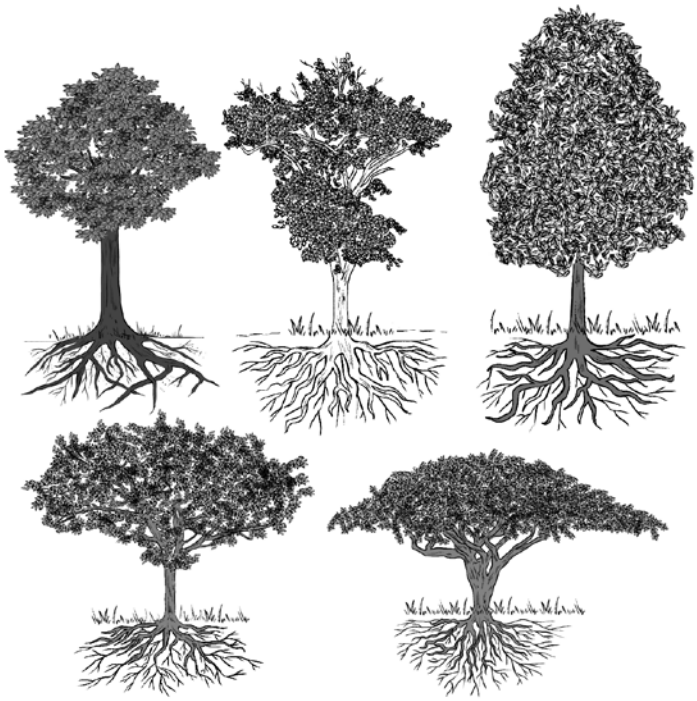


Figure 2. Variation in crown shape in tree species

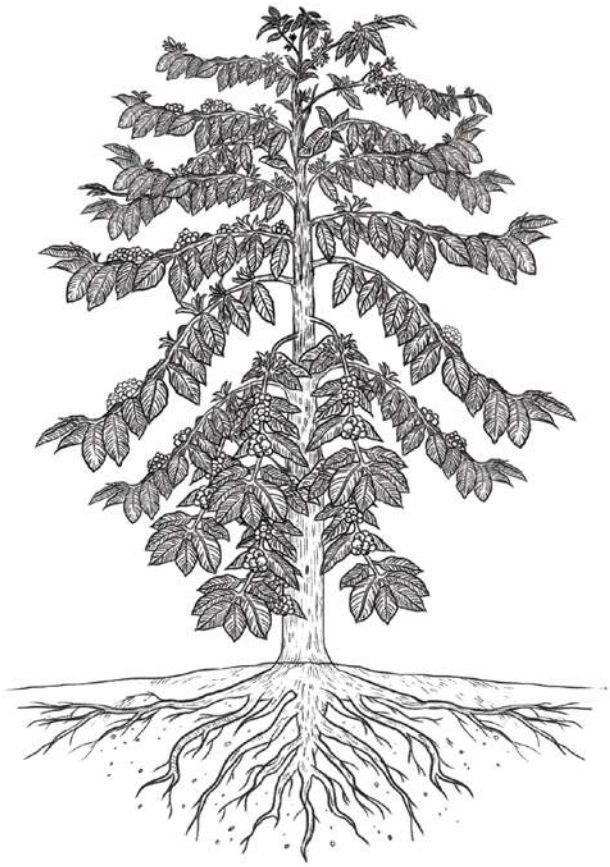
The crown of a tree consists of the live branches and leaves. The shape and density of the crown affect the amount of light reaching flagship species in agroforestry systems, and the choice of crown type and positioning of trees is critical in managing light. Tree crowns take many forms (Figure 2) but can be described simply using measures such as crown shape, size (diameter and height) and density.

Knowledge of crown characteristics of different species is needed to decide on 'the right tree'. If it is not possible to choose tree species with crown types that best suit the rest of the agroforestry system, then crown shape, size and density must be actively managed (see Chapter 7, **Management of trees in agroforestry systems).**

Deciduousness and leaf size

Whether and when a tree loses its leaves is also important: in particular, the timing and duration of leaflessness in relation to crop growth and to seasonal variations in rainfall and temperature. Some tropical tree species are leafless for extended periods at the hottest time of the year.

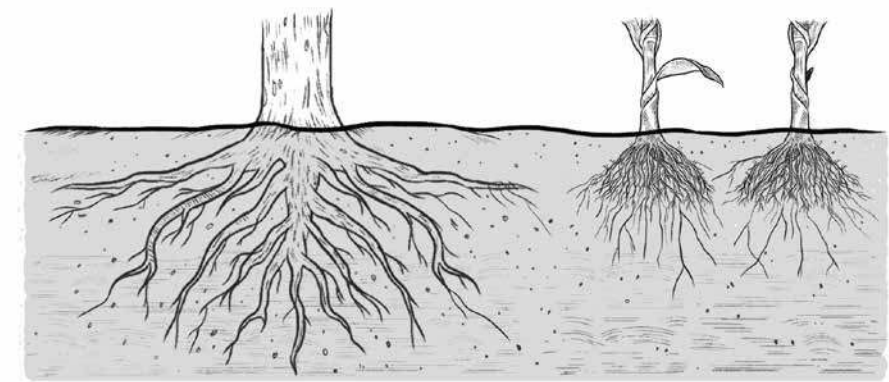
Some tree species naturally produce less shade than others. Such trees may have small leaves, or leaves that tend more to a vertical, rather than horizontal orientation.



Root characteristics

In general, trees have much larger root systems than annuals and other non-woody plants. They can extend for tens of metres – both in depth and horizontally.

However, although they often have deep roots, trees – like other plants – get most of their nutrients from soil layers near the surface, where their roots will compete with crop roots. Woody roots near the surface can also interfere with ploughing.



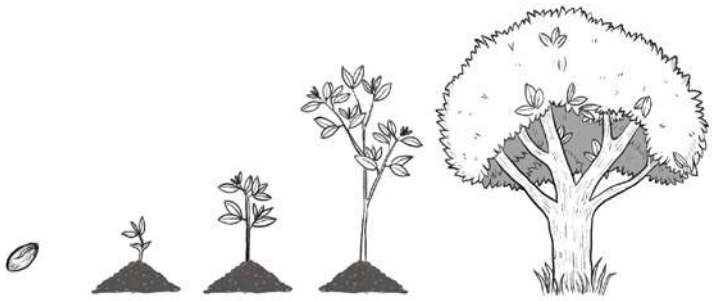
Some types of root systems combine very well with flagship species and other system components, particularly those that are deep enough to bring water and nutrients from layers that crops can't reach, and that have limited horizontal reach – and therefore compete little with crops. Although there is little scientific information about characteristics of root systems, farmers are often well aware of differences between species. This knowledge should be tapped in species selection (see Chapter 5**, Co-design and establishment of agroforestry systems**). However, local soil characteristics, such as density, structure, texture, moisture content and nutrient availability, also affect what the root system of an individual tree looks like.

Growth rate and invasiveness

In general, rapid growth of trees in agroforestry systems is an advantage. It speeds establishment of the system and reduces the time that farmers must wait for financial returns.

However, fast-growing trees use a lot of nutrients, and some species are so vigorous that they may be a poor choice for certain agroforestry systems. For example, all the commonly planted eucalypts are very fast growing, and when moisture or nutrients are not in abundant supply, they will often reduce the growth of flagship and other species. Many other species grow very quickly, but most eucalypts are adapted to survive and grow in nutrient- and water-scarce places, so they require special caution.

Fast-growing exotic species may also produce large quantities of easily spread seeds. The combination of fast growth and abundant seed enables them to invade natural vegetation and farmland, causing ecological and economic damage. Some invasive species, such as mahogany, do not grow particularly quickly. Farmers or their advisers should check candidate tree species against the **global invasive species database** before making final decisions.



Agroecological contribution

The central role of agroforestry in responding to global challenges is very much based on the agroecological contributions of trees. Some of these are outlined in Chapter 3, **Agroforestry systems as circular systems**. It is important to note that the environmental benefits of trees in agroforestry systems go beyond the scale of individual agroforestry systems, providing benefits at the scales of **landscape** (water regulation, **habitat connectivity**), region (climate regulation) and planet (climate-change mitigation).

Soil life



The soil is more than simply the earth in which plants are anchored. In a healthy agroforestry system, it should contain millions of living organisms that play vital roles in the agroecosystem. Without these, the nutrients

that plants need to be healthy and productive are not available, and there is almost no biomass circulation in fields. Table 1 lists five important functional groups of soil life; organisms in the same functional group contribute to the same ecosystem function.

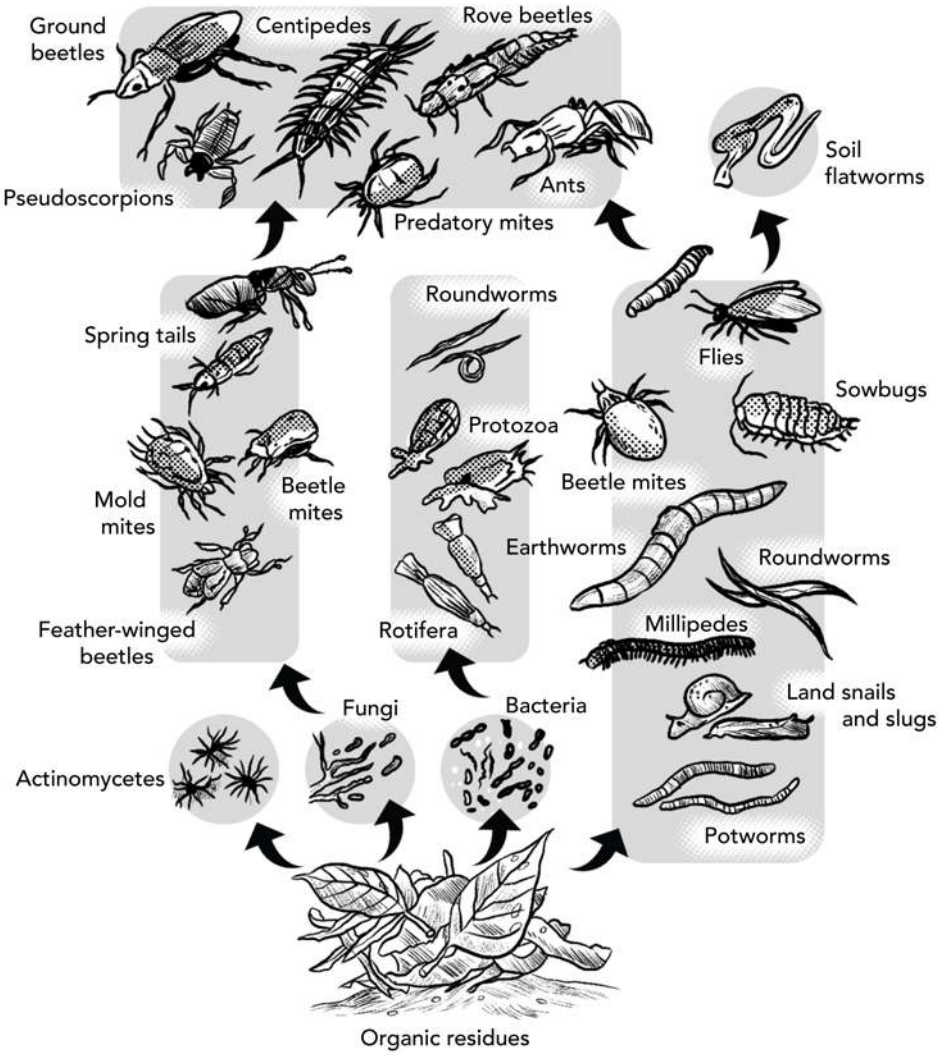
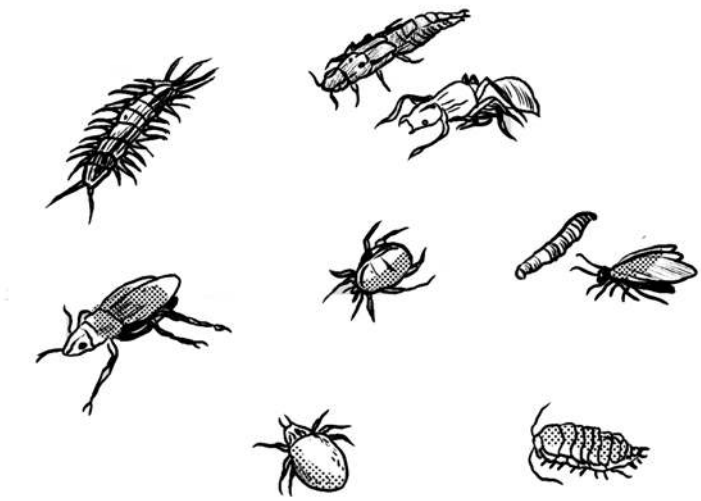


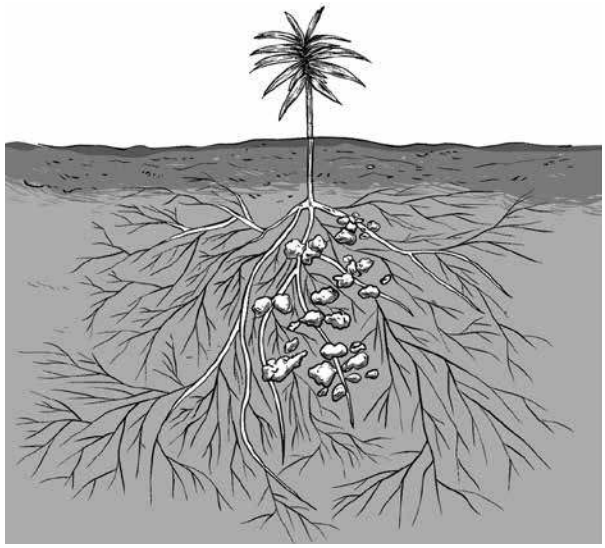
Table 1. Functional groups of soil organisms and the agroecological services they provide[a]

Functional group	Agroecological services provided
Microsymbionts	Plant nutrition and water supply Resistance to pests and diseases
Soil engineers	Soil structure (regulation and maintenance)
Decomposers	Decomposition Formation of soil organic matter
Elemental transformers	Nutrient supply

[a] Based on Swift et al. 2004. Table 1: Relationship between key functional groups of organisms, the ecosystem level functions they perform and the ecosystem goods and services they provide. *In* Biodiversity and ecosystem services in agricultural landscapes – Are we asking the right questions? *Agriculture Ecosystems & Environment*. 104(1):132.



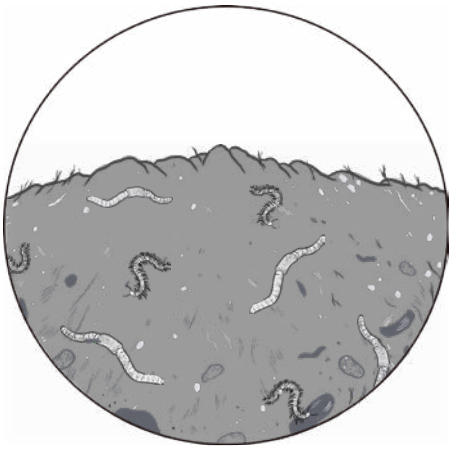
Microsymbionts



Microsymbionts include mycorrhizal fungi and nitrogen-fixing bacteria. Their importance is difficult to exaggerate. Roots are not 'just roots'; in most species they form close, mutually beneficial relationships called 'mycorrhizae' with certain types of fungi. These mycorrhizal fungi greatly increase the volume of soil that plant roots reach, help to make phosphorus available to plants, improve drought tolerance, and help prevent root infections. In addition, increasing evidence suggests that nutrients can move directly between trees through mycorrhizal connections – even between trees of different species.

Nitrogen-fixing bacteria form relationships with so-called nitrogen-fixing trees (see Chapter 3, **Agroforestry systems as circular systems**). In reality, the bacteria, not the trees, are responsible for the fixation process. This is the process by which the nitrogen in the air – where most of the planet's nitrogen is found – is 'fixed' in compounds that, unlike atmospheric nitrogen, can be absorbed by plant roots. These bacteria are responsible for about two-thirds of nitrogen fixation on Earth. Before the invention of efficient methods to manufacture nitrogen fertilizer they fixed almost 100%.

Soil engineers



Soil engineers include earthworms, termites and other invertebrates. They form and maintain the structure of the soil by burrowing, transporting soil particles and transforming these particles into soil lumps of different sizes.

Decomposers and elemental transformers

Decomposition is the gradual breakdown of dead animal and plant matter. Invertebrates start the process; the end point is the release of energy, water, carbon dioxide and nutrients as a result of the activities of bacteria. Elemental transformers are bacteria that obtain energy from simple substances; they convert the complex molecules in organic matter into forms that plants can use as nutrient sources. They are important in cycling important nutrients like carbon, nitrogen and sulphur.

3

AGROFORESTRY SYSTEMS AS CIRCULAR SYSTEMS





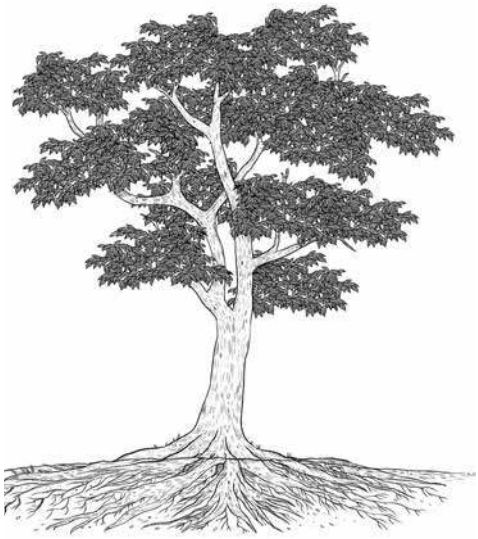
In agriculture, sustainability is often synonymous with increased efficiency of production. This implies increasing yield while reducing, or not increasing, the area of land used and consumption of water and fertilizer. Modern agriculture, which depends on the regular use of fertilizers, is a linear production process, whereby nutrients and water (agricultural inputs) are added to the system and converted into biomass, which is then exported in the form of animal or crop produce. Integrated farming systems, on the other hand, are circular systems that combine a diversity of crops, animals and trees with varying spatial and seasonal arrangements. They mimic natural water and nutrient-flow processes, with less need for artificial inputs like fertilizers, herbicides, and pesticides. In addition, agroforestry systems – one form of integrated farming system – are unmatched by other land uses in their ability to provide multiple goods and services at the same time.

To understand what these advantages are based on, it is useful to explore how agroforestry systems work. The following sections describe how agroforestry contributes to nutrient and water management, soil conservation and erosion control.

Nutrients and water

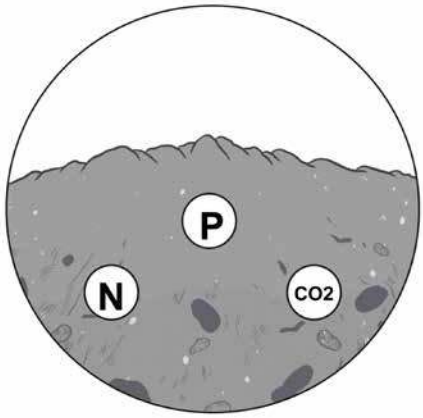
Plant nutrients are the chemical elements that make up the food that plants need to grow and thrive. These nutrients are found in the soil, and come principally from the rotting of leaves, twigs, stems and animal material; the very slow breakdown of minerals in the soil; and fertilizers applied by farmers. Soils that have low levels of nutrients are referred to as having low fertility and are not very productive.

Most soil nutrient sources are found in the upper soil layers. If the soil is in good health – with sufficient air, water and organic matter – it retains the nutrients, which can then be gradually absorbed by growing plants. If the soil is exposed, its upper layers – and the nutrients in them – can be washed away in water runoff. In poor quality soils, particularly those that are sandy textured and free draining, nutrients can also be washed rapidly into deeper soil layers, below the level of crop roots. This is called 'leaching'.



Trees can play an important role in keeping nutrients available by preventing runoff of water and improving its retention in the soil. Their deep roots can also pull up water and nutrients from far down in the ground. Nutrients make up the 'building blocks' of stems and leaves, or are used by the trees in other ways. After the old leaves, stems, twigs and branches fall as litter, they rot and provide more nutrients needed by plants. Trees are therefore important nutrient-recycling 'pumps' that maintain good soil conditions for plant growth (Figure 3).







Nitrogen and phosphorus are two of the most important minerals for plant growth. Lack of both is one of the main reasons for resorting to artificial fertilizers, which is why effective nutrient recycling is so important. Nitrogen-fixing trees, which come mostly from the very large legume (pea) family, can substantially improve the fertility of soils, and can be used in several ways in agroforestry systems. One approach is to grow them in rows, with crops planted in between, so that the falling leaf litter directly fertilizes the soil. Young tree branches can also be cut and mixed into the soil. Sometimes 'tired' soils are left in fallow – that is, without crops – to allow trees and bushes to grow back naturally. Fallowing allows soils to recover from overuse, and nitrogen-fixing shrubs and trees, such as some leguminous species as well as alder and casuarina, can be grown on the fallows to speed up that process.

Efficient fixation of nitrogen requires a minimum level of phosphorus in the soil; fixation can be insignificant in soils that are low in phosphorus, and this is often the limiting growth factor. Animal manure is a good source of phosphorus, which is one reason why livestock are an important part of numerous agroforestry systems. Many trees provide shoots and leaves that can be fed to animals; the resulting nutrient-rich manure can then be applied to crops in the system, including by carrying manure to fields from animal pens (see **Livestock with trees** in Chapter 8).

Soil health depends not only on the presence of nutrients and water, but also on its physical quality. Farmers can maintain that quality by ensuring their soil always contains sufficient organic matter. Soils that are low in organic matter are easily washed or blown away, lose water and nutrients, and can become compacted and difficult to cultivate.



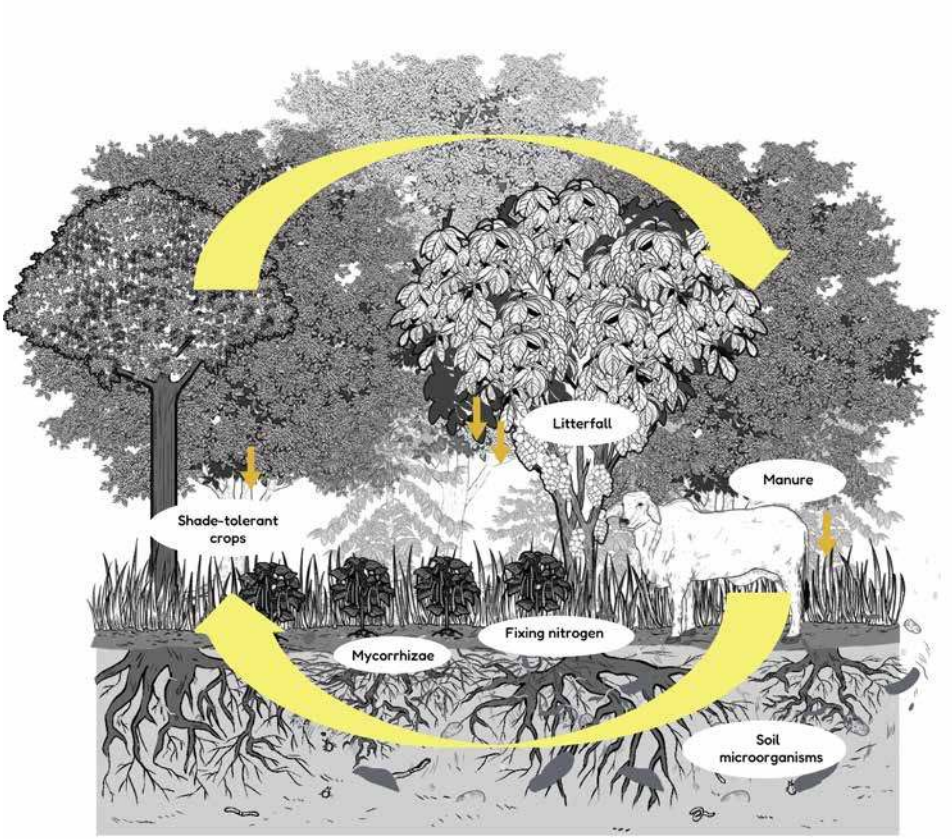


Figure 3. The nutrient cycle

Trees pull up water from the soil through their roots. Some of this water is then released into the air by a process called 'transpiration'. Water that falls as rainfall either evaporates (due to the sun's heat), infiltrates the soil, or flows as surface runoff into watercourses, ponds and lakes. The combined process of transpiration and evaporation is known as 'evapotranspiration'. It is the total amount of water that is released to the atmosphere (Figure 4). Water infiltration into the soil depends on the structure of the soil's surface and lower layers, which is strongly influenced by its organic matter content.

Trees also help to regulate temperature. Their shade reduces daytime temperatures, which can protect crops from extreme heat. At night, the tree canopy increases temperatures by trapping heat, which can prevent frost damage, but may also reduce yield. In **multistrata systems**, the

shade cast by upper-storey trees cools the lower storeys, which reduces evapotranspiration and helps to keep water in the soil.[3]

Trees usually consume more water than crops; they may therefore compete with crops for water. Fortunately, crops mostly take up water from the upper soil, while trees usually have deeper-reaching roots and can access water from deeper soil layers, which can reduce competition between the two components. However, the level of potential competition varies depending on the tree species, and expert advice from farmers or professionals is needed when deciding which trees to plant with which crops.

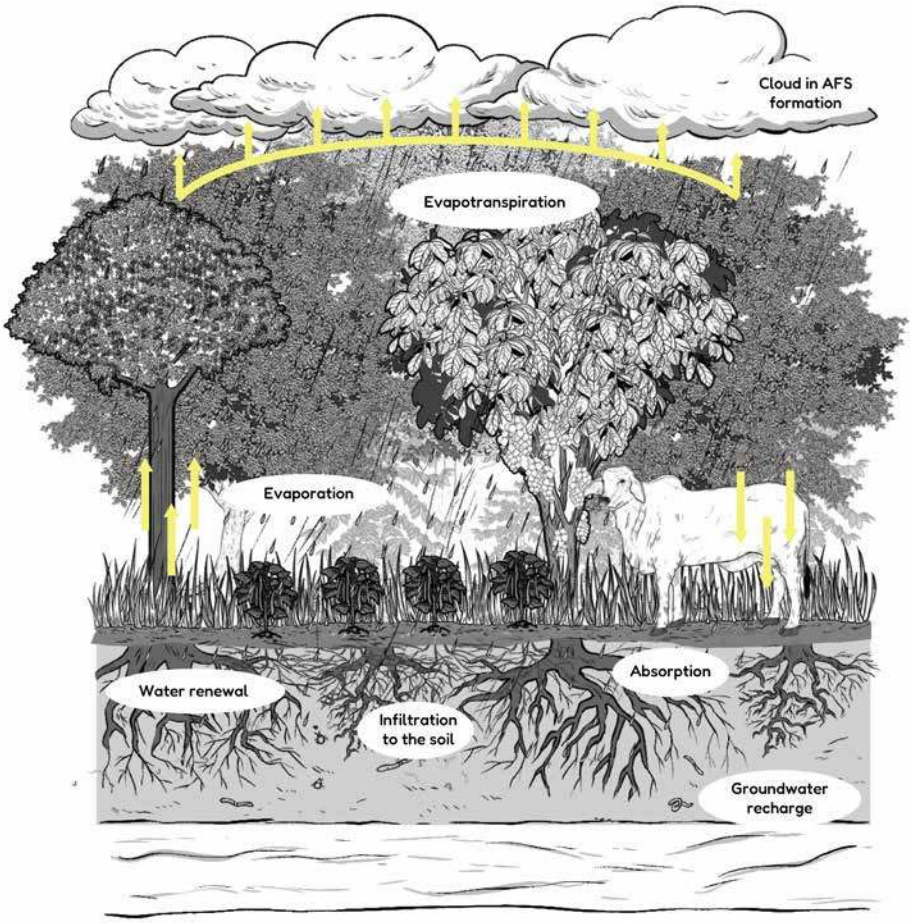
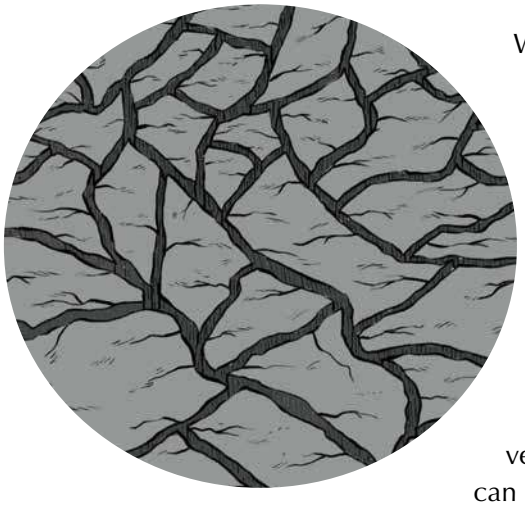


Figure 4. Water cycling in an agroforestry system

[3] For more information on this complex topic see Gosme M, Dufour L, Inurreta-Aguirre H, Dupraz C. 2016. Microclimatic effect of agroforestry on diurnal temperature cycle. Paper presented at the 3rd European Agroforestry Conference, Montpellier, 23–25 May 2016. http://www.repository.utl.pt/bitstream/10400.5/11690/1/EURAF2016%20-%20Book%20of%20Abstracts%20final.pdf.

Soil conservation and erosion control

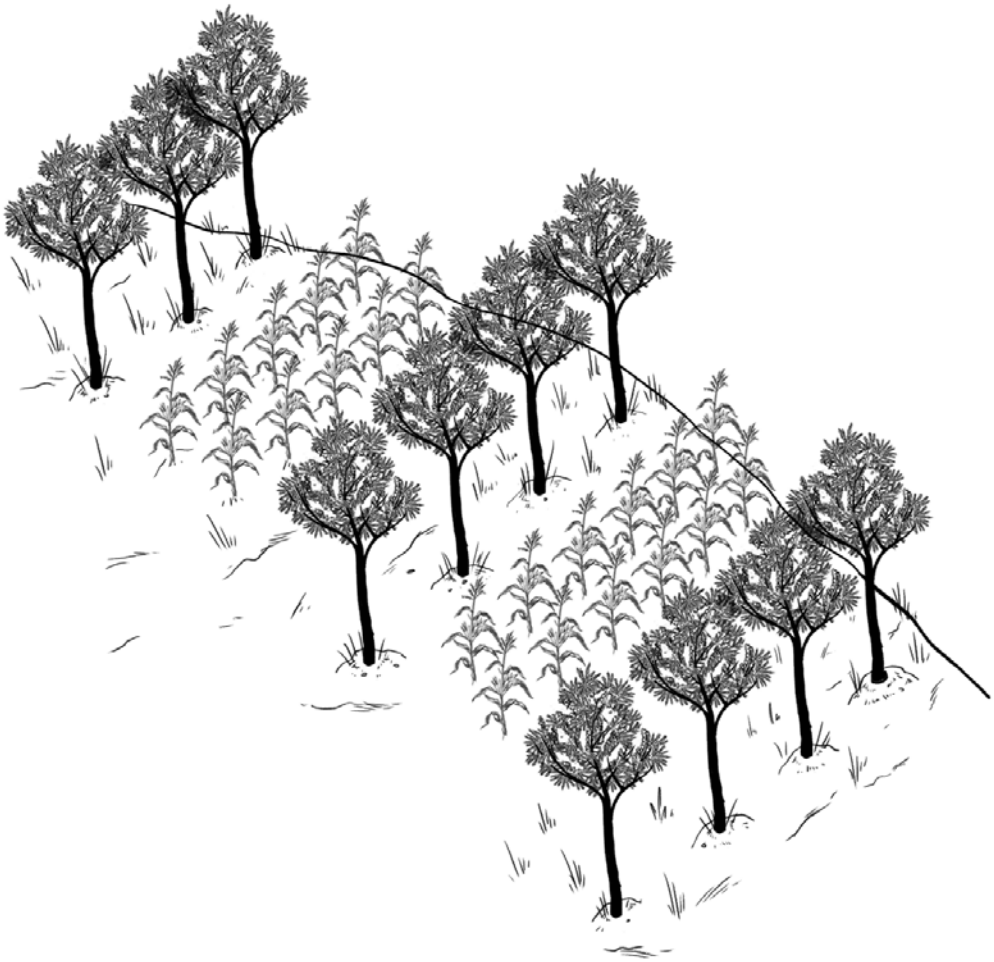


When soil is exposed by the preparation of land for crop planting, it is vulnerable to being blown away by the wind (wind erosion) or washed away by the rain (water erosion). Such soil loss can seriously reduce a farm's capacity to grow crops. The risk of erosion is particularly high in windy locations; in places where the bare soil has been exposed by removal of all vegetation; and on slopes, where rain can quickly wash the soil downhill.

The first way of preventing erosion is to ensure that the land is managed in a way that retains as much plant cover as possible (the cover approach). Then, erosion can be controlled by preventing water flow from carrying away the soil (the barrier approach). Barriers can hold water long enough for it to penetrate the soil, but if local conditions lead to an excessive build-up of water, the barriers should allow controlled runoff to minimize erosion damage and allow the water to reach natural streams and lakes.



In agroforestry systems, trees provide continuous vegetation cover, and their roots play an important role in binding the soil and protecting it from erosion. If planted on slopes, they create barriers that prevent rapid water flow.[4] Grasses like Napier and Vetiver can be planted between tree rows to make the barrier more effective, while providing food for livestock.



[4] Zomer, Robert J, Bossio DA., Trabucco A, Yuanjie Li, Gupta DC, Singh VP. 2007. *Trees and water: Smallholder agroforestry on irrigated lands in Northern India*. IWMI Research Report 122. Colombo, Sri Lanka: International Water Management Institute. 41p. https://doi.org/10.3910/2009.122.

4

PRINCIPLES OF AGROFORESTRY DESIGN





Agroforestry design involves deciding what will be produced, selecting the components of the system, and determining how they will be arranged. These decisions also define how and when the components will be established, and how they will be managed. It also includes other considerations, such as how to support the enabling environment for agroforestry.

Farmers have been designing agroforestry systems for millennia – sometimes consciously, sometimes instinctively. Professionals can support this process in several ways:

- by advising farmers who haven't managed agroforestry systems before, or farmers who are recent migrants to a given agroecological zone
- by helping to ensure that the systems meet wider goals (particularly environmental ones) in such a way that farmers' interests are not negatively affected
- by introducing farmers to new scientific knowledge
- by organizing or supporting collaborative design processes that include all relevant stakeholders.

In this section, we outline three principles of agroforestry design: farmer-centredness; aptness to people, place and purpose; and synergy. Successful agroforestry interventions follow these design principles. Interventions that ignore them have a high chance of failure.

The principle of farmer-centredness



In designing agroforestry projects and programmes, put the goals and aspirations of farming families first, before considering how you will meet other, wider goals.

Farmers' goals and aspirations

Smallholders and their families are as diverse as the land and living things under their stewardship, but they also often have characteristics and constraints in common. These may include the following:

- Very often, farming is not their sole or even their main source of income.
- Their other commitments may leave limited time to spend on their own farms.
- They tend to avoid risk because they are not wealthy enough to be able to absorb financial losses without losing the ability to satisfy basic needs.
- They tend to favour economic activities with short-term returns.
- They will not necessarily occupy their current property forever; for example, they may want to leave for the city or to acquire a more productive property.
- They may have a deep knowledge of local plants, animals, land and agriculture; on the other hand, if they have recently migrated from another region, they may have little local knowledge of their new surroundings.

These characteristics shape the goals and expectations that lead farmers to grow trees on their land. Their main motivations are to help meet their needs for food and income, and to protect the fertility and productivity of their land. Trees may also have other benefits, such as marking boundaries or establishing the household's rights over land.



Applying the principle of farmer-centredness



Many institutions promote agroforestry to support other goals – particularly environmental ones, such as biodiversity conservation, water management, climate-change mitigation or land restoration. All of these are valid goals, but they cannot override farmers' priorities. Farmers should not be 'steamrollered' by persuasive and enthusiastic professionals into installing agroforestry systems that do not meet their needs.

Rather, institutions should first ensure that their proposed interventions meet the following requirements:



They should generate income or provide useful products.



They should reduce the risks faced by farm households, such as food and nutritional insecurity, threats to land tenure and sovereignty, falling prices, droughts, pests and diseases.

Once these requirements are met, institutions can consider how to further their own goals without reducing the benefits to farmers. For example, to support biodiversity goals, a programme to install shelterbelts in pastureland might prioritize native species, especially **keystone species** – providing that these generate the same benefits to farmers as the alternatives. When these alternatives have higher establishment costs, then farmers should be compensated for the extra expense (see **Incentives** in Chapter 5).



The principle of aptness to place, people and purpose

In designing agroforestry projects and programmes, remember that 'one size fits all' does not apply: all agroforestry systems need to be customized to local conditions.



The principle of farmer-centredness guides the overall approach to the design of agroforestry systems by putting farmers' goals first. The principle of aptness is about delivering on these goals. This requires careful consideration of the specific characteristics of place, people and purpose. Such locally tailored interventions contrast starkly with 'off-the-shelf' interventions based on inflexible technical packages.

Aptness to place



All agroforestry designs must be adapted to each farm's specific agroecological conditions: a system cannot simply be transferred from one farm to the next without considering the characteristics of each farm – and each planting location within that farm. For example, although shade trees can increase the biodiversity of cacao and coffee plantations, as well as fulfilling valuable agroecological roles, the natural shadiness of the site must be considered when deciding how many shade trees to plant per hectare – which depends on which way it is facing, on the surrounding landscape and on the cloudiness of the location. Our sections **From principles to practice: Key systems** (Chapter 8) and **Stories from the front line** (Chapter 9) provide other examples of how agroforestry systems must match specific sites.

Aptness to people



The agroforestry system that is chosen must be adapted to local knowledge, the capacity of the farmer, and the availability of labour. Farmers are usually highly skilled and often have detailed knowledge of local soils, climate, indicator species, pests and diseases, and crop management. However, their knowledge is not boundless, and their skills and intuition may be limited to those crops with which they are most familiar. Also, many farmers are migrants, and they may have moved from areas that are very different to their new locations. In practice, this means either that agroforestry systems should centre on crops or products with which farmers are familiar, or that supporting agencies must provide ongoing support (extension and training), or both of the above. The more integrated the components of an agroforestry system, the more labour is generally required, especially in the first few years. For this reason, it is important to design and implement systems that are not too large to be manageable given the amount of labour that is available to the household or landowner.

Aptness to purpose



In the past, agroforestry has sometimes been promoted to farmers as a 'silver bullet' technology. In this approach, more emphasis has been placed on adoption of agroforestry itself than on specific products and species choice. By contrast, aptness to purpose means that the agroforestry design, particularly the species used and the products they yield, must correspond closely and rigorously to farmers' expectations. It is not enough for a system to be technically effective. The product or products are central, while the agroforestry system is viewed not as an end in itself but as a way of producing them.

Farmers' species preference is strongly affected by available markets. Trees take a long time to grow and to produce, and it is important to consider whether markets are likely to be similar once the product becomes available.

Feasibility



The principle of aptness incorporates the concept of feasibility. Biological, economic, legal, logistic and sociocultural constraints may make it difficult to shift from an existing farming system to agroforestry. Planners must analyse these constraints carefully and realistically, and make sure that the plans they make take them into account.



The design process (see **Design process** in Chapter 5) requires consideration of all these aspects of aptness. Together, they make up the local agroecological, human and economic context to which interventions must be adapted. Design often includes a component of adjustment of generic options (for example, multistrata cacao agroforestry) to specific local contexts.

The principle of synergy

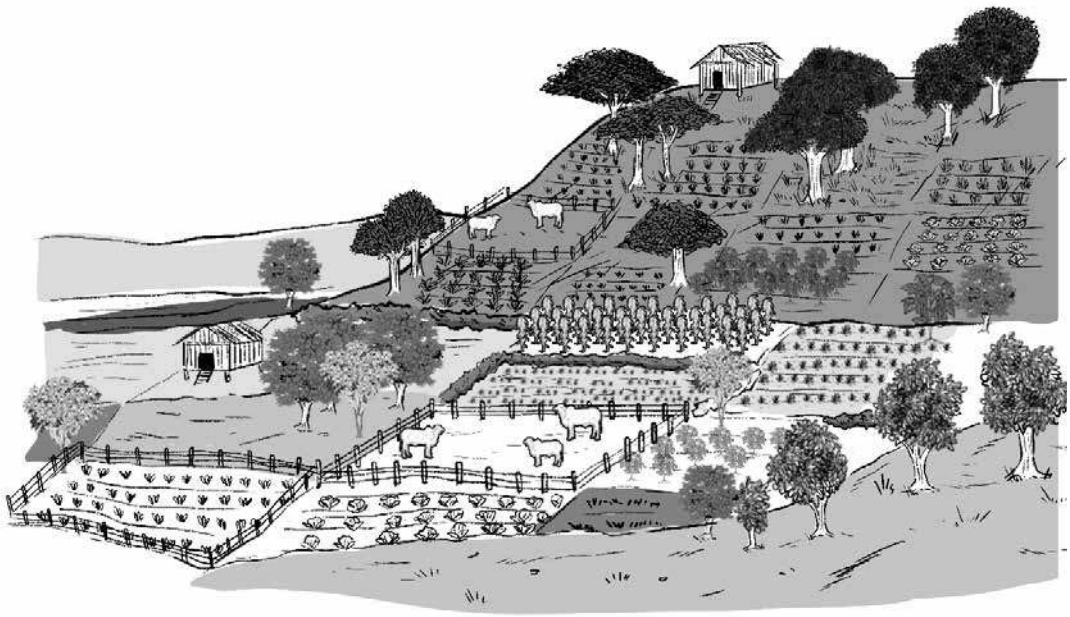
In designing agroforestry projects and programmes, harness the full potential of agroforestry by ensuring that trees, crops and livestock interact in mutually beneficial ways that sustain multiple ecosystem services.

To realize the full potential of agroforestry, systems must be designed so that the components work together in a complementary way. This is referred to as 'synergy'. Basic levels of synergy can be achieved even in simple agroforestry systems in which one crop or a few crops dominate, and trees and livestock are added to provide additional products for the household, income and ecosystem services. To achieve synergy, trees, crops and livestock are arranged to make the best use of nutrients, water and energy within systems, while managing the competition for them.

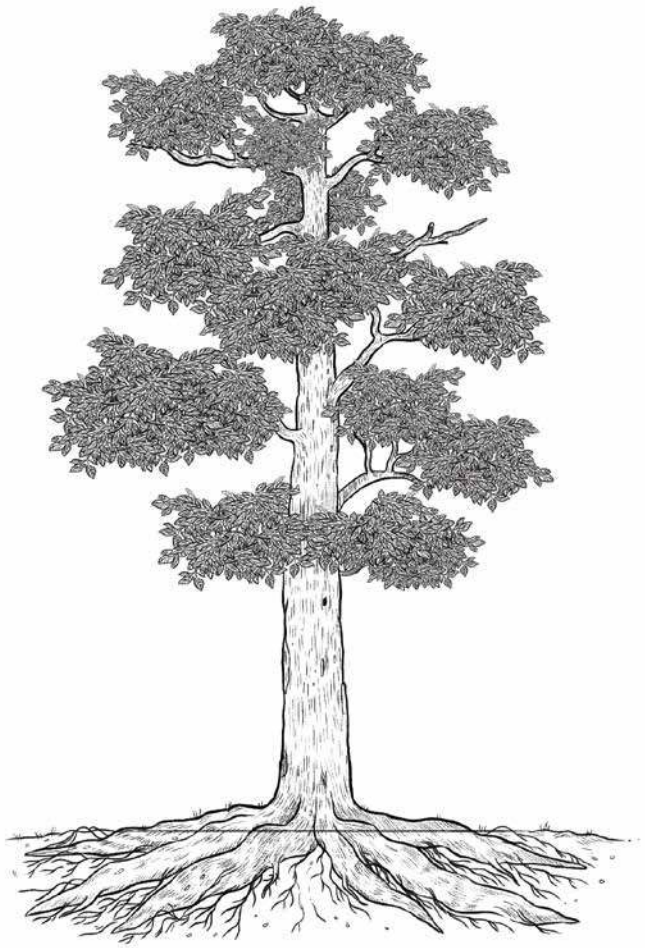
Full synergy is achieved when the productivity from an agroforestry system is greater than it would be if the components were established in separate **monocultures**. These agroforestry systems are 'more than the sum of their parts'. This is usually achieved by arranging trees, crops and animals in

such a way that they have maximum interaction. Competition in these systems is controlled by carefully selecting species that support each other. For example, in **successional systems**, resource sharing between different species is optimized by planting species with different life cycles, which succeed each other over time. These systems are designed to mimic the nutrient and water flows in natural ecosystems, such as forests, and minimize the need to bring fertilizer in from outside the system.

Where possible, agroforestry systems should have a high diversity of species. These not only provide a range of food and income opportunities for farmers, but also increase ecological complexity and support ecosystem services such as pollination, shade, soil fertility and biodiversity conservation. Diverse systems benefit farmers by producing food and products for sale, and improving the long-term productivity of the land – while also contributing to local and global environmental benefits. Some plant species produce substances that reduce the germination and



growth of neighbouring plants. This is referred to as 'allelopathy'. Trees have more apparent allelopathic effects than other plant types. Local communities usually have knowledge about which plant combinations should be avoided. If that knowledge has been lost, or local people are migrants to the area and do not have it, it is essential to consult a plant specialist before introducing new, unknown combinations.[5]



[5] The science of allelopathy is quite complex. See for example Rizvi SJH, Tahir M, Rizvi V, Kohli R, Ansari A. 1999. Allelopathic interactions in agroforestry systems. *Critical Reviews in Plant Sciences*. 18(6): 773–796. https://doi.org/10.1080/07352689991309487.

5

CO-DESIGN AND ESTABLISHMENT OF AGROFORESTRY SYSTEMS



Farmers have been designing agroforestry systems for thousands of years. They are the users and stewards of the land, and they often have a deep knowledge of the local plants, animals, the soil, weather, and the interactions between them. We use the term 'co-design' to describe the process of bringing scientific and technical expertise together with this local knowledge. It is an important step in creating agroforestry land-use systems that deliver optimal outcomes. The role of outside experts is to listen and learn from the farmer; to share their specialist knowledge, including experience they have gained from working in other regions or countries; and to facilitate the overall design process.

In this chapter, we provide guidance on:

- the kinds of information or knowledge needed to co-design agroforestry systems; and
- how to gather and use this information.

If both these components of co-design are done well, then the systems we design will reflect the principles of farmer-centrality, aptness and synergy.

What kind of knowledge is needed for co-design of agroforestry systems?



In co-designing agroforestry systems, the most important types of knowledge needed are the following:

- knowledge about the needs, aspirations and capacities of farmers and their families
- knowledge about the profitability of different agroforestry products
- knowledge about the local conditions that might affect the profitability or feasibility of such products
- knowledge about the different agroforestry systems in which selected products can be grown.

In most cases, no one person or group is the holder of all these types of knowledge. For example:

- Individual farmers know best about their own needs and aspirations.
- The local community will know about any customary limitations on land use.
- Marketing specialists may have information on current and future demand for products.
- Agroforestry specialists will know how much labour is required to manage different systems.

The information needed for agroforestry design will be different in each case. However, many of the same questions come up often. Table 2 lists the most common ones, and the most relevant knowledge holder(s) in each case. Additional specific questions to be addressed will vary from place to place.



Table 2. Core questions for agroforestry design

Question	Knowledge holders
What does the farming family want to produce on the farm, and how can trees contribute to that?	Farming family
How important is the farm for the farming family, compared to other tasks and any other work that require time away from the farm?	Farming family
Are the farmer and family members interested in planting more trees on the farm? If so, then why? (e.g., to provide better growing conditions for crops, increase fodder production, sell timber)	Farming family
What adult labour is available for maintaining any new trees that are planted on the farm?	Farming family
What previous experience does the farmer have in managing agroforestry systems?	Farmer
What is the farmer's level of technical understanding, and how willing are they to master a new system?	Farmer
What is the soil like in the parts of the farm where the farmer plans to plant more trees?	Farmer
Who is the owner of the farm? If it is not the farmer, what rights do they assert or have over the land? (Trees are a long-term investment and may be incompatible with short-term or insecure land tenure)	Farmer, technical experts
What are the main features of the climate of the area, particularly the mean annual rainfall and the length and timing of any dry season(s)?	Community, technical experts
What risks are associated with farming in the area, including pests and diseases?	Community, technical experts

Table 2. Core questions for agroforestry design *cont...*

Which farming systems and practices are used in the area?	Community, technical experts
Which crop species and varieties, livestock types and breeds, and tree species grow well in the area?	Community
Which tree species with the desired characteristics are available locally?	Farmers
Which crop species and varieties, livestock types and breeds, and tree species can be grown together?	Farmers, agroforesters
What are the most profitable agricultural products in the area?	Community
Can value chains be made to work better for farmers and other actors? (Farmers need to be connected with buyers and producers in order to sell their agroforestry products)	Community, technical experts
Does poor transport infrastructure (e.g., poor roads or absence of roads) make it difficult to get products to market?	Community, technical experts
What laws and government policies regulate production, harvesting and transport of livestock, tree and crop products?	Community, technical experts
Which species characteristics, such as biological traits, income-earning potential, and food value, would allow farmers to best address their objectives? (Several techniques are available to help participants rank their preferences for tree characteristics. It is also important to explore which characteristics are not valued)	Farmers, agroforesters
What are the main agroforestry options suitable for the area?	Farmers, community, agroforesters
What are the labour requirements of different agroforestry options, and how do they fit with households' other time and work obligations, considering the interests of families?	Community, agroforesters
How easy is it to make the transition to agroforestry? (A given agroforestry system may seem to offer what farmers are looking for, but it may be that the challenges of conversion from current land uses are too great for the system to truly be apt to people and place)	Farmers, community, agroforesters

Design process

Finding the optimal agroforestry system for a community or farmer(s) is a journey rather than a straight-line, one-way process from knowledge, through analysis, to formulation of interventions. In a participatory procedure, such a regimented approach would be problematic. The role of those facilitating the journey is to ensure that knowledge and views are sought, obtained and shared; those views are well informed; and that decisions are consensual, even when that means 'agreeing to differ'. Facilitators or extension service providers who work with farmers to develop agroforestry systems need to be:

- well versed in local conditions;
- equipped with approaches to elicit information from other knowledge holders;
- receptive to local views;
- able to facilitate the sharing of information between stakeholders.

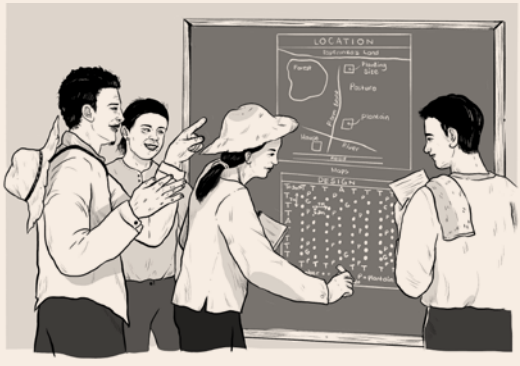


Box 2. Participatory techniques for gathering village- and household-level information of relevance to agroforestry design

Participatory mapping involves drawing maps with community members, and can help to develop common understanding. Community representatives could add to a map of their village drawn with markers on a large sheet of paper, or scratched into the earth with sticks, to show the distribution of land and social relationships, including common resources and any land-use conflicts.

Transects, also called **group walks**, can help get a sense of how people interact with their production systems and environment. A walk across the farm, combined with group discussions, can help outsiders gain valuable insights and information about the interplay between social and environmental factors.

Role-playing games: a landscape or farm can be represented simply as a board game in which participants are asked to make decisions about management. They can test out different courses of action and their potential consequences; exchange experiences, knowledge and perspectives; and build understanding about why others make particular choices. Subsequent discussion about how well the game represented reality provides further information about how players make decisions in real life.[6]



[6] A lot of useful information on participatory approaches can be found from many sources on the internet. A useful reference for this sort of game playing is Speelman EN, Rodela R, Doddema M, Ligtenberg A. 2019. Serious gaming as a tool to facilitate inclusive business: A review of untapped potential. *Current Opinion in Environmental Sustainability* 41: 31–37. https://doi.org/10.1016/j.cosust.2019.09.010.

Various approaches to participatory procedures of this sort exist (see Box 2 for examples). They need to consider local culture and society, including gendered interests and needs, as well as levels of literacy and broader education. A formal Participatory Learning and Action approach can be appropriate, but supporting agencies should select approaches they are familiar with and that they have capacity to implement.

The time needed for different approaches depends on the depth of knowledge required and the farmers' capacity. Some processes are completed in a day, while others require deep trust and involve long-term relationships between the external agents and community.

Independently of the specific methodology used, facilitators and interviewers should take care to avoid common pitfalls by adhering to the following guidelines:

- Keep in mind that a truly participatory process is not designed to deliver a particular technique to a farmer, but is rather a series of activities in which the expert and farmer learn from each other and develop interventions together.

- Be aware that interviewing farmers is not a participatory approach in itself. In a question-and-answer interview, often used in farmer surveys, many participants will give answers that they think the questioner wants – or indeed will say anything just to get rid of them!

- Consider that farmers are usually not used to discussing their land-use decisions or livelihood aspirations with other people.

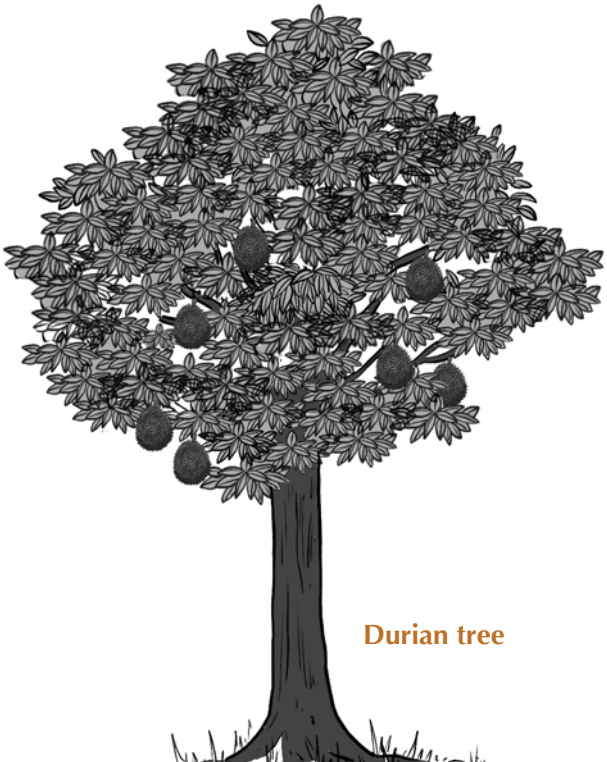
- Ensure engagement with those who carry out the physical tasks of farming and those who make decisions – whether these are women or men. Other community members should also be involved: in particular, youth should be included, as they may one day manage the systems.

Species selection

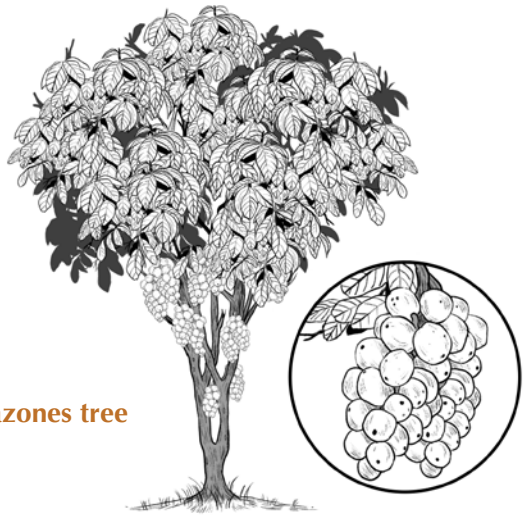
Species selection is a particularly important part of agroforestry design because the species choice determines which goods and services will be produced by the system. The design principles of aptness and synergy are particularly important in species selection. Species should be at least compatible with each other, and preferably capable of mutually beneficial interaction, and their products should have high market potential or other uses important to the farmer.



In some cases, specific methodologies or criteria may be used in species selection – for example, the fruit tree portfolio approach (Box 3), or use of native species (Box 4).



Durian tree



Lazones tree

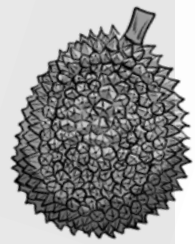
Box 3. Fruit tree portfolios: A targeted approach to species selection

In places where diets are heavily based on starchy staple crops, farming families – and especially women and children – are prone to debilitating nutrient deficiency diseases. The fruit tree portfolio approach to species selection responds to this problem by enabling the year-round production of fruits that are rich in vital nutrients and vitamins. It does so by taking advantage of the varying harvest times of tree fruits.

1 **The first step** in building a fruit tree portfolio is to determine whether a given community experiences food insecurity and serious nutrient gaps. For example, in many parts of the world, farming households experience 'lean seasons' before the main harvest, when stored household food supplies are running short.

2 **The second step** is to identify additional fruit tree species that can compensate for food and nutrition gaps, especially during the gap periods identified in the first step. Indigenous and underutilized tree species have an important role to play in local farm systems because they are often well adapted to the local soil and climatic conditions, and less vulnerable to variation in rainfall.

Both steps are highly participatory processes. This ensures that the portfolio responds to real needs, and that the species selected are locally available and acceptable to farming families. The portfolios can be further adapted to include other food trees – those that provide nuts, leafy vegetables, seeds and oils – and other crops such as vegetables, pulses and staples.[7]



[7] The fruit tree portfolio approach can be used with virtually any agroforestry system. It can be applied at the individual farm or community level. More details on the approach can be found in McMullin S, Njogu K, Wekesa B, Gachuiri A, Ngethe E, Stadlmayr B, Jamnadass R, Kehlenbeck K. 2019. Developing fruit tree portfolios that link agriculture more effectively with nutrition and health: A new approach for providing year-round micronutrients to smallholder farmers. *Food Security* 11: 1355–1372. https://doi.org/10.1007/s12571-019-00970-7.

Box 4. Trees and biodiversity conservation

Trees-on-farms contribute directly to the conservation of biodiversity by:

- increasing agrobiodiversity – the variety and spatial variability of animals, plants and microorganisms in an agricultural landscape
- conserving indigenous, rare or threatened tree species (in situ conservation)
- providing foraging or breeding opportunities for wild or farmland-adapted animals.

Almost all agroforestry systems lead to an increase in agrobiodiversity (the planted species and livestock), but they do not necessarily contribute to the conservation of naturally occurring species or habitat connectivity. To do so, more native trees and species – those that occur naturally in the planting zone – need to be included in the system. However, from the farmer's viewpoint, native species often have both opportunities and challenges:

Opportunities

- Native tree species are well adapted to the local soil, water and climatic conditions.
- Farmers and local markets are often familiar with the species and their products.

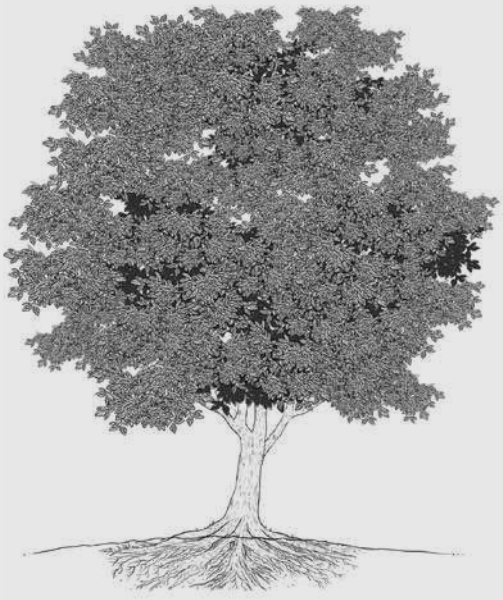
Challenges

- It may be difficult to obtain seed or plants of native species, particularly if they are rare or threatened.
- Many countries control the harvesting and transport of native tree species. These regulations, designed to protect natural forest from overexploitation, often apply (counterproductively) to planted trees on private land. The administrative procedure of applying for felling and transport licences can be complicated.

Box 4. Trees and biodiversity conservation *cont...*

- Farmers often prefer widely available exotic species due to their rapid growth and – in many cases – high demand for their products (poles, timber, fruits, etc.).
- The native species of highest biodiversity value may provide no direct benefit at all to the farmer.

If the challenges to farmers growing native species outweigh the opportunities, then governments or others interested in biodiversity conservation will need to provide appropriate incentives and remove disincentives (see **Incentives** in Chapter 5). Farmers or landowners should be considered as private sector partners in conservation, who will need to be compensated for allocating land, resources and labour. Such compensation should cover not only the planting of the tree, but also its subsequent management and protection.



From design to action: The planting plan and its implementation

Importance of the planting plan

The final stage of design is to draw up a planting plan. The planting plan is important for three reasons:

1. Its preparation provides an opportunity to document the results of the design process. This ensures that everyone involved is 'on the same page' and avoids misunderstandings.

2. The planting of agroforestry systems requires more care than the planting of conventional agricultural, horticultural or forestry plantings, because the layout – the planting locations of trees and crop plants – is more complicated than in a plantation with just one species. A plan is needed to ensure that everything goes smoothly.

3. It helps ensure that the implementation – and not just the design – takes account of the agroforestry design principles.



What needs to be in the planting plan?

The planting plan needs to cover the 'where, what, when and who' of the process. It should be a short document that everyone involved can easily understand. It may be useful to draw up a simple template. Figure 5 is an example of a planting plan prepared in this way. Whatever form the plan takes, site visits are essential, because the plan must be based on specific local conditions of soil, vegetation, aspect, and so on.

For **sequential systems**, it is important to map out the life cycle and productive lives of the different species, so that planting can be timed to avoid productivity gaps and resulting cash flow problems. These can occur between the end of one component's lifespan and the beginning of the next one's productive period.

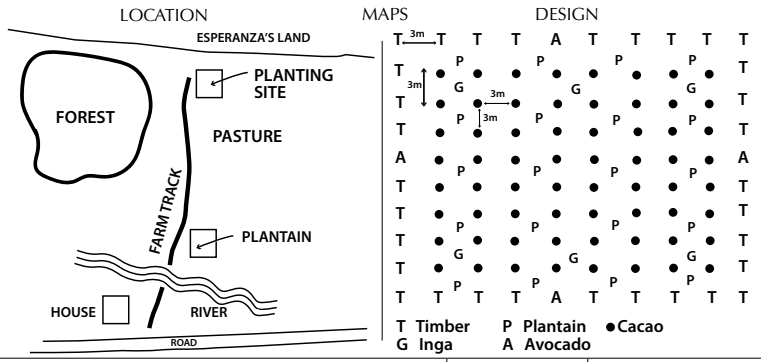
Co-design is guided by the **agroforestry principles**. However, while following these principles, wider considerations should not be neglected. Foremost among these is the safety of all concerned, particularly during establishment of agroforestry systems. Organizations involved in implementing agroforestry activities should ensure the following guidelines are adhered to:

- At the start of the establishment process, do a risk assessment to identify hazards and risks, and measures to eliminate or reduce them. 'Hazard' means the source of danger (for example, snakebite) and its potential severity (high, in the case of snakebite). 'Risk' means the probability of this happening (low, medium or high).
- Make sure that the planting plan specifies responsibilities (who does what). Consider individuals' wishes, skills and physical condition. Some tasks demand strength; others, attention to detail.
- Ensure that edged tools are sharp. Sharp tools speed up all operations and are safer if properly used.
- During establishment, follow the measures laid out in the risk management plan, including the wearing of any safety equipment.

Like agriculture and forestry, agroforestry is a relatively high-risk activity, and adhesion to safety guidelines will avoid accidents that could lead to injury or even be life threatening.

Planting Plan Description

Owner: *Nicholás Torres*	**Date of plan:** *November 2021*
Supporting organization: *Futuroverde (NGO)*	
System: *Multistrata, cacao with Inga, timber trees and avocado*	
Planned date of planting: *September 2022*	
Where:	
Description: *old shifting cultivation parcel of 0.5ha, on slope that runs up to the boundary with Esperanza Valverde's land*	



What	When	Who
Place order for 50 grafted avocado plants	*December 2021*	*Nicholás*
Arrange for help with site preparation (cleaning, marking, hole-digging) and planting of temporary shade	*February*	*Nicholás*
Place order for timber seedlings and Inga (see attached list) and 550 cacao seedlings. Check up on avocado plants	*March 2022*	*Nicholás*
Extract plantain suckers from parcel near river (225) and transport to site	*March*	*Nicholás (transport using Furturoverde's pickup)*
Ensure that all tools needed for preparation are available on site	*start of April*	*Miguel*
Clean site	*April*	*Nicholás and helpers or workers*
Mark site with guide stakes for temporary shade (plantain)	*April*	*Nicholás and helpers or workers*
Dig planting holes for plantation	*April*	*Nicholás and helpers or workers*
Plant temporary shade (plantain)	*April*	*Nicholás and helpers or workers*
Arrange for availability of grafting material (scions of fine flavour cacao varieties)	*July*	*Miguel from Futuroverde*
Clean site	*August*	*Nicholás and helpers or workers*
Dig planting holes for cacao, avocado, Inga, and timber trees	*August*	*Miguel with Nicholás and workers*
Transport all nursery plants to site	*September*	*Miguel*
Plant cacao, Inga, timber trees and avocado	*September*	*Nicholás and workers, helped by Miguel*
Grafting of fine flavour scions on cacao seedlings	*Start of dry season*	*Nicholás and Futuroverde cacao specialist*

Figure 5. A simple planting plan prepared using a template

Incentives



Farmers' main expectation from their land is for it to support their livelihoods, whether or not the farm is their main source of income. In recent years, however, wider environmental agendas have started to affect farmers. They face ever-increasing expectations from the international community and local authorities, who are looking to them to be custodians of the landscape as well as managers of productive small businesses. Sometimes – when agroforestry is both the best economic and environmental option – farmers will need little encouragement to invest

in it. But if farmers are to take on additional responsibilities for the landscape, especially if this involves costs to them, then they will need incentives.



Governments who wish to promote agroforestry for livelihoods, income, conservation or restoration should put in place effective incentive systems. Incentives might be financial (such as grants, loans, tax breaks, favourable interest rates for borrowing or access to insurance), or might involve improving the enabling conditions for agroforestry (such as nurseries, high-quality planting materials or improved markets for agroforestry products). They could include schemes of payment for ecological services, or social recognition for practising skilled and responsible farming. Where farmers are expected to contribute to global goals such as combating climate change, conservation or restoration, then they have a right to expect access to global funding – either directly through projects or via carbon-financing schemes (see also Box 4: Trees and biodiversity conservation).

Removing incentives to ecologically damaging agricultural practices, and disincentives to environmentally friendly practices, is also part of the picture. Around the world, conventional high-input agriculture has been subsidized, while farmers have been penalized for harvesting or transporting indigenous trees that they have cultivated on their own land (Box 4). If farmers are to become land stewards, as many policymakers expect, then incentive structures will need to change radically.



6

PLANTING MATERIAL IN AGROFORESTRY



When farmers plant trees, they use a variety of planting material (PLM), including seed, seedlings, rooted cuttings, grafted plants and branch stakes. Planners of agroforestry interventions need to ensure that farmers can get PLM that is apt for purpose and readily available. These two requirements are linked, because insufficient supply of PLM of acceptable quality may lead to use of poor-quality material. In this chapter, we provide answers to two important questions:

- What makes PLM apt for purpose?
- How can we ensure that farmers have access to it in sufficient quantity?

In this overview, we concentrate on seedlings, particularly bagged seedlings, and seed.

Quality of planting material and aptness for purpose

PLM that is apt for purpose has two characteristics. First, after it is planted or sown it survives and grows normally. Second, it yields products in the quantity and of the quality required. These two characteristics usually go together.

Nursery plants



An apt-for-purpose nursery plant should have the following characteristics:

- a single stem, woody at the base and at least partially woody up to about three-quarters of its height
- dark green leaves, without any yellow patches and without signs of pests or disease
- new leaves at the tip
- roots that are not penetrating the base of the bag or other container.



Tree seedlings in containers, such as polyethylene bags, are generally ready to plant out when they are 30–40 centimetres tall, but – in some cases – farmers may have good reason to prefer larger plants. Plants need to be sturdy enough to withstand transport, handling and transplanting. The sturdiness coefficient is a useful measure of plant quality. It is calculated by dividing the plant height in centimetres by the basal diameter in millimetres. Its value should not exceed six. Farmers who regularly plant trees will learn by experience to avoid plants that are too straggly and to always plant sturdy seedlings.

Plants can fail to be apt for purpose in various ways. Those that are too small may be smothered by weeds before the farmer has a chance to carry out the first weeding, or their root systems may be too small to withstand a snap drought after planting. Root systems can also be deformed: as a young tree grows, roots that were severely bent and twisted in the nursery usually get worse, not better, and end up strangling themselves. This leads to slow growth or death. Root spiralling, by contrast, can be corrected by pruning at the base and sides. However, it is better if farmers use plants with no defects.



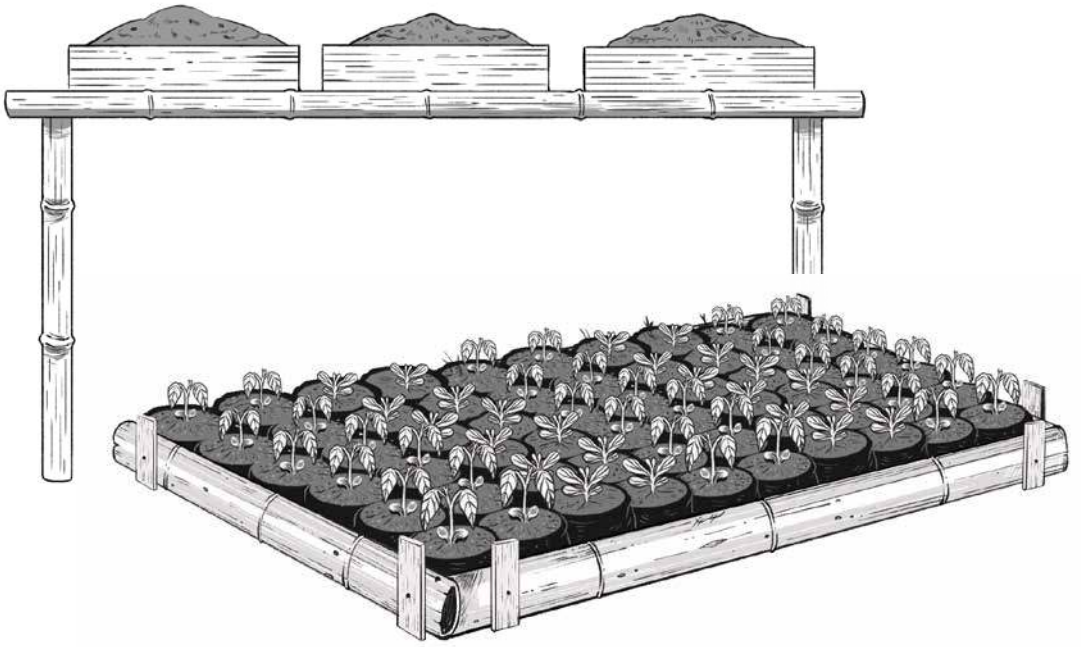


Seed



Farmers and the agencies that support them use seed either to produce seedlings in a nursery, or to sow directly in the agroforestry system. To be apt for purpose, a seed must be capable of germination and, following germination, must produce a seedling that grows normally. Seed should also be free of pests and diseases that might spread to other seeds or plants. Such seed meets the requirements of the following dimensions of seed quality:

- **physical quality** (seed is undamaged, whole, within normal size ranges, clean, and free of soil, plant parts, etc.)
- **physiological quality** (seed is living and otherwise 'in full working order')
- **health** (seed is free of external and internal pests and diseases).



Genetics and aptness for purpose

PLM that meets all the requirements listed above can still fail to be apt for purpose if it is not genetically adapted to the planting site. In a nursery, it is easy to produce healthy, apparently high-quality plants that, however, are not well adapted to nearby planting sites. For example, seedlings of rainforest species can flourish in a nursery in a seasonally dry area because they are watered by the nursery staff, but they would be unlikely to survive a long dry season once planted out.

To be adapted means that the genetic makeup of the PLM equips it to withstand the range of environmental conditions that it will be subjected to when planted out. This includes variation due to climate, soil, competition and management practices. Well-adapted PLM survives and grows normally under this range of environmental conditions. Trees raised from material that is not locally adapted usually have lower growth rate; in extreme cases, poor adaptation leads to complete failure of a plantation.

'Well adapted' does not mean 'improved'. A productive agroforestry system can usually be established with unimproved PLM, **if** it is well adapted and has been produced using **best practices** – including those for seed collection. Fruit species can be an exception to this rule, for the following main reasons:

- Although an unimproved fruit tree may survive and grow well, it might not produce fruit for many years. Trees of grafted improved varieties will usually produce fruit within 2–5 years, depending on the species.

- Some markets might demand fruit of specific improved varieties, such as Hass avocado or Haden mango.

Access to planting material

Farmers have access to apt-for-purpose PLM if:

- best practices have been followed in seed collection and handling and in nursery management;
- appropriate supply systems for PLM are in place.

A full explanation of these conditions is outside the scope of this publication. However, we outline some important points below.

Best practices

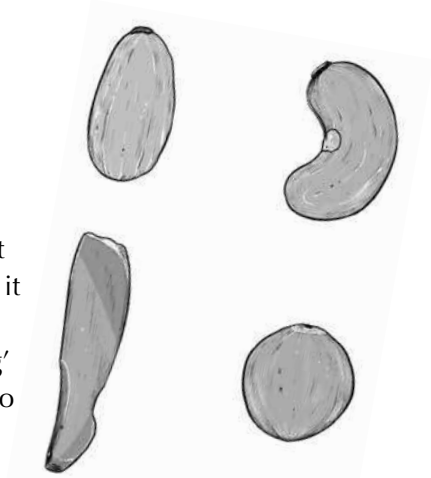
Ensuring genetic adaptation
Plants will usually be well adapted if the seeds they were raised from were collected in an area that has a similar climate to where they will be planted. Therefore, to find out whether seedlings or seed are likely to be well adapted, it is necessary to know their provenance (where they come from). Exact information is best, but even approximate information may be enough to know whether the provenance has a similar climate to the planting zone.





Lack of genetic diversity may also lead to poor adaptation, even if the seed is local. Inbreeding depression is one problem: if a farmer collects all their seed from one tree, the resulting plants may all be healthy and well adapted, but if these interrelated trees eventually mate among themselves, then the seed they produce will be inbred, with slower growth and lower survival. A small sample (like the single tree just mentioned) is also a risky sample, as it could be highly unrepresentative, with a far-below-average genetic makeup. For this reason, seed should be sourced from at least 30 mother trees, which themselves should be part of a larger population.

For a few tree species, genetically improved seed or plants may be available. Although not usually essential, improved PLM can increase productivity, quality and profitability. However, caution is advisable, because smallholder conditions may be very different from the conditions for which the improved sources have been developed. For example, improved eucalypts developed for highly fertilized and clean-weeded pulpwood plantations in, say, southern Brazil, may not be suitable for smallholder agroforestry systems in, say, Central America. There is no point in paying a premium for plants that are not improved for the site in question.



Seed handling and storage

Poor physiological and health quality have particularly serious effects on seed germination and subsequent growth. Often, these result from incorrect seed storage or other seed handling problems. If seed has not been stored in the right conditions, or has been stored for too long, then it will probably be of poor physiological or health quality. What 'the right conditions' and 'too long' mean depends on the species and, particularly, to which **seed storage category** it belongs (Box 5).

Most tropical tree species have recalcitrant seed. However, many of the commonly planted species have orthodox seed. Recalcitrant or intermediate seed should not be offered for sale if it has been stored for more than a few days – unless, in the case of intermediate seed, it is known from experience that it is likely to be viable.

The surest way to know whether seed is of good health and physiological quality is to test it. Seed merchants should carry out germination tests before offering the seed for sale. Unfortunately, while the legal sale of crop seeds is often strictly regulated, the sale of tree seeds is frequently poorly regulated, so merchants may not offer this information. For this reason, seed buyers may wish to carry out their own germination tests. For large purchases, seed merchants may be willing to supply a free sample from the same seedlot, or a small sample can be bought for testing. If the seed germinates to the buyer's satisfaction, then a larger quantity can be bought with confidence. A flotation test is a quicker but less accurate test. A sample of seed – ideally, about 100 – is placed in a container with water for 24 hours. Seed that is of high physiological quality will usually sink and will often begin to swell.



Box 5. Seed storage categories: Orthodox, recalcitrant and intermediate

Seed specialists categorize seeds by how they respond to drying. If drying is possible without killing the seed, seed can be stored for longer.

Orthodox seed can be dried so that it contains very little water. After drying, it can be stored at low temperature for many years or even decades without important decline in health or physiological quality. Orthodox seed dries out naturally before it is dispersed from the tree, but must be further dried prior to long-term storage.

Recalcitrant seed dies when it is dried. In some cases, it can be stored for short periods (days to weeks) if kept in moist conditions that stop it from drying out. Recalcitrant seed does not dry out naturally before it is dispersed from the tree. Some people use the term 'recalcitrant' loosely, to mean any seed that is difficult to germinate. This is incorrect.

Intermediate seed is between orthodox and recalcitrant. It can be partially dried, but not as much as orthodox seed. Therefore, like recalcitrant seed, intermediate seed cannot be stored for long periods of time.



To find out what category a species belongs to, consult the databases of the international society for seed science: https://seedscisoc.org/.

Nursery practices

Many manuals describing best nursery practices are available. Some commonly neglected elements of good practice are listed below:



Where possible, sow seed directly in the plant container, rather than in a germination bed: 'pricking out' is a common cause of root deformity.



Use a soil mix with enough clay content to permit formation of a root-plug that, while not hard, remains intact during removal from the container.



Include forest soil or humus to ensure development of mycorrhizae.



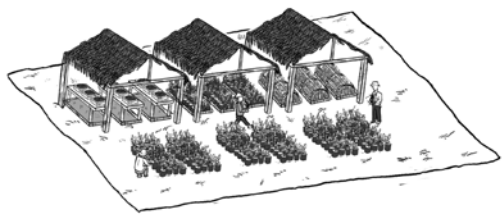
Harden off seedlings before planting or sale by reducing watering and increasing exposure to the sun.



Cull (throw away) poor quality seedlings, including those that are past their 'sell-by date' (sturdiness coefficient greater than six, or roots growing through base of container and into the soil).



Keep records to facilitate control and to allow accurate information to be given to PLM users.

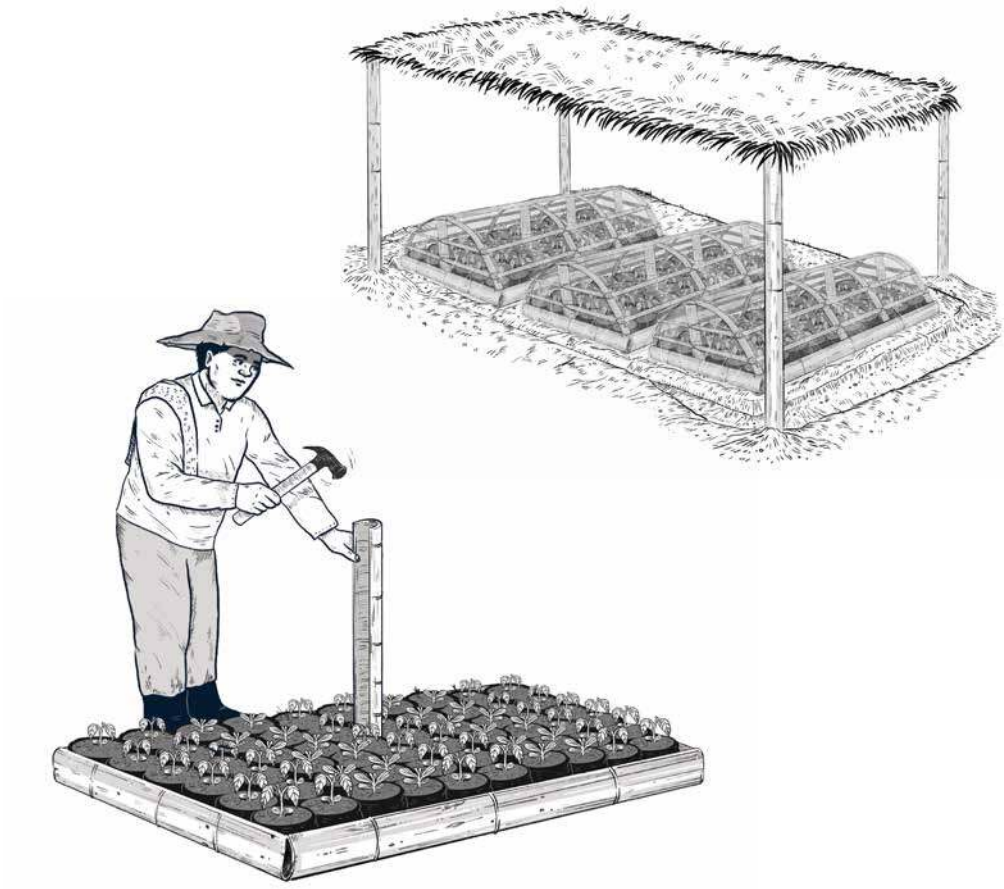


Nursery customers should be wary of nurseries that don't follow these practices.

Use of apt-for-purpose seed in nurseries is also an aspect of best practice. In the case of physical, sanitary and physiological quality, this is a matter of economics, efficiency and feasibility (plants can't be produced from seed that doesn't germinate). To secure genetic adaptation, nursery operators should obtain seed from seed sources that are located nearby, in similar climatic conditions, or that prior experience (including any formal provenance testing) has shown to be adapted.

Choosing a seed merchant or nursery

Several aspects of seed and plant quality cannot be evaluated simply by inspecting seed or seedlings. For example, seed of poor physiological quality can look normal, and poor adaptation of plants and seed may not be visible until the first years of growth in the field. Faced with this problem, farmers and those who support them have two alternatives. The first is to make their own seed collections and operate their own nurseries, in both cases following best practices. The second option is to ensure that the seed merchant or nursery operator is both trustworthy and competent. Trustworthiness can be verified by talking to other clients and by first-hand experience. Competence can be checked in various ways (such as by asking about the practices followed). In the case of gaps in competence and knowledge, supporting agencies may be able to provide training. Such training benefits both producers and consumers of PLM.



Planting material supply systems

A planting material supply system consists of the institutions, installations, policies, laws and value-chain actors that control and affect the access to PLM by users. A full account of approaches to developing context-appropriate supply systems is outside the scope of this publication. However, in the following sections we outline some important concepts and describe seven broad interventions that supporting agencies can make.

Access requires two conditions to be met:

- PLM must be available: this means that apt-for-purpose seed sources of the species must exist, and nurseries must stock plants produced from the seed.
- Farmers must be able to acquire the material (usually, plants), that is, they have the resources to travel to the nursery, buy the plants, and transport them to their farms – or an agency must either support them to do this, or provide the PLM directly.



In supporting farmers' access to high-quality PLM, two perspectives must be considered. The first perspective is that of the project manager or project participant, who needs material for a specific intervention – usually, one identified in an agroforestry design process – to be carried out in a specific place in the near future. The second perspective is broader; it is concerned with the seed and nursery subsector in a specific geographical area, rather than with a specific project. Both perspectives are considered below.

The project perspective

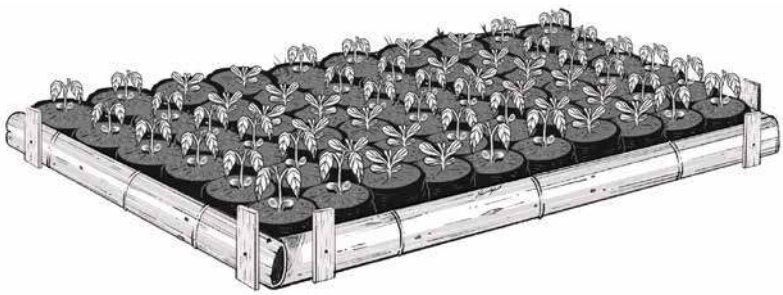
Selection of species should consider the availability of PLM; even if farmers express strong preference for a given species, there is no point in selecting it if no material is available within the relevant time frame – which often means at the start of a project. This time frame also means that, from the project perspective, farmers' short-term access, rather than longer-term sustainability of access or availability, is the main concern.

There are many ways in which support agencies can ensure that farmers are able to access PLM, particularly plants. One way is by distributing plants for free. Two arguments against free distribution are sometimes made:

- Farmers will not value trees that they have not paid for.
- Free distribution of PLM undermines local private sector nurseries.

1 **The first argument** is true in some cases. However, if farmers do not value trees that they have planted on their land, then the fault is with the agroforestry design process, not the farmers. If this argument matches common experience, then it is because projects often have not truly met farmers' needs or aspirations.

2 **The second argument** is stronger. Free distribution of plants in a given region will unfairly affect local nurseries' sales of plants of the same or similar species. If this results in closure of nurseries, then it will also affect longer-term sustainability of plant supply. In most cases, the solution is not to stop free distribution of plants, but rather for the distributing agency to obtain them from local nurseries, at market prices. Such agencies can also provide training to local nurseries, where needed, so that they can fulfil this role.



The broader perspective: developing PLM supply systems

It is easy to design PLM supply systems on paper. However, if these don't reflect common realities, then they will remain 'castles in the air'. These realities include the following:

- There is a need, at least in large countries, for decentralized supply of PLM.
- Regulatory personnel, who often have an agricultural background, often have no expertise in agroforestry PLM quality.
- Regulating the whole supply chain, from seed source to planting site, is complex.
- Drafting, enacting and implementing effective seed laws is difficult, particularly for seed of multiple species with different collection and storage requirements.
- Tree planters and planting material producers often lack financial resources and technical capacity.

The strategic need is to identify farmers' needs and plan how to respond to them, rather than to attempt to construct perfect systems according to preconceived criteria.



In Table 4, we provide a 'menu' of seven broad areas in which supporting agencies may help enhance farmers' access to fit-for-purpose PLM. These are guided by four principles: farmer-centredness (as for agroforestry interventions in general); self-sufficiency (where possible, supply of PLM should be financed by sale of PLM); openness to intervention when markets fail; and a 'no harm' principle that intervention should not conflict with self-sufficiency.

Table 4. Menu of intervention options to enhance planting material supply systems

Intervention	Role
1. Implement a diagnostic study to guide interventions and actions	Ensures that substantive interventions and actions are appropriate and evidence based
2. Support the design, reform and application of laws and regulations	Supports other interventions
3. Promote or establish agroforestry nurseries	Appropriate when planting stock is unavailable due to lack of nurseries
4. Support the establishment and management of seed sources	Appropriate when seed sources are lacking or when they are unproductive
5. Support nursery operators' access to fit-for-purpose seed	Appropriate when nursery operators cannot obtain seed from existing seed sources
6. Support improvements in the quantity or quality of production by existing nurseries	Appropriate when nurseries do not produce sufficient apt-for-purpose plants of the species that farmers require
7. Support farmers' access to nursery plants	Appropriate when farmers are unable to acquire the planting stock that is available in nurseries

7

MANAGEMENT OF TREES IN AGROFORESTRY SYSTEMS



Many tree establishment efforts focus entirely on the planting process, and fail to consider the years of time and effort needed to care for the planted trees and to manage their growth. It is true that the cultivation of trees usually requires less work than crop cultivation. **But trees planted on farms do require management, whatever the system, and this management can be very time consuming.** The design process must ensure that farmers are aware of the demands of management and that they consider these before deciding to establish an agroforestry system. Growing trees requires a long-term commitment, sufficient resources and sound knowledge about how trees interact with their environments, including other components of agroforestry systems. In this section, we provide an overview of how to manage trees in agroforestry systems.

Agroforestry systems differ, but broad management goals are similar

The expertise, knowledge and time needed to manage trees in agroforestry systems depend on the complexity of the system or practice. For example, farmers are likely to spend more time managing the trees in a multistrata system than those in a living fence. They will also need more knowledge and expertise for the multistrata system. In each case, however, management aims at achieving one or more of the following goals:

- System components have adequate supplies of nutrients and water.
- System components, particularly flagship species, are free of pests, diseases and other agents of damage, or are not critically affected by them.
- System components, particularly flagship species, have optimum light conditions.
- Productivity and profitability of the system are maximized.
- Trees grow to the right shape and size for their location and roles.
- Specific environmental goals, if any, are met.

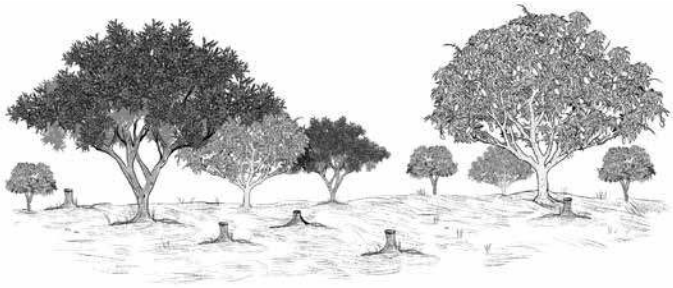
Some tree management activities resemble those used to optimize growth, health and quality in orchards or forestry plantations. Other measures are also needed because of the special characteristics of agroforestry systems.

Thinning

Thinning is the removal of individual trees. It is done by felling the trees at ground level, usually with a chainsaw. The main aim of thinning is to manage competition, particularly between trees of the same species. It increases both the productivity and quality of the remaining trees.

In conventional timber plantations, which usually have trees of one species planted at 3.0 metre or 2.5 metre square spacing, thinning is a very important management practice. By gradually removing most of the trees that were initially planted, it concentrates production on a smaller number of large-diameter, well-formed and valuable final harvest trees. In forestry, thinning is a planned activity: foresters plant many more trees than will be needed in the end, because high initial density improves stem quality, helps to control weeds, and ensures that enough high-quality individuals can be selected for the final crop. Sometimes it is also possible to sell some of the thinned trees for poles, small-scale construction material, charcoal or firewood.

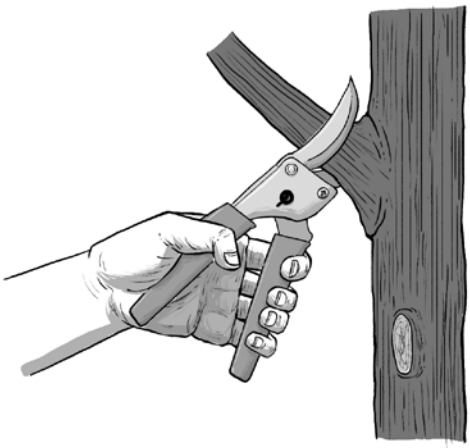
Some agroforestry systems, or parts of them, may resemble traditional timber plantations – for example, woodlots and simple successional systems such as **taungya**. In these cases, thinning practices will resemble those used in forestry. In other agroforestry systems, each tree is intended to be either a permanent component of the system, or – in systems with a successional component – to die naturally due to shade or because it has reached the end of its life cycle. In these cases, thinning is less common. Its intensity and timing will depend on each specific case.



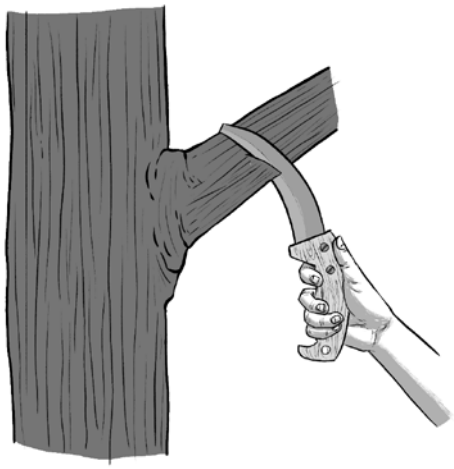
Crown management

Crown reduction is the removal of some or all of the above-ground parts of the tree – that is, its stem or branches. Four types of crown reduction are used in agroforestry: pruning, lopping, coppicing and pollarding. Crown management also includes the decision **not** to reduce the crown when more shade is needed.

Pruning



Pruning is the agroforester's main tool for managing competition. Selective removal of branches from one tree can prevent it from overshading and crowding out another. The removal of multiple branches from the crown may be referred to as 'crown thinning'. A pruned tree is less able to capture sunlight through its leaves, reducing the amount of energy it



captures, which leads to the dieback of some roots. Thus, pruning is also an important tool to control root competition.

Pruning to regulate shade also affects microclimate, including temperature and air circulation, which have important effects on pests and diseases. Shade favours some pests and diseases, whereas others are favoured by unshaded conditions, so crown management will depend on which pathogens are of concern.

Pruning is also used to enhance the productivity, or quality of production, of the tree itself – particularly in the case of timber trees, fruit trees, cacao and coffee. When a timber tree increases in girth, its side branches remain embedded in the wood; they are visible as knots in sawn timber. Pruning of dead and living side branches improves quality by preventing the formation of knots – an essential activity in the production of high-quality timber.

However, no more than one-third of the live crown should be removed in any given year. The belief that removing nearly all the branches of a tree will concentrate growth on the main stem is mistaken; rather, the tree's height and diameter growth will slow down, because it will **photosynthesize** less. Some timber species are self pruning, and should not be pruned artificially, except when live branches of flotilla trees are interfering with flagship trees. Pruning should also be used in all woody species to remove diseased branches.

For cacao, coffee and many fruit species, highly specific pruning techniques – which are beyond the scope of this booklet – have been developed to maximize productivity and quality.

Pruning, especially for high-value tree crops, is a skill that needs training and a good understanding of the physiology of a tree, as some species have highly specific requirements. It must be carried out with care, particularly on high-value trees. The best time to prune timber trees and young fruit trees is usually at the end of the dry season, because airborne fungal spores are likely to be less frequent, and because growth will soon start again and cover pruning wounds. In the case of mature (bearing) fruit trees, pruning is usually carried out after harvest.

Lopping

Lopping is used when a 'rough-and ready' approach to crown reduction is acceptable – for example, in pruning live fences or in some forms of **alley cropping**. Lopping may sometimes be used instead of pruning when competition for light can be reduced by removing just part of a branch, particularly when the branch base is difficult to reach.



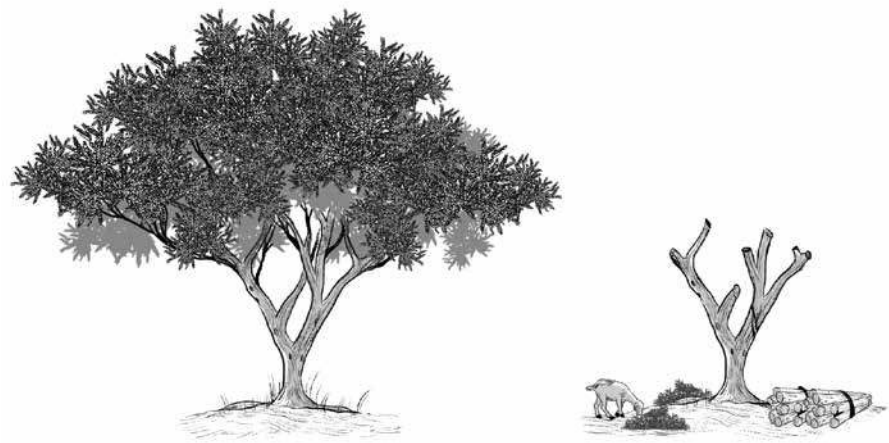
Coppicing

Many tree species produce new shoots from their stump or roots when cut, and coppicing takes advantage of this. The shoots, once woody and large enough, are harvested for poles, rods and fuelwood. The tree component in alley cropping is also usually managed by coppicing (or lopping). Coppicing can also be used to manage trees on soil conservation structures. In these cases, development of a large crown may affect the stability of the structure. Coppicing will lead to the death of some roots, but the stability-enhancing larger roots will remain.



Pollarding

Historically, pollarding was used instead of coppicing in situations where browsing animals would otherwise have fed on coppice shoots. In agroforestry, it is important as a shade management technique, particularly in some simpler multistrata systems, such as coffee with erythrina pollards and laurel in Costa Rica.



a. acacia tree before pollarding

b. acacia tree after pollarding

Weeding

Weeding is the cutting or removal of herbaceous plants, grasses or climbers that grow near or on tree seedlings. Selective weeding targets plants that are visibly affecting the development of the seedling or the adult tree. Sometimes the negative effects of other plants are not very easily visible, but the growth rate of tree seedlings can be strongly impeded, for example by competition from grasses or other aggressive vegetation. A good standard weeding practice, especially when trees are to be established in pastures, is to 'clean weed' a circle around each seedling to a distance of 0.5 metres from the stem. To prevent erosion and drying, the bare soil should be protected by **mulching** with the uprooted plants and plant parts.





Fertilization

Fertilization is the addition of external sources of nutrients to replace those that have been lost to the system and that, as a result, are deficient. In agroforestry, the right spatial and temporal arrangement of trees, crops and livestock allows **organic matter** and nutrients to be circulated on the farm, so that the use of fertilizers can be reduced without losing yield. However, new agroforestry systems often need to be 'kick-started' with initial fertilization, and many will benefit from regular applications afterwards. In many countries, guidelines are available for fertilization of commercial agroforestry systems (for example, cacao and coffee) and fruit tree orchards.

Prunings may be used as mulch. However, woody material, particularly large branches, takes longer to decompose. To speed its decay, it should be placed in direct contact with the soil, and then covered with leaves and other non-woody parts. In some systems, plants such as bananas or plantains, Mexican sunflower and panic grass are deliberately planted to be later cut for mulch.



Some organic matter that accumulates in the fields, such as fresh animal manure or ripe fruit, can attract pests and diseases. These should be composted off-site and later added or returned to the system.



8

FROM PRINCIPLES TO PRACTICE: KEY SYSTEMS



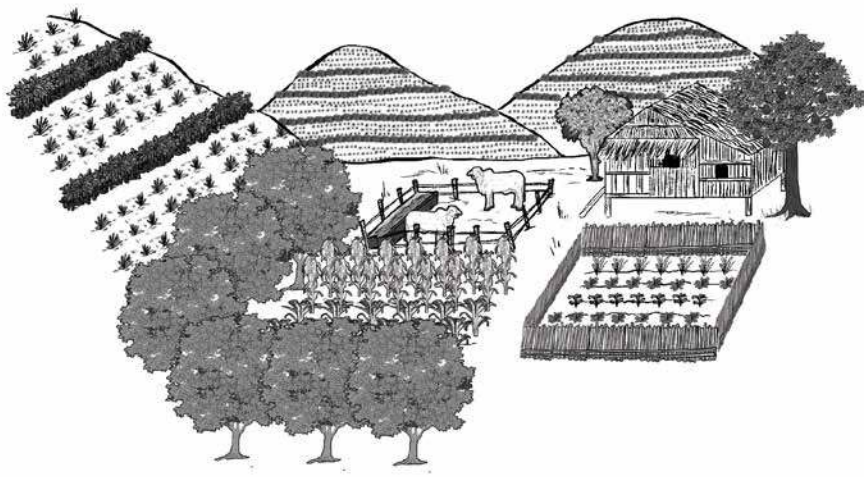
When applying the three principles of agroforestry design (farmer-centredness; aptness to place, people and purpose; and synergy), it is important to recognize that each landscape is unique and should therefore have its own unique agroforestry systems. Trees in these systems can have very different functions, ranging from being flotilla species in systems that focus on food production, to being the flagship species in systems that are designed to restore degraded forested landscapes.

This chapter describes a small selection of featured systems to illustrate the different functions of trees and the resulting design and management implications. The featured systems should not be read as blueprints for direct application, but as examples of common and robust systems that might be used in specific locations after appropriate modifications.

Annual crops with trees (ACT)

In these systems, a tree component provides ecosystem services that increase the productivity of the flagship component, an annual crop. The selected tree species often also provide useful products. Because annual crops are usually sun-loving, the design and management of these systems seek to increase the numbers of trees on the farm without creating competition for light.

ACT systems follow two main approaches. In intercropping, trees are planted among the crops. In niche planting, the trees are planted in 'niche' locations, such as farm boundaries, places with low soil fertility, riparian sections and home compounds. Niche planting is often preferred when cropping areas are very small, or when the work needed to manage competition would be out of proportion to the benefits gained from the trees.



ACT systems are most appropriate for farms that have low productivity, particularly when caused by the following factors:

- low soil fertility
- vulnerable soils – whether due to low organic matter or high risk of erosion
- climates in which prolonged drought or irregular rainy seasons are common.

Trees help to counteract these causes of low productivity by increasing soil fertility, keeping moisture in the soil for longer periods, and preventing soil erosion (see Chapter 3, **Agroforestry systems as circular systems**). These pros need to be weighed against the following cons:

- loss of physical space for the flagship species (because trees will occupy space where crops could be)
- reduction of crop yield due to shading
- in semi-arid regions, potential water competition between trees and crops.

Design guidelines

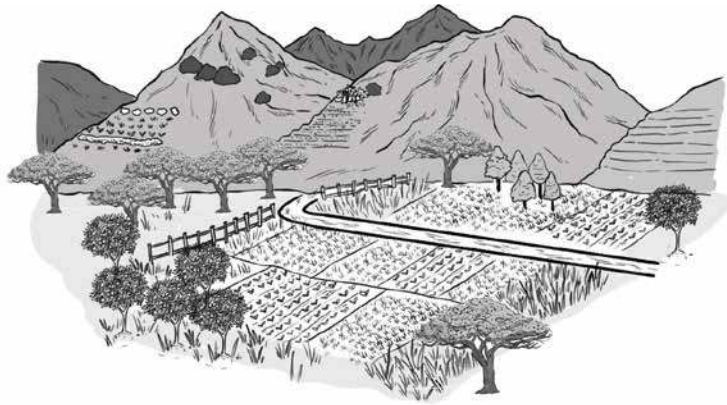
ACT systems are often introduced to farmers as part of agroforestry programmes that aim to restore or improve soil fertility, increase farm productivity, and boost household income and well-being. ACT systems are very variable and context specific. Institutions and individuals supporting farmers should work with them, following guidelines such as those outlined in Chapter 5, **Co-design and establishment of agroforestry systems**. This will ensure that the system is aligned with the farmer's goals and their household's needs.

The benefits of tree flotilla species to the farm, crop and farmer will be highest if the following rules are followed:

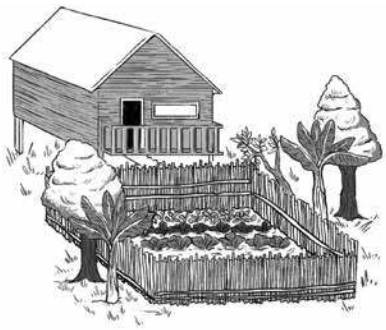
- select the right tree for the farmer
- select the right place for the tree
- select the right type of tree for the system, considering growth rate, crown type and root systems.

On sloping land, trees should be planted along contour lines to stabilize the slopes and prevent the soil from eroding. Tall trees and shrubs mixed with grasses can be a good combination, because these each bind a different layer of soil. Having species of different heights also reduces shade. Where erosion is already a problem, trees should be planted at close spacing so that as much as possible of the area benefits from the soil-binding effect of their roots. Later, the trees can be thinned to manage the canopy size and shade.

On flat areas, especially small fields, trees are usually scattered in with crops. This allows the crops to directly benefit from the trees – particularly nitrogen-fixing species – without receiving too much shade.







Planting niches should be identified with the farmer and other household members, making sure that the practical aspects of management are considered. For example, fodder trees should be close to livestock pens. Fruit trees should be planted close to the house, both for ease of harvest and for security reasons – the same applies to other high-value species.

Livestock are an essential part of some ACT systems because their manure serves as organic fertilizer. On steep slopes, zero-grazing strategies are more appropriate, as they prevent livestock from damaging both the soil and the system.



The components (crops, trees, livestock) and how they are best arranged differ from farm to farm, but we list some general guidelines below:

- Trees are usually less demanding of nutrients than are crops. It is generally desirable that trees growing with crops should have deep roots, with fewer roots near the soil surface where most crop roots are found.
- Competition effects can be reduced by watering, fertilizing and mulching at competition zones.
- Farmers can correct excessive shade by pruning or thinning the trees responsible.
- To minimize shading effects, tree rows should be planted parallel to the path of the sun.
- Some tree species naturally produce less shade. Such trees may have small leaves, have leaves that tend to go upwards rather than outwards, or be bare or partly bare (the latter trait is particularly useful if it coincides with the cropping season).



- The chosen species should have deep roots, grow quickly, preferably fix nitrogen, and have a light crown that allows sunlight to pass through easily.
- Highly productive, fast-growing types of trees such as eucalypts and acacias use a lot of water, and they can reduce water availability for other components in a system. These kinds of trees can best be used on farm as a block or woodlot.
- High-yielding perennials that need a lot of light, such as oil palm, are not suitable to be grown under trees, but they make excellent shade trees for smaller plants such as cacao and coffee (see **Multistrata cacao agroforestry systems in Central America and the Philippines** in this chapter).
- In semi-arid areas, termites can cause serious damage to seedlings and mature trees, so mitigation measures like selecting termite-resistant species and employing protection strategies are very important. Strategies include removing dead and damaged wood from trees quickly after damage, mulching to provide alternative food sources, and careful management of the tree (especially when it is young) to keep it healthy and able to resist termites. Pesticides can be used, but they tend to kill natural enemies, and are most effective when used to kill termite colonies rather than to protect individual trees, which is undesirable. Several botanical extracts can be used to control termites.[8]

[8] For more information on controlling termites in agroforestry, see Nkunika P, Sileshi G, Nyejeo P, Ahmed BM 2013. *Termite management in tropical agroforestry*. Lusaka: University of Zambia Press. http://dspace.unza.zm/handle/123456789/6496.

Management guidelines

Many farmers deliberately maintain mature trees in their fields and/or protect naturally occurring seedlings so that they grow into mature trees (a technique known as 'farmer-managed natural regeneration'). However, farmers usually have little experience of managing trees that have been reintroduced into landscapes dominated by annual crops. Unlike farmers who specialize in tree crops such as fruit species, coffee or cacao, or that have timber plantations, they may be unfamiliar with tree growth in height, girth and roots over time, and may not have been trained in how to maximize growth or productivity.

The following points are often key elements of successful management:

- The most important activity in managing ACT systems is the management of the trees' crowns to minimize shading of sun-loving annual crops.

- Several factors affect the amount of time it takes before trees begin to interfere with crop growth, including soil fertility, climate, crown shape, root types and the shade tolerance of the crops in question. Farmers need to be trained to understand how the various elements of an agroforestry system respond to tree pruning and thinning (see Chapter 7, **Management of trees in agroforestry systems**).

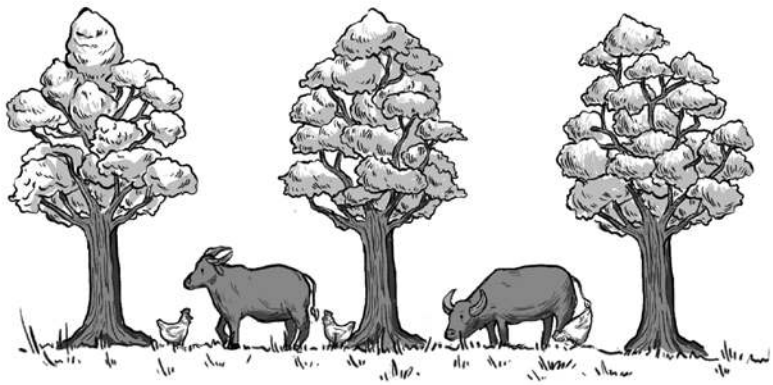
- Warning signs of light competition include yellowing of the crop leaves – which suggests low photosynthesis and chlorophyll levels – and, in cereal crops, stunting and thinness. Trees that are grown too close together usually respond by putting all their energy into height growth, rather than growth in diameter.

- Pruning, coppicing and pollarding are important in reducing competition. They also provide useful products such as firewood, mulch, green manure, fodder and stakes for climbing plants.

Livestock with trees

Description of the system

We are focusing here specifically on livestock in mixed farming systems. For the integration of trees in intensive cattle ranging systems, see Box 7. As an integrated part of the agroforestry system, livestock offer opportunities for increased profitability, risk reduction through diversification, better human nutrition, improved land use and higher agroecological efficiency. Trees are planted to provide ecosystem services to the livestock; however, they also introduce some environmental and health risks that must be mitigated by informed and careful management.



Box 7. The untapped potential of living fences



An important use of linear planting of trees is as live fences to control livestock. Fencing is very expensive, but live fences are cheaper to establish than their dead post counterparts. Live fencing is widely practiced on ranches around the world, including in Honduras and Nicaragua, where the system has recently been studied in depth.[9]

Fencing prevents animals from straying and allows land to be divided into grazing blocks for rotational grazing. Large-scale cattle producers generally have low tolerance of trees in open pasture, and in Honduras they usually remove them when canopy cover reaches 20%. However, they actively plant trees around field boundaries.

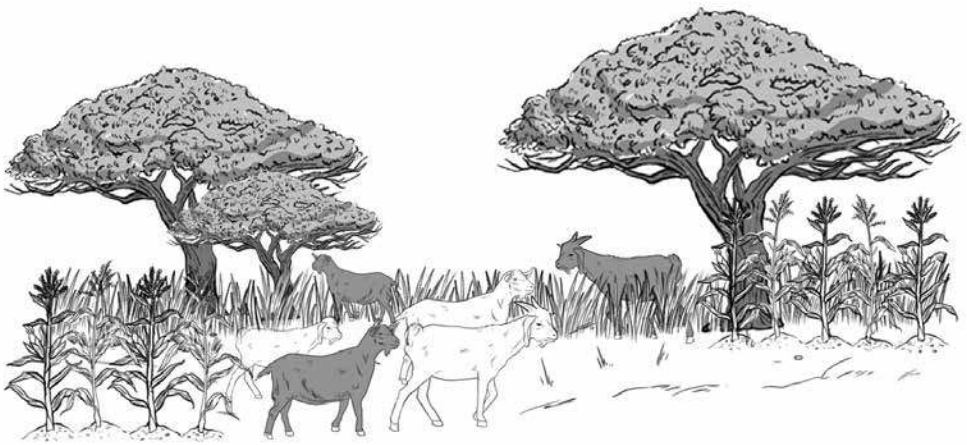
Live fences are very productive. An inventory of cattle ranches across 25,000 hectares in Catacamas municipality, Honduras, identified more than 10,000 fence segments spanning 1,730 kilometres, with a linear density of almost 70 metres per hectare, covering 6.4% of the land. The potential for timber production from live fences in the 2.9 million hectares of pasture in Honduras was estimated to be equivalent to 200,709 hectares of conventional timber plantations, given that one kilometre of live fence is equivalent to the yield of one hectare of plantation.

[9] Somarriba E and Quesada F. 2009. *Agroforestry farm planning: manual for farming families*. 1st Edition. Costa Rica: CATIE.
Somarriba E, Orozco-Aguilar L, Cerda R, López-Sampson A. 2018. Analysis and design of the shade canopy of cocoa-based agroforestry systems. *In* Umaharan P. ed. *Achieving sustainable cultivation of cocoa*. London: Burleigh Dodds Science Publishing, 1–31. https://doi.org/10.1201/9781351114547.

Livestock, especially browsers like goats, can cause serious damage to crops and trees if not controlled. In sloping landscapes, free-grazing livestock can seriously affect soil stability, and thereby undermine the sustainability of the entire farm.

Zero-grazing systems, as opposed to free grazing, are those in which livestock are not free to roam the farm to access grazing, browsing or other feed resources. Usually, this means that animals are housed in buildings or pens, to which their feed is carried. The system is particularly appropriate for smallholder farmers that own livestock but have limited lands and income. Livestock are often kept on small farms to provide food for the farming household and income from milk, eggs or meat. The animals that are kept are usually cows, sheep, goats, pigs, rabbits, and, in some places, guinea pigs.

Although small-scale farmers often tether animals where feed can be found, such as along farm boundaries, on roadsides or in other accessible land, the limited space available on farm usually leads to animals being kept under a 'zero grazing' regime. Trees and shrubs in the system are selected to provide high protein alternatives to add to the basic crop-residue animal feed. This improves the condition of the livestock and avoids using all the available crop residues – some of which should be left on the field as mulch and for soil erosion control – or buying costly commercial feed supplements.



Zero grazing offers farmers several advantages. It can improve livestock management or make it easier, in the following ways:

- It can allow livestock to be kept on farms that would otherwise be too small.
- Because zero-grazed animals are usually kept close to the homestead, household members can more easily manage them as part of their general activities.
- Farmers are more likely to notice pests and diseases.
- More accurate management of the nutrition of individual animals is possible, by feeding them controlled amounts of different foods. This can improve feed use and is an important first step towards market orientation (for example, engagement in the small-scale dairy sector).

It also offers advantages in overall farm management and other important aspects:

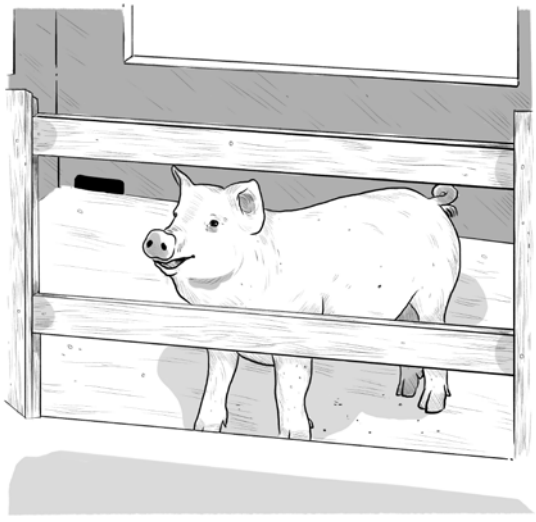
- Manure is easily collected and recycled to crop fields, increasing resource-use efficiency across the whole system. Animal manure is rich in nitrogen and is a particularly important source of phosphorus.



- Confinement of the livestock prevents adverse effects on soil and natural vegetation, and damage to the crops of farmers and their neighbours is also minimized.
- Housed animals are less vulnerable to theft and predation.

Zero-grazing systems also have some potential disadvantages. They are labour intensive, and availability of labour may limit the number of animals that can be kept. Furthermore, buildings and equipment cost money. The investments needed may be recouped by greater future profits, but nevertheless may be unacceptable to more risk-averse farmers. Although the monetary costs may be reduced by using local materials and building methods, they will rarely be zero.

Zero grazing can also have adverse effects on the health of animals and their keepers. For the farming family, close contact with livestock brings an increased risk of zoonotic disease. The livestock, too, are at greater risk of disease and transfer of parasites. For the animals, however, the effects on health and welfare can be more far reaching. Unfortunately, zero-grazed animals are often kept in inhumane conditions, with inadequate space, lack of bedding and insufficient shelter from the weather. Confinement is not healthy for any animal, and a lack of exercise can lead to long-term discomfort and psychological impacts. Where possible, steps should be taken to minimize these animal welfare problems. Those supporting and advising farmers can contribute to this by pointing out the advantages of avoiding adverse effects like foot damage, joint damage, skin damage, increased aggression, other behavioural changes and poor reproductive performance. All of these are distressing to the animal and result in reduced longevity and lower productivity.

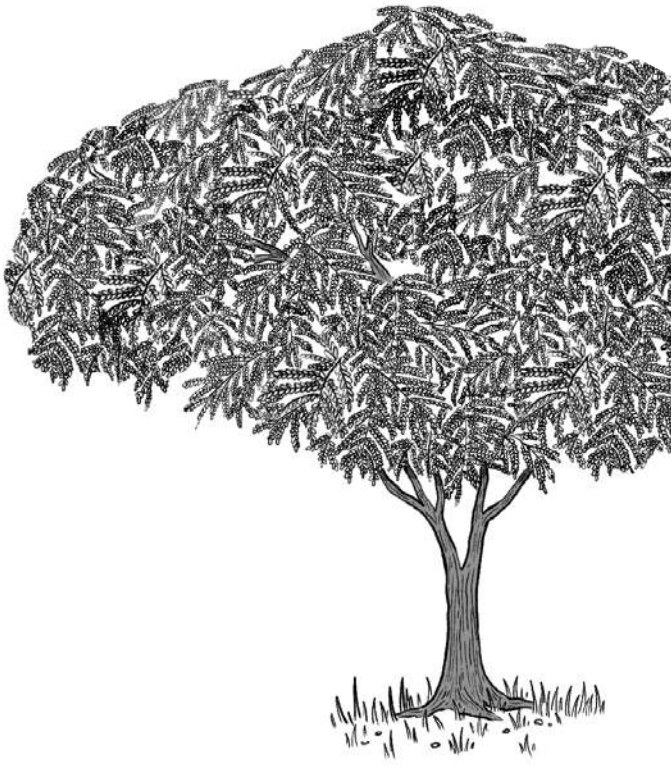


Design guidelines

Fodder tree production

Fodder for zero-grazed animals is often provided in the form of grasses, fodder legumes, and branches cut from shrubs and trees. There are many ways to include fodder trees for zero-grazed livestock in agroforestry systems, such as in ACT systems or as living fences. Good candidate species produce large amounts of high-quality, palatable fodder; grow relatively fast; remain productive during the dry season when fodder grass is scarcer; and tolerate frequent coppicing or pollarding. Species that are going to be planted with the main food crops should have small, sparse crowns (to reduce shade) and roots deeper than those of the crop (to reduce competition for nutrients and water). Some leguminous species offer the additional benefit of improving soil fertility by fixing nitrogen.

On sloping fields, as in the aforementioned ACT systems, trees and shrubs should be planted along contour lines to prevent soil erosion. In locations such as field boundaries, riparian stretches and small vacant plots, trees and shrubs can be grown more densely in fodder banks, which are permanent plots that maximize foliage production. Associated grass species should be fast growing; highly productive; well adapted to frequent cutting or defoliation; able to provide good soil cover; palatable to livestock; high in nutritional value; and either easy to conserve as hay or silage, or able to stay green in the dry season.

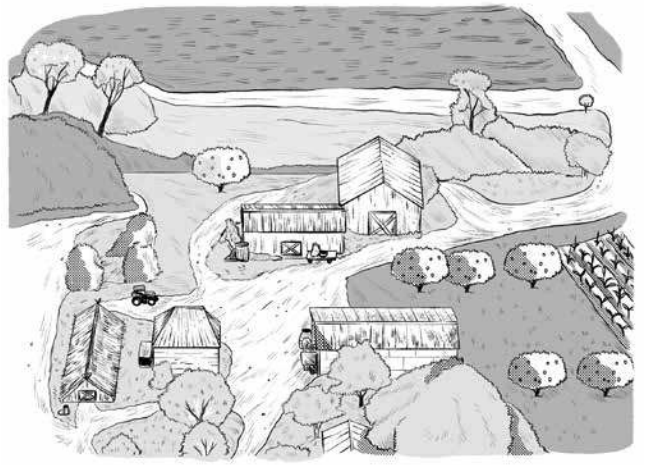


Livestock type

Any type of livestock species or breed may be kept under a zero-grazing system. However, zero grazing requires relatively high capital investment and ongoing managerial and material inputs. Therefore, farmers are more likely to use the system for animals that generate substantial returns, such as dairy cows or sheep reared for meat that will be marketed. Eggs, which can be produced in large numbers from improved hen breeds, often have ready demand and can command a high price. On the other hand, the ease and efficiency of feeding small livestock, such as rabbits, can make them an attractive simple option, both for the family diet and for sale.

Investment in improved breeds (such as crossbred dairy animals or specialized sheep breeds) allows farmers to make the most of the opportunities for higher incomes that zero grazing offers. However, these animals do not automatically produce more: they require much improved feed management (see below) to reach their potential, as well as careful attention to their health and well-being.





Location

The siting of a zero-grazing unit will be influenced by several factors, which need to be balanced. Closeness to the homestead makes monitoring of animals simple, but if many animals are kept, and feed and water sources are not close, the labour needed may be beyond the capacity of the farming family. Other considerations include road access, electric power and availability of local building materials. Each farmer needs to consider their own situation with respect to location and try to identify the most appropriate compromise.

Housing and associated infrastructure

A variety of housing designs can be used for zero-grazing livestock, depending on the farmer's wider objectives. For example, slatted floors can greatly improve manure recycling efficiency, if this is an important consideration. However, slatted housing for large ruminants must be extremely robust, and is therefore relatively expensive to construct. The importance of well-designed equipment, such as feed troughs, should not be underestimated. Inadequate feeders can cause wastage of up to 30 percent of offered feed, and this wastage will reduce profitability.

Management guidelines

Tree and shrub management

Effective management for animal feed of trees, shrubs and other fodder plants can be quite complex: it requires experience and access to good, reliable information – for example from extension services. Indonesia's three-strata forage zero-grazing system, developed for dry areas, is a good example. It combines fodder plants, grasses, ground legumes, fodder shrubs and fodder trees. Its components have differing life cycles or growth patterns, so high-quality fodder is available all year round. Grass and ground legumes are harvested during the wet season, shrub legumes during the dry season, and fodder trees during the late dry season when little else grows.[10] In Rwanda, fodder banks are less complex, as farmers prefer to plant leguminous shrubs such as calliandra with the crops to support soil fertility. Fodder banks are usually designed and planted around a main fodder grass, such as Napier grass, but in rotation with other grass species that are used for other purposes. They are also grown in permanent positions on land not used for cropping, for example along rivers.



[10] For an example of planting for year-round fodder see Nitin IM, Lana K, Sukanten W, Suarna M, Putra S. 1989. The concept and development of three-strata forage systems. *In* Proceedings of the IDRC workshop Shrubs and Tree Fodders for Farm Animals, Denpasar, Indonesia, 24–29 July 1989. 92–102 https://idl-bnc-idrc.dspacedirect.org/bitstream/handle/10625/18656/IDL-18656.pdf.

Feed management

Decisions regarding the appropriate feeding of livestock always involve a compromise between production objectives (which can be multiple and conflicting) and the range and quantities of feeds that can be made available. Some cut-and-carry systems operate efficiently on feed resources that are collected locally from common land (such as roadside grasses), while others may be wholly dependent on planted forages and other feed or food-feed[11] crops. Trees and shrubs, particularly legumes, provide high-quality, nitrogen-rich feed to complement crop residues.

Manure collection and handling

Zero grazing offers an important opportunity for farmers to make more effective use of the manure produced by their animals. In some cases, it may be used as fuel, but recycling nutrients by applying it to croplands is also a very widespread practice. In addition to improving resource use efficiency through the system, manure recycling can deliver direct financial benefits by reducing expenditure on purchased fertilizer. Direct application to crops should be avoided. Farmers need knowledge on good composting practice to maximize the value of manure.

Watering

Watering is often overlooked, but is of critical importance for housed livestock, which are unable to find water for themselves. Water requirements have been discussed in the livestock section under the High needs for care and maintenance subsection. Common sense is also helpful: if a farmer expects a dairy cow to produce 20 litres of milk each day, but only offers her 10 litres of water, disappointment is going to be the only outcome!



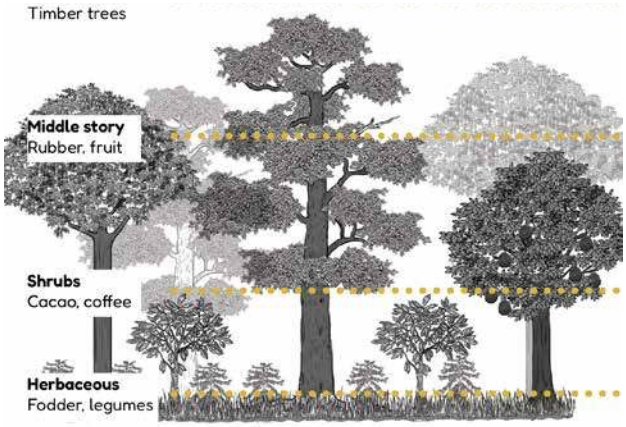
[11] Food-feed crops are those consumed by both people (food) and animals (feed).

Multistrata perennial agroforestry

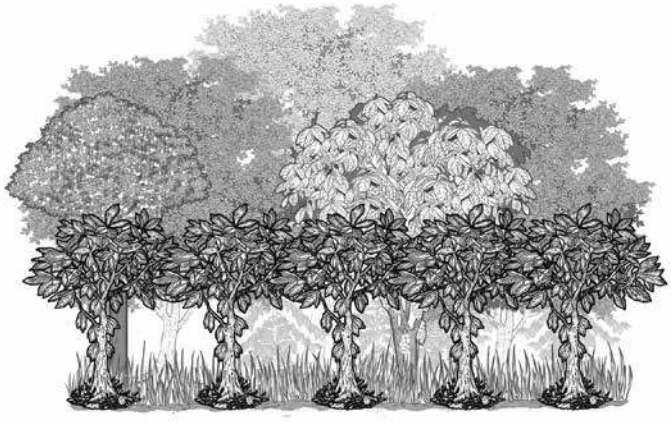
Description of the system

Multistrata perennial agroforestry systems are the most complex of all kinds of agroforestry. They include both simultaneous and successional systems. In a simultaneous system, the same species, with their horizontal and vertical arrangements, are maintained for the system's entire productive lifetime. In a successional system, species composition and structure change over time.

Both systems take advantage of species' differing needs for water, light and nutrients. Simultaneous systems deliberately combine species that reach different heights, dividing into different layers the vertical space between soil and sun. Plants and trees are selected in such a way that, at any given moment, each species or group of species ideally occupies a specific layer based on its relative height and its need for sunlight (see Chapter 7, **Management of trees in agroforestry systems**). However, as tree height



changes over time, management is needed to maintain the different layers. In successional systems, this resource-sharing strategy is taken further by planting species with different life cycles. Over time, the species succeed each other; plants with shorter lives develop alongside longer-lived ones. When they are pruned or complete their life cycle, they leave a beneficial legacy: organic matter in the soil, and the results of their interactions with other plant, animal and microbial species. These improve the soil's structure, fertility and moisture content.



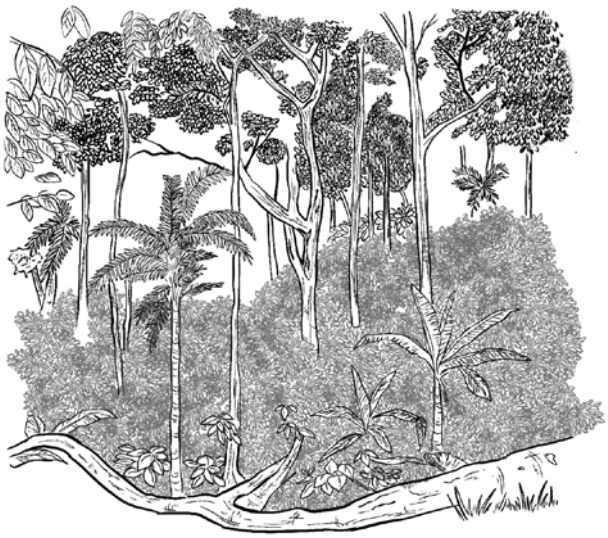
Multistrata systems vary greatly in their design and species composition, depending on the region and the type of farming community. However, they are usually designed around one flagship crop that represents the main production objective for the farmer or the farming community. Often, the production objectives change over time as different products become available, as different components of the system mature, and as market conditions and product values change. Because of the high investment required in labour and planning, it is essential that the farmer is closely involved in design and planning. Only farming communities that fully understand the time and commitment needed to manage these systems will be successful in reaping their benefits.

Design of multistrata systems entails optimization of the use of space by arranging plants in space and time, thereby ensuring maximum productivity of the flagship crop through careful management of the density of the different layers. Well-designed multistrata systems take advantage of existing seed stocks by supporting the natural regeneration of native tree species, while deliberately adding new components with important livelihood potential, such as fruit trees or commodities like oil palm and rubber.

Design guidelines

A carefully designed canopy structure is required to optimize the use of horizontal and vertical space. This is done based on the productive life cycles and heights of different species.

In hilly terrain, trees should be planted following contour lines, to minimize soil erosion.



The system is initially designed around optimization of the productivity of the flagship species (which varies from place to place), with flotilla species chosen to maximize synergies. This requires detailed knowledge of the flagship's physiological needs and productive stages.

Trees that produce a lot of foliage, and nitrogen-fixing species, enhance nutrient circulation in the system. In successional systems, these functional groups of species should be represented in each stage of the succession to reduce dependency on commercial fertilizers. The life cycles and productivity ranges of the different species should be mapped out to time

the planting of the successional stages. This avoids productivity gaps and cash flow shortages. It is important to avoid production gaps between the end of one succession's lifespan and the productive period of the next.

Components need to be managed continuously throughout the productive cycle of the system to regulate their population density, minimize competition, and enhance complementarity. Farmers need to be aware that managing such a system is a full-time job. It is very demanding in terms of labour requirements at all stages (planning, planting and management). These demands need to be compatible with labour availability and the farmer's knowledge, experience, learning capacity and aspirations.

Livestock can be included in the system, but plant species then need to be protected from damage and overgrazing. In uplands with steep slopes, zero-grazing strategies are more appropriate, as they prevent livestock damage to both the land and the system.

Management guidelines

The basic principle for managing multistrata systems is to manipulate the species composition and the spatial and temporal structure to provide the right amount of shade at the right time for the plants in the system. This requires detailed knowledge of the phenology of each species – that is, about the timing of leaf loss, production of new leaves and shoots, flowering and fruiting.

Light triggers plant growth and, therefore, extraction of nutrients from the soil. In low-fertility sites, and if fertilizer has not been added, excess light will stress the plant, leading to poor growth and low yield. Signs of stress include discoloration or death of leaves, and general weakness; highly stressed plants can die. Farmers can achieve maximum production for a given level of soil fertility by managing light penetration.

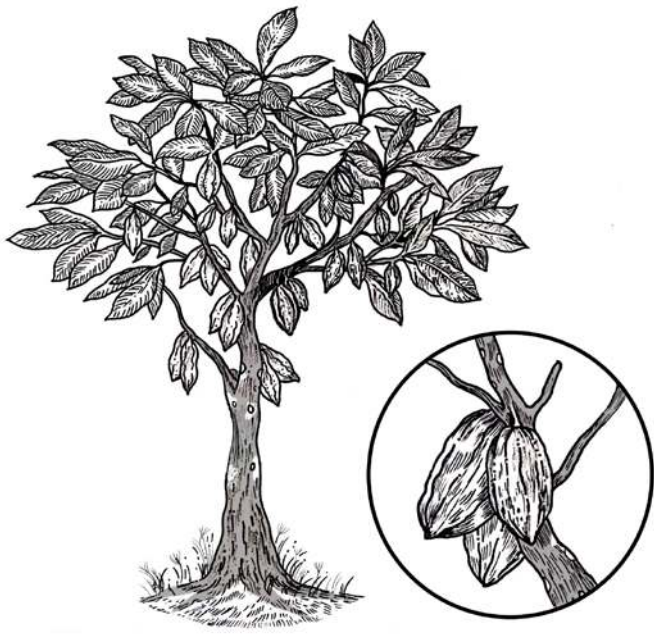
To assess whether the shading of the system is optimal, extension workers can follow an easy, three-step procedure:

1. Assess the exposure to sunlight of the system at different times of the day. Even on the same farm, different areas can receive very different amounts of sunlight, depending on their position in the landscape.

2. Walk through the system at different times of the day and observe the light penetration on the ground. This will tell you whether the system is too dense and if it requires thinning or pruning.

3. Discuss with the farmer the level of productivity that they can manage. How fertile is the soil? Can they afford to buy and apply additional fertilizer? Based on the available sunshine hours, decide on an optimal shade regime to achieve that productivity level.

Using the assessments made in the above three steps, work with the farmer to develop an optimal monthly routine for managing the canopy and stock density.



Multistrata cacao agroforestry systems in Central America and the Philippines

Description of the system

Cacao was first cultivated as a shade crop under the canopies of native trees. In many countries, however, the availability of full-sun varieties, coupled with pressure to increase production, eventually led to replacement of these systems. Yield in unshaded systems can be five times greater than in shaded systems, but they also demand a lot of nutrients, leading to soil exhaustion. As a result, cacao farmers tend to open new fields when establishing plantations, often leading to deforestation.

Because of its shade tolerance, cacao is one of the crops most suited to agroforestry systems, especially for farming families who prefer diversified income streams to reliance on one crop. There is also renewed interest among large chocolate producers in agroforestry, because climate change is beginning to affect plantations in traditional cacao growing areas. Cacao plants are highly sensitive to variation in climate, particularly temperature, but also changes in rainfall and in the number of hours of sunlight. Concern is also growing over deforestation caused by cacao expansion.

The diversity, botanical composition and structural complexity of cacao agroforestry systems varies widely between geographical regions, between farms within a region, and even between sections within a plantation. Across regions, design and management are often deficient, leading to suboptimal shade (excessive, deficient or uneven).

Design guidelines

In this multistrata system, cacao is grown as an understorey of the canopy and emergent layers (Figure 6). In a well-managed system, ground-cover crops such as forage legumes and grasses can also be grown beneath the cacao.

The layers optimize the use of light, water, nutrients and space. The emergent and canopy layers maximize light utilization and provide shade to the cacao trees, which absorb both direct and diffused light.

Below ground, the range in root depth allows the system to exploit more water and nutrients, and protects the soil against erosion and landslides. Deep rooted trees bring up nutrients such as nitrate, phosphate and organic acids from deep in the soil, and make these available to shallow-rooted trees, including cacao, through their litter.

Management guidelines

To ensure maximum health and productivity of the plants, shade levels must be carefully planned, monitored and managed over the lifetime of a shade cacao system.

Inappropriate shade levels have dramatic effects. Under-shaded trees will show stunting, signs of water stress, scalded leaves and stems, flower cushions damaged by scalding, fewer leaves and low yield. Over-shaded trees will also show stunted growth, together with elongated stems and leaves, low flowering rates, fewer pods, higher incidence of disease (including *Phytopthora*, cherelle rot, pod rot and swollen-shoot virus), and higher incidence of pests such as pod borers.

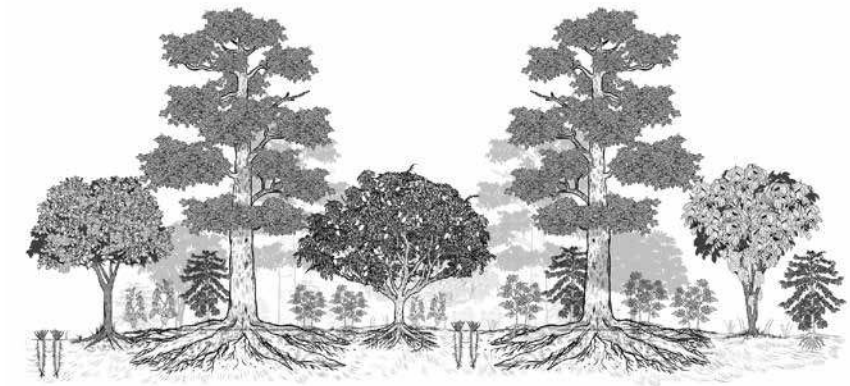
The amount of shade required in a particular plantation depends on the degree of self shading, phenological stage, site conditions and canopy characteristics, as explained below.

Self shading

In the crown of a cacao tree, upper leaves and branches cast shade on lower leaves, while neighbouring cacao trees cast shade on each other. Both are examples of self shading. If self shading is high, then less overstorey shade is needed. If this happens, fewer shade trees will be required and the farmer loses the option of producing additional goods (timber, fruit, etc.) and services (cultural, environmental) in the cacao plot.

Self shading is determined by factors related to the form and size of the cacao tree, and to planting configurations and spacings, alone and in combination. Important ones include:

- age of cacao plants (young plants have small crowns, hence low self shading – overstorey shade is needed)
- pruning frequency and intensity (infrequent and light pruning results in tall cacao trees, crown overlap between neighbouring cacao trees and high self shading)
- whether the trees are grafted or grown from seed (grafted cacao trees tend to be shorter and more open crowned than trees grown from seed, resulting in less self shading if the plantation is well managed)
- tree spacing (triangular planting arrangements allow more cacao trees per hectare than square or rectangular configurations. Self shading is higher in such 'packed' planting configurations).



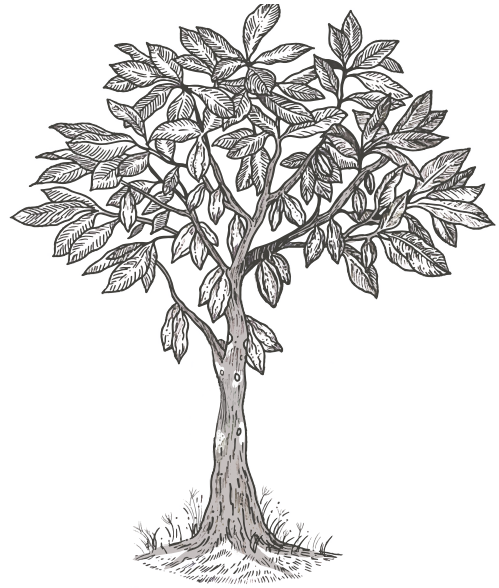
Temporal dynamics

In addition to the changes in self shading that occur as cacao plants grow older and bigger, the light requirements of a cacao tree vary according to its annual cycle: light is particularly needed at the flowering and pod-filling stages. For optimal cacao performance, shade must be adjusted to the monthly rhythms of the cacao plant by timely pruning or pollarding of the shade trees.

Site conditions

Several site factors influence the amount of light reaching the cacao plot and the optimal level of shade required:

- the latitude, exposure and slope of the land determine the amount of sunlight reaching it
- high prevalence of cloud reduces sunlight
- the topography of the surrounding land and the height of the surrounding vegetation affect the degree of lateral shade cast on a plot
- soil fertility: in infertile soils, farmers can only grow low-shade cacao if they apply fertilizers. If they are unable to buy fertilizers, then more shade is needed, because it reduces sunlight and demand for soil nutrients
- rainfall: for each location, it is important to establish the annual rainfall below which the introduction of trees to unshaded cacao would result in poor crop performance due to competition for water
- altitude: cacao can be planted from 50–1,200 metres above sea level. The optimum elevation is around 400–800 metres. Above 500 metres, cacao requires less shade than at lower elevations
- temperature: the higher the temperature, the more shade is required, so cacao trees generally require more shade during the dry season. This has implications for pruning practice: shade trees should be pruned at the onset of the rainy season and allowed to regrow during the dry season to provide the required shade during this period.



Canopy characteristics

The shade cover should be similar in the different parts of a cacao plantation. In practice, however, many cacao plantations have patches with either too much shade or no shade at all. In part, this results from variation in the vertical and horizontal distribution of the canopy cover and its botanical composition. Tall trees cast less intense shade than short ones, and tree species differ both in the time of the year when they become leafless and in the duration of leafless periods.

Case studies

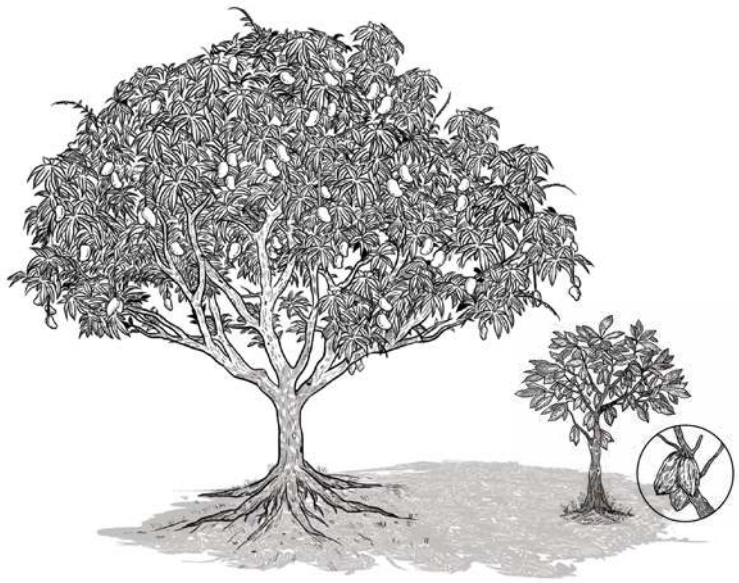
Cacao agroforestry in Central America

Several Indigenous Peoples of the Mesoamerican Biological Corridor, including the Chibchas and Mayas, cultivate shade cacao. These cacao plantations usually harbour between 125 and 145 tree species at densities of 85–166 trees per hectare. Most trees are planted (except in rustic cacao systems, which make use of thinned natural forest), although some are derived from natural regeneration. Shade trees are used for timber (*Cordia alliodora*, *Cedrela odorata*), fruit (*Musa* spp., citrus, avocado, coconut, peach palm, mango), and shade (*Inga* spp., gliricidia, leucaena). Tree canopies have three vertical strata (low <10 metres, medium 10–20 metres and high >20 metres), containing about 50%, 30% and 20% of total tree density, respectively.

In these systems, cacao is cultivated at 100–800 metres of elevation in small plots (1.2 hectares per farm). Cacao trees are typically spaced at 4 × 4 metres (625 plants per hectare) in most countries. They are usually produced using hybrid seed from either controlled pollination or from pods selected on local farms. Grafted cacao has been used only since around 2010 and is still scarce. Most farmers have two or more cacao plots per farm.

Five basic types of shade canopy are used in the region:

1. cacao under shade of one species, either a timber species (for example, *Cordia alliodora*) or a nitrogen-fixing legume (for example, *Inga* species)
2. cacao under two shade strata composed of a mixture of fruit (for example, citrus, *Nephelium lappaceum*, peach palm), legume, and timber species (*C. alliodora*)
3. cacao intercropped with banana and fruit species under one shade species (timber or legume)
4. diversified home gardens with cacao, fruit, timber and legume species
5. rustic cacao ('cabruca'): cacao planted under thinned natural forest.



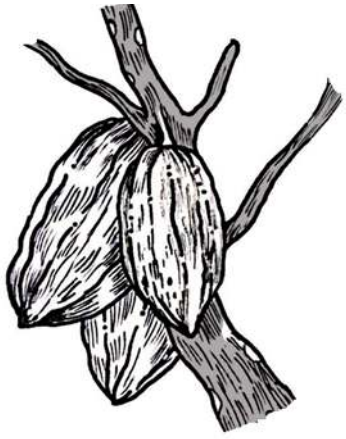
Cacao agroforestry in Claveria, the Philippines

Claveria is an upper-watershed town in eastern Misamis Oriental Province on the island of Mindanao. Land-use practices in the region affect lives and livelihoods of people living in the 13 downstream coastal municipalities: soil erosion from poor land management contributes to the siltation and pollution of downstream water bodies, and affects the quality and availability of marine resources.

In this context, cacao agroforestry is particularly important, because it provides permanent soil cover and reduces erosion, landslides and runoff. Due to its multiple components, the system acts as a climate adaptation and mitigation measure, ensuring climate-resilient, sustainable streams of products and services. It also helps to buffer farmers against fluctuations in commodity prices.

Cacao combined with rubber, fruits and forage legumes is common in sloping lands in the region. In this system, contour lines with 0.5-metre natural vegetative filter strips are laid out every 10 metres. Rubber trees are planted at a spacing of two metres apart, approximately 25 centimetres above the grass strips; about 500 such trees can be accommodated on one hectare. Two rows of cacao trees at three-metre spacing are planted in the alleys (between the grass strips), 3.5 metres from the rubber rows, resulting in around 660 cacao trees per hectare. Bananas are then planted between the rubber trees at a density of 500 per hectare. Fruit trees such as durian or langsat are then planted in the middle of each cacao row, spaced at 10 metres apart (100 per hectare). Pinto peanut, a forage legume, can also be planted as a ground-cover crop.

Cacao with coconut and fruit trees is also used on sloping lands, while cacao with rubber, vanilla and banana is used on flatter areas. Cacao is also used in parkland systems, both on sloping and flat areas.

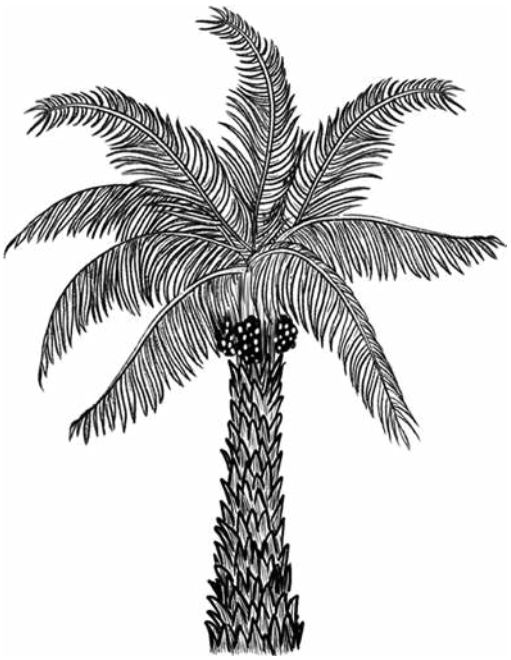


Oil palm agroforestry

Description of the system

Agroforestry systems for commercial oil palm are multistrata systems that offer an alternative to farmers and companies that want to produce palm oil, but do not want to specialize in just one crop – or who would like their oil to have a smaller environmental footprint.

The system was developed with farmers in Brazil. Individuals had a strong influence on the spatial design of the systems and the selection of the species, so no two systems are the same. They require agricultural technology packages different from conventional oil palm models. Instead of maximizing palm oil yields per hectare, they aim to diversify livelihood options by combining oil palm with other crops of commercial value, allowing simultaneous production of food crops. They also aim to improve nutrient and water flows by taking advantage of the synergies between the different components. To provide space for the other components, palm density varies from 60 to 100 trees per hectare, whereas the density in conventional systems is 144 per hectare.



Management of oil palm agroforestry systems requires considerable knowledge, so training and technical assistance for farmers are important elements of creating, implementing and maintaining systems that will be successful and profitable.

Design guidelines

Oil palm agroforestry is suitable for all soil types and land types that are suitable for oil palm – that is, well-drained soil in flat areas.

The components and spacing are designed around the flagship crop, oil palm. Flotilla species are selected based on how each farmer prioritizes objectives such as incomes, food provision, and contributions to soil health and biodiversity. In Brazil, farmers selected cash crops such as cacao and fertilizer species, including inga, gliricidia and Mexican sunflower.

To simplify the harvesting and transport of fresh fruit bunches from the oil palm, it is advisable to plant oil palm rows in sections, with light-loving crops and trees interspersed between the sections. In Brazil, farmers have opted for double, triple or quadruple rows of oil palm, keeping the conventional 7–9 metre spacing between the palms.

Oil palm is the most productive oil producing plant and has very high light and nutrient requirements. It is therefore important to avoid creating competition for these resources. As in all multistrata systems, competition for light is managed by selecting species that reach different heights, effectively dividing the horizontal space between soil and sun into different layers (see **Multistrata perennial agroforestry** in this chapter), and by actively managing the system once the components grow. Oil palm trees should not receive more than 10–15% shade, so high-strata trees should be placed at maximum distance from the oil palm rows and be carefully chosen for a crown shape that is as porous to light as possible.

Shade-loving crops like cacao, ginger, turmeric, some woody ornamentals and – possibly – coffee can be grown underneath oil palm trees. As oil palm seedlings have very fast-growing, aggressive, shallow roots, perennial components such as cacao must be planted simultaneously to make sure that they establish good root systems and are able to thrive.

The components need to be arranged both to avoid shading the oil palm, and to ensure appropriate light levels for other components in the system. Sun-loving plants should not be planted directly under oil palm. However, it is generally advisable to err on the side of planting too many at the start: it is easier to get rid of surplus plants than to establish them later.

When integrating food crops, it is important to take advantage of the oil palm's life cycle. The palms take five years to reach maximum height and crown cover. During this period, they do not cast much shade, and sun-loving crops like maize and cassava can be easily grown between the oil palm and the other perennials in the system. Beyond this time frame, and depending on the system design and spacing, many understorey crops – particularly the annuals and biennials – will only have adequate light to grow for a few more years. Once the system matures, perennial and semi-perennial food crops like bananas can be planted and can stay in the system for about 8–10 years.

Oil palm agroforestry is relatively labour intensive, and, as in all agroforestry systems, the more integrated the components, the more labour is generally required – especially in the first few years. This adds to the labour demand for the oil palm itself, which tends to increase over time, in contrast to many other crops. As such, it is important to design and implement the system at a scale that is manageable, given the amount of labour that is available to the household or landowner.



Management guidelines

Some of the shade-loving crops that can be grown in an oil palm agroforestry system, such as cacao and cut flowers, may be financially almost as important as the oil palm itself, so their needs and light requirements must be carefully managed. At times when oil palm prices are low and those for other cash crops are higher, farmers can prune their palms more intensively than usual to provide more light and increase the yield of the understorey crops.

Pruning is an important element in any kind of oil palm cultivation, and farmers need to be well trained in the basic principles of managing productive adult oil palms.

Because oil palm is a 'hungry' crop, it will require fertilizer, especially during the first five years, to ensure good yields and prevent stunting – particularly when farmers are working on degraded soils. Organic fertilizer can be created on-site by composting available biomass, though the system will not be able to supply adequate on-site fertilizer for the first few years.

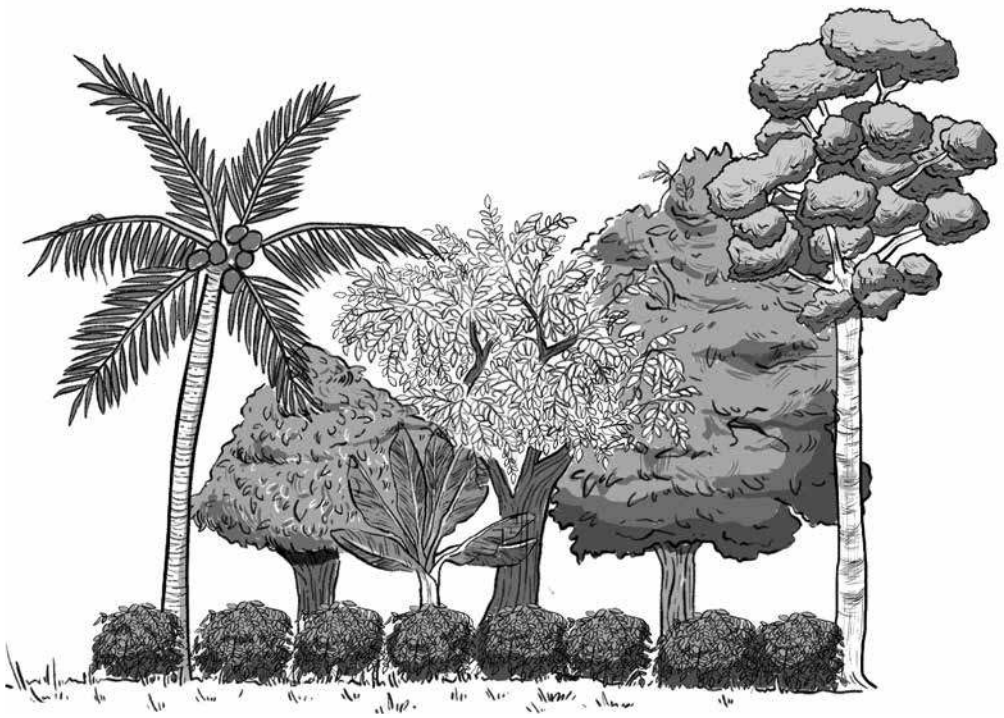
A biodiverse, successional oil palm agroforest can reduce or eliminate the need for pesticides, though particular organic inputs and agroecological techniques will be required to address these issues, particularly in the shorter-term.

'Slash-and-mulch' of fertilizer species (which produce biomass very efficiently) is an important element of soil health management. This feeds the system on a continuing basis.



Rainforestation farming

System description



Rainforestion farming is a successional, multistrata agroforestry system developed to restore native forest cover and forest diversity, while providing diverse income streams to farmers from multiple crop and tree products, including timber. Such systems are designed to shift the livelihood component gradually from crops to trees, ensuring that farmers have an incentive to contribute to forest restoration.



Rainforestion farming was developed in the Philippines in the 1990s as a solution for community forest areas. Exploitation of the natural mixed dipterocarp forests had turned these areas into highly degraded biological and agronomical 'deserts'. Upland areas without trees were dominated by highly invasive grasses such as *Imperata cylindrica*, a land cover type that does not adequately support the needs of local communities.

Initially, fast-growing exotic species were used to reforest government land, but these proved ill-suited to withstand the typhoons that regularly hit the archipelago. So, researchers – together with communities – began exploring the use of more resilient pioneer and hardwood native tree species, particularly those from the dipterocarp family. The rainforestion farming systems they developed proved to be relatively resilient to environmental shocks such as extreme weather events, as well as livelihood challenges such as market fluctuations and food insecurity.

Rainforest farming is often used on government land, so specific attention needs to be given to tenure rights. On land where the community does not have the right to harvest the timber, farmers must be compensated for their labour in planting and managing native timber trees. In this way, the community becomes a partner in a private–public restoration effort.



Seedling sales have also provided a major source of income for many rainforestion farming communities, especially if governments actively promote procurement from community managed nurseries in national reforestation programmes.

Design guidelines

- Rainforestation farming systems build on traditional community knowledge. Soil types, whether of ultramafic, sedimentary or volcanic origin, affect what species can be grown in the system. A good knowledge of the site, the forest species, and their interaction is essential.

- It can also be extremely valuable to involve all family members – including children – in the development of a rainforestion farming system, because the management needs and benefits from a successful system will span generations.

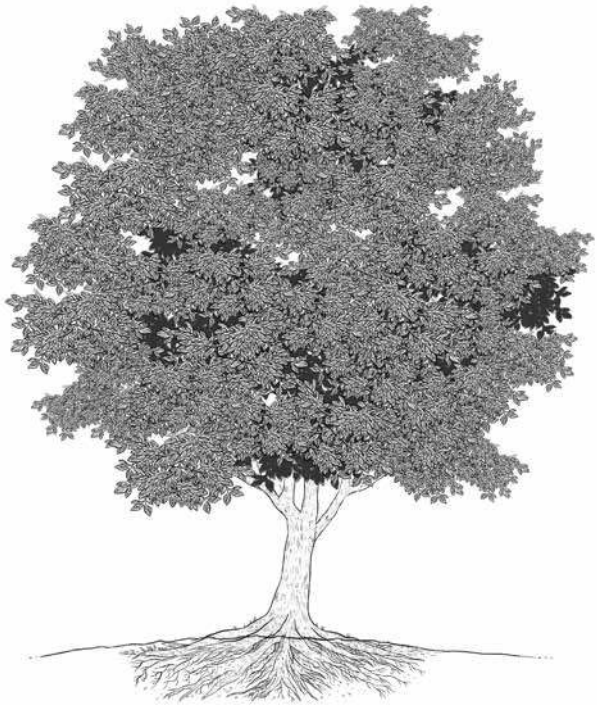
- For areas with little tree cover, a successional model in which agricultural crops give way to fruit trees, which in turn give way to timber trees is common. The spatial layout must be adjusted to the individual growth of the trees to ensure that fruit trees aren't too quickly overtopped by the timber trees, which could lead to an unacceptable decline in production. In open areas, native tree spacing of 2,500 trees per hectare is currently recommended, as this allows quick canopy closure and site capture.

- Many dipterocarp seedlings are sensitive to light and temperature. For this reason, they are grown under a canopy of nurse trees. This canopy is made up of fast-growing, native, pioneer species planted together with the fruit trees. The shade of any pre-existing exotic tree species, such as mahogany or mangium, can be taken advantage of, but these should be slowly replaced by native trees. Existing trees can be used as supports for other crops such as yams and black pepper.

- Sun-loving root and ground crops, such as cassava, sweet potato, taro, yam and pineapple are planted at the early stages, but once trees grow taller and denser, and light infiltration is less than 70%, they need to be replaced with shade-tolerant crops such as coffee, cacao and ginger.

- On Leyte Island in the Philippines, abaca has been a particularly successful cash crop for rainforestion farming. This endemic relative of bananas is harvested for its high-quality fibre, which is widely used and valued internationally in products such as teabags. Traditionally, it is grown under nitrogen-fixing trees such as *Erythrina fusca* and narra. It is fast growing and quick to recover after extreme weather events such as typhoons, and is most productive in an environment with around 50% shade. The fibre extracted from the abaca's pseudostems amounts to only around 2% of the plant's biomass; the rest is left on the farm as mulch, so abaca can be cultivated in one spot for around 40 years. One option for integrating abaca into a rainforestion farming system is to plant it a couple of years before planting canopy trees, so that it then provides useful shade for the tree seedlings when they are small.

- Integration of animals such as chickens and small ruminants is encouraged to enable nutrient cycling from their manure.



Management guidelines

- The guiding principle for managing rainforestion systems is the same as for all other multistrata systems: that is, to manipulate the canopy structure through spacing, thinning and pruning to provide the right amount of shade at the right time for all the plants in the system.

- The strong emphasis on timber trees is a distinguishing characteristic of rainforestation systems. Consequently, silvicultural treatments such as thinning and pruning to remove overhead shade, and cutting of vines and climbing bamboo, need to be done continuously.

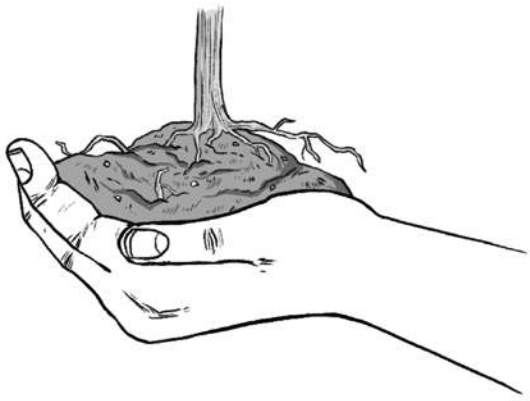
- Naturally occurring tree seedlings need to be actively managed and protected by controlling competing grasses, shrubs and climbers (assisted natural regeneration).

- On degraded sites, the natural dipterocarp seedling stock is usually very scarce or absent. Dipterocarps therefore must be planted, a process called 'enrichment planting', using seeds and seedlings collected from or near mother trees in natural forest.

- Dipterocarps do not produce seeds every year. Rather, seed is produced every 2–10 years. In such 'mast years', all trees of a species produce their seeds at the same time. Dipterocarp seeds are recalcitrant, so seed storage is not possible.



- In years when there is no seed production, wildlings (small seedlings from natural regeneration) can be extracted from the natural forest. This practice is not harmful in places where natural regeneration is abundant (for example, under mature trees). Wildlings are often damaged during extraction and transport, but if they are placed in a shaded recovery chamber – a small, closed structure of bamboo and acetate plastic sheets, with relatively constant temperature and high humidity – their survival rate will be close to 100%.





9

STORIES FROM THE FRONT LINE





In this chapter, we illustrate how the application of the design principles is used to adapt generic agroforestry models, like those in the previous chapter, so that these work for individual farmers and their contexts. We also show how neglect of the principles can cause problems. We do this through a series of synthetic case studies, based on real-life situations.[12]

[12] The specific situations portrayed are fictional, and any resemblance to real locations, real persons or real institutions is completely coincidental.

Hidden hunger and land degradation: An NGO responds to a village's request for assistance

Agroforestry systems exemplified:
Zero-grazing livestock production, trees integrated with seasonal agriculture

The farmers' needs and the response



Villagers in a mixed farming area in Kenya have been worried about erosion and declining productivity. This is not their only problem. For example, many of the mothers worry that their children are not getting all the vitamins they need for healthy growth, especially towards the end of the dry season when food can be scarce.

Mary, one of the villagers, heard about agroforestry when visiting her sister in a neighbouring village, where an NGO called TreesFP was carrying

out a project. She was able to arrange a visit by the project leader, and eventually the NGO began activities in her village.

Mary comments, "We weren't sure about accepting help from TreesFP. We liked the name – it means Trees for People – but some people have had bad experiences with agroforestry projects. But we decided to listen to what they had to say."

Rose, an extension worker, was appointed as team leader by TreesFP.

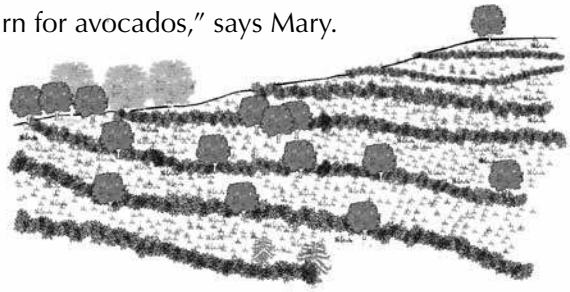
"It was difficult at the start, as some of the villagers expected us to come with ready-made solutions," she says. "There were comments like 'What use are you, if you don't know what we should do?' It took a while for me to explain that we had to share our different sorts of knowledge in order to find solutions."

Mary says, "It is true what Rose says. TreesFP were able to help my neighbour Judith to install what they call a zero-graze system for her cattle. That didn't suit me, but it was fine for Judith because she has grown-up children who can help her with the work. Mine are small, but TreesFP helped me plant my fruit tree portfolio."

Her neighbour Judith comments, "We now have our cattle in feeding pens, close to the house. TreesFP taught us how to plant fodder banks of Napier grass and agroforestry trees, which we cut and carry to the cows. This means that they are no longer overgrazing our sloping land and causing erosion. We collect and carry manure to the fields and fodder banks to fertilize them. It is a bit more work, but it is worth it. And we also give Mary some fodder for her cow."



"But only in return for avocados," says Mary.



How the design principles were used and applied

Rose explains:
"The zero graze and the fruit tree portfolio are just two of the interventions that we helped farmers with. That really illustrates how seriously we take the **principle of farmer-centredness**. We don't come with ready-made solutions. You have to work them out with the farmers.

"The **principle of aptness to place, people and purpose** is nicely illustrated by the cases of Judith and Mary. We quickly realized that Mary didn't have the time or the family labour to dedicate to a cut-and-carry system, but she was keenly interested in having a wider choice of fruits, year round. Through careful selection of the species to be planted, we were able to ensure that the trees would not interfere with the crops, particularly the maize, which would be seriously affected by any shade. In fact, we went beyond species – we were able to get some dwarf varieties of both mango and avocado that work for her. At the level of the village, too, the high participation we had in the portfolio process will ensure that we have a good match with local conditions and needs.



"Judith and some others really wanted to minimize the labour needed for the cut-and-carry system, so in most cases we established fodder banks of Napier grass close to the feeding pens. Also, in many cases we were able to plant the fodder shrubs like calliandra quite close to the pens. But the majority of the fodder trees are in the contour plantings that we established on the former grazing land. To choose the species, we asked the farmers to rank their preferences, and then we discussed in a workshop which species would work best together.

"With respect to the **principle of synergy**, finding the species that work best together was part of that. Even clearer, though, is the nutrient cycle that the villages now have going with the cut-and-carry system. The manure goes straight back on to the same fields that have previously been degraded by the cattle. That means that they are indirectly transferring nutrients from the fodder trees and grass back to the pastureland. In a few years, it should be possible to use the pastureland for crops, or to reintroduce livestock again, but with sustainable stocking rates.

"The principle of synergy also helps people to think out of the box. Mary is now growing some leafy vegetables and medicinal species in the partial shade cast by some of the fruit trees."



Land restoration for livelihoods and biodiversity conservation

Agroforestry systems exemplified: Simplified cacao multistrata system, sequential biodiverse fertilizer-species system, complex cacao system

The farmers' needs and the response

Ignacio, his son Julio, and Ignacio's cousin Rafaela are smallholder farmers in the Peruvian Amazon. Like others in their village, Santo Domingo, which is at 500 metres above sea level, they wanted both to increase their incomes and recover some of the degraded pastureland on their farms. For this reason, when they heard that people from FuturoVerde, a local NGO, were going to make a presentation in the local primary school about farm diversification and land restoration, they decided to attend.

Rafaela comments, "A few years back, we heard that FuturoVerde had received some money from Europe that they wanted to invest in sustainable agriculture, so that got us interested. I wasn't expecting much, but the agronomist from FutureVerde, Miguel, comes from near here, and he convinced me that this was something we shouldn't miss out on."

Miguel from FuturoVerde explains more about the opportunity: "The funding comes from an impact investor based in Switzerland. They invest in land restoration projects that also strengthen local livelihoods and contribute to biodiversity conservation. They contacted us to help them in building a portfolio here. As you probably know, Santo Domingo is in the buffer zone of one of our most important national parks."

Ignacio comments: "Attending that meeting was one of the best decisions we ever made. We were able to work with FuturoVerde to come up with solutions that really suited us. Not like that other NGO, who only wanted us to stick to their technological package. So, in my case – as you can see, I'm not as young as I was – they helped me install a cacao system that I could manage, same with my cousin Rafaela and her rehabilitated pastureland. And with my son Julio …"

"… I can explain Dad," says Julio. "I told them that cacao is the future, but a cacao that promotes biodiversity and is good for the environment. They knew that I was willing and able to put the time in … I think that my cacao plot is now one of the most diverse in the region. All this was possible because there is money to help in the establishment process, which is the most difficult part for us."



How the design principles were used and applied

Miguel explains: "You can see some differences and similarities in what we ended up doing with Ignacio, Rafaela and Julio. Ignacio's plot is fairly simple: a ground cover of centro[13] that helped to control weeds, fix nitrogen and add organic matter to the soil in the first years, cacao at 4 metres spacing, inga shading, and an upper storey of mahogany."

"I'm very proud of the mahogany," says Ignacio. "When I was young, mahogany was abundant here. Now it's all gone. So, with these mahogany trees, I'm helping to restore things to what they were before." Julio, his son, adds: "I think they also add to the sale value of the property, in case one day we want to move on."

"As I was explaining," continues Miguel, "what we did with each farmer depended on the situation. For example, in Rafaela's case, she had a lot of degraded pastureland that she wanted to recover. She decided to opt for a sequential system, which means that the components change in a planned way as time goes on. First, we planted fertilizer species in five-metre-wide strips along the contour lines. Some of these just work by producing a lot of biomass, which then goes back to the soil after being cut back: for example, highly productive grasses like panic grass. We also planted Mexican sunflower, which is a great accumulator of phosphorus and potassium. Plus the leguminous species, of course – trees like gliricidia and inga, both of which tolerate these acid soils, and shorter-lived legumes like stylo and pigeon pea. Between the fertilizer strips, we planted one-metre-wide beds with annual crops and the fruit trees and plantain that you can now see. Eventually these are going to shade out the fertilizer species, and the organic matter will come from the trees themselves. If you come back in five years, it will look a bit like a forest – but one that will be full of productive species, what we call an agroforest."

[13] *Centrosema molle*, a ground cover plant widely planted in Latin America.

"I suppose we didn't treat the land very well over the years," says Rafaela. "Also, it is quite steep. Before I installed my restoration system you could see the earth, all dry and hard, through the blades of grass. Like that field you can see over there." She points to a neighbour's land: the criss-crossing tracks of the cattle on the gentle slope can be clearly seen. "I'm really happy that my land will be healthy again."

"And then there's Julio," says Miguel. "His cacao plot is a lot more complex than his dad's …"

"Yes," says Julio, "with the cacao, the ginger, the turmeric, the black pepper, the annato, not to mention the timber species in the overstorey, it is complex. Managing this plot is a bit like managing a herd of cattle. You get to know each animal – each tree, in this case. You need to know when the flowering is coming, when the tree most needs light; you need to know how to read the signs that they give you. I pollard the gliricidias when the cacao is coming up to flowering time, because that's when it needs more light. Later on, when the cacao trees get bigger and start to self shade, I'll probably thin out some of the gliricidias and prune the other trees in the second storey. Sure, it's labour intensive, but I'm a cacao farmer. It's what I do. The more I put in, the greater the rewards."



"So," says Miguel, "you can see that the farmers' interests were paramount, even though FuturoVerde and our funding partner have other goals. We explained all that to the community. Transparency first. Don't forget too that what you see with these three cases is only a sample of what we did. We've been able to negotiate a cacao export deal with one of the big cacao trading houses, and we've also given training in agribusiness management. We don't just come here to 'do agroforestry'. In some cases, we thought that other types of production were more suitable. That's what the **principle of farmer-centredness** means in practice.

"I think you can see how seriously we've taken the matching of these systems to what the farmers wanted … to what their land is suitable for, too. But if we're talking about the **principle of fitness to people, place, and purpose**, then you have to take into account the wider objectives. I mentioned before that our funding partner is interested not just in livelihoods, but also in enhancing biodiversity. If all we'd done had been to establish systems like Ignacio's simple cacao, then I don't think we'd have met those goals. What we've done is to diversify the structure of the landscape here, as well as to increase connectivity with remnant patches of forest. Simple cacao on its own wouldn't be enough, but when it's combined with agroforests, complex cacao, and other actions (like the boundary and riparian trees we've been helping people put in), what you have is a much more uneven landscape, with lots of different niches for wildlife. It's a far cry from the old mix of degraded pastureland and forest remnants.



"It's pretty clear that the **principle of synergy** is what underlies these systems. For example, in Ignacio's system, the cacao and inga give strong lateral shade to the young mahogany trees. That reduces the incidence and severity of mahogany shoot borer. That's why almost all Ignacio's mahoganies have just one, straight stem.

"Then look at Julio's system. We gave it a kick-start using organic fertilizers – manure, rock phosphate – but now you have a closed system where the only things coming out are

the cacao seeds. He composts the pulp and the pods, and it goes back to the plot. The soil is moist, even in the dry season, because of the trees and the organic matter. Julio is like a manager of synergy, helping to channel light and nutrients where they are needed. Or we can talk about how Rafaela's fertilizer species prepared the ground – literally – for what she has now.

"But it's more than that, because our intervention was designed to have an effect greater than the sum of its parts. The different interventions complement each other, as far as biodiversity is concerned. Then the biodiversity value helps make demand for the cacao beans more stable, with the different kinds of certification we have. And one of Julio's sisters is talking about opening a tourist lodge, so people going to the national park can also learn about how agroforestry and trees-on-farms can help with conservation. So, we have synergy of different sorts and at different levels – agroecological at the plot level, ecological at the landscape level … even economic synergy."



Forest restoration through 'rainforestation'

Agroforestry systems exemplified: Complex sequential multistrata system

The farmers' needs and the response

Historically, management of the once magnificent dipterocarp forests near the community of La Pacífica in Leyte Province, Philippines had not served community members well. A village elder explains what happened: "When I was young, the land was covered with forest. Then the logging companies came in, and they left a wasteland. There were a few low-value trees left, but all the upland areas were covered by this kugon[14] grass. A wasteland is what they left, in spite of all the promises. We wanted to bring the forest back, but no one really knew how."

[14] *Imperata cylindrica*, a highly invasive grass native to tropical and subtropical areas worldwide (except the Americas).

The villagers were at first sceptical when Angel, an agronomist who works with the local municipality, visited them to tell them about what he called 'rainforestation'. Angel comments, "People are right to be sceptical, but they heard me out. I told them that we had technical back up from an international research centre and a local NGO called ReForesta, and I invited a group to go and see the work that they've done in one of the other partner municipalities."

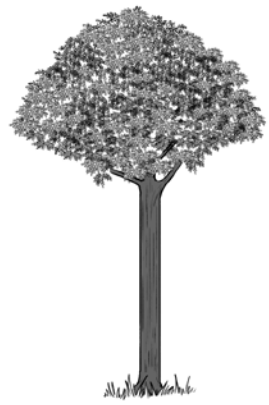
Gabriela, a community leader adds, "When we made that visit, we talked to other villagers, and we could see there was something in this rainforestation. That was ten years ago. Now you can see the results. Over that time, we've used agroforestry to bring back the forest. We have 50 hectares in total, and the time and energy we've put in are paying off. On top of that, the community nursery is making money from selling seedlings."



How the design principles were used and applied

Villagers in the area have a lot of experience in cultivation of abaca[15] so they were pleased when the agronomists from ReForesta told them that it would work well in the rainforestation system. Angel explains: "Some of the dipterocarps grow best under light shade. Traditionally, that's also how people have cultivated abaca. When we explained about rainforestation in the agroforestry design workshop, the people at one suggested that abaca can provide some of the shade that the dipterocarps need. We also use some fast-growing native trees that fix nitrogen, especially agoho and narra.[16] We start off with some staple crops. Cereals can be used in the first year but we also plant some root crops that are a little more shade tolerant. The villages chose sweet potato and taro. Also some pineapple – not a root crop of course, but they do well in the kind of shade that the system has over the first two years. We also plant the abaca at the very start."

"At the end of the first year, the villagers start harvesting the root crops and replacing them with shade crops. They chose ginger as a cash crop, with some robusta coffee, just for their own use. By the end of the second year, all the root crops and the pineapple have been harvested, and we put in the dipterocarp seedlings. We also thin the agoho and the narra to reduce competition with all the other components. We do a second thinning much later, when we thin out some of the dipterocarps. That also allows enough light for continued abaca production, or even to put some cacao into the system. In the long term, the villagers will be able to selectively harvest the dipterocarps, as they have full tenure rights over the forest and its products. But I think that right now they are just glad to have the forest coming back."



Community leader Gabriela also had the idea of installing a community tree nursery. She comments, "Of course,

[15] *Musa textilis*, a species of banana native to the Philippines and grown for its fibre.
[16] *Casuarina equisefolia* and *Pterocarpus* spp.

the nursery is not an agroforestry system – we understand that – but it is an important part of **our** system. We collect local seeds of all the species, as well as what we call wildlings: orphan seedlings that we take in and look after until they are in good condition for planting. Sometimes it's the best way to get plants of the most valuable species, as many of them only produce seed once every few years. We can't wait that long!



"We received training in nursery operation from ReForesta, so we know all about things like root quality, sturdiness index and mycorrhizae. Here, we are experts. We don't sell the plants, but the municipality pays us and some other community and private nurseries to produce them, and then they give them to the projects. It's a win-win, a bit like trees and mycorrhizae."

Manolo, lead agroforester with ReForesta, takes up the story: "The rainforestation idea came from a partnership between Visayas State University and the German Technical Cooperation Agency, and has proved to be highly successful.[17] However, it won't work on the ground unless what we do is **farmer-centred**. So, we explain that part of our interest is to restore the dipterocarp forest, but that it has to be done in a way that benefits them from the start. It has to be farmer-centred, whatever global challenges you have in mind. In fact, that's one of the benefits of a sequential system – if it's done right. There are no gaps, no time when the income dries up. Here, you get the cereals and root crops, then the abaca and the ginger kick in. I think Angel didn't mention that most of the villagers also plant various other fruit species, either at the same time as the narra, or when the dipterocarps go in. Here, there will always be something to harvest. I was happy when the villagers suggested abaca, as it's a very thrifty species, perfect for agroforestry. The fibre that's extracted from it only makes up about two percent of the biomass – the rest stays in the system as mulch. It's **synergy** in action – abaca loves the shade that the narra and agoho give, and I guess the nitrogen that they fix … but it all goes back to them and the other components."

[17] For more information, see: Rainforest Information Portal. 2010. Manila: Rain Forest Restoration Initiative. Accessed 2 June 2022. http://www.rainforestation.ph/index.html.

When things haven't gone right

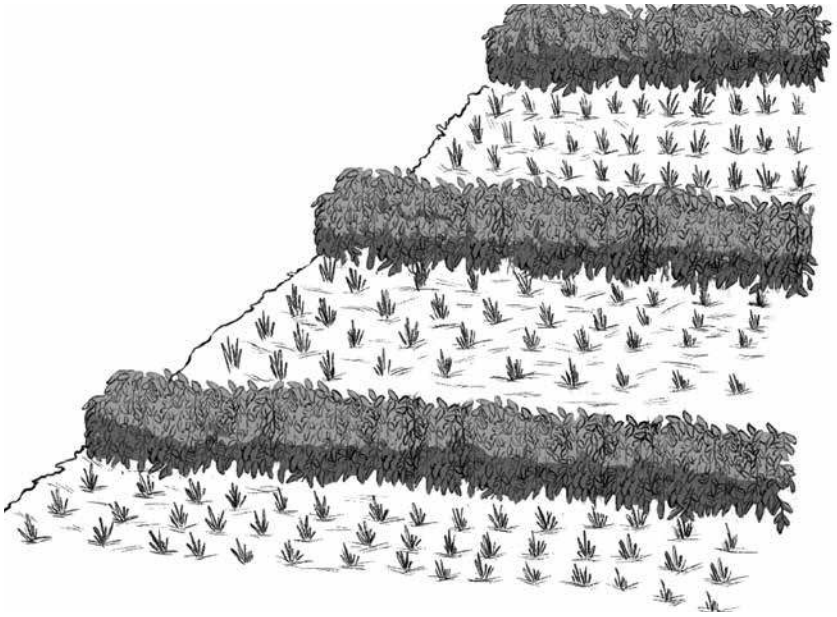
The farmers' needs and the response

The villagers in the Kenyan community we described in the first story have become agroforestry enthusiasts, but it hasn't always been that way. Mary explains: "As I said before, some people have had bad experiences with agroforestry. That was a long time ago, when I was quite little. But you know what they say, 'once bitten, twice shy'. And even those who were not bitten can be shy of the dog."

Extension worker Rose takes up the story. "After Mary told me what she could remember, I tried to find out what happened. I have a friend who works for the organization that led that agroforestry project, and she found some old reports and talked to some of the people who were involved. And, in the reports, I also found the names of some of the participants, who are still here in the village. In the final report there is even a nice picture of Mary when she was little, standing by one of the eucalyptus trees that they planted. She looks very happy in that photo."

"I think that they'd just given me a very big mango," says Mary, laughing.

"Let me tell you what happened," says Rose. "They planted trees in two different places, the same as we did with TreesFP: in the lowland near the houses, like where Mary has her fruit trees, and in the pastureland, like where Judith used to have her cows. But that is where the similarities end …



"They knew that people were short of cattle fodder and timber, so on all the boundaries of the farms they planted hedgerows with species that cattle can eat, like calliandra, gliricidia, leucaena, with timber trees like teak and pine every six metres. If you look, you can still see a few of the teak trees. But people didn't like the hedgerows, because they had to be pruned all the time and – since they're all spread out – it took too long to collect the leaves for fodder. And then there were the mice …"

"I can still remember them squeaking as I walked along our hedgerow," interrupts Mary.

"… If you read the report," continues Rose, "it explains how in the end they had to recommend not sowing maize seed too close to the hedgerows, because the mice would eat it all. After the project, everyone grubbed out the hedges. The timber trees have been cut down one-by-one over the years. They've been quite useful, but no one has planted any new ones.

"They tried alley cropping too. They knew that here the soils are quite poor and that most people can't afford fertilizer, so that was the solution, they said. They brought in some workers to plant gliricidia in rows in some of the fields. They were supposed to pump up the nutrients from below. There were four metres between the rows, and people planted their maize in the alleys. Every three months they were supposed to prune back the trees and leave the foliage to decompose, like green manure. But the maize didn't grow any faster, and the trees took up space that we used for crops before. The report doesn't say what happened in the end, but everyone says they mostly had to poison the trees to get rid of them.

"Then, on the hilly pastures, they said it was important that the cattle had some shade, and that trees could also help to prevent the grass from drying out so much in the dry season. So, people planted some trees – I think they got them from South America – and they had to protect them with little stockades around the groups of trees until they were too tall for the cattle to reach the leaves. To be fair, the trees were very fast growing, and after just one growing season they were almost two metres tall. So, they said that the people could remove the stockades to put the cattle in. I think that everybody knew what would happen. The cows couldn't reach the leaves, but they liked to scratch themselves on the trunks of the trees, and they soon pushed most of them over and broke the stems."



How the principles were misapplied or ignored: What went wrong?

"I think I can answer that question quite quickly," says Rose. "You know, the report talks a lot about participatory methods. But I think they thought that participation was just about asking people questions. Also, if you look at some of their tables of results, you can see that most of the people they talked to were men. I think that the women would have given them another story. So, this wasn't really a **farmer-centred process**, because either they didn't talk to the real farmers, or they didn't listen. As for the **principle of aptness** – as you can see, it is quite dry here and, as I have said, the soils are poor. That means that we can't have trees here that are going to take the nutrients and water that the crops need. Solutions that work elsewhere won't work here. You know that photo I mentioned, of Mary with the eucalypt tree? Well, you know, that tree is still there. It is because someone planted it in a very dry and rocky place that is not good for anything else. Almost all the others are gone.

"As for **synergy**, these systems were synergistic in theory, but in practice the different components were not connected, except negatively. The hedgerows competed with the maize and provided habitat for mice, the gliricidia just took up space, and the trees in the pastures were a bit of a waste of time."

Rose smiles. "So, nought out of three. You know, we are lucky. We've had the chance to learn from these kinds of mistakes."

Scientific names of species and genera

abaca: *Musa textilis*
acacia: Acacia species, especially A. *mangium*
açaí: *Euterpe oleracea*
African mahogany: *Khaya* species
agoho: *Casuarina equisetifolia*
alder: *Alnus* species
andiroba: *Carapa guianensis*
annatto: *Bixa orellana*
arrowroot: *Maranta arundinacea*
avocado: *Persea americana*
bananas: *Musa x paradisiaca* species
beans: *Phaseolus* species, especially *P. vulgaris*
beet: *Beta vulgaris*
black pepper: *Piper nigrum*
cacao: *Theobroma cacao*
calliandra: *Calliandra calothyrsus*
cassava: *Manihot esculenta*
casuarina: *Casuarina* species
centro: *Centrosema* species
chard: *Beta vulgaris*
chilli peppers: *Capsicum* species
durian: *Durio* species
erythrina: *Erythrina* species
eucalypt: *Eucalyptus* and *Corymbia* species
falcata (Moluccan albizia): *Falcataria falcata*
ginger: *Zingiber officinale*
gliricidia: *Gliricidia sepium*
hog plum: *Spondias mombin*
imperata: *Imperata* species
inga: *Inga* species, especially *I. edulis*
ipê: *Handroanthus* species
kudzu: *Pueraria* species
laurel: *Cordia alliodora*
lazones: *Lansium domesticum*
leucaena: *Leucaena leucocephala*
mango: *Mangifera indica*
Mexican sunflower: *Tithonia diversifolia*
musizi: *Maesopsis eminii*
napier grass: *Pennisetum purpureum*
narra: *Pterocarpus indicus*
panic grass: *Panicum* spp.
pigeon pea: *Cajanus cajan*
pine: *Pinus*
stylo: *Stylosanthes* species
sugarcane: *Saccharum officinarum*
taperebá: *Spondias mombin*
taro: *Colocasia esculenta*
tea: *Camellia sinensis*
teak: *Tectona grandis*
turmeric: *Curcuma longa*
vanilla: *Vanilla planifolia*
Vetiver grass: *Chrysopogon zizanioides*
yam: *Dioscorea* species

Glossary

The glossary includes only terms not defined in the main text.

Agrobiodiversity: biological diversity on farms, including, but not limited to species and varieties of domesticated plants and livestock

Agroecological services: positive contributions of one component of an agroforestry system (or other agroecosystem) to the growth, productivity or sustainability of other components (for example, shade, nitrogen-fixation)

(Agroforestry) intervention: any action or set of actions aimed at promoting or improving the practice of agroforestry

Alley cropping: an agroforestry practice in which crops are grown in "alleys" between rows of trees. The trees are regularly coppiced to produce nutrient-rich mulch

Biomass: plant or animal material, including dead and decomposing material

Climate change adaptation: in the agroforestry context, refers to actions taken to enable farming families to cope better with the effects of climate change

Climate change mitigation: actions taken to reduce current or future greenhouse gas emissions, or to reduce existing greenhouse gas concentrations

Climate-smart agriculture: agriculture that is adapted to, or that mitigates climate change

Domestication: the processes by which farmers or scientists change the genetic characteristics of wild plants and animals so that they are more useful for agriculture

Ecosystem services: the benefits environmental assets such as land, water, vegetation and atmosphere provide to people, especially in the form of essential goods and services (for example, clean air, water and food)

Enabling environment: factors, other than sound agroforestry practice and the natural environment, that determine whether a given agroforestry

intervention is likely to be feasible or successful: for example, government policies, the availability of credit, or the quality of extension services (Enabling environments can be favourable or unfavourable.)

Flagship species: the main species (one or more) in an agroforestry system: that is, the crop, livestock, or tree species that the farmer considers the most important component

Flotilla species: species (usually trees or other plants) included in an agroforestry system primarily for the agroecological services they provide to the system and to the flagship species

Habitat connectivity: the degree to which animals, pollen and seeds are able to move between larger blocks of habitat, particularly via smaller habitat "bridges" (for example, riparian corridors) or "stepping-stones" (for example, scattered copses or individual trees)

Keystone species: a species that plays a very important ecological role due to other species' dependence on it

Landscape: a sizeable area of land (of the order to tens to hundreds of square kilometres) in which humans and natural ecosystems interact. (The interactions often drive changes in land cover, land use, livelihoods and demographics.)

Microsymbiont: a microorganism that forms a mutually beneficial relationship with another organism

Modern variety: relatively recent varieties of staple crops, particularly the dwarf varieties associated with the major yield increases of the "Green Revolution"

Monoculture: a crop field, orchard or forestry plantation consisting of only one species

Mulch: material used to cover the ground in order to control weed growth and maintain soil moisture. (In agroforestry, the term refers especially to dead leaves and branches, although other organic and inorganic materials can also be used.)

Multistrata system: an agroforestry system with distinct layers (strata) formed by the crowns of trees and understorey components.

Organic matter: animal and plant material (mostly dead or decomposing) in the soil

Photosynthesis: the process by which plants use light energy to make carbohydrates

Rotation length: in forestry, the number of years between regeneration and final harvesting of a stand of trees

Sequential system: see Successional system

Successional system: an agroforestry system, usually multistrata, that changes in a planned way over time as components reach the end of their life cycles or are shaded out by other components.

Technological package: a predetermined, standard set of cultivation practices for a given species or product, particularly one that growers are obliged to follow as a condition for receiving credit, subsidies or technical support

Traditional varieties: varieties of crops and trees, genetically differing from their wild relatives, that have been developed over long periods of time by farmers, and that are well suited to local conditions and farming practices

Taungya: the practice of growing staple crops in young timber plantations, typically as a means of plantation establishment

AGROFORESTRY:
A PRIMER

Conventional agriculture is very productive. But high productivity comes at a cost: soil that is depleted or eroded, watercourses that are polluted or drying up, and a food system that produces 20–40% of greenhouse gas emissions. Many people now agree that we urgently need to transform the food system, including agriculture. Agroforestry, as a nature-based approach to production and land use, will play an important role in this transformation. Agroforestry is not new; farmers have practised it for thousands of years, and scientists have recognized it since the 1970s as a productive and ecologically sustainable form of agriculture and land use. But now agroforestry is suddenly at centre stage; it is promoted as a land-use strategy to support climate change mitigation and climate change adaptation, biodiversity conservation, sustainable agriculture and other goals.



Many organizations recommend or use it as a tool for restoring ecosystems – not only agricultural ones, but also forest landscapes.

Although not a cure-all, agroforestry has great potential to contribute to all the goals mentioned above. However, agroforestry is not just a matter of adding trees to farms. To realize its potential, practitioners need to understand its principles. *Agroforestry: A primer* is a guide to agroforestry principles and concepts – and how to use them effectively.

Editors: Anja Gassner and Philip Dobie

Foreword by Fergus Sinclair

Authors: Marlito Bande, Brian Chiputwa, Richard Coe, Jonathan P. Cornelius, Philip Dobie, Anja Gassner, Rhett D. Harrison, Hanna J. Ihli, Clement A. Okia, Andrew Miccolis, J. David Neidel, Stepha McMullin, Agustin Mercado, Athanase Mukuralinda, Caroline Pinon, Eduardo Somarriba, Peter Thorne and Etti M. Winter.

CIFOR

World Agroforestry

Federal Ministry for the Environment, Nature Conservation and Nuclear Safety

ADB

ISBN 978-9-96-610861-6

Table of Contents

1.	**Wheat**	1
2.	**Rice**	3
3.	**Maize**	9
4.	**Sorghum**	12
5.	**Pearl millet**	14
6.	**Oat**	15
7.	**Chick Pea**	15
8.	**Red Gram/Pigeon Pea**	17
9.	**Pea**	18
10.	**Lentil**	20
11.	**Groundnut**	21
12.	**Soybean**	23
13.	**Mustard**	23
14.	**Sesamum**	24
15.	**Linseed**	25
16.	**Castor**	25
17.	**Safflower (Kusum)**	26
18.	**Sunflower**	27
19.	**Sugarcane**	28
20.	**Potato**	30
21.	**Cotton**	32
22.	**Jute**	35
23.	**Tobacco**	38
24.	**Berseem or Egyptian clover**	41

Agriculture Wallah

Subject: Agronomy

Wheat



- S.N : Triticum astivam
- Family : Poaceae
- Origin: Turkey / S.W. Asia.
- Inflorosence spike
- 2n = 42
- fruit type caryopsis
- Temperate crop.
- Most important crop cultivated throughout
- the world.
- DBW 187 Variety was developed by : IIWBR, Kernal
- India rank second both area and production after rice crop.
- WB 2 is biofortified variety released in india (Fe and Zn content)
- It is called king of Cereals
- Wheat protein is : Gluten
- Gluten protein responsible for baking quality.
- 8 – 12% of gluten protein having in wheat.
- Inflorescence of wheat is called : Ear/Spike
- Seed of wheat is called : Cryopsis
- Seed treated by Hot water/Systematic fungicide/Solar energy/Vitavax(To prevent from Loose smut disease
- Permanent adventetitious root below the soil surface after appearance of the first leaves is called **Clonal root.**
- Gene responsible for daffiness in wheat is **Norin-10**
- **Sharbati sonara** was first Indian dwarf amber grained variety of wheat made from Sonara 64 by γ – gamma rays.
- **Isolated** by **Dr.N.E. Borlaug** in 1960 at maxico
- The gene responsible for dwarfness in : Norin-10 **(Rht_1 & Rht_2)** sd_1, sd_2
- **Green revolution** in India is said to be mostly **Wheat revolution**. (HD – 2329) varity
- Variety were responsible for **Green revolution** in india is : **Lerma** (100 kg seed)
- **srojo, sonara 63, sonara 64 and mayo 64**
- Atlas 66 is a biofortified hybrid (nutrient rich) wheat variety which has been used as a donor for improving cultivated wheat and is rich in **proteins.**
- **Interspecific cross** between bread **wheat** and **rye**: Rimpu (1st in crops)
- **Triticale** is cross between Wheat & Rye.
- Imported from Mexico (1965) by Minister of Agriculture Shri C. Subramanian,
- Test weight of wheat grain is about **40g**
- Ear cockle disease of wheat is caused by **Anguina tritici**.
- Test weight of phaloris minor 2 gm

1. **Triticum aestivam 2n = 6x = 42**
   - Hexaploidy
   - Mexican dwarf wheat (Bread wheat)
   - Most widely cultivated in India
   - 87% area
2. **Triticum Vulgare 2n = 42**



   - It is a **common bread** wheat
   - Grown in rainfed condition
3. **Triticum durum 2n = 28**



   - Also known macaroni wheat
   - Use Suji formation

4. **Triticum dicoccum 2n = 28**



- Also known **Emmer wheat** and Winter wheat
- Suitable for preparation of south Indian dish "**Uppumav**"

5. **Triticum spherococum 2n = 14**



- Indian dwarf wheat / Club wheat.

6. **Triticum monocoum 2n = 14**



- Einkorn wheat.

Distribution;-

World

- Production- China>India
- Area- India>China
- Productivity- UK>Germany

India

- Production- Uttar Pradesh> Madhya Pradesh> Punjab
- Area- Uttar Pradesh > Madhya Pradesh> Punjab
- Productivity- Punjab 47.12q/hac

Climate :

- It is a C_3 and long day plant.
- Optimum temperature for sowing : 12 – 34°C
- Optimum temperature for sprouting : 20-25°C
- Optimum temperature for growth : 20°C
- Base temperature of wheat : 4°C

Soil :

- Loam to clay loam is best.
- pH 6.7
- Soil should be free from clods.
- Most sensitive crop to poor soil aeration

Seed Rate :

1. Bradcasting – 120-125 kg/hac.
2. Dibbling method: 25 – 30 kg/hac.
3. FIRB method: 75 – 80 kg/hac.
4. Spacing: R × P– 22 – 5 × 10cm.

Nutrient management :

- Irrigated condition (NPK): N : 120 – 180 kg/hac.
  P : 60 – 70 kg/hac.
  K : 40 – 60 kg/hac.
- Application of $ZnSO_4$ – 25 kg/hac.
- Biofrtilizer for wheat is : Azotobactor

Irrigation :

- Wheat crop required 4 to 6 irrigation.
- Transition zone between root and shoot is known as crown root initiation.
- CRI word given by : **B.L. Bhardwaj**

1. **CRI :** 21 DAS. (Day after sowing)
2. **Tillering :** 40-45 DAS.
3. **Jointing :** 60-65 DAS.
4. **Flowering :** 80-85 DAS.
5. **Milking :** 100-105 DAS.
6. **Dough stage:** 115-120 DAS.

Method of Irrigation:

- Most common irrigation method is : **Border strip method > Check basin**

Weed Management :

1. **Dicot Weed :** Chenopodium album, Anagalis arvensis.
2. **Monocot weed :** Phalaris minor, Avina fatua.
3. **Sedges :** cyperus rotandus / Esculantus.

- Most common weed in wheat field : **Phaloris minor**.
- Which chemical control phaloris minor : **Isoproturon / Sulphosulphuron**
- Also use weed management

1. Pendimethaline 1 kg a.i/hac.
2. 2, 4, D (Ethyl ester)

- Sensitive variety to 2, 4, D is : **HD 2009 (Arjun)**

Varieties :

Single gene dwarf

- Lerma Rojo 64-A
- Sujata
- Giriraj (for nothen hills)
- Sonalika

Double gene dwarf

- Kalyan Sona
- Sonaro-64

- **HD-2009, HD-2329**

Mutant varieties: released from IARI,New Delhi

- Sharbati Sonora
- **Pusa lerma** (MP SADO-2021)

Other varieties

- Lok-1 (suitable for late sowing)
- **Malavia-12** (15% protein content)

Yield :

1. Irrigated condition: 40 – 45 q/hac.
2. Rainfed condition : 20 – 25 q/hac

- Pearling Index :
- Denotes resistance of weed seeds and breaking
- Kernel Hardness
- Project directorate of wheat research. Karnal Haryana. India Institute of Wheat Research **(IIWR)**

Rice

1. Rice (Paddy)

- Scientific Name: Oryza sativa
- Family: Poaceae
- 2n = 24
- Fruit type caryopsis
- Origin Place: Southeast Asia/SEA (Indo-Burma)
- Protein oryzenin
- Test weight: 25g (Basmati rice: 21g)
  - Most cereals have three stamens, but rice has **six**.
  - **Andhra Pradesh** is known as the **"Rice Bowl of India"** due to the Krishna-Godavari delta region. The same term is also used for **Chhattisgarh**.

Distribution and Importance

- Rice is the world's most important food crop, followed by wheat.
- It is a staple food for more than 60% of the world's population and more than 50% of Indian population.
- After maize and wheat, rice is third most important cereal in the world (production).
- About 90% of the world's rice cultivation area is in Asia, and nearly 90% of the rice is produced and consumed in Asia.
- In India, rice ranks first among cereals in terms of both area and production.

World

- Production: China > India > Bangladesh > Indonesia
- Area: India > China > Bangladesh
- Productivity: Australia > Egypt > USA
- Largest importers of rice: Indonesia > Philippines

India

- Production: Telangana (16.87)> Uttar Pradesh(15.99) > West Bengal(15.69)
- Area: Uttar Pradesh > West Bengal
- Productivity: Punjab (64 quintal/ha)

Rice Composition and Nutritional Values

- Rice contains less protein among cereals, with 6–7% in white rice, 7.8% in brown rice, 70% carbohydrates, and 2–2.5% fat.
- Rice grain is relatively rich in lysine, which is 4% of total protein fraction.
- **Rice Bran:** The bran is a thin, brown, nutrient-rich aleurone layer tightly attached to the endosperm, beneath the husk. It contains the average 20% oil and 12–15% protein.

Plant



- The stem (culm) is mostly hollow, except at the nodes.
- Two auricles (one on each side of the leaf base) and one ligule are present.
- The appearance of flag leaf marks the end of vegetative phase.
- The swelling of the flag leaf sheath is called **booting**.
- Agronomically, the 50% panicle emergence stage is known as **heading**.
- The effective root zone depth for rice is **60 cm**, whereas it is **120 cm** for sorghum, pearl millet, and maize.

Note: The jointed stem of cereals is known as culm (haulm).

Classification of Rice

- **Genus**: *Oryza*
- **Total Species**: 24 species (22 wild & 2 cultivated).
- **Cultivated Species**:
  1. Oryza sativa – Asian rice
  2. Oryza glaberrima – African rice

Cultivated Species

1. Oryza sativa (Asian rice)



- It is originated from Oryza nivara (annual) and Oryza rufipogon (perennial).

The three ecotypes or subspecies (races) of Asian rice (Oryza sativa) are:

1. Indica – India
2. Japonica – Japan
3. Javanica (Balu rice) – Indonesia

Particulars	1. Indica	2. Japonica	3. Javanica
Geographic origin	Tropical and subtropical Asia	Temperate East Asia	Indonesia
Photoperiod sensitivity	Sensitive	Insensitive	Insensitive
Tillering capacity	High	Low	Intermediate
Seed dormancy	Present	Limited	Limited
Input responsiveness	Poor response to higher inputs	High input responsive & highest productivity	

- The japonica race grows in both tropical and temperate regions. However, it thrives well in temperate regions with seasonal temperature fluctuations (e.g., China, Japan, South Korea, Italy, Egypt, and Russia).

2. Oryza glaberrima (African rice)



- Cultivation area: **West Africa only**.
- Characteristics: Prone to lodging and panicle shattering.
- Notable Variety: **CG-14**

Key Developments in Rice Research

1. **1946**: Central Rice Research Institute (CRRI) was established at **Cuttack, Orissa**.
2. Notable Indica-Japonica hybrids were developed: **ADT 27** (Norin 10 × GEB 24) and **Mahsuri** (India), **Malinga** (Malaysia).
3. **1960**: International Rice Research Institute (**IRRI**) established at **Los Banos, Philippines**.
4. **1965**: All India Coordinated Rice Improvement Project (**AICRIP**) established at Rajendranagar, Hyderabad.

Varietal Development:

Variety	Details
TN-1 (Taichung Native 1)	• Developed in Taiwan in 1949. TN-1 was the world's first semi-dwarf rice variety. Introduced to India in 1964 by G.V. Chalam. • It is first introduced variety of rice in India.
IR S	• Developed by Henry M. Beachel at IRRI, Philippines in 1964 through crossing Dee-gee-woo-gen Peta. Known as Miracle Rice globally. First high-yielding semi-dwarf variety of rice. It is a photo-insensitive variety. • It is the second introduced rice variety in India, introduced in 1966. •
IR 36	• Developed by Dr. G.S. Khush at IRRI, Philippines; a semi-dwarf variety resistant to multiple diseases and pests.
Jaya	• Developed by Dr. Shastri at CRRI, Cuttack (Orissa) in 1968 by crossing

	TN-1 and T-141. First high-yielding semi-dwarf rice variety in India, better yield than IR 8 and TN-1. Therefore, known as Miracle Rice in India.
Padma	• A reverse cross of Jaya, developed by crossing T-141 and TN-1.
Ratna	• Developed by crossing TKM-6 × IR-8. (Pusa Ratna: Mung bean

Rice Hybrids

- **Hybrid Rice:** The first hybrid rice was developed in China in 1974 by Prof. Yuan Long Ping, who is known as the Father of Hybrid Rice, India was the second country to develop hybrid rice.
- **Prof. E. A. Siddiq:** A renowned professor of biotechnology and genetics, referred to as "Mr. Rice of India" and the Father of Hybrid Rice in India. He developed several rice varieties, including Pusa Basmati-1, Pusa 2-21, Pusa 169 and, Pusa 203.

Basmati Varieties

- **Pusa Basmati-1:** It is the world's first high yielding semi- dwarf variety of basmati (aromatic) rice, developed by IARI in 1989.
- **Pusa Basmati 1979 and Pusa Basmati 1985:** Non-GM herbicide-tolerant, contains mutated ALS gene, tolerant to Imazethapyr.
- Pusa Basmati 1121: Known for long grains.
- Ambemohar: Mango blossom scent
- PRH-10: The first hybrid basmati (aromatic) rice variety.

Stress-Tolerant Varieties

Variety	Stress Tolerance
Pusa Basmati 1509	Drought tolerant and early maturing
Sahbhagi Dhan	Drought tolerant
Swarna-Sub 1	Submergence tolerant
CR Dhan 801	Tolerant to submergence, drought, and heat
CSR 36	Salt tolerant
Jalamagna	Deep water tolerant
Himali	Cold tolerant, suitable for high altitudes
Phalguna	Tolerant to iron toxicity (browning).

Other Rice Varieties

- Norin8, Norin 18, Sita, Pankaj (waterlogged), Sabarmati, Garima, Rajeshwari, Hema, Swarna, IR 64, BPT 5204 (Samba Mahsuri), Gobindobhog (West Bengal), Kala Namak (Black husk, UP).
- DRR Dhan 45 (IET 23832): First biofortified rice variety (Zn rich: 22.3 mg/kg).
- Pokali: Saline-tolerant variety from Kerala, thrives in waterlogged coastal areas.
- Mutant varieties: Jagannath, Prabhavati and Sattari. 2

Key Facts

- The first public-sector hybrid of rice: PHB-71.
- The first super rice variety: Lunishree (salt-tolerant).
- The first mutant variety of rice: Jagannath (Orrisa).
- Varieties suitable for direct seeding: Bala, Cauvery, Nagina-22, Pusa 2-21.
- BLB (bacterial leaf blight) tolerant variety of rice: TKM-6.
- The first protein-rich variety of rice: CRR Dhan-310.
- Aerobic rice varieties: Pusa Hybrid 10, Pusa Sugandh 3, Pusa Sugandh 5.

Climate

- Rice is a C_3 plant, a short-day plant (SDP), and requires a hot and humid climate. optimum temperature for rice germination is 27°C.
- Optimum temperature for growth and flowering is 25-30°C and at maturity 20-25 °C.

Soil

- Clay and clay loam soils are best suited for rice cultivation. pH range between 4-6 (5.5).

Note

A. **Lowland Rice:** Rice grown in flooded, puddled, and saturated soils by transplanting seedlings from a nursery. It faces less weed problems compared to upland rice.

B. **Upland Rice:** Rice grown in well-drained, non-flooded areas is called upland rice. Puddling in Rice: Described in Tillage.

Sowing

- Row orientation: North- South orientation is often recommended for row crops because it allows for more uniform sunlight distribution throughout the day.

- Growing seasons: The terms Aus, Aman, and Boro refer to three major types of rice cultivation cycles primarily practiced in South Asia, particularly India and Bangladesh.
- India mainly produces Aman (means peace) rice, accounts 84% of total area.

Aspect	Aus	Aman	Boro
Sowing Time	May–June (March–April)	June–July	December–January
Harvesting Time	September–October	November–December	March–April
Known as	Autumn rice	Winter rice	Spring/summer rice
Major Regions	Assam	Entire country	West Bengal, Orissa, Bihar
Productivity	Lower than Aman and Boro	Moderate	Highest among the three

- Aus + Aman rice = Kharif rice.

S. No.	Particulars	Seed Rate
1	Broadcasting (direct seeding) 	**100 kg/ha** (80–100 kg/ha)
2	Drilling (direct seeding) 	**60 kg/ha** (60–80 kg/ha)
3	Hybrid rice	**15 kg/ha** (15–20 kg/ha)
		
4	Dapog nursery 	3–4 kg/m^2 or 75–120 kg/ha
5	SRI system 	5–8.0 kg/ha (7.5 kg/ha)
6	Aerobic Rice System 	30–35 kg/ha
7	Transplanted basmati rice 	35–40 kg/ha
8	Drum seeder 	30–40 kg/ha

- **Spacing:** 20 x 20 cm or 20 x 10 cm
- Seedlings/hill: 2-3 seedlings /hill (Hybrids: 1-2 seedlings/hill)
- **Depth of transplanting:** 2-4 cm.

Nursery Techniques

- Nursery area of 1000 m² (0.10 ha or 10% of the main field) is required.
- In wet bed nursery, seed rate is 40-50 kg/ha, and seedlings are transplanted at 20-25 DAS.
- The transplanting of rice seedlings should be done at 3-4 (4) leaf stage.

Dapog nursery or Mat method



- The Dapog nursery or Mat method of raising rice seedlings was developed in the Philippines in 1966.

Seed Bed Material	• It can be a soil-less nursery (material like banana leaves, plastic sheets, or cemented floor) or a mixture of soil, well-decomposed organic matter, and rice hull.
Transplanting Time	• 11-14 days
Required Area	• 25 – 30 m^2 / ha or $1m^2$ = $200m^2$
Seed Quantity	• 3 – 4kg / (m^2) use pre-germinated seeds).

System of Rice Intensification (SRI)

- Developed by **Henri de Laulanie** in Madagascar (Africa) in 1983.
- It is accepted that SRI practices began to be adopted in India between 2000 and 2003. 1st in tamilnadu
- Norman Uphoff is a key figure associated with the promotion & development of SRI
- The critical LCC (Leaf Colour Chart) value for nitrogen application: 3-4

Benefits

- 20–100% or more yield compared to conventional methods.
- Average yield with SRI is 6–7 tons/ha (compared to 2–3 tons/ha in conventional methods).
- Seed saving up to 90% and water saving up to 50%.

Water Management

- Water requirement- 1500-200 mm.
- **21 days after flowering**. Submergence of **5 cm** of water in the rice field during the transplanting, reproductive, and grain formation stages is adequate.

Advantage of Submergence

- Less weed problem
- Fixation of nitrogen by Blue green algae.
- Increased availability of nutrients such as P, Fe, Mn, Zn and silicon.

1. Increase in the pH of acidic soils and a decrease in the pH of sodic soils.
2. Reduction in redox potential (lower oxygen concentration).
3. Reduction of nitrates to **N_2O or N_2** (denitrification).
4. Reduction of sulphate to sulphide, causing sulphide toxicity (known as **Akiochi disease**, reported in highly reduced soils in Japan).

Critical Stages for Irrigation

- About 50 to 60% of applied water to rice is lost by deep percolation.

1. **Tillering stage:** Occurs 2–20 days after transplanting.
2. **Booting or late panicle initiation stage:** This is the most critical stage for irrigation.

Beushening in Rice

- Beushening is a traditional practice in direct-seeded lowland rice, performed 4–6 weeks after sowing. It involves cross-ploughing of the young rice crop using a light country plough in 5–10 cm of standing water. It is done once or twice to control weeds, optimize plant spacing, and improve soil aeration.

Nutrient/ Fertilizer Management

- For high-yielding varieties, apply 130 kg N, 60 kg P_2O_5, and 60 kg K_2O per hectare. For hybrids, the N dose is 180–200 kg/ha.
- As a thumb rule, 20 kg N is required to produce one ton of rice.

Soil Zones in Water-Logged Paddy Fields

- Oxidized Zone: A thin surface layer (10 mm thick) containing oxygen.
- Reduced Zone: Lies immediately below the oxidized layer, lacking oxygen.

Mechanisms of N loss from paddy fields

1. **Denitrification (60%)**
   - Facultative anaerobic bacteria utilize nitrate (NO_3) as a source of oxygen, reducing it to nitrogen (N) or nitrous oxide (N_2O); this process is known as denitrification. It is a major cause of nitrogen loss in rice fields.

- **Denitrification bacteria:** Pseudomonas, Bacillus, Micrococcus, Paracoccus, Achromobacter, and some Sulphur-oxidizing bacteria (Thiobacillus thiooxidans, Thiobacillus denitrificans).

2. **Leaching (8–14%) Nutrient loss**
   - Nitrate fertilizers are more prone to leaching due to their high mobility in soil. It can lead to groundwater pollution.
3. **Volatilization (1–4%) Gaseous loss**
   - Nitrogen loss through volatilization in the form of ammonia (NH_3) is common in alkaline soils and under high-temperature conditions.

Nitrogen Use Efficiency (NUE)

- **Nitrogen Use Efficiency in rice is 30–40%** (50–60% in other crops).
- Deep placement of ammoniacal nitrogen fertilizers (e.g., ammonium sulphate) in the **reduced zone** is more efficient than surface application.
- Use of nitrification inhibitors and slow-release N fertilizers improve NUE.

Micro-nutrients

1. **Khaira disease in rice**
   - Khaira disease in rice is caused by zinc deficiency.
2. **Browning of Rice (Iron Toxicity)**
   - **Iron toxicity** occurs in waterlogged, acid-sulphate soils, particularly in Ultisols and Oxisols.
3. **Akiochi Disease in Rice**
   - It is due to the accumulation of hydrogen sulphide (H_2S) gas by reduction of sulphate by sulphur-reducing bacteria (Desulphovibrio and Desulfotomaculum) in waterlogged conditions. It was first reported from Japan.
4. **Role of Silicon in Rice**
   - It enhances plant resistance by forming a thick cuticle-silica layer, which acts as a barrier against insect pests and diseases.

Modern tools of N management

1. **Leaf Colour Chart (LCC)**
   - A durable, pocket-sized tool made of high-quality plastic with 6 green strips of varying shade (1: pale green to 6: dark green).
2. **SPAD meter (chlorophyll meter)**
   - SPAD stands for Soil Plant Analysis Development portable device that measures leaf chlorophyll content non-destructively, indicating plant nitrogen status.
3. **Green Seeker**
   - A handheld sensor that measures crop vigour by detecting reflected light (NDVI).

Biological Nitrogen Fixation

- The blue-green algae (BGA) or cyanobacteria (species: Anabaena, Nostoc) fix nitrogen in flooded rice fields and improve soil fertility.
- BGA can reduce the nitrogen requirement of rice by 25–30%, and also increases crop yield by 25–30%.
- It fixes 20–30 kg N/ha. The seed rate of BGA is 10 kg/ha.
- Azolla (Azolla pinnata) is a biofertilizer but it is also used as an organic manure.
- Non-symbiotic (free-living) nitrogen-fixing bacteria like *Azotobacter* and *Azospirillum* also contribute to biological nitrogen fixation (BNF) in rice fields.

Biological Nitrogen Fixation

- The blue-green algae (BGA) or cyanobacteria (species: *Anabaena, Nostoc*) fix nitrogen in flooded rice fields and improve soil fertility.
- BGA can reduce the nitrogen requirement of rice by 25–30%, and also increases crop yield by 25–30%.
- It fixes 20–30 kg N/ha. The seed rate of BGA is 10 kg/ha.
- BGA is applied @ 10 kg/ha after 7 days of transplanting with standing water in the field. A water fern Azolla harbours Anabaena azollae in symbiosis, widely used as biofertilizer in rice fields. Azolla (Azolla pinnata) is a biofertilizer but it is also used as an organic manure. Non-symbiotic (free-living) nitrogen-fixing bacteria

Environmental Concerns

- Low land paddy fields emit methane (CH_4) gas due to reduction of CO_2 under anaerobic conditions in

water-logged soils, which is a significant greenhouse gas.

Harvesting

- Paddy crop should be harvested at 21–23% moisture content in the grain.
- The average harvest index of rice is 40%.
- The rice husk contains about 18% Silica. Rice husk has 20% ash, 90% of which is silica. The silica content in rice straw generally ranges from 5% to 10%.

Post-harvesting operations

Hulling

- The process of removing the husk or hull from the paddy is known as hulling.
- The average hulling percentage is about 66% or 2/3 of paddy.

Milling

- The removal of the bran layer and embryo (germ) from the brown rice is called milling. Milling produces white rice by removing the outer layers.

Polishing

- Further refining of milled rice to give it a glossy appearance is called polishing.

Parboiling:

- A pre-milling hydrothermal treatment in which paddy is soaked, steamed, and dried to improve its milling quality and nutritive value. Vitamins and micronutrients are generally present in the aleurone layer of the rice grain and move into the grain during parboiling. It conserves vitamin B_{12}.

Order of different operations

- Paddy (rough rice) → Brown rice (after hulling) → White rice (after milling) → Polished rice (after polishing).

Maize



- Botanical Name : *Zea mays*
- 2n = 20
- Family : Poaceae / Gramineae
- Origin from Central America (Mexico)
- USA Rank First area & production in Maize
- India 7th rank in production
- Annual teosinte type : *Zea Mexicana*
- 40% of the Maize production occurring in USA
- India counts only 2% of the world production

Area & distribution in India:

- Highest area under India : Karnataka > M.P > Rajasthan.
- Production : Karnataka > M.P
- Productivity : T.N
- Source : Directorate of Economic & Statistics, (During year 2017-18)

Classification:

On the basis of endosperm characters

1. Flint corn	Zea Mays indurate	(Most widely grown spp. in India)
2. Dent corn	Z. mays indentata	Most widly grown spp. In USA
3. Sweet corn	Z. mays saccharta	Producing kernals for canning industry
4. Pop corn	Z. mays everta	corn flakes, corn chip and popcorn
5. Pod corn	Z. mays tunicate	
6. Flour corn	Z. mays amylaca	Soft corn consist of starch
7. Waxy corn	Z. mays ceratina	produce starch

- Maize is a C_4 plant
- Cross pollinated plant
- Yellow colure of maize due to – **Cryptoxanthin**
- Maize protein is called – **Zein protein** (10%)
- And carbohydrate – 70%
- Maize possesses the **higher yield potential** among the cereals that is why known as **queen of cereals.**

Roots :

1. **Seminal roots:** Embryo produce seminal roots.to nourished seedling plants. Temporary root system. (RAEO-2017)
2. **Adventitious root/Crown root:** Emerge from above seed below the ground and Emerge from basal node is called adventitious root.
3. **Lateral Roots/Aerial root:** Emerge above the ground surface and below the stem.

Climate:

1. Optimum temperature for germination – 21°C
2. Growth and development – 32°C
3. Emergence stop – <13°C
4. Base temperature for maize – 10°C
5. Retard growth – >32°C

Soil :

1. Soil pH should be 6.5 – 7.5
2. Maize is particularly sensitive to **water logging**.
3. If can be practice all type of soil but best is **loam, & clay loam**.

Seed Rate:

1. Hybrid seed: 20 – 25 kg/hac. (CG ADA-2020)
2. Composite varieties: 18 – 20 kg/hac.
3. Fodder purpose: 35 – 45 kg/hac. (IBPS AFO-2016)
4. Sweet corn : 8 kg/hac.
5. Baby corn : 25 kg/hac.
6. Pop corn : 12 kg/hac.

Spacing:

1. General spacing : 60 × 20 cm
2. Sweet corn : 75 × 20 cm
   75 × 30 cm
3. Baby corn : 60 × 15 cm
   60 × 20 cm
4. Fodder : 30 × 10 cm

Sowing method:

1. Ridge & farrow method followed
2. Conventional method: plat bed method.
3. Maize is sowing in east-west direction it helps to augment the grain productivity due to greater availability of light.

Nutrient management:

1. NPK: 120 : 60 : 40 kg/hac.
2. Hybrid: 120 : 60 : 40 kg/hac.
3. Long duration: 120 : 60 : 40 kg/hac.
4. Irrigated condition 120 : 60 : 40 kg/hac.
5. Rainfed condition: 80 : 40 : 20 kg/hac.

- 1/3 Nitrogen Basel dose other knee high stage, Tasseling & sillking.
- Tassel = Male part
- Sillking = Female part
- ZnSo4 25kg/hac. Basel dose.
- Maize crop deficiency ZnSO4 caused **white bud.** (BHU-2020)
- **Phosphorus deficiency** in maize characterized by stunted growth, leaf tip, margins are purplish and abnormally dark green leaf colour

Water Management: (Critical Stages)

1. Knee high stage
2. Flowering (between **tesseling** and **silking satge**) stage.

Water requirement for maize is about 500 – 800mm.

- It we have four irrigation:

1. 6 leaf stage
2. Late knee high stage
3. 50% silking stage
4. Dough stage.

Weed Management :

- Most critical stage is : 4 – 6 weeks
- Pre. Emergence herbicide: Atrazin & Simazin 1 – 1.5 kg a.i/hac.(BHU-2014)
- Post Emergence herbicide: paraquat (protected spray for weed control)

Yield:

- 18-20 Quintals/ha (MP RAEO-2021)

Vareiteis:

- First time in India (1961) double cross hybrids **Gnaga safed-1, Ganga safed-2**, **Daccan, Ranjeet** were released under all India co-operdinated research project on maize.
- **African tall** is a variety used for fodder purpose.

QPM (Quality protein maize):

- Quality protein maize in enough to tryptophan & lysine protein
- QPM deficient in leucine & isoleucine

Varieties:

Sakti, Rattan, Profiena, Shaktiman-1 & Shaktiman-2 (By using Opaque-2 gene)

Baby Corn :

- Baby corn is a unfertilized cob.
- Length of cob is 8 – 10 cm.